Calculus

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Contents

Chapter 1 Calculus (微積分)

1 Function (函數)

I Variable (變數) or indeterminate

A variable or an indeterminate is a symbol, typically a letter, that refers to an unspecified mathematical object.

II Function, map, or mapping (映射)

A function, a map, or a mapping is formed by three sets, the domain (定義域) X, the codomain (對應域) Y, and the graph R that satisfy the three following conditions:

$$R \subseteq \{(x, y) \mid x \in X, y \in Y\},$$

$$\forall x \in X, \exists y \in Y, (x, y) \in R,$$

$$(x, y) \in R \wedge (x, z) \in R \implies y = z.$$

We say a function is from, over, or on its domain (or sometimes a superset of its domain) and into, to, or in its codomain. We also say a function is between its domain (or sometimes a superset of its domain) and its codomain.

A function f that satisfies the above is denoted as:

$$f : X \to Y.$$

in which the domain X is also denoted as D_f.

The range (值域) or image (像), denoted as R_f or $f(X)$, is defined as:

$$\{y \mid \exists x \in X, (x, y) \in R\}.$$

If $x \in X$ and $(x, y) \in R$, we write $y = f(x)$, in which $f(x)$ is called the image of x under f, x is called the independent variable (自變數 or 獨立變數) or input, and y is called the dependent variable (應變數 or 依賴變數) or output; and the function f is also denoted as:

$$f : X \to Y; x \mapsto y,$$

and f and y is used interchangeably.

If I is a subset of the domain of f, $f(I)$ is defined as

$$f(I) := \{y \mid \exists x \in I \text{ s.t. } f(x) = y\}.$$

A function whose domain is a subset of $\mathbb{R}$ is called a real-domain function. A function whose range is a subset of $\mathbb{R}$ is called a real-valued function. A function whose domain and range are both subsets of $\mathbb{R}$ is called a real function.

III Types of Functions

Consider a function f:

$$f: X \to Y;\ x \mapsto y.$$

i Injection (單射) or Injective function or One-to-one (一對一) function

$$\forall a, b \in X \text{ s.t. } f(a) = f(b):\ a = b$$

ii Many-to-one (多對一) function

$$\exists a \neq b \in X:\ f(a) = f(b)$$

iii Surjection (滿射 or 蓋射) or Surjective function or Onto function

$$f(X) = Y$$

iv Bijection (對射) or Bijective function or One-to-one (一對一) function or One-to-one correspondence (一一對應)

Injective and surjective function.

IV Partial Function (偏函數)

A partial function from X to Y is a function from a subset of X to Y. The notation of a partial function is the same as a function except the fact that it is a partial function should be specified.

V Arity (元數), valency, or valence

The number of arguments taken by a function. A function with arity of zero is called nullary, a function with arity of one is called unary, and a function with arity of two is called binary.

A function of n real variables $x_1, x_2, \dots, x_n$ is used interchangeablely with a function of a vector variable $\mathbf{x} = (x_1, x_2, \dots, x_n)$ in $\mathbb{R}^n$.

VI Restriction and extension (延拓)

We say that a function $f: X \to Y$ is an extension of another function $g: U \subseteq X \to Y$ to X and that g is a restriction of f to U iff $\forall u \in U:\ f(u) = g(u)$. If Y has zero element 0, the trivial extension of g to X is $h: X \to Y$ defined as

$$h(x) = \begin{cases} g(x), & x \in U \\ 0, & x \in X \setminus U \end{cases}.$$

Let $f: X \to Y$ be a function and I be a set. $f|_I$ or $f \restriction I$ is defined as a function $f|_I: I \cap X \to Y$ such that for every $x \in I \cap X$, $f|_I(x) = f(x)$.

VII Property in a variable

Unless otherwise specified, given a property P defined for a function $f(w_1, \dots w_n, x, y_1, \dots y_n)$ in which $w_1, \dots w_n, x, y_1, \dots y_n$ are independent variables, we say f is with property P in x if the function $g(x)$ that is defined to map x to $f(w_1, \dots w_n, x, y_1, \dots y_n)$ for any x such that $w_1, \dots w_n, x, y_1, \dots y_n$ is in the domain of f for any fixed $w_1, \dots w_n, y_1, \dots y_n$ is with property P.

Unless otherwise specified, given a property P defined for a function $f(w_1, \dots w_n, x, y_1, \dots y_n)$ in which $w_1, \dots w_n, x, y_1, \dots y_n$ are independent variables, we say f is with property P in x for a fixed $w_1, \dots w_n, y_1, \dots y_n$ if the function $g(x)$ that is defined to map x to $f(w_1, \dots w_n, x, y_1, \dots y_n)$ for any x such that $w_1, \dots w_n, x, y_1, \dots y_n$ is in the domain of f is with property P.

Unless otherwise specified, given a property P defined for a function $f(w_1, \dots w_n, x, y_1, \dots y_n)$ in which $w_1, \dots w_n, x, y_1, \dots y_n$ are independent variables, we say f is with property P in x at a point $w_1, \dots w_n, x_0, y_1, \dots y_n$ in the domain of f if there exists a neighbourhood X of x_0 such that the function $g(x)$ that is defined to map x to $f(w_1, \dots w_n, x, y_1, \dots y_n)$ for any $x \in X$ such that $w_1, \dots w_n, x, y_1, \dots y_n$ is in the domain of f is with property P.

VIII (Indexed) family

An indexed family is a function f from a set I called index set to a set. $f(i)$ for any $i \in I$ is denoted as x_i. The image of f is denoted as $(x_i)_{i \in I}$ or $\{x_i\}_{i \in I}$, or (x_i) or $\{x_i\}$ if I is assumed to be known, often $\mathbb{N}_0$ or $\mathbb{N}$.

IX Indicator function (指示函數或示性函數) or characteristic function (特徵函數)

An indicator function or a characteristic function of a set A is a function that maps elements of A to 1, and all others to 0, denoted as 1_A, $\mathbf{1}_A$, or $\mathbb{I}_A$.

X Sequences (數列)

i Sequence

A sequence in X is a function of which the domain is a set $\{x \in \mathbb{Z} \mid l \leq x \leq m\}$ or $\{x \in \mathbb{Z} \mid l \leq x\}$ and the codomain is a topological space X, in which l is an integer, usually 0 or 1, and m is a integer, denoted as $\langle a_n \rangle$, $\{a_n\}$, (a_n), $\langle a_n \rangle_{n=l}^m$, $\{a_n\}_{n=l}^m$, or $(a_n)_{n=l}^m$, with $m = \infty$ when its domain is $\{x \in \mathbb{Z} \mid l \leq x\}$, where the subscript n refers to the nth element of the sequence, that is, the function value when the independent variable is n.

ii Finite Sequence (有限數列)

A finite sequence is a sequence with finite terms, i.e. $\langle a_n \rangle_{n=l}^{m}, \quad m \in \mathbb{Z}$.

iii Infinite sequence (無窮數列)

An infinite sequence is a sequence with infinite terms, i.e. $\langle a_n \rangle_{n=l}^{\infty}$. Unless otherwise specified, sequences refer to infinite sequences.

iv Sequence of partial sums

The sequence of partial sums of a sequence $\langle a_n \rangle_{n=l}^{\infty}$ is the sequence $\left\langle s_n = \sum_{i=l-1}^{n} a_i \right\rangle_{n=l-1}^{\infty}$ where $s_{l-1} = 0$ or $\left\langle s_n = \sum_{i=l}^{n} a_i \right\rangle_{n=l}^{\infty}$.

XI Series (級數)

i Series

The sequence of partial sums of a sequence.

ii Finite Series (有限級數)

The sequence of partial sums of a finite sequence.

iii Infinite Series (無窮級數)

The sequence of partial sums of an infinite sequence.

iv Cauchy product

$$\left(\sum_{i=0}^{\infty} a_i\right)\left(\sum_{j=0}^{\infty} b_j\right) = \sum_{k=0}^{\infty}\sum_{l=0}^{k} a_l b_{k-l}.$$

v Power series

A power series (in x) is a series of the form

$$\sum_{n=0}^{\infty} c_n x^n,$$

where c_n are constants called coefficients and x is variable.

$$\sum_{n=0}^{\infty} c_n (x-a)^n$$

is called a power series in $(x-a)$, a power series centered at a, or a power series about a.

XII Evaluation

Let f be a function. We denote $f(b) - f(a)$ as $f(x)\Big|_a^b$, $\Big[f(x)\Big|_a^b$, $\Big(f(x)\Big|_a^b$, $\Big\{f(x)\Big\}\Big|_a^b$, $\Big[f(x)\Big]\Big|_a^b$, $\Big(f(x)\Big)\Big|_a^b$, $f(x)\Big]_a^b$, $\Big[f(x)\Big]_a^b$, or $\Big(f(x)\Big]_a^b$.

XIII Tests of real functions

i Vertical line test (垂線測試)

For a graph on xy plane, if any vertical line intersects it, y is not a function of x.

ii Horizontal line test (水平線測試)

Given a graph of a real function $y = f(x)$ on xy plane, if any horizontal line intersects it more than once, $f(x)$ is not bijective.

XIV Fixed point, fixpoint, or invariant point (不動點 or 定點)

A fixed point, fixpoint, or invariant point of a function $f:\ X \subseteq Y \to Y$, is a point $x \in X$ such that $f(x) = x$.

i Increasing (遞增) Decreasing (遞減), and Monotone (單調)

Let X and Y be a strictly totally ordered sets and $f:\ X \to Y$ be a function.

- f is called strictly increasing on an interval $I \subseteq X$ if

$$\forall a, b \in I:\ a < b \implies f(a) < f(b).$$

- f is called strictly increasing if

$$\forall a, b \in X:\ a < b \implies f(a) < f(b).$$

- f is called strictly decreasing on an interval $I \subseteq X$ if

$$\forall a, b \in I:\ a < b \implies f(a) > f(b).$$

- f is called strictly decreasing if

$$\forall a, b \in X:\ a < b \implies f(a) > f(b).$$

Let X and Y be a totally ordered sets and $f:\ X \to Y$ be a function.

- f is called non-decreasing, monotone increasing, or monotonically increasing on an interval $I \subseteq X$ if

$$\forall a, b \in I:\ a < b \implies f(a) \leq f(b).$$

- f is called non-decreasing, monotone increasing, or monotonically increasing if

$$\forall a, b \in X: \ a < b \implies f(a) \le f(b).$$

- f is called non-increasing, monotone decreasing, or monotonically decreasing on an interval $I \subseteq X$ if

$$\forall a, b \in I: \ a < b \implies f(a) \ge f(b).$$

- f is called non-increasing, monotone decreasing, or monotonically decreasing if

$$\forall a, b \in X: \ a < b \implies f(a) \ge f(b).$$

- f is called monotone or monotonic on an interval $I \subseteq X$ if it is either monotone increasing or monotone decreasing on I.
- f is called monotone or monotonic if it is either monotone increasing or monotone decreasing.

The terms increasing and decreasing are define to be either monotone or strictly definition in different sources. Here, we use the monotone definition.

XV Homogeneous function

i General homogeneity

A function F with domain $\Omega \subseteq \mathbb{R}^n$ is called a homogeneous function of degree $\alpha \in \mathbb{R}$ if and only if for any nonzero real number t and $\mathbf{x}, t\mathbf{x} \in \Omega$, $F(t\mathbf{x}) = t^\alpha F(\mathbf{x})$.

ii Positive homogeneity

A function F with domain $\Omega \subseteq \mathbb{R}^n$ is called a positively homogeneous function of degree $\alpha \in \mathbb{R}$ if and only if for any positive real number t and $\mathbf{x}, t\mathbf{x} \in \Omega$, $F(t\mathbf{x}) = t^\alpha F(\mathbf{x})$.

XVI Symmetry

Given a function $f: \ X \to Y$ such that $\forall x \in X: \ -x \in X$.

i Even Function (偶函數)

f is an even function if for all $x \in X$, $f(x) = f(-x)$.

ii Odd Function (奇函數)

f is an odd function if for all $x \in X$, $-f(x) = f(-x)$.

XVII Transformation (變換)

i Translation, transition, or shift (平移)

For any function $f: \mathbb{R} \to \mathbb{R}$, shifting $y = f(x)$ right by h units and up by k units on the xy coordinate plane yields $y = f(x-h)+k$.

$f(x-a)$ is also denoted as $(\tau_a f)(x)$, where τ_a is translation operator.

ii Scaling (伸縮 or 縮放 or 拉伸)

For any function $f: \mathbb{R} \to \mathbb{R}$, on the xy coordinate plane, expand $y = f(x)$ vertically by a times the original value with the x axis as the reference line, and expand $y = af\left(\frac{x}{b}\right)$ horizontally by b times the original value with the y axis as the reference line, to obtain $y = af\left(\frac{x}{b}\right)$.

$f(-x)$, called reflection of f (through $x=0$), is also denoted as $\tilde{f}$ or $\hat{R}f$, where $\tilde{}$ or $\hat{R}$ is reflection operator.

XVIII Function composition (函數合成)

For two functions $f: X \to Y$ and $g: V \to W$ such that $g(V) \subseteq X$, the composition of them, denoted as $(f \circ g)$, is defined as:

$$(f \circ g): V \to Y;\ x \mapsto= f(g(x))$$

XIX Inverse function (反函數)

For a bijective function $f: X \to Y$, the inverse of it, denoted as f^{-1}, is defined as:

$$f^{-1}: Y \to X;\ f(x) \mapsto x.$$

XX Inverse of Composition

Suppose $f_1, \ldots, f_n$ are bijective,

$$\left(\operatorname*{bigcirc}_{i=1}^{n} f_i\right)^{-1} = \operatorname*{bigcirc}_{i=1}^{n} {f_{n-i+1}}^{-1}$$

XXI Preimage (像原) or Inverse image

Given an open subset B of the codomain of a function f the preimage or inverse image of f on B, denoted as $f^{-1}(B)$ is defined as

$$f^{-1}(B) = \{x \in X | f(x) \in B\}.$$

XXII Translation invariance

For a map M defined for some functions, translation invariance is defined as: for all a such that $M(f(x))$ and $f(x+a)$ are both defined,

$$M(f(x+a)) = M(f(x)).$$

XXIII Piecewise function (分段函數)

A piecewise function is a function defined of the form:

$$f(x) = \begin{cases} f_1(x), & x \in A_1, \\ f_2(x), & x \in A_2, \\ \dots & \\ f_n(x), & x \in A_n \end{cases},$$

where

$$\bigcup_{i=1}^{n} A_i = D_f \wedge \forall i \neq j \wedge i, j \in \mathbb{N} \wedge i, j \leq n : \ A_i \cap A_j = \varnothing.$$

XXIV Elementary functions

A constants, a power function, a logarithmic function, an exponential function, a trigonometric function, an inverse trigonometric function, or any function that can be built up with finite defined operations of addition, subtraction, multiplication, division, and composition of functions on any finite set of elementary functions is called an elementary function.

An integral of which the integrand is an elementary function is called an elementary integral.

XXV Some Types of Real Functions

A function in $\mathbb{R}^n$ is of a type below if all its components are of that type.

i Power Function (冪函數)

A power function if a function $f(x) = kx^a$ where $k \in \mathbb{C}$, called coefficient, and $a \in \mathbb{R}$, called exponent, are constants (some requires $k \in \mathbb{R}$ or $k = 1$).

ii Root Function (根式函數)

A root function is a power function of which the exponent is $\frac{1}{n}$ where $n \in \mathbb{N}$.

iii Reciprocal Function

The reciprocal function is the function $f(x) = \frac{1}{x}$.

iv Polynomial (多項式)

A polynomial is a function $P(x) = \sum_{k=0}^{n} a_k x^k$ where $a_k \in \mathbb{R}$ or $\mathbb{C}$ are constants called coefficients and n is a nonnegative integer called the degree of P.

A polynomial of degree 1 is called a linear function. A polynomial of degree 2 is called a quadratic function.

v Rational Function (有理函數)

A function f is called a rational function if there exists two polynomials $P(x)$ and $Q(x)$ such that $f(x) = \dfrac{P(x)}{Q(x)}$ for all x in the domain.

If deg$(P) \geq$ deg(Q), $f(x)$ is improper; otherwise $f(x)$ is proper.

vi Algebraic Function (代數函數)

A function f is called an algebraic function if there exists a polynomial $P(y, x)$ in two variables such that $P(f(x), x) = 0$ for all x in the domain.

vii Transcendental Function (超越函數)

A real function is called a transcendental function if it is not an algebraic function.

XXVI Periodic function (週期函數)

i Periodic function (週期函數)

Let f be a function with $D_f \subseteq I$. If there exists an interval I and positive real number T such that

- Either I is unbound or $|I| = nT$ for some $n \in \mathbb{N}$,
-
  $$t \in D_f \wedge t + T \in I \implies t + T \in D_f,$$
  $$t \in D_f \wedge t - T \in I \implies t - T \in D_f,$$
  and
- For any $t \in D_f$ such that $t + T \in D_f$
  $$f(t + T) = f(t),$$

then, we say that f is periodic and that any such T is a period (週期) of f, and in contexts where its period is referred to as a single value, it's the smallest positive T.

ii Translation-invariant map property

If f is a periodic function with period T and M is a map defined for f that satifies translation invariance, then for any interval $I \subseteq D_f$ that is either unbounded or of length nT for some $n \in \mathbb{N}$ such that $M(f_I)$ is defined, $M(f) = M(f_I)$.

iii Phase (相位) and phase angle (相位角)

The phase $\varphi(t)$ and phase angle $\varphi(t)$ of a periodic function $f(t)$ with period T given a reference start of a period t_0 are defined as

$$\varphi(t) = \frac{t - t_0}{T} - \left\lfloor \frac{t - t_0}{T} \right\rfloor,$$

$$\varphi(t) = 2\pi\varphi(t),$$

which are both functions of period T and are zero at $t_0 + nT$ for all $n \in \mathbb{N}$.

Some sources define phase as φ and/or phase angle as φ, which is not used here.

For two periodic functions $f(t)$ and $g(t)$ such that $g(t) = Af(t+\tau)$ for some real number A, τ, the phase shift (相位移) $\Delta\varphi$ and phase angle shift (相位角移) $\Delta\varphi$ of $g(t)$ relative to $f(t)$ are defined as

$$\Delta\varphi = \frac{\tau}{T} - \left\lfloor \frac{\tau}{T} \right\rfloor$$

$$\Delta\varphi = 2\pi(\Delta\varphi).$$

$g(t)$ is called to lag $f(t)$ by phase τ or phase angle $2\pi(\Delta\varphi)$. $f(t)$ is called to lead $g(t)$ by phase τ or phase angle $2\pi(\Delta\varphi)$. The phase difference (相位差) and phase angle difference (相位角差) between them are defined as the magnitude of $\Delta\varphi$ and $\Delta\varphi$ respectively.

If the phase difference of $f(t)$ and $g(t)$ is 0, they are called in phase (同相); otherwise they are called out of phase (異相). If the phase difference of $f(t)$ and $g(t)$ is 0.5 and $\alpha f(t) + g(t) = 0$, they are called completely out of phase (反相).

2 Limit (極限)

I Convention

i Extended real number system

When using the extended real number system as the codomain of limits:

$$\forall a \in \mathbb{R} \wedge a > 0 : \ a/0 = \infty,$$

$$\forall a \in \mathbb{R} \wedge a < 0 : \ a/0 = -\infty,$$

when the numerator 0 is a limit and the result is to be put in a limit or assigned as a result of a limit by limit definition.

When using the extended real number system as the codomain of limits, any function f with limit at $a \in \{-\infty, \infty\}$ being $\pm\infty$ is extended by $f(a) = \pm\infty$ when the input a is a limit and the result is to be put in a limit or assigned as a result of a limit by limit definition.

ii Notation and reading

$\lim\limits_{x\to a} f(x) = L$ is also denoted as $f(x) \to L$ as $x \to a$, and read $f(x)$ approaches L (for $L = \infty$ also $f(x)$ gets arbitrarily large and for $L = -\infty$ also $f(x)$ gets arbitrarily small) as x approaches (or gets arbitrarily close to) a (for $a = \infty$ also x gets arbitrarily large and for $a = -\infty$ also x gets arbitrarily small).

II Of Real Sequence

i Limit

For a real sequence $\langle a_n \rangle_{n=l}^{\infty}$, the limit of it (as n approaches infinity) is denoted as $\lim_{n\to\infty} a_n$ or $\lim_n a_n$. For $L \in \mathbb{R}$:

$$\lim_{n\to\infty} a_n = L \equiv \forall \varepsilon > 0 : \exists M \in \mathbb{N} \text{ s.t. } n \in \mathbb{N} \land n \geq M \implies |a_n - L| < \varepsilon.$$

In other words, as n becomes arbitrarily large, a_n gets arbitrarily close to L.

If such M exists for some $L \in \mathbb{R}$, we say the limit exists and the sequence converges (收斂) to L; otherwise, we say the limit does not exist (DNE) and the sequence diverges (發散).

$$\lim_{n\to\infty} a_n = \infty \equiv \forall M > 0, \exists M \in \mathbb{N} \text{ s.t. } n \in \mathbb{N} \land n \geq M \implies a_n > M.$$

$$\lim_{n\to\infty} a_n = -\infty \equiv \forall M < 0, \exists M \in \mathbb{N} \text{ s.t. } n \in \mathbb{N} \land n \geq M \implies a_n < M.$$

ii Limit inferior, infimum limit, limit infimum, liminf, inferior limit, lower limit, or inner limit, and limit superior, supremum limit, limit supremum, limsup, superior limit, upper limit, or outer limit

For a real sequence $\langle a_n \rangle_{n=l}^{\infty}$, the limit inferior is defined as

$$\liminf_{a\to\infty} a_n \coloneqq \sup_{n\geq l} \inf_{m\geq n} a_m,$$

and the limit superior is defined as

$$\limsup_{a\to\infty} a_n \coloneqq \inf_{n\geq l} \sup_{m\geq n} a_m.$$

III Of Real Series

i Limit

For a real series

$$S_n = \sum_{i=l}^{n} a_i,$$

where a_i are terms of a real sequence $\langle a_n \rangle_{n=l}^{\infty}$. The limit of S_n is denoted as $\lim_{n\to\infty} S_n$ or $\sum_{i=1}^{\infty} a_i$ and defined as the limit of the sequence $\langle S_n \rangle_{n=l}^{\infty}$. If the limit exists and equal to L, we say the series converges to L; otherwise, we say the series diverges.

ii Absolute convergence and conditional convergence

We say a real series $S_n = \sum_{i=1}^{\infty} a_i$ converges absolutely to L iff the series $\sum_{i=1}^{\infty} \left|a_i\right|$ converges to L. If a real series is convergent but not convergent absolutely, we say it is convergent conditionally.

iii Radius and interval of convergence for power series

For a power series $S_n(x) = \sum_{n=0}^{\infty} c_n(x-a)^n$, the interval of x such that $S_n(x)$ is convergent is called interval of convergence, and if $S_n(x)$ is convergent for $|x-a| < R$ and divergent for $|x-a| > R$, for some $R \in \bar{\mathbb{R}}_{>0}$, then R is called radius of convergence.

IV Of real-valued net

Below, we will discuss limits of nets in the set of real number.

i Limit

For a real-valued net $\langle x_a \rangle_{a \in A}$ in which A is a directed set, the limit of $\langle x_a \rangle_{a \in A}$ is denoted as $\lim_a x_a$ or $\lim_{a \to \infty} x_a$. For $L \in \mathbb{R}$:

$$\lim_a x_a = L \equiv \forall \varepsilon > 0 :\ \exists a_0 \in A \text{ s.t. } a \in A \wedge a \geq a_0 \implies |x_a - L| < \varepsilon.$$

If such a_0 exists for some $L \in \mathbb{R}$, we say the limit exists or the net converges to L; otherwise, we say the limit does not exist (DNE).

$$\lim_a x_a = \infty \equiv \forall M > 0, \exists a_0 \in A \text{ s.t. } a \in A \wedge a \geq a_0 \implies x_a > M.$$

$$\lim_a x_a = -\infty \equiv \forall M < 0, \exists a_0 \in A \text{ s.t. } a \in A \wedge a \geq a_0 \implies x_a < M.$$

ii Limit inferior, infimum limit, limit infimum, liminf, inferior limit, lower limit, or inner limit, and limit superior, supremum limit, limit supremum, limsup, superior limit, upper limit, or outer limit

For a real-valued net $\langle x_a \rangle_{a \in A}$, the limit inferior is defined as

$$\liminf_x x_n := \sup_{n \geq l} \inf_{m \geq n} x_m,$$

and the limit superior is defined as

$$\limsup_x x_n := \inf_{n \geq l} \sup_{m \geq n} x_m.$$

V Of Real-Domain Function

i Limit at a point

Let I be an interval containing the point a. Let $f(x)$ be a function defined on I, except possibly at a itself. The limit of $f(x)$ as x approaches a is denoted as $\lim_{x \to a} f(x)$. For $L \in \mathbb{R}$:

$$\lim_{x \to a} f(x) = L \equiv \forall \varepsilon > 0 :\ \exists \delta > 0 \text{ s.t. } 0 < |x-a| < \delta \implies |f(x) - L| < \varepsilon.$$

In other words, as x becomes arbitrarily close to a, $f(x)$ gets arbitrarily close to L.

If such δ exist for some $L \in \mathbb{R}$, we say the limit exists; otherwise, we say the limit does not exist (DNE).

$$\lim_{x \to a} f(x) = \infty \equiv \forall N > 0 : \exists \delta > 0 \text{ s.t. } 0 < |x - a| < \delta \implies f(x) > N.$$

$$\lim_{x \to a} f(x) = -\infty \equiv \forall N < 0 : \exists \delta > 0 \text{ s.t. } 0 < |x - a| < \delta \implies f(x) < N.$$

ii Limit at Infinity

Let I be a left-bounded, right-unbounded interval with the point a being its endpoint on the left. Let $f(x)$ be a function defined on I. The limit of $f(x)$ as x approaches ∞ is denoted as $\lim_{x \to \infty} f(x)$. For $L \in \mathbb{R}$:

$$\lim_{x \to \infty} f(x) = L \equiv \forall \varepsilon > 0 : \exists M > a \text{ s.t. } x > M \implies |f(x) - L| < \varepsilon.$$

In other words, as x becomes arbitrarily large, $f(x)$ gets arbitrarily close to L.

If such M exists for some $L \in \mathbb{R}$, we say the limit exists and the function converges to L as x becomes arbitrarily large; otherwise, we say the limit does not exist (DNE) and the function diverges as x becomes arbitrarily large.

Let I be a right-bounded, left-unbounded interval with the point a being its endpoint on the right. Let $f(x)$ be a function defined on I. The limit of $f(x)$ as x approaches $-\infty$ is defined as follows:

$$\lim_{x \to -\infty} f(x) = L \equiv \forall \varepsilon > 0 : \exists M < a \text{ s.t. } x < M \implies |f(x) - L| < \varepsilon.$$

In other words, as x becomes arbitrarily small, $f(x)$ gets arbitrarily close to L.

If such M exists for some $L \in \mathbb{R}$, we say the limit exists and the function converges to L as x becomes arbitrarily small; otherwise, we say the limit does not exist (DNE) and the function diverges as x becomes arbitrarily small.

$$\lim_{x \to \infty} f(x) = \infty \equiv \forall N > 0 : \exists M > 0 \text{ s.t. } x > M \implies f(x) > N.$$

$$\lim_{x \to \infty} f(x) = -\infty \equiv \forall N > 0 : \exists M > 0 \text{ s.t. } x > M \implies f(x) < N.$$

$$\lim_{x \to -\infty} f(x) = \infty \equiv \forall N > 0 : \exists M < 0 \text{ s.t. } x < M \implies f(x) > N.$$

$$\lim_{x \to -\infty} f(x) = -\infty \equiv \forall N > 0 : \exists M < 0 \text{ s.t. } x < M \implies f(x) < N.$$

$\lim_{x \to \infty} f(x)$ is sometimes denoted as $f(\infty)$; $\lim_{x \to -\infty} f(x)$ is sometimes denoted as $f(-\infty)$.

iii One-sided Limits

Right-hand Limit (右極限): Let I be a left-open interval with the point a being its endpoint on the left. Let $f(x)$ be a function defined on I. The right-hand limit of $f(x)$ as x approaches a is denoted as $\lim_{x \to a^+} f(x)$. For $L \in \mathbb{R}$:

$$\lim_{x \to a^+} f(x) = L \equiv \forall \varepsilon > 0 : \exists \delta > 0 \text{ s.t. } 0 < x - a < \delta \implies |f(x) - L| < \varepsilon.$$

In other words, as x becomes arbitrarily close to a and is greater than a, $f(x)$ gets arbitrarily close to L.

$$\lim_{x\to a^+} f(x) = \infty \equiv \forall N > 0 :\ \exists\delta > 0 \text{ s.t. } 0 < x - a < \delta \implies f(x) > N.$$

$$\lim_{x\to a^+} f(x) = -\infty \equiv \forall N < 0 :\ \exists\delta > 0 \text{ s.t. } 0 < x - a < \delta \implies f(x) < N.$$

For a function $f(x)$, $\lim\limits_{x\to a^+} f(x)$ can also be denoted as $f(a^+)$.

$\lim\limits_{x\to-\infty^+}$ is the same as $\lim\limits_{x\to-\infty}$.

Left-hand Limit (左極限): Let I be a right-open interval with the point a being its endpoint on the right. Let $f(x)$ be a function defined on I. The left-hand limit of $f(x)$ as x approaches a is denoted as $\lim\limits_{x\to a^-} f(x)$. For $L \in \mathbb{R}$:

$$\lim_{x\to a^-} f(x) = L \equiv \forall\varepsilon > 0 :\ \exists\delta > 0 \text{ s.t. } 0 < a - x < \delta \implies |f(x) - L| < \varepsilon.$$

In other words, as x becomes arbitrarily close to a and is less than a, $f(x)$ gets arbitrarily close to L.

$$\lim_{x\to a^-} f(x) = \infty \equiv \forall N > 0 :\ \exists\delta > 0 \text{ s.t. } 0 < a - x < \delta \implies f(x) > N.$$

$$\lim_{x\to a^-} f(x) = -\infty \equiv \forall N < 0 :\ \exists\delta > 0 \text{ s.t. } 0 < a - x < \delta \implies f(x) < N.$$

For a function $f(x)$, $\lim\limits_{x\to a^-} f(x)$ can also be denoted as $f(a^-)$.

$\lim\limits_{x\to\infty^-}$ is the same as $\lim\limits_{x\to\infty}$.

iv Horizontal asymptote (水平漸近線)

We say $y = L$ is a horizontal asymptote of $y = f(x)$ iff $\lim\limits_{x\to\infty} f(x) = L$ or $\lim\limits_{x\to-\infty} f(x) = L$.

v Slant asymptote (斜漸近線)

We say $y = mx + b$ with $m \neq 0$ is a slant asymptote of $y = f(x)$ iff $\lim\limits_{x\to\infty} (f(x) - (mx + b)) = 0$ or $\lim\limits_{x\to-\infty} (f(x) - (mx + b)) = 0$.

vi Vertical asymptote (鉛直漸近線)

We say $x = a$ is a vertical asymptote of $y = f(x)$ iff $\left|\lim\limits_{x\to a^+} f(x)\right| = \infty$ or $\left|\lim\limits_{x\to a^-} f(x)\right| = \infty$.

VI Of Real-valued Functions from Topological Space

i Limit

Let (X, τ) be a topological space, $a \in E \subseteq X$, and f be a function definited on E except possibly at a with codomain $\mathbb{R}$. The limit of f as x approaches $a \in E$ is denoted as $\lim\limits_{x\to a} f(x)$. For $L \in \mathbb{R}$:

$$\lim_{x\to a} f(x) = L \equiv \forall\varepsilon > 0 :\ \exists U \in \tau \wedge a \in U \text{ s.t. } x \in E \cap U \setminus \{a\} \implies |f(x) - L| < \varepsilon.$$

If such U exist for some $L \in \mathbb{R}$, we say the limit exists; otherwise, we say the limit does not exist (DNE).

$$\lim_{x \to a} f(x) = \infty \equiv \forall N > 0 : \exists U \in \tau \land a \in U \text{ s.t. } x \in E \cap U \setminus \{a\} \implies f(x) > N.$$
$$\lim_{x \to a} f(x) = \infty \equiv \forall N < 0 : \exists U \in \tau \land a \in U \text{ s.t. } x \in E \cap U \setminus \{a\} \implies f(x) < N.$$

ii Limit inferior, infimum limit, limit infimum, liminf, inferior limit, lower limit, or inner limit, and limit superior, supremum limit, limit supremum, limsup, superior limit, upper limit, or outer limit

Let (X, τ) be a topological space, $a \in E \subseteq X$, and f be a function definited on E except possibly at a with codomain $\mathbb{R}$. The limit inferior of f as x approaches $a \in E$ is defined as

$$\liminf_{x \to a} f(x) \coloneqq \sup\{\inf\{f(x) \mid x \in E \cap U \setminus \{a\}\} \mid a \in U \in \tau\},$$

and the limit superior of f as x approaches $a \in E$ is defined as

$$\limsup_{x \to a} f(x) \coloneqq \inf\{\sup\{f(x) \mid x \in E \cap U \setminus \{a\}\} \mid a \in U \in \tau\}.$$

VII Of function between normed spaces

i Definition

Let V, W be two normed spaces. Let I be an open subset of V containing the point a. Let f be a function with codomain W defined on I, except possibly at a itself. The limit of $f(x)$ as x approaches a is denoted as $\lim_{x \to a} f(x)$ or $\lim_{\|x-a\|_V \to 0} f(x)$, and for $x = (x_1, x_2, \ldots, x_n)$ and $a = (a_1, a_2, \ldots, a_n)$ also $\lim_{(x_1, x_2, \ldots, x_n) \to (a_1, a_2, \ldots, a_n)} f(x_1, x_2, \ldots, x_n)$, $\lim_{\substack{x_1 \to a_1 \\ x_2 \to a_2 \\ \vdots \\ x_n \to a_n}} f(x_1, x_2, \ldots, x_n)$. For $L \in W$:

$$\lim_{x \to a} f(x) = L \equiv \forall \varepsilon \exists \delta > 0 \text{ s.t. } 0 < \|x - a\|_V < \delta \implies \|f(x) - L\|_W < \varepsilon.$$

If such M exist for some $L \in W$, we say the limit exists; otherwise, we say the limit does not exist (DNE).

VIII Of Nets of Fuctions

i (Pointwise) limit

Let X be a set and Y be a topological space. A net of functions $\langle f_a \rangle_{a \in A}$ all having the same domain X and codomain Y is said to converge pointwise (逐點收斂) to a function $f : X \to Y$ or have a pointwise limit function $f : X \to Y$, denoted as

$$\lim_a f_a = f \text{ pointwise}$$

or

$$\lim_a f_a = f,$$

iff

$$\forall x \in X : \lim_a f_a(x) = f(x).$$

ii Uniform limit

Let X be a set and (Y, d) be a metric space. A net of functions $\langle f_a \rangle_{a \in A}$ all having the same domain X and codomain Y is said to converge uniformly (一致收斂) to a function $f: X \to Y$ or have a uniform limit function $f: X \to Y$, denoted as

$$\lim_a f_a = f \text{ uniformly},$$

iff

$$\forall \varepsilon \in \mathbb{R}_{>0}: \exists N \in A: a \in A \wedge N \leq a \implies \sup_{x \in X} d(f_a(x), f(x)) < \varepsilon.$$

IX Of Nets of Fuctions on Discrete Points to Function on Connected Space

i (Pointwise) limit

Let X be a connected topological space and Y be a topological space. A net of functions $\langle f_a \rangle_{a \in A}$ having domains $X_a \subseteq X$ and codomain Y is said to converge pointwise to a function $f: X \to Y$ or have a pointwise limit function $f: X \to Y$, denoted as

$$\lim_a f_a = f \text{ pointwise}$$

or

$$\lim_a f_a = f,$$

iff

$$\forall x \in X \forall \langle x_a \rangle_{a \in A}: \left(\left(\forall a \in A: x_a \in X_a \right) \wedge \lim_a x_a = x \right) \implies \lim_a f_a(x_a) = f(x).$$

ii Uniform limit

Let X be a connected topological space and (Y, d) be a metric space. A net of functions $\langle f_a \rangle_{a \in A}$ having domains $X_a \subseteq X$ and codomain Y is said to converge uniformly to a function $f: X \to Y$ or have a uniform limit function $f: X \to Y$, denoted as

$$\lim_a f_a = f \text{ uniformly},$$

iff

$$\forall \varepsilon \in \mathbb{R}_{>0} \exists N \in A \forall b \in A \forall \langle x_a(x) \rangle_{a \in A}: \left(\forall a \in A: x_a(x) \in X_a \right) \wedge \left(\forall x \in X: \lim_a x_a(x) = x \right) \wedge N \leq b) \implies \sup_{x \in X} d$$

X Big O notation and little notation

i Big O notation at a point

For a point a in a topological space and real-valued functions f and g defined on some neighborhood of a except possibly at a, $f(x) = O(g(x))$ as $x \to a$ if there exists some neighborhood I of a and positive real number M such that

$$|f(x)| \leq M\, |g(x)|.$$

ii Little o notation at a point

For a point a in a normed space and real-valued functions f and g defined on some neighborhood of a except possibly at a, $f(x) = o(g(x))$ as $x \to a$ if

$$\lim_{x\to a} \frac{f(x)}{g(x)} = 0.$$

iii Big O notation at infinity

For real-valued functions f and g defined on some interval $[a, \infty)$, $f(x) = O(g(x))$ as $x \to \infty$ if there exists some real number T such that for all $x \geq T$,

$$|f(x)| \leq |g(x)|.$$

For real-valued functions f and g defined on some interval $(-\infty, b]$, $f(x) = O(g(x))$ as $x \to -\infty$ if there exists some real number T such that for all $x \leq T$,

$$|f(x)| \leq |g(x)|.$$

iv Little o notation at infinity

For real-valued functions f and g defined on some interval $[a, \infty)$, $f(x) = o(g(x))$ as $x \to \infty$ if

$$\lim_{x\to\infty} \frac{f(x)}{g(x)} = 0.$$

For real-valued functions f and g defined on some interval $(-\infty, b]$, $f(x) = o(g(x))$ as $x \to -\infty$ if

$$\lim_{x\to-\infty} \frac{f(x)}{g(x)} = 0.$$

3 Continuity (連續性)

I Continuity

i Definition of continuity of real functions

- For a point a in the domain of a function f, if and only if $\exists \lim\limits_{x\to a} f(x)$ and $\lim\limits_{x\to a} f(x) = f(a)$, we say $f(x)$ is continuous (連續的) at a.
- For a point a in the domain of a function f, if and only if $\exists \lim\limits_{x\to a^+} f(x)$ and $\lim\limits_{x\to a^+} f(x) = f(a)$, we say $f(x)$ is continuous from the right or right-continuous at a; if and only if $\exists \lim\limits_{x\to a^-} f(x)$ and $\lim\limits_{x\to a^-} f(x) = f(a)$, we say $f(x)$ is continuous from the left or left-continuous at a.
- For an open interval I that is a subset of the domain of a function f, if and only if $f(x)$ is continuous at all points in I, we say $f(x)$ is continuous on I.
- For an right-open interval $[a, b)$ that is a subset of the domain of a function f, if and only if $f(x)$ is continuous on (a, b) and continuous from the right at a, we say $f(x)$ is continuous on $[a, b)$.

- For an left-open interval $(a, b]$ that is a subset of the domain of a function f, if and only if $f(x)$ is continuous on (a, b) and continuous from the left at b, we say $f(x)$ is continuous on $(a, b]$.
- For an closed interval $[a, b]$ that is a subset of the domain of a function f, if and only if $f(x)$ is continuous on both $[a, b)$ and $(a, b]$, we say $f(x)$ is continuous on $[a, b]$.
- if and only if $f(x)$ is continuous at all points in its domain, we say $f(x)$ is continuous.
- f is called continuous at ∞ if $\lim\limits_{x\to\infty} f(x) \in \mathbb{R} \cup \{-\infty, \infty\}$.
- f is called continuous at $-\infty$ if $\lim\limits_{x\to-\infty} f(x) \in \mathbb{R} \cup \{-\infty, \infty\}$.

ii Definition of continuity of functions between topological spaces

- A function $f : I \subseteq X \to Y$ where X and Y are topological spaces is continuous at a point $x \in I$ if and only if for any neighborhood V of $f(x)$ in Y, there is a neighborhood U of x such that $f(U) \subseteq V$.
- A function $f : I \subseteq X \to Y$ where X and Y are topological spaces is continuous if and only if for any open subset V of Y, the preimage of f on V is an open subset of X.
- A function $f : I \subseteq X \to Y$ where X and Y are topological spaces is continuous on a subset J of I if and only if for any open subset V of Y, the joint set of the preimage of f on V and J is open in the subspace topology of X in J.

iii Discontinuity (不連續（點）)

A point is a discontinuity of a function f if it is in the domain of f but f is not continuous on it.

iv Type of discontinuity of real functions

For a function f with domain $U \subseteq \mathbb{R}$ and a point $a \in U$ that f is discontinuous at, the discontinuity a of f can be classified as below:

- **Removable (可去) discontinuity**: If $\exists \lim\limits_{x\to a} f(x) \land \lim\limits_{x\to a} f(x) \neq f(a)$, we call $f(a)$ a removable discontinuity. A discontinuity that is not a removable discontinuity is called a non-removable discontinuity.
- **Jump (跳躍) discontinuity**: If $\exists \lim\limits_{x\to a^-} f(x) \land \exists \lim\limits_{x\to a^+} f(x) \land \lim\limits_{x\to a^-} f(x) \neq \lim\limits_{x\to a^+} f(x)$, we call $f(a)$ a jump discontinuity.
- **Infinite (無窮) discontinuity**: If at least one of $\lim\limits_{x\to a^-} f(x)$ and $\lim\limits_{x\to a^+} f(x)$ does not exist, and that those in $\lim\limits_{x\to a^-} f(x)$ and $\lim\limits_{x\to a^+} f(x)$ that do not exist are either ∞ or $-\infty$, we call $f(a)$ an infinite discontinuity.
- **Type I discontinuity**: A discontinuity of f that is either a removable discontinuity or a jump discontinuity is called a type I discontinuity of f.

- **Essential (本質) discontinuity or type II discontinuity**: A discontinuity of f that is not a type I discontinuity is called an essential discontinuity or a type II discontinuity of f.

v Removable discontinuity of functions between topological spaces

Let $f : I \subseteq X \to Y$ be a function where X and Y are topological spaces. We say that $f(a)$ is a removable discontinuity of f iff f is discontinuous at a and there exists $L \in Y$ such that the function

$$g(x) = \begin{cases} L, & x = a \\ f(x), & x \in I \setminus \{a\} \end{cases}$$

is continuous at a. A discontinuity that is not a removable discontinuity is called a non-removable discontinuity.

vi Singularity or singular point (奇點)

A point is a singularity or a singular point of a function f over a topological space iff it is either in the relative complement of D_f in the closure of D_f or a discontinuity of f.

vii Isolated singularity

For a function f over a topological space and a singularity a of f, a is an isolated singularity iff there exists an open set $U \subseteq D_f \cup \{a\}$ such that there is no singularity of f in $U \setminus \{a\}$.

viii Removable singularity

An isolated singularity a of a function f over a topological space is a removable singularity iff there exists a function g defined on $D_f \cup \{a\}$ such that $f(x) = g(x) \forall x \in D_f \setminus \{a\}$ and a is not a singularity of g.

ix Examples of continuous real functions

The following types of functions are continuous at every number in their domains: algebraic functions, trigonometric functions, inverse trigonometric functions, exponential functions, logarithmic functions.

II Piecewise continuity

i Piecewise continuity of real functions

Let $f : B \subseteq \mathbb{R} \to \mathbb{R}$ be a function and D be subset of B. If there exists an indexed family $\{[a_i, b_i] \mid i \in I\}$ of closed intervals such that:

- for every closed interval V, the set $\{i \in I \mid V \cap [a_i, b_i] \neq \varnothing\}$ is finite (some sources require instead that $n(I)$ is finite, which is stronger and not used here),
- $D \subseteq \bigcup_{i \in I} [a_i, b_i] \subseteq B$,
- function $f\big|_{a_i}^{b_i} : (a_i, b_i) \to Y$ is continuous for every $i \in I$, and

- there exist finite $\lim_{x \to a_i^+} f(x)$ and $\lim_{x \to b_i^-} f(x)$ for every $i \in I$.

, we say f is piecewise continuous on D.

We say f is piecewise continuous iff it is piecewise continuous on B.

ii Piecewise continuity of functions between topological spaces

Let X and Y be topological spaces, $f : B \subseteq X \to Y$ be a function, and D be a subset of B. If there exists a locally finite (some sources require instead finite, which is stronger and not used here) indexed family $\{U_i \mid i \in I\}$ of closed subsets of X such that:

- $D \subseteq \bigcup_{i \in I} U_i \subseteq B$, and
- there exists a function $g_i : U_i \to Y$ that is continuous on U_i such that $\forall x \in \operatorname{int}(U_i) : f(x) = g_i(x)$ for every $i \in I$.

, we say f is piecewise continuous on B.

We say f is piecewise continuous iff it is piecewise continuous on B.

III Uniform continuity (一致連續)

i Uniform continuity of functions between metric spaces

Let (X, d_1) and (Y, d_2) be metric spaces, and $f : D \subseteq X \to Y$ be a function. f is called to be uniformly continuous if for every real number $\varepsilon > 0$ there exists a real number $\delta > 0$ such that $x, y \in D$ with $d_1(x, y) < \delta$ implies $d_2(f(x), f(y)) < \varepsilon$.

Let (X, d_1) and (Y, d_2) be metric spaces, and $f : O \subseteq X \to Y$ be a function. f is called to be uniformly continuous on $D \subseteq O$ if for every real number $\varepsilon > 0$ there exists a real number $\delta > 0$ such that $x, y \in D$ with $d_1(x, y) < \delta$ implies $d_2(f(x), f(y)) < \varepsilon$.

ii Uniform continuity of functions between topological vector spaces

Let X and Y be topological vector spaces, and $f : D \subseteq X \to Y$ be a function. f is called to be uniformly continuous if for every neighborhood V of the zero vector in Y, there exists a neighborhood U of the zero vector in X such that $x, y \in D$ with $x - y \in U$ implies $f(x) - f(y) \in V$.

Let X and Y be topological vector spaces, and $f : O \subseteq X \to Y$ be a function. f is called to be uniformly continuous on $D \subseteq O$ if for every neighborhood V of the zero vector in Y, there exists a neighborhood U of the zero vector in X such that $x, y \in D$ with $x - y \in U$ implies $f(x) - f(y) \in V$.

IV Absolute continuity (AC) (絕對連續) of functions from an interval to a metric space

Let (X, d) be a metric space and $I \subseteq \mathbb{R}$ be an interval. A function $f : I \to X$ is absolutely continuous on I if for every positive number ε, there exists a positive number δ such that for any finite sequence of disjoint subintervals $[x_k, y_k]$ of I,

$$\sum_k |y_k - x_k| < \delta$$

implies

$$\sum_k d\left(f(y_k), f(x_k)\right) < \varepsilon.$$

The collection of all absolutely continuous functions from I into X is denoted $AC(I; X)$, or, if X is known in the context, $AC(I)$.

Let (X, d) be a metric space and $I \subseteq \mathbb{R}$ be an interval. A function $f : I \to X$ is absolutely continuous on $J \subseteq I$ if for every positive number ε, there exists a positive number δ such that for any finite sequence of disjoint subintervals $[x_k, y_k]$ of J,

$$\sum_k |y_k - x_k| < \delta$$

implies

$$\sum_k d\left(f(y_k), f(x_k)\right) < \varepsilon.$$

V Lipschitz continuity (利普希茨連續)

i (Golbal/regular) Lipschitz (continuity) of functions between metric spaces

Let (X, d_1) and (Y, d_2) be metric spaces, and $f : D \subseteq X \to Y$ be a function. f is called to be Lipschitz continuous if there exists a real constant K such that $\forall x, y \in D$, $d_2(f(x), f(y)) \leq Kd_1(x, y)$.

Let (X, d_1) and (Y, d_2) be metric spaces, and $f : O \subseteq X \to Y$ be a function. f is called to be Lipschitz continuous on $D \subseteq O$ if there exists a real constant K such that $\forall x, y \in D$, $d_2(f(x), f(y)) \leq Kd_1(x, y)$.

If a function $f : D \subseteq X \to Y$ between metric spaces (X, d_1) and (Y, d_2) is such that there exists a real constant K such that $\forall x, y \in D$, $d_2(f(x), f(y)) \leq Kd_1(x, y)$, then, any such K is called a Lipschitz constant (利普希茨常數) of the function f, and f is called to be K-Lipschitz or Lipschitz continuous with constant K. The smallest such K is called (best) Lipschitz constant or dilation of f.

ii Local Lipschitz (continuity) of functions between metric spaces

Let (X, d_1) and (Y, d_2) be metric spaces, and $f : D \subseteq X \to Y$ be a function. f is called to be locally Lipschitz continuous at $u \in D$, if there exists a neighborhood U of u and a real constant K such that $\forall x, y \in U$, $d_2(f(x), f(y)) \leq Kd_1(x, y)$.

Let (X, d_1) and (Y, d_2) be metric spaces, and $f : D \subseteq X \to Y$ be a function. f is called to be locally Lipschitz continuous if for any $u \in D$, it is locally continuous at u.

Let (X, d_1) and (Y, d_2) be metric spaces, and $f : O \subseteq X \to Y$ be a function. f is called to be locally Lipschitz continuous on $D \subseteq O$ if for any $u \in D$, it is locally continuous at u.

iii Short map

A short map is a function between metric spaces that is 1-Lipschitz.

iv Contraction mapping, contraction map, contraction, contractive mapping, contractive map, or contractor (壓縮映射)

A contraction mapping is a function f from a metric space to itself such that there exists a real number $0 \leq K < 1$ such that f is K-Lipschitz, in which K is sometimes called a contraction constant and the smallest such K is sometimes called the best contraction constant or the contraction constant.

4 Derivative (導數)

I Notation

i Leibniz's notation (萊布尼茲符號) for derivatives

Suppose a dependent variable y represents a function f of an independent variable x, that is,

$$y = f(x).$$

Then, the derivative function of y or f (with respect to (w.r.t.) x) can be written as

$$\frac{\mathrm{d}y}{\mathrm{d}x}, \quad \frac{\mathrm{d}}{\mathrm{d}x}y, \quad \frac{\mathrm{d}\left(f(x)\right)}{\mathrm{d}x}, \quad \text{or } \frac{\mathrm{d}}{\mathrm{d}x}\left(f(x)\right),$$

in which $\frac{\mathrm{d}}{\mathrm{d}x}$ is called a differential operator (微分運算子) or a derivative operator (導數運算子); the nth derivative function of y or f (with respect to x) can be written as

$$\frac{\mathrm{d}^n y}{\mathrm{d}x^n}, \quad \frac{\mathrm{d}^n}{\mathrm{d}x^n}y, \quad \frac{\mathrm{d}^n\left(f(x)\right)}{\mathrm{d}x^n}, \quad \text{or } \frac{\mathrm{d}^n}{\mathrm{d}x^n}\left(f(x)\right),$$

in which $\frac{\mathrm{d}^n}{\mathrm{d}x^n}$ is called a differential operator or a derivative operator.

ii Leibniz's notation for partial derivatives

Suppose a dependent variable y represents a function f of an independent variable vector $\mathbf{x} = (x_1, x_2, \dots x_n)$, that is,

$$y = f(\mathbf{x}).$$

Then, the partial derivative function of y or f with respect to x_i can be written as

$$\frac{\partial y}{\partial x_i}, \quad \frac{\partial}{\partial x_i}y, \quad \frac{\partial\left(f(x)\right)}{\partial x_i}, \quad \text{or } \frac{\partial}{\partial x_i}\left(f(x)\right),$$

in which $\frac{\partial}{\partial x_i}$ is called a partial differential operator (偏微分運算子) or a partial derivative operator (偏導數運算子);

the nth partial derivative function of y or f with respect to x_i can be written as

$$\frac{\partial^n y}{\partial {x_i}^n}, \quad \frac{\partial^n}{\partial {x_i}^n} y, \quad \frac{\partial^n (f(x))}{\partial {x_i}^n} \text{or } \frac{\partial^n}{\partial {x_i}^n} (f(x)),$$

in which $\frac{\partial^n}{\partial {x_i}^n}$ is called a partial differential operator or a partial derivative operator;

the nth mixed partial derivative function of y or f m_{i_k} times with respect to x_{i_k}, and then $m_{i_{k-1}}$ times with respect to $x_{i_{k-1}}$, …, and then m_{i_1} times with respect to x_{i_1}, in which $\sum_{j=1}^{k} m_{i_j} = n$, can be written as

$$\frac{\partial^n y}{\partial^{m_{i_1}} {x_{i_1}}^{m_{i_1}} \partial^{m_{i_2}} {x_{i_2}}^{m_{i_2}} \ldots \partial^{m_{i_k}} {x_{i_k}}^{m_{i_k}}},$$

$$\frac{\partial^n}{\partial^{m_{i_1}} {x_{i_1}}^{m_{i_1}} \partial^{m_{i_2}} {x_{i_2}}^{m_{i_2}} \ldots \partial^{m_{i_k}} {x_{i_k}}^{m_{i_k}}} y,$$

$$\frac{\partial^n f(\mathbf{x})}{\partial^{m_{i_1}} {x_{i_1}}^{m_{i_1}} \partial^{m_{i_2}} {x_{i_2}}^{m_{i_2}} \ldots \partial^{m_{i_k}} {x_{i_k}}^{m_{i_k}}}, \quad \text{or}$$

$$\frac{\partial^n}{\partial^{m_{i_1}} {x_{i_1}}^{m_{i_1}} \partial^{m_{i_2}} {x_{i_2}}^{m_{i_2}} \ldots \partial^{m_{i_k}} {x_{i_k}}^{m_{i_k}}} f(\mathbf{x}),$$

in which $\frac{\partial^n}{\partial^{m_{i_1}} {x_{i_1}}^{m_{i_1}} \partial^{m_{i_2}} {x_{i_2}}^{m_{i_2}} \ldots \partial^{m_{i_k}} {x_{i_k}}^{m_{i_k}}}$ is called a partial differential operator or a partial derivative operator.

iii Lagrange's notation (拉格朗日符號) or Prime notation for derivatives

Suppose a dependent variable y represents a function f of an independent variable x, that is,

$$y = f(x).$$

Then, the derivative function of y or f (with respect to x) can be written as

$$y', \quad \text{or } f'(x);$$

the nth derivative function of y or f (with respect to x) can be written as

$$y^{(n)}, \quad \text{or } f^{(n)}(x),$$

in which $^{(n)}$ can be replaced with n primes (′) for usually $n < 4$ (For example, y'' is equivalent to $y^{(2)}$, and f'' is equivalent to $f^{(2)}$.).

iv Newton's notation (牛頓符號), dot notation, flyspeck notation, or fluxions for derivatives

Suppose a dependent variable y represents a function f of an independent variable t, that is,

$$y = f(t),$$

where t usually represents time.

Then, the derivative function of y or f (with respect to t) of the function f can be written as

$$\dot{y},$$

the second derivative function of y or f (with respect to t) of the function f can be written as

$$\ddot{y},$$

and so on.

v Subscript notation for partial derivatives

Suppose a dependent variable y represents a function f of an independent variable vector $\mathbf{x} = (x_1, x_2, \dots x_n)$, that is,

$$y = f(\mathbf{x}).$$

Then, the partial derivative function of y or f with respect to x_i can be written as

$$y_{x_i}, \quad y'_{x_i}, \quad f_{x_i}, \quad \text{or } f'_{x_i};$$

the nth partial derivative of the function f with respect to x_i can be written as

$$y_{x_i x_i \dots x_i}, \quad {y^{(n)}}_{x_i x_i \dots x_i}, \quad f_{x_i x_i \dots x_i}, \text{or} \quad {f^{(n)}}_{x_i x_i \dots x_i},$$

in which the subscript are n x_is and $^{(n)}$ can be replaced with n primes (′) for usually $n < 4$ (For example, y'' is equivalent to $y^{(2)}$, and f'' is equivalent to $f^{(2)}$.);

the nth mixed partial derivative of the function f m_{i_1} times with respect to x_{i_1}, and then m_{i_2} times with respect to x_{i_2}, …, and then m_{i_k} times with respect to x_{i_k}, in which $\sum_{j=1}^{k} m_{i_j} = n$, can be written as

$$y_{x_{i_1} x_{i_1} \dots x_{i_2} x_{i_2} x_{i_2} \dots x_{i_2} \dots x_{i_k} x_{i_k} \dots x_{i_k}}, \quad {y^{(n)}}_{x_{i_1} x_{i_1} \dots x_{i_2} x_{i_2} x_{i_2} \dots x_{i_2} \dots x_{i_k} x_{i_k} \dots x_{i_k}},$$
$$f_{x_{i_1} x_{i_1} \dots x_{i_2} x_{i_2} x_{i_2} \dots x_{i_2} \dots x_{i_k} x_{i_k} \dots x_{i_k}}, \quad \text{or } {f^{(n)}}_{x_{i_1} x_{i_1} \dots x_{i_2} x_{i_2} x_{i_2} \dots x_{i_2} \dots x_{i_k} x_{i_k} \dots x_{i_k}},$$

in which the subscript are m_{i_j} x_{i_j}s for all x_{i_j}.

vi Euler's notation (歐拉符號) for derivatives

Suppose a dependent variable y represents a function f of an independent variable x, that is,

$$y = f(x).$$

Then, the derivative function of y or f (with respect to x) can be written as

$$Df(x), \quad (Df)(x), \quad D_x f(x), \quad (D_x f)(x), \quad \mathrm{d}f(x), \quad (\mathrm{d}f)(x), \quad \mathrm{d}_x f(x), \quad \text{or } (\mathrm{d}_x f)(x),$$

in which D, D_x, d, or d_x is called a (unary) differential operators or a derivative operator, and Df, $D_x f$, $\mathrm{d}f$, or $\mathrm{d}_x f$ are called the differential (微分子) of y or f (with respect to x);

the nth derivative function of y or f (with respect to x) can be written as

$$D^n f(x), \quad (D^n f)(x), \quad D^n{}_{xx\ldots x} f(x), \quad \left(D^n{}_{xx\ldots x} f\right)(x), \quad \mathrm{d}^n f(x), \quad (\mathrm{d}^n f)(x), \quad \mathrm{d}^n{}_{xx\ldots x} f(x), \quad \text{or } \left(\mathrm{d}^n{}_{xx\ldots x}$$

in which subscript are n xs, D^n, $D^n{}_{xx\ldots x}$, d^n, or $\mathrm{d}^n{}_{xx\ldots x}$ is called a (unary) differential operators or a derivative operators, and $D^n f$, $D^n{}_{xx\ldots x} f$, $\mathrm{d}^n f$, or $\mathrm{d}^n{}_{xx\ldots x} f$ is called the differential of the $(n-1)$th derivative function of y or f.

vii Euler's notation for partial derivatives

Suppose a dependent variable y represents a function f of an independent variable vector $\mathbf{x} = (x_1, x_2, \ldots x_n)$, that is,

$$y = f(\mathbf{x}).$$

Then, the partial derivative function of y or f with respect to x_i can be written as

$$\partial_{x_i} f(\mathbf{x}), \quad \left(\partial_{x_i} f\right)(\mathbf{x}), \quad \mathrm{d}_{x_i} f(\mathbf{x}) \text{or } \left(\mathrm{d}_{x_i} f\right)(\mathbf{x}), \quad D_{x_i} f(\mathbf{x}) \text{or } \left(D_{x_i} f\right)(\mathbf{x}),$$

in which ∂_{x_i}, d_{x_i}, or D_{x_i} is called a partial differential operator or a partial derivative operator;

the nth partial derivative function of y or f with respect to x_i can be written as

$$\partial_{x_i x_i \ldots x_i} f(\mathbf{x}), \quad \partial^{(n)}{}_{x_i x_i \ldots x_i} f(\mathbf{x}), \quad D_{x_i x_i \ldots x_i} f(\mathbf{x}), \quad D^{(n)}{}_{x_i x_i \ldots x_i} f(\mathbf{x}), \quad \mathrm{d}_{x_i x_i \ldots x_i} f(\mathbf{x}), \quad \text{or } \mathrm{d}^{(n)}{}_{x_i x_i \ldots x_i} f(\mathbf{x}),$$

in which subscript are n x_is, ${}^{(n)}$ can be replaced with n primes ($'$) for usually $n < 4$ (For example, ∂'' is equivalent to $\partial^{(2)}$, and D'' is equivalent to $D^{(2)}$.), and $\partial_{x_i x_i \ldots x_i}$, $\partial^{(n)}{}_{x_i x_i \ldots x_i}$, $D_{x_i x_i \ldots x_i}$, $D^{(n)}{}_{x_i x_i \ldots x_i}$, $\mathrm{d}D_{x_i x_i \ldots x_i}$, or $\mathrm{d}^{(n)}{}_{x_i x_i \ldots x_i}$ is called a partial differential operator or a partial derivative operator;

the nth mixed partial derivative function of y or f m_{i_1} times with respect to x_{i_1}, m_{i_1} times with respect to x_{i_2}, …, m_{i_k} times with respect to x_{i_k}, in which $\sum_{j=1}^{k} m_{i_j} = n$, can be written as C

$$\partial_{x_{i_1} x_{i_1} \ldots x_{i_2} x_{i_2} x_{i_2} \ldots x_{i_2} \ldots x_{i_k} x_{i_k} \ldots x_{i_k}} f(\mathbf{x}), \quad \partial^{(n)}{}_{x_{i_1} x_{i_1} \ldots x_{i_2} x_{i_2} x_{i_2} \ldots x_{i_2} \ldots x_{i_k} x_{i_k} \ldots x_{i_k}} f(\mathbf{x}),$$

$$D_{x_{i_1} x_{i_1} \ldots x_{i_2} x_{i_2} x_{i_2} \ldots x_{i_2} \ldots x_{i_k} x_{i_k} \ldots x_{i_k}} f(\mathbf{x}), \quad D^{(n)}{}_{x_{i_1} x_{i_1} \ldots x_{i_2} x_{i_2} x_{i_2} \ldots x_{i_2} \ldots x_{i_k} x_{i_k} \ldots x_{i_k}} f(\mathbf{x}),$$

$$\mathrm{d}_{x_{i_1} x_{i_1} \ldots x_{i_2} x_{i_2} x_{i_2} \ldots x_{i_2} \ldots x_{i_k} x_{i_k} \ldots x_{i_k}} f(\mathbf{x}), \quad \text{or } \mathrm{d}^{(n)}{}_{x_{i_1} x_{i_1} \ldots x_{i_2} x_{i_2} x_{i_2} \ldots x_{i_2} \ldots x_{i_k} x_{i_k} \ldots x_{i_k}} f(\mathbf{x}),$$

in which subscript are m_{i_j} x_{i_j}s for all x_{i_j}, ${}^{(n)}$ can be replaced with n primes ($'$) for usually $n < 4$ (For example, ∂'' is equivalent to $\partial^{(2)}$, and D'' is equivalent to $D^{(2)}$.), and $\partial_{x_i x_i \ldots x_i}$, $\partial^{(n)}{}_{x_i x_i \ldots x_i}$, $D_{x_i x_i \ldots x_i}$, $D^{(n)}{}_{x_i x_i \ldots x_i}$, $\mathrm{d}_{x_i x_i \ldots x_i}$, and $\mathrm{d}^{(n)}{}_{x_i x_i \ldots x_i}$ is called a partial differential operator or a partial derivative operator.

II Ordinary Derivative of Functions with Real Domain

Let W be a topological vector space. The ordinary derivative (常導數), total derivative (全導數), or derivative (導數) of a function $f(u): U \subseteq \mathbb{R} \to W$ (with respect to u) at $x \in U$, denoted as $f'(x)$, is defined as

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

if the limit exists. If such limit exists, we say f is differentiable (可微的) at x.

We define the (first(-order)) ordinary derivative (function) (常導（函）數) or derivative (function) (導（函）數) of f (with respect to u) as a function f' with codomain W such that for any $x \in U$ at which f is differentiable, f' maps x to the derivative of f at x.

The derivative of the kth(-order) ($k \in \mathbb{N}$) derivative function of f at $x \in U$ is called the $(k+1)$th(-order) derivative of f (with respect to u) at $x \in U$. The derivative function of the kth(-order) ($k \in \mathbb{N}$) derivative function of f is called the $k+1$th(-order) derivative function of f (with respect to u).

If for any $n \in \mathbb{N}$, the nth(-order) derivative of a function f at a point x in its domain exists, we say f is infinitely differentiable at x.

If f is differentiable at all point in $I \subseteq U$, we say f is differentiable on I; if f is differentiable on U, we say f is differentiable. If $f^{(n-1)}$ exists and is differentiable at all point in $I \subseteq U$, we say f is n-times differentiable on I; if $f^{(n-1)}$ exists and is differentiable on U, we say f is n-times differentiable.

The operation of finding the derivative or derivative function is called ordinary differentiation (常微分) or differentiation (微分).

Specifically, the 0th(-order) derivative of f is f itself.

The derivative of a f at x represents the slope (斜率) at x or the lineal element at x (a miniature tangent line at x).

III One-sided Derivative of Functions with Real Domain

Let W be a topological vector space and $f : U \subseteq \mathbb{R} \to W$ be a function.

The left-hand derivative of f at $x \in U$, denoted as $f'_-(x)$, is defined as

$$f'_-(x) = \lim_{h \to 0^-} \frac{f(x+h) - f(x)}{h}$$

if the limit exists.

The right-hand derivative of f at $x \in U$, denoted as $f'_+(x)$, is defined as

$$f'_+(x) = \lim_{h \to 0^+} \frac{f(x+h) - f(x)}{h}$$

if the limit exists.

IV Ordinary derivative of Functions with Real Vector Domain

Let W be a normed vector space. A function $f(u) : U \subseteq \mathbb{R}^n \to W$ with $n \in \mathbb{N}$ is differentiable at $x \in U$ if there exists a bounded linear operator $A : \mathbb{R}^n \to W$ such that

$$\lim_{\|h\| \to 0} \frac{\|f(x+h) - f(x) - A(h)\|_W}{\|h\|} = 0.$$

If there exists such an operator A, it is unique, so we define the (first(-order)) ordinary derivative, total derivative, or derivative of f (with respect to u) at x, denoted as $Df(x)$, as A.

We define the (first(-order)) ordinary derivative (function) or derivative (function) of f (with respect to u) as a function Df with codomain $B(\mathbb{R}^n, W)$, in which $B(\mathbb{R}^n, W)$ is the space of all bounded linear operators from $\mathbb{R}^n$ to W, such that for any $x \in U$ at which f is differentiable, Df maps x to the derivative of f at x.

The derivative of the kth(-order) ($k \in \mathbb{N}$) derivative function of f at $x \in U$ is called the $k+1$th(-order) derivative of f (with respect to u) at $x \in U$, denoted as $(D^{k+1}f)(x)$. The derivative function of the kth(-order) ($k \in \mathbb{N}$) derivative function of f is called the $k+1$th(-order) derivative function of f (with respect to u), denoted as $D^{k+1}f$.

If f is differentiable at all point in $I \subseteq U$, we say f is differentiable on I; if f is differentiable on U, we say f is differentiable. If $D^{(n-1)}f$ exists and is differentiable at all point in $I \subseteq U$, we say f is n-times differentiable on I; if $D^{(n-1)}f$ exists and is differentiable on U, we say f is n-times differentiable.

The operation of finding the derivative or derivative function is called ordinary differentiation or differentiation.

Specifically, the 0th(-order) Fréchet derivative of f is f itself.

V Partial derivative of Functions with Real Vector Domain

Let W be a topological vector space and $f(\mathbf{x}) : U \subseteq \mathbb{R}^n \to W$ be a function with $\mathbf{x} = (x_1, x_2, \dots, x_n)$, and X be the set $\{x_1, x_2, \dots, x_n\}$.

The (first(-order)) partial derivative (偏導數) $\dfrac{\partial f}{\partial x_i}$ of f with respect to $x_i \in X$ at $u \in U$ is defined as

$$\begin{aligned}\frac{\partial f}{\partial x_i} &= \lim_{h\to 0} \frac{f(u + h\mathbf{e}_i) - f(u)}{h} \\ &= \frac{\mathrm{d}}{\mathrm{d}h} f(u + h\mathbf{e}_i)\Big|_{h=0}\end{aligned},$$

in which $\mathbf{e}_i \in \mathbb{R}^n$ is the unit vector in the direction of x_i.

The partial derivative of the kth(-order) ($k \in \mathbb{N}$) partial derivative (function) (偏導（函）數) of f with respect to $x \in X$ with respect to $x \in X$ at $u \in U$ is called the $k+1$th(-order) partial derivative of f with respect to x at $u \in U$. The partial derivative function of the kth(-order) ($k \in \mathbb{N}$) partial derivative function of f with respect to $x \in X$ with respect to $x \in X$ is called the $k+1$th(-order) partial derivative function of f with respect to x.

The partial derivative of the kth(-order) partial derivative function of f with respect to $x_1 \in X$ with respect to $x_2 \in X$ at $u \in U$ is called the $(k+1)$th(-order) mixed partial derivative of f with respect to $x_1, x_1, \dots x_1$ (k times) and x_2 at $u \in U$. The partial derivative function of the kth(-order) partial derivative function of f with respect to $x_1 \in X$ with respect to $x_2 \in X$ at $u \in U$ is called the $(k+1)$th(-order) mixed partial derivative function of f with respect to $x_1, x_1, \dots x_1$ (k times) and x_2.

The partial derivative of the kth(-order) mixed partial derivative function of f with respect to $x_1, x_2, \dots x_k \in X$ with respect to $x_{k+1} \in X$ at $u \in U$ is called the $(k+1)$th(-order) mixed partial derivative of f with respect to $x_1, x_2, \dots x_k, x_{k+1}$ at $u \in U$. The partial derivative function of the kth(-order) mixed partial derivative function of f with respect to $x_1, x_2, \dots x_k \in X$ with respect to $x_{k+1} \in X$ is called the $(k+1)$th(-order) mixed partial derivative function of f with respect to $x_1, x_2, \dots x_k, x_{k+1}$.

The operation of finding the partial derivative or partial derivative function is called partial differentiation (偏微分).

Specifically, the 0th(-order) partial derivative of f is f itself.

VI Fréchet derivative (弗蘭歇導數)

i Fréchet derivative

Let V and W be normed vector spaces and U be an open subset of V. A function $f(u): U \to W$ is Fréchet differentiable (弗蘭歇可微的) or differentiable at $x \in U$ if there exists a bounded linear operator $A: V \to W$ such that

$$\lim_{\|h\|_V \to 0} \frac{\|f(x+h) - f(x) - A(h)\|_W}{\|h\|_V} = 0.$$

If there exists such an operator A, it is unique, so we define the (first(-order)) Fréchet derivative, ordinary derivative, or derivative of f (with respect to u) at x, denoted as $Df(x)$, as A.

We define the (first(-order)) Fréchet derivative (function) (弗蘭歇導（函）數), ordinary derivative (function), or derivative (function) of f (with respect to u) as a function Df with codomain $B(V, W)$, in which $B(V, W)$ is the space of all bounded linear operators from V to W, such that for any $x \in U$ at which f is Fréchet differentiable, Df maps x to the Fréchet derivative of f at x.

The Fréchet derivative of the kth(-order) ($k \in \mathbb{N}$) Fréchet derivative function of f at $x \in U$ is called the $k+1$th(-order) Fréchet derivative of f (with respect to u) at $x \in U$, denoted as $(D^{k+1} f)(x)$. The Fréchet derivative function of the kth(-order) ($k \in \mathbb{N}$) Fréchet derivative function of f is called the $k+1$th(-order) Fréchet derivative function of f (with respect to u), denoted as $D^{k+1} f$.

If f is Fréchet differentiable at all point in $I \subseteq U$, we say f is Fréchet differentiable on I; if f is Fréchet differentiable on U, we say f is Fréchet differentiable. If $D^{(n-1)} f$ exists and is Fréchet differentiable at all point in $I \subseteq U$, we say f is n-times Fréchet differentiable on I; if $D^{(n-1)} f$ exists and is Fréchet differentiable on U, we say f is n-times Fréchet differentiable.

The operation of finding the Fréchet derivative or Fréchet derivative function is called Fréchet differentiation (弗蘭歇微分), ordinary differentiation, or differentiation.

Specifically, the 0th(-order) Fréchet derivative of f is f itself.

VII Smoothness (光滑性 or 平滑性)

i Continuously differentiable functions

A function is called (n-times) continuously differentiable iff it is (n-times) differentiable and its (nth) derivative is continuous.

ii Smoothness

A function f with real domain that has a derivative that is continuous on its domain is said to be of class C^1 or continuously differentiable, denoted as $f \in C^1$, or be a C^1-function.

A function f that has a derivative that is continuous on a subset I of its domain is said to be of class C^1 on I, of class $C^1(I)$, or continuously differentiable on I, denoted as $f \in C^1(I)$.

A function f with real domain that has a kth derivative that is continuous on its domain is said to be of class C^k or k-times continuously differentiable, denoted as $f \in C^k$, or be a C^k-function.

A function f that has a kth derivative that is continuous on a subset I of its domain is said to be of class C^k on I or of class $C^k(I)$, denoted as $f \in C^k(I)$.

Generally, the term smooth function refers to a C^∞-function, that is a function that is of class C^k for any $k \in \mathbb{R}$. However, it may also mean "sufficiently differentiable" for the problem under consideration.

iii Piecewise smoothness of real functions

Let $f: D \subseteq \mathbb{R} \to \mathbb{R}$ be a function. If there exists a family $\{[a_i, b_i] \mid i \in I\}$ of closed intervals such that:

- for every closed interval V, the set $\{i \in I \mid V \cap [a_i, b_i] \neq \varnothing\}$ is finite (some sources require instead that $n(I)$ is finite, which is stronger),
- $D \subseteq \bigcup\limits_{i \in I} [a_i, b_i]$,
- function $f\big|_{a_i}^{b_i}: (a_i, b_i) \to Y$ is of class $C^k((a_i, b_i))$ for every $i \in I$, and
- there exist finite $\lim\limits_{x \to a_i^+} f(x)$ and $\lim\limits_{x \to b_i^-} f(x)$ for every $i \in I$.

, we say f is of class piecewise C^k (or of class piecewise $C^k\left(\bigcup\limits_{i \in I} [a_i, b_i]\right)$); when $k = \infty$ or sufficiently large, we say f is piecewise smooth (on $\bigcup\limits_{i \in I} U_i$).

iv Piecewise smoothness of functions between topological spaces

Let X and Y be topological spaces, and $f: D \subseteq X \to Y$ be a function. If there exists a locally finite (some sources require instead finite, which is stronger) family $\{U_i \mid i \in I\}$ of closed subsets of X such that:

- $D \subseteq \bigcup_{i \in I} U_i$, and
- there exists a function $g_i : U_i \to Y$ that is of class $C^k\left(U_i\right)$ such that $\forall x \in \operatorname{int}\left(U_i\right)\ f(x) = g_i(x)$ for every $i \in I$.

, we say f is of class piecewise C^k (or of class piecewise $C^k\left(\bigcup_{i \in I} U_i\right)$); when $k = \infty$ or sufficiently large, we say f is piecewise smooth (on $\bigcup_{i \in I} U_i$).

v With compact support

$C_c^k(U)$ for $k \in \mathbb{N} \cup \{\infty\}$ denotes the vector space of $C^k(U)$-functionals with compact support. For any compact subset K of U, $C^k(K; U)$ denotes the vector space of $C^k(U)$-functionals with compact support on K.

VIII Gateaux Differentiation (加托微分)

i Gateaux derivative (加托導數) and partial derivatives

Let V be a locally convex topological vector spaces (LCTVS), W be a topological vector space, U be an open subset of V, and $f : U \to W$ be a function.

The (first(-order)) Gateaux derivative $df(u; \psi)$ of f at $u \in U$ in the direction $\psi \in V$ is defined to be

$$\begin{aligned} df(u; \psi) &= \lim_{\tau \to 0} \frac{f(u + \tau\psi) - f(u)}{\tau} \\ &= \frac{\mathrm{d}}{\mathrm{d}\tau} f(u + \tau\psi)\Big|_{\tau=0} \end{aligned}$$

if the limit exists. If for any $\psi \in V$, the Gateaux derivative exists, then it is said that f is Gateaux differentiable (加托可微的) at u.

We define the (first(-order)) Gateaux derivative (function) (加托導（函）數) of f in the direction $\psi \in V$, denoted as df, as a function $df : U \to W$, such that for any $u \in U$ at which f is Gateaux differentiable, df maps u to the Gateaux derivative of f at u in the direction $\psi \in V$.

The Gateaux derivative of the kth(-order) ($k \in \mathbb{N}$) Gateaux derivative function of f in the direction $\psi \in V$ in the direction $\psi \in V$ at $u \in U$ is called the $k+1$th(-order) Gateaux derivative of f in the direction ψ at $u \in U$. The Gateaux derivative function of the kth(-order) ($k \in \mathbb{N}$) Gateaux derivative function of f in the direction $\psi \in V$ in the direction $\psi \in V$ is called the $k+1$th(-order) Gateaux derivative function of f in the direction ψ.

The Gateaux derivative of the kth(-order) Gateaux derivative function of f in the direction $\psi_1 \in V$ in the direction $\psi_2 \in V$ at $u \in U$ is called the $(k+1)$th(-order) mixed Gateaux derivative of f in the direction $\psi_1, \psi_1, \dots \psi_1$ (k times) and ψ_2 at $u \in U$. The Gateaux derivative function of the kth(-order) Gateaux derivative function of f in the direction $\psi_1 \in V$ in the direction $\psi_2 \in V$ is called the $(k+1)$th(-order) mixed Gateaux derivative function of f in the direction $\psi_1, \psi_1, \dots \psi_1$ (k times) and ψ_2.

The Gateaux derivative of the kth(-order) mixed Gateaux derivative function of f in the direction $\psi_1, \psi_2, \dots \psi_k \in V$ in the direction $\psi_{k+1} \in V$ at $u \in U$ is called the $(k+1)$th(-order) mixed Gateaux derivative of f in the direction $\psi_1, \psi_2, \dots \psi_k, \psi_{k+1}$ at $u \in U$. The Gateaux derivative function of the kth(-order) mixed Gateaux derivative function of f in the direction $\psi_1, \psi_2, \dots \psi_k \in V$ in the direction $\psi_{k+1} \in V$ is called the $(k+1)$th(-order) mixed Gateaux derivative function of f in the direction $\psi_1, \psi_2, \dots \psi_k, \psi_{k+1}$.

The operation of finding the Gateaux derivative or Gateaux derivative function is called Gateaux differentiation (加托微分).

Specifically, the 0th(-order) Gateaux derivative of f is f itself.

ii Partial derivative

Let V be a locally convex topological vector spaces (LCTVS), W be a topological vector space, U be an open subset of V, and $f : U \to W$ be a function, $\mathbf{x}$ be the independent variable vector of f, and X be the set of all independent variables of f.

The (first(-order)) partial derivative of f with respect to $x_i \in X$ at $u \in U$ is defined as the Gateaux derivative of f in the direction of x_i at u if it exists.

The partial derivative of the kth(-order) ($k \in \mathbb{N}$) partial derivative (function) of f with respect to $x \in X$ with respect to $x \in X$ at $u \in U$ is called the $k+1$th(-order) partial derivative of f with respect to x at $u \in U$. The partial derivative function of the kth(-order) ($k \in \mathbb{N}$) partial derivative function of f with respect to $x \in X$ with respect to $x \in X$ is called the $k+1$th(-order) partial derivative function of f with respect to x.

The partial derivative of the kth(-order) partial derivative function of f with respect to $x_1 \in X$ with respect to $x_2 \in X$ at $u \in U$ is called the $(k+1)$th(-order) mixed partial derivative of f with respect to $x_1, x_1, \dots x_1$ (k times) and x_2 at $u \in U$. The partial derivative function of the kth(-order) partial derivative function of f with respect to $x_1 \in X$ with respect to $x_2 \in X$ at $u \in U$ is called the $(k+1)$th(-order) mixed partial derivative function of f with respect to $x_1, x_1, \dots x_1$ (k times) and x_2.

The partial derivative of the kth(-order) mixed partial derivative function of f with respect to $x_1, x_2, \dots x_k \in X$ with respect to $x_{k+1} \in X$ at $u \in U$ is called the $(k+1)$th(-order) mixed partial derivative of f with respect to $x_1, x_2, \dots x_k, x_{k+1}$ at $u \in U$. The partial derivative function of the kth(-order) mixed partial derivative function of f with respect to $x_1, x_2, \dots x_k \in X$ with respect to $x_{k+1} \in X$ is called the $(k+1)$th(-order) mixed partial derivative function of f with respect to $x_1, x_2, \dots x_k, x_{k+1}$.

The operation of finding the partial derivative or partial derivative function is called partial differentiation.

Specifically, the 0th(-order) partial derivative of f is f itself.

IX Applicational interpretation

i Rate of change

If $y = f(x)$, then $f'(a)$ where defined is called the instantaneous rate of change of y with repsect to x at a. A function $f: U \to W$ is Fréchet differentiable (弗蘭歇可微的) or differentiable at $x \in U$ if there exists a bounded linear operator $A: V \to W$ such that

$$\lim_{\|h\|_V \to 0} \frac{\|f(x+h) - f(x) - A(h)\|_W}{\|h\|_V} = 0.$$

If there exists such an operator A, it is unique, so we define the (first(-order)) Fréchet derivative, ordinary derivative, or derivative of f (with respect to u) at x, denoted as $Df(x)$, as A.

We define the (first(-order)) Fréchet derivative (function) (弗蘭歇導（函）數), ordinary derivative (function), or derivative (function) of f (with respect to u) as a function Df with codomain $B(V, W)$, in which $B(V, W)$ is the space of all bounded linear operators from V to W, such that for any $x \in U$ at which f is Fréchet differentiable, Df maps x to the Fréchet derivative of f at x.

The Fréchet derivative of the kth(-order) ($k \in \mathbb{N}$) Fréchet derivative function of f at $x \in U$ is called the $k+1$th(-order) Fréchet derivative of f (with respect to u) at $x \in U$, denoted as $(D^{k+1} f)(x)$. The Fréchet derivative function of the kth(-order) ($k \in \mathbb{N}$) Fréchet derivative function of f is called the $k+1$th(-order) Fréchet derivative function of f (with respect to u), denoted as $D^{k+1} f$.

If f is Fréchet differentiable at all point in $I \subseteq U$, we say f is Fréchet differentiable on I; if f is Fréchet differentiable on U, we say f is Fréchet differentiable. If $D^{(n-1)} f$ exists and is Fréchet differentiable at all point in $I \subseteq U$, we say f is n-times Fréchet differentiable on I; if $D^{(n-1)} f$ exists and is Fréchet differentiable on U, we say f is n-times Fréchet differentiable.

The operation of finding the Fréchet derivative or Fréchet derivative function is called Fréchet differentiation (弗蘭歇微分), ordinary differentiation, or differentiation.

Specifically, the 0th(-order) Fréchet derivative of f is f itself.

If $y = f(x)$ is defined on some interval $[a, b]$ with $b > a$, then $\dfrac{f(b) - f(a)}{b - a}$ is called the average rate of change of y with repsect to x over $[a, b]$.

ii Physical interpretation

Let f be a function with domain being $\mathbb{R}$ or $\mathbb{R}_{\geq 0}$. If we interprets $f(t)$ as the position of y at time t, then:

- f is called the position function of y,
- if $f(x)$ is defined on some interval $[a, b]$ with $b > a$, then $f(b) - f(a)$ is called the displacement of y over $[a, b]$,
- f' is called the velocity function of y,
- $f'(a)$ where defined is called the instantaneous velocity of y at a,

- if $f(x)$ is defined on some interval $[a, b]$ with $b > a$, then $\dfrac{f(b) - f(a)}{b - a}$ is called the average velocity of y over $[a, b]$,
- f'' is called the acceleration function of y,
- $f''(a)$ where defined is called the instantaneous acceleration of y at a,
- if $f'(x)$ is defined on some interval $[a, b]$ with $b > a$, then $\dfrac{f'(b) - f'(a)}{b - a}$ is called the average acceleration of y over $[a, b]$,
- f''' is called the jerk function of y,
- $f'''(a)$ where defined is called the instantaneous jerk of y at a, and
- if $f''(x)$ is defined on some interval $[a, b]$ with $b > a$, then $\dfrac{f''(b) - f''(a)}{b - a}$ is called the average jerk of y over $[a, b]$.

iii Biological interpretation

Let f be a function with domain being $\mathbb{R}$ or $\mathbb{R}_{\geq 0}$. If we interprets $f(t)$ as the number of some population y at time t, then:

- f is called the growth (function) of y,
- $f'(a)$ where defined is called the instantaneous rate of growth of y at a, and
- if $f(x)$ is defined on some interval $[a, b]$ with $b > a$, then $\dfrac{f(b) - f(a)}{b - a}$ is called the average rate of growth of y over $[a, b]$.

iv Economical interpretation

Let f be a function with domain being $\mathbb{R}$ or $\mathbb{R}_{\geq 0}$. If we interprets $f(t)$ as the benefit (aka revenue) or cost of consuming or producing y of quantity t, then:

- f is called benefit (aka revenue) or cost (function) of y, and
- $f'(a)$ where defined is called the margin benefit (aka revenue) or cost of y at a.

X Differential

For a normed vector-valued function f of an independent variable x, we define the differential of it, $\mathrm{d}f$, in terms of the differential of x, denoted as $\mathrm{d}x$, as

$$\mathrm{d}f = f' \,\mathrm{d}x\,.$$

For a normed vector-valued function f of independent variables $x_1, x_2, \ldots, x_n$, we define the (total) differential of it, $\mathrm{d}f$, in terms of the differentials of $x_1, x_2, \ldots, x_n$, denoted as $\mathrm{d}x_1, \mathrm{d}x_2, \ldots, \mathrm{d}x_n$, as

$$\mathrm{d}f = \sum_{k=1}^{n} f_{x_k} \,\mathrm{d}x_k\,.$$

XI Antiderivative (反導函數), inverse derivative, primitive function, primitive integral, or indefinite integral (不定積分)

i Definition

An antiderivative (aka inverse derivative, primitive function, primitive integral, or indefinite integral) of a function $f(x)$ on subset I of the domain of f is a differentiable function $F(x)$ such that

$$F'(x) = f(x)$$

for all $x \in I$.

The process of solving for antiderivatives is called antidifferentiation (反微分) or indefinite integration (不定積分).

ii Theorem

If a function $F(x)$ with codomain Y is an antiderivative of a function $f(x)$ on subset I of the domain of f, then the family of all antiderivatives of $f(x)$ on I is

$$\{F(x) + C \mid C \in Y\}.$$

iii Notation

If a function $F(x)$ with codomain Y is an antiderivative of a function $f(x)$ on subset I of the domain of f, we write

$$\int f(x)\,\mathrm{d}x = F(x) + C,$$

where C denotes an arbitrary constant in Y that does not depend on x, called constant of integration. The symbol $\int$ is called integral sign, introduced by Leibniz.

An indefinite integral of the form

$$\int f(x)\,\mathrm{d}x = \begin{cases} F(x) + C_1, & x \in A_1 \\ F(x) + C_2, & x \in A_2 \\ \vdots\ F(x) + C_n, & x \in A_n \\ (\dots) \end{cases}$$

with an interval A such that $\bigcup\limits_{i=1}^{n} A_i \subseteq A$ and that $A \setminus \bigcup\limits_{i=1}^{n} A_i$ is a set B of measure zero and that $F(x)$ is not defined on C and defined on $\bigcup\limits_{i=1}^{n} A_i$, is usually abbreviated as

$$\int f(x)\,\mathrm{d}x = \begin{cases} F(x) + C, & x \in A \\ \dots \end{cases}$$

where C is to be understood as notation for a locally constant function of x.

When computing indefinite integral, we usually absorb constants into C without changing its symbol, that is,

$$\int f(x)\,\mathrm{d}x = \ldots = F(x) + a + C = F(x) + C.$$

XII Taylor series

i Taylor series (泰勒級數) or Taylor expansion (泰勒展開)

Let $f : D \subseteq \mathbb{R} \to \mathbb{R}$ be a function and $I = (a - R, a + R) \subseteq D$ be an interval such that $f \in C^\infty(I)$, then the Taylor series of f at $a \in I$ (or about a, near a, or centered at a) is

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n,$$

of which convergence is not guaranteed, and convergence to f is not guaranteed even if it's convergent.

Taylor series of at 0 is called Maclaurin series (馬克勞林級數) or Maclaurin expansion (馬克勞林展開).

ii Taylor polynomial

Let $f : D \subseteq \mathbb{R} \to \mathbb{R}$ be a function and $I = (a - R, a + R) \subseteq D$ be an interval where f is k times differentiable, then the k-th degree (or order) Taylor polynomial (or approximation) of f at a (or about a, near a, or centered at a) is

$$T_n(x) = \sum_{n=0}^{k} \frac{f^{(n)}(a)}{n!}(x-a)^n,$$

in which the first degree one is also called linear approximation, linearization, or tangent line (approximation), and

$$R_n(x) = f(x) - T_n(x)$$

is called remainder.

Taylor polynomial of at 0 is called Maclaurin polynomial.

iii Real analyticity (實解析性)

A real-valued function f is real analytic at a point x_0 in its domain, if it is infinitely differentiable at x_0 and that the Taylor expansion of f at x_0 converges to $f(x)$ pointwise for any x in a neighborhood of x_0.

A real-valued function is called to be real analytic on an open set I that is a subset of its domain if it is real analytic at any point in I.

A real-valued function is called to be real analytic if it is real analytic at any point in its domain.

XIII Differential equation (微分方程)

i Differential Equation

An equation containing the derivatives of one or more unknown functions or dependent variables, with respect to one or more independent variables, is said to be a differential equation (DE).

ii Ordinary differential equation (ODE) (常微分方程)

If a differential equation contains only ordinary derivatives of one or more unknown functions with respect to a single independent variable, it is said to be an ordinary differential equation (ODE).

iii Partial differential equation (PDE) (偏微分方程)

If a DE contains partial derivatives of one or more unknown functions of two or more independent variables is called a partial differential equation (PDE).

iv Order of a DE

The order of a differential equation is the order of the highest derivative in the equation.

XIV Graph-related

i Critical point (臨界點)

Let f be a function with domain being a subset of a LCTVS and codomain $\mathbb{R}$ and c be a point in the domain of f, if $f'(c) = 0$ or f' does not exist at c, then c is a critical point, of f, and $f(c)$ is called a critical value of f.

If the domain of f is a subset of $\mathbb{R}$ or $\mathbb{C}$, a critical point is also called a critical number.

ii Relative extremum (相對極值), local extremum (局部極值), or extremum (極值)

Let $f: D \subseteq X \to \mathbb{R}$ be a function with X being a topological space. It is said that a relative maximum (相對極大值) or maximum (極大值) $f(c)$ of f occurs at $c \in D$ if there exists an open subset $I \ni c$ of X such that $\forall x \in I \cap D: f(c) \geq f(x)$.

Let $f: D \subseteq X \to \mathbb{R}$ be a function with X being a topological space. It is said that a relative minimum (相對極小值) or minimum (極小值) $f(c)$ of f occurs at $c \in D$ if there exists an open subset $I \ni c$ of X such that $\forall x \in I \cap D: f(c) \leq f(x)$.

Relative maximum and relative minimum are collectively called relative extreme.

Let $f: D \subseteq X \to \mathbb{R}$ be a function with X being a topological space. It is said that a strict relative maximum or strict maximum $f(c)$ of f occurs at $c \in D$ if there exists an open subset $I \ni c$ of X such that $\forall x \in I \cap D: f(c) > f(x)$.

Let $f: D \subseteq X \to \mathbb{R}$ be a function with X being a topological space. It is said that a strict relative minimum or strict minimum $f(c)$ of f occurs at $c \in D$ if there exists an open subset $I \ni c$ of X such that $\forall x \in I \cap D: f(c) < f(x)$.

Strict relative maximum and strict relative minimum are collectively called strict relative extreme.

iii Absolute extremum (絕對極值 or 最值) or global extremum (全域極值)

Let $f: D \to \mathbb{R}$ be a function. It is said that a absolute maximum (絕對極大值 or 最大值) $f(c)$ of f occurs at $c \in D$ if $\forall x \in D: f(c) \geq f(x)$.

Let $f: D \to \mathbb{R}$ be a function. It is said that a absolute minimum (絕對極小值 or 最小值) $f(c)$ of f occurs at $c \in D$ if $\forall x \in D: f(c) \leq f(x)$.

Absolute maximum and absolute minimum are collectively called absolute extreme.

iv Saddle point (鞍點) or minimax

Let f be a function with codomain $\mathbb{R}$ and c be a point in the domain of f , if $f'(c) = 0$ and c is not a local extremum of f, then c is a saddle point of f.

v Concavity (凹性)

Let V be a vector space and function $f: J \subseteq V \to \mathbb{R}$ be continuous on a convex set $I \subseteq J$.

- If for any $x, y \in I$ and $\alpha \in (0, 1)$, $f((1-\alpha)x + \alpha y) < (1-\alpha)f(x) + \alpha f(y)$, often called "$f$ lies strictly below all of its secant lines on I", then f is called to be strictly concave up/upward/upwards (CU), strictly convex, strictly convex up/upward/upwards, or strictly cup-shaped.
- If for any $x, y \in I$ and $\alpha \in (0, 1)$, $f((1-\alpha)x + \alpha y) > (1-\alpha)f(x) + \alpha f(y)$, often called "$f$ lies strictly above all its secant lines on I", then f is called to be strictly concave, strictly concave down/downward/downwards (CD), strictly convex down/downward/downwards, or strictly cap-shaped.
- If for any $x, y \in I$ and $\alpha \in (0, 1)$, $f((1-\alpha)x + \alpha y) \leq (1-\alpha)f(x) + \alpha f(y)$, often called "$f$ lies weakly below all of its secant lines on I", then f is called to be weakly concave up/upward/upwards (CU), weakly convex, weakly convex up/upward/upwards, or weakly cup-shaped.
- If for any $x, y \in I$ and $\alpha \in (0, 1)$, $f((1-\alpha)x + \alpha y) \geq (1-\alpha)f(x) + \alpha f(y)$, often called "$f$ lies weakly above all of its secant lines on I", then f is called to be weakly concave, weakly concave down/downward/downwards (CD), weakly convex down/downward/downwards, or weakly cap-shaped.

The terms without weakly and strictly are define to be either weakly or strictly in different contexts; here we define them to be strictly.

vi Point of inflection or inflection point (反曲點 or 拐點)

Let V be a vector space, $f: J \subseteq V \to \mathbb{R}$ be a function, and $I \subseteq J$ be a convex set. If f is concave upward on (a, b) and concave downward on (b, c), or, concave downward on (a, b) and concave upward on (b, c), then $(b, f(b))$ is called an inflection point of f.

vii Kink

A kink is a point x on the graph of a function f such that f is continuous but not differentiable at x.

viii Corner or sharp corner

A corner or sharp corner is a point x on the graph of a real function f such that f is continuous but not differentiable at x and that the left-hand derivative and the right-hand derivative of f at x both exist and are finite.

ix Vertical tangent

A vertical tangent is a point x on the graph of a real function f such that f is continuous at x and that the left-hand derivative and the right-hand derivative of f at x are both ∞ or are both $-\infty$.

x Cusp

A cusp is a point x on the graph of a real function f such that:

- f is continuous at x,
- at least one of the left-hand derivative and the right-hand derivative of f at x are ∞ or $-\infty$,
- x is not a vertical tangent of f, and
- the left-hand derivative and the right-hand derivative of f at x are both either finite, ∞, or $-\infty$.

5 Function of Bounded Variation

I Of real functions

i Partition of an interval

A partition $P(x, n)$ of a compact interval $[a, b]$ is a finite sequence of numbers of the form

$$P(x, n) := \{x_i : x_0 = a \wedge x_n = b \wedge \forall 1 \leq i < j \leq n : x_i < x_j\}_{i=0}^{n}.$$

Each $[x_i, x_{i+1}]$ is called a subinterval in the partition. The length of a closed interval $[c, d]$ is defined as $d - c$. The mesh or norm of a partition is defined as

$$\max_i (x_{i+1} - x_i)$$

for every integer $i \in [0, n-1]$.

ii Total variation

Let $f: X \subseteq \mathbb{R} \to \mathbb{R}$ be a function. The total variation of f over interval $[a, b] \subseteq X$, denoted as $V_a^b(f)$, $V_{[a,b]}(f)$, $V(f, [a, b])$, or $V(f; [a, b])$, is

$$V_a^b(f) = \sup_{P(x,n)\in\mathcal{P}} \sum_{i=0}^{n-1} \left| f\left(x_{i+1}\right) - f\left(x_i\right) \right|,$$

where $\mathcal{P}$ is the set of all partitions of f over $[a, b]$.

iii Function of bounded variation (BV)

f is called of bounded variation or called a (globally) BV function over $[a, b]$ iff $V_a^b(f)$ is finite.

iv Function of locally bounded variation

f is called of locally bounded variation or called a locally BV function over an open interval (a, b) iff $V_c^d(f)$ is finite for all $[c, d] \in (a, b)$.

v Space of functions of bounded variation

The space of all (globally) BV functions over $[a, b]$ is denoted as BV$([a, b])$. Banach? PLACEHOLDE

vi Space of functions of bounded variation

The space of all locally BV functions over (a, b) is denoted as $\mathrm{BV}_{\mathrm{loc}}((a, b))$ Banach? PLACEHOLDER

vii Induced measure

For a BV function $g: (a, b) \subseteq \mathbb{R} \to \mathbb{R}$, the signed measure $\mu_g(x)$ induced by g, is defined for any interval $(c, d) \subseteq (a, b)$, as

$$\mu_g((c, d)) = g(d) - g(c).$$

If g is non-decreasing, then μ_g is non-negative. If g is non-increasing, then μ_g is non-positive.

viii Jordan decomposition of BV function

The Jordan decomposition g^+ and g^- of a function g of bounded variation on interval $[a, b]$ is defined as

$$g^+(x) = V_a^x(g), \quad g^-(x) = g(x) - g^+(x),$$

and are both non-decreasing.

II Of real fields

PLACEHOLDER

III In measure theory

PLACEHOLDER

6 Definte Integration (定積分)

I Notation

The integral of a function $f(x)$ with domain being a subset of $\mathbb{R}$, called integrand (被積函數), with respect to x, called integration variable (積分變數), from a to b, which the open interval between a and b is called the domain of integration (積分域) or interval of integration (積分區間) and a, b are called limits of integration (積分極限 or 積分上下限), is denoted as

$$\int_a^b f(x)\,\mathrm{d}x\,,$$

in which when $a > b$, the integral is defined as

$$\int_a^b f(x)\,\mathrm{d}x \coloneqq -\int_b^a f(x)\,\mathrm{d}x\,.$$

The integral of a function $f(\omega)$ with respect to ω over Ω, called the domain of integration and which the limit points of Ω are called are called limits of integration (積分極限), is denoted as

$$\int_\Omega f(\omega)\,\mathrm{d}\omega\,.$$

II (Proper) Riemann integral (黎曼積分) and Darboux integral (達布積分)

i Premise

(Proper) Riemann integral and Darboux integral are two equivalant definitions of definite integral of functions over compact intervals of $\mathbb{R}$ to $\mathbb{R}$.

ii Partition of an interval

A partition $P(x, n)$ of a compact interval $[a, b]$ is a finite sequence of numbers of the form

$$P(x, n) := \{x_i : x_0 = a \wedge x_n = b \wedge \forall 1 \le i < j \le n : x_i < x_j\}_{i=0}^n.$$

Each $[x_i, x_{i+1}]$ is called a subinterval in the partition. The length of a closed interval $[c, d]$ is defined as $d - c$. The mesh or norm of a partition is defined as

$$\max_i \left(x_{i+1} - x_i\right)$$

for every integer $i \in [0, n-1]$.

iii Tagged partition

A tagged partition $P(x, n, \xi)$ of a interval (a, b) is a partition together with a choice of a sample point or tag within each of all n subintervals, that is, numbers $\{\xi_i\}_{i=0}^{n-1}$ with $\xi_i \in [x_i, x_{i+1}]$ for each integer $i \in [0, n-1]$. The mesh of a tagged partition is the same as that of an ordinary partition.

Suppose that two tagged partitions $P(x, n, \xi)$ and $Q(y, m, \zeta)$ are both partitions of the interval $[a, b]$. We say that $Q(y, m, \zeta)$ is a refinement of $P(x, n, \xi)$ if for each integer $i \in [0, n-1]$, there exists an integer $r(i) \in [0, m-1]$ such that $x_i = y_{r(i)}$ and that $\forall i \in [0, n-1] : \exists j \in [r(i), r(i+1)]$ s.t. $\xi_i = \zeta_j$. That is, a tagged partition breaks up some of the subintervals and adds sample points where necessary, "refining" the accuracy of the partition.

We can turn the set of all tagged partitions into a directed set by saying that one tagged partition is greater than or equal to another if the former is a refinement of the latter.

iv Riemann sum

Let f be a real-valued function defined on a interval $[a, b]$. The Riemann sum of f over a tagged partition $P(x, n, \xi)$ of $[a, b]$ is defined to be

$$R(f, P) := \sum_{i=0}^{n-1} f(\xi_i) \left(x_{i+1} - x_i \right).$$

Each term in the sum is the product of the value of the function at a given point and the length of an interval. Consequently, each term represents the signed area of a rectangle with height $f(\xi_i)$ and width $x_{i+1} - x_i$. Thus the Riemann sum is the signed area of all the rectangles.

v Darboux sum

Let f be a real-valued function defined on a interval $[a, b]$. The lower and upper Darboux sums, $L(f, P)$ and $U(f, P)$, of f over a partition $P(x, n)$ of $[a, b]$, are two specific Riemann sums of which the tags are chosen to be the infimum and supremum (respectively) of f on each subinterval:

$$L(f, P) := \sum_{i=0}^{n-1} \inf_{\xi \in [x_i, x_{i+1}]} f(\xi)(x_{i+1} - x_i),$$
$$U(f, P) := \sum_{i=0}^{n-1} \sup_{\xi \in [x_i, x_{i+1}]} f(\xi)(x_{i+1} - x_i).$$

vi Riemann left, right, and midpoint sums

Let f be a real-valued function defined on a interval $[a, b]$. The Riemann left (aka left-sided), right (aka right-sided), and midpoint sums, $R_L(f, P)$, $R_R(f, P)$, and $R_M(f, P)$, of f over a partition $P(x, n)$ of $[a, b]$, are three specific Riemann sums of which the tags are chosen to be the left endpoint, right endpoint, and midpoint (respectively) on each subinterval:

$$R_L(f,P) := \sum_{i=0}^{n-1} f\left(x_i\right)(x_{i+1}-x_i),$$

$$R_L(f,P) := \sum_{i=0}^{n-1} f\left(x_{i+1}\right)(x_{i+1}-x_i),$$

$$R_M(f,P) := \sum_{i=0}^{n-1} f\left(\frac{x_i+x_{i+1}}{2}\right)(x_{i+1}-x_i).$$

vii Riemann sums over partition that divides interval equally

A Riemann sum of f over a tagged partition $P(x,n,\xi)$ that divides an interval into subintervals of equal length is denoted as $R_n(f)$.

A lower Darboux sum, an upper Darboux sum, a Riemann left sum, a Riemann right sum, and a Riemann midpoint sum of f over a partition $P(x,n)$ that divides an interval into subintervals of equal length are sometimes denoted as $L_n(f), U_n(f), R_{Ln}(f), R_{Rn}(f)$, and $R_{Mn}(f)$ respectively.

viii Riemann integral

Let f be a real-valued function defined on a interval $[a,b]$. The Riemann integral of f on $[a,b]$ exists and equals s if for all $\varepsilon > 0$, there exists $\delta > 0$ such that for any tagged partition $P(x,n,\xi)$ whose mesh is less than δ,

$$|R(f,P)-s| < \varepsilon.$$

ix Darboux integral

Let f be a real-valued function defined on a interval $[a,b]$. The Darboux integral of f on $[a,b]$ exists and equals s if for all $\varepsilon > 0$, there exists $\delta > 0$ such that for any partition P whose mesh is less than δ,

$$|U(f,P)-s| < \varepsilon \wedge |L(f,P)-s| < \varepsilon.$$

III Extensions of Riemann integral

i Improper (Riemann) integral (瑕（黎曼）積分)

An integral $\displaystyle\int_a^b f(x)\,\mathrm{d}x$ is an improper integral if it satisfies one or more of the below conditions:

1. $a = -\infty$,
2. $b = \infty$,
3. $f(x)$ is unbounded or undefined somewhere in $[a,b]$.

The improper integrals are defined by limits recursively as:

- For $a = -\infty$:

$$\int_{-\infty}^{b} f(x)\,\mathrm{d}x = \lim_{a\to-\infty}\int_a^b f(x)\,\mathrm{d}x,$$

called improper integrals of type 1.

- For $b = \infty$:
$$\int_a^{\infty} f(x)\,\mathrm{d}x = \lim_{b\to\infty} \int_a^b f(x)\,\mathrm{d}x\,,$$
called improper integrals of type 1.

- For $f(x)$ that is unbounded or undefined at a:
$$\int_a^b f(x)\,\mathrm{d}x = \lim_{c\to a^+} \int_c^b f(x)\,\mathrm{d}x\,,$$
called improper integrals of type 2.

- For $f(x)$ that is unbounded or undefined at b:
$$\int_a^b f(x)\,\mathrm{d}x = \lim_{c\to b^-} \int_a^c f(x)\,\mathrm{d}x\,,$$
called improper integrals of type 2.

- For $f(x)$ that is unbounded or undefined at $c \in (a, b)$:
$$\int_a^b f(x)\,\mathrm{d}x = \lim_{t\to c^-} \int_a^t f(x)\,\mathrm{d}x + \lim_{t\to c^+} \int_t^b f(x)\,\mathrm{d}x\,,$$
called improper integrals of type 2.

In which if any of the terms diverge or is undefined, the improper integral diverges and is undefined; otherwise, the improper integral exists finitely and converges to a finite value.

We say an improper integral of $f(x)$ over interval I converges absolutely to L iff the improper integral of $|f(x)|$ over I converges to L. If an improper integral is convergent but not convergent absolutely, we say it is convergent conditionally.

ii Cauchy principal value (柯西主值) or PV integral

The Cauchy principal value or PV integral, denoted as p.v., is a weaker notion of convergence, defined by taking symmetric limits.

The p.v. of $\int_{-\infty}^{\infty} f(x)\,\mathrm{d}x$, denoted as P.V.$\int_{-\infty}^{\infty} f(x)\,\mathrm{d}x$ or $\mathcal{P}\int_{-\infty}^{\infty} f(x)\,\mathrm{d}x$, is defined with:
$$\text{P.V.}\int_{-\infty}^{\infty} f(x)\,\mathrm{d}x \coloneqq \lim_{R\to\infty} \int_{-R}^{R} f(x)\,\mathrm{d}x\,,$$
if the limit exists.

The p.v. of $\int_a^b f(x)\,\mathrm{d}x$, in which $f(x)$ is unbounded or undefined at $c \in (a, b)$, denoted as P.V.$\int_a^b f(x)\,\mathrm{d}x$ or $\mathcal{P}\int_a^b f(x)\,\mathrm{d}x$, is defined with:
$$\text{P.V.}\int_a^b f(x)\,\mathrm{d}x \coloneqq \lim_{\varepsilon\to 0} \left(\int_a^{c-\varepsilon} f(x)\,\mathrm{d}x + \int_{c+\varepsilon}^b f(x)\,\mathrm{d}x \right),$$
if the limit exists.

If an improper integral converges, the p.v. of it converges.

IV Riemann–Stieltjes integral

i Definition

Below, we will define the Riemann–Stieltjes integral of a function $f : X \subseteq \mathbb{R} \to \mathbb{R}$ over a BV function $g : Y \subseteq \mathbb{R} \to \mathbb{R}$ over an interval $[a, b] \subseteq (X \cap Y)$, denoted as

$$\int_{x=a}^{b} f(x)\, \mathrm{d}g(x)\,,$$

or

$$\int_{a}^{b} f(x)\, \mathrm{d}g(x)\,.$$

The Riemann–Stieltjes sum of f with respect to g over a tagged partition $P(x, n, \xi)$ is defined as

$$S(f, g, P) := \sum_{i=0}^{n-1} f(\xi_i) \left(g\left(x_{i+1}\right) - g(x_i)\right).$$

The Riemann–Stieltjes integral

$$\int_{x=a}^{b} f(x)\, \mathrm{d}g(x)$$

exists and equals s if for all $\varepsilon > 0$, there exists $\delta > 0$ such that for any tagged partition $P(x, n, \xi)$ whose mesh is less than δ,

$$|S(f, g, P) - s| < \varepsilon.$$

V Extensions of Riemann–Stieltjes integral

An integral $\int_{x=a}^{b} f(x)\, \mathrm{d}g(x)$ is an improper integral if it satisfies one or more of the below conditions:

1. $a = -\infty$,
2. $b = \infty$,
3. $f(x)$ is unbounded or undefined somewhere in $[a, b]$.

The definition of improper integral and Cauchy principal value as extensions of the Riemann–Stieltjes integral are the same as those for the Riemann integral.

VI Lebesgue integral (勒貝格積分)

A definition of definite integral.

i Premise

Let (E, Σ, μ) be a measure space with possibly signed measure μ. Below, we will define the Lebesgue integral of measurable functions on E to $\mathbb{R} \cup \{-\infty, \infty\}$.

ii Of an indicator functions

The integral of an indicator function 1_S of a measurable subset S of E is defined to be

$$\int 1_S \,\mathrm{d}\mu = \mu(S).$$

iii Of a nonnegative simple function

A simple function s is a finite real linear combinations of indicator functions of disjoint measurable subsets of E, that is,

$$s: \sum_k a_k 1_{S_k},$$

where the coefficients a_k are real numbers and S_k are disjoint measurable sets. When the coefficients a_k are positive real numbers, s is called nonnegative.

The integral of a nonnegative simple function $s = \sum_k a_k 1_{S_k}$ over E is defined to be

$$\int_E s \,\mathrm{d}\mu = \sum_k a_k \int 1_{S_k} \,\mathrm{d}\mu = \sum_k a_k \mu(S_k),$$

where this sum can be finite or ∞.

The integral of a nonnegative simple function $s = \sum_k a_k 1_{S_k}$ over a subset B of E is defined to be:

$$\int_B s \,\mathrm{d}\mu = \sum_k a_k \mu\left(S_k \cap B\right).$$

iv Of a nonnegative measurable function

Let f be a nonnegative measurable function on some measurable subset B of E into $\mathbb{R} \cup \{-\infty, \infty\}$. We define

$$\int_B f \,\mathrm{d}\mu = \sup\left\{\int_B s \,\mathrm{d}\mu \mid \forall x \in B: \ 0 \leq s(x) \leq f(x) \wedge s \text{ is a nonnegative simple function}\right\}.$$

v Of a measurable function

Let f be a measurable function on some measurable subset B of E into $\mathbb{R} \cup \{-\infty, \infty\}$. We first define

$$f^+(x) = \begin{cases} f(x), & \text{if } f(x) > 0 \\ 0, & \text{otherwise} \end{cases}, \quad \text{and}$$

$$f^-(x) = \begin{cases} -f(x), & \text{if } f(x) < 0 \\ 0, & \text{otherwise} \end{cases}.$$

Note that both f^+ and f^- are nonnegative and that

$$f = f^+ - f^-, \quad |f| = f^+ + f^-.$$

Then we define the Lebesgue integral of f to exist if

$$\min\left(\int f^+ \,\mathrm{d}\mu, \int f^- \,\mathrm{d}\mu\right) < \infty.$$

In this case we define

$$\int f \,\mathrm{d}\mu = \int f^+ \,\mathrm{d}\mu - \int f^- \,\mathrm{d}\mu.$$

vi Integrability

Let f be a function on some measurable subset B of E into $\mathbb{R} \cup \{-\infty, \infty\}$.

If

$$\int |f| \,\mathrm{d}\mu < \infty,$$

we say that f is Lebesgue integrable.

vii Connection with improper Riemann integral

An integral as Lebesgue integral is finite if it is either proper Riemann integrable or absolute convergent as improper Riemann integral, and the Lebesgue integral and proper or improper Riemann integral agree on the result.

An integral as Lebesgue integral is not finite if it is conditionally convergent as improper Riemann integral.

viii L^p spaces (L^p 空間) or Lebesgue space

For set X and measure μ, the vector space $L^p(X, \mu)$ consists of the equivalence class under the equivalence relation of equality μ-a.e. of measurable functions $f : X \to \mathbb{R}$ (or $\mathbb{C}$) that

$$\int_X |f(x)|^p \,\mathrm{d}\mu(x) < \infty.$$

For $1 \leq p < \infty$, define L^p norm:

$$\|f\|_p \coloneqq \left(\int_X |f(x)|^p \,\mathrm{d}\mu(x)\right)^{1/p}.$$

$L^p(X, \mu)$ is also denoted as $L^p(X)$ when μ is clear.

ix Locally integrable L^p space

For set X and measure μ, the vector space $L^p_{\mathrm{loc}}(X, \mu)$ consists of the equivalence class under the equivalence relation of equality μ-a.e. of measurable functions $f : X \to \mathbb{R}$ (or $\mathbb{C}$) that for all compact subset $K \subseteq X$, $f \in L^p(K, \mu)$. $L^p_{\mathrm{loc}}(X, \mu)$ is also denoted as $L^p_{\mathrm{loc}}(X)$ when μ is clear.

VII Weak derivative

Let $U \subseteq \mathbb{R}$ and $u \in L^1_{\mathrm{loc}}(U)$. We say that $v \in L^1_{\mathrm{loc}}(U)$ is a weak derivative of u if

$$\int_U u\varphi' \,\mathrm{d}\mu = -\int_U v\phi \,\mathrm{d}\mu$$

for all $\varphi \in C_c^\infty(U)$.

Let $U \subseteq \mathbb{R}^n$, $u \in L^1_{\mathrm{loc}}(U)$, and α be a muti-index. We say that $v \in L^1_{\mathrm{loc}}(U)$ is a αth weak derivative of u if

$$\int_U u D^\alpha \varphi \,\mathrm{d}\mu = (-1)^{|\alpha|} \int_U v \phi \,\mathrm{d}\mu$$

for all $\varphi \in C_c^\infty(U)$.

VIII Sobolev space

For $k, p \in \mathbb{N} \cup \{\infty\}$, the Sobolev space $W^{k,p}(\mathbb{R})$ is defined as the subset of functions f in $L^p(\mathbb{R})$ such that f and its weak derivatives up to order k have a finite L^p norm.

The Sobolev spaces are equipped with a norm

$$\|f\|_{k,p} = \left(\sum_{i=0}^{k} \left\| f^{(k)} \right\|_p^p \right)^{\frac{1}{p}}, \quad p \in \mathbb{N},$$

and

$$\|f\|_{k,\infty} = \max_{i=0,1,\ldots,k} \|f^{(i)}\|_\infty.$$

IX Lebesgue–Stieltjes integral

i Definition

Below, we will define the Lebesgue–Stieltjes integral of a function $f : X \subseteq \mathbb{R} \to \mathbb{R}$ over a BV function $g : Y \subseteq \mathbb{R} \to \mathbb{R}$ over an interval $[a, b] \subseteq (X \cap Y)$, denoted as

$$\int_{x=a}^{b} f(x) \,\mathrm{d}g(x),$$

or

$$\int_a^b f(x) \,\mathrm{d}g(x).$$

The Lebesgue–Stieltjes integral of f with respect to g over $[a, b]$ is defined as the Lebesgue integral

$$\int_a^b f(x) \,\mathrm{d}\mu_g(x).$$

ii Connection with improper Riemann–Stieltjes integral

An integral as Lebesgue–Stieltjes integral is finite if it is either proper Riemann–Stieltjes integrable or absolute convergent as improper Riemann–Stieltjes integral, and the Lebesgue–Stieltjes integral and proper or improper Riemann–Stieltjes integral agree on the result.

An integral as Lebesgue–Stieltjes integral is not finite if it is conditionally convergent as improper Riemann–Stieltjes integral.

X Bochner integral (博赫納積分)

i Premise

Let (E, Σ, μ) be a measure space and X be a Banach space with norm $\|\cdot\|_X$. Below, we will define the Bochner integral of measurable functions on E to X.

ii Of an indicator functions

The integral of an indicator function 1_S of a measurable subset S of E is defined to be

$$\int 1_S \,\mathrm{d}\mu = \mu(S).$$

iii Of a simple function

PLACEHOLDER

iv Of a measurable function

PLACEHOLDER

v Bochner spaces (博赫納空間)

Let (E, Σ, μ) be a measure space and X be a topological space.

The Bochner space $L^p(E; X)$ with $p \in \mathbb{N} \cup \{\infty\}$ is defined to be the quotient space of the space of all Bochner measurable functions $f(t) : E \to X$ such that the norm $\|f(t)\|_{L^p(E;X)}$ of it, defined with

$$\|f(t)\|_{L^p(E;X)} \coloneqq \left(\int_E \|f(t)\|_X{}^p \,\mathrm{d}\mu \right)^{1/p}, \quad p \in \mathbb{N},$$

and

$$\|f(t)\|_{L^\infty(E;X)} \coloneqq \operatorname{ess\,sup}_{t \in E} \|f(t)\|_X,$$

is finite by the equivalence relation of equality almost everywhere.

7 Differential Geometry (微分幾何)

I Curve

i Definition

A (parameterized or parametric) (topological) curve in a topological space X is (the image or graph of) a continuous map $\gamma(t) : I \to X$ for some interval I, where t is called parameter. $\gamma(t)$ is also called the parametrization of the curve.

If $I = [a, b]$, $\gamma(a)$ is called initial point, and $\gamma(b)$ is called terminal point.

ii Closed curve or loop

A curve is closed or is a loop if $I = [a, b]$ and $\gamma(a) = \gamma(b)$.

For $X = \mathbb{R}^n$, the equations of each component being equal to a function of t are called parametric equations.

iii Path or arc

A curve is a path or an arc if I is closed and bounded.

iv Simple curve

A curve is simple if $I = [a, b]$ and for all $s \neq t \in [a, b)$, $\gamma(s) \neq \gamma(t)$.

v Curve in Enclidean space

A plane curve is a curve on a plane. A space curve is a curve in a three-dimensional space.

A Jordan curve is a plane simple closed curve. And the region enclosed by a Jordan curve is called Jordan domain.

vi Winding number

The winding number or winding index of a closed curve in the plane around a given point is an integer representing the total number of times that the curve travels counterclockwise around the point.

vii Orientation

Two parameterized curves with same range $\gamma : T \to X$ and $\zeta : U \to X$ with T, U being intervals, are called of different orientations if there exists $t_1 < t_2 \in T, u_1 < u_2 \in U$ such that

$$\gamma(t_1) = \zeta(u_2) \wedge \gamma(t_2) = \zeta(u_1) \wedge \gamma(t_1) \neq \gamma(t_2).$$

viii Mechanical term

The terms in mechanics are sometimes used to describe derivatives of curve, e.g., position $\mathbf{r}$, velocity $\mathbf{r}'$, speed $\|\mathbf{r}'\|$, acceleration $\mathbf{r}''$, and jerk or jolt $\mathbf{r}'''$.

II Arc length

i Definition

Given a curve $\mathbf{r}(t) = (x_1(t), x_2(t), \quad , x_n(t)), \quad t \in [a, b]$. Let $P(t, k)$ be a partition of interval $[a, b]$. Define the polygonal length of $\mathbf{r}$ along $P(t, k)$ as

$$s(P(t, k)) = \sum_{i=1}^{k} \|\mathbf{r}(i) - \mathbf{r}(i-1)\|.$$

The arc length of the curve is

$$s = \sup_{P \in \mathcal{P}} s(P),$$

where $\mathcal{P}$ is the set of all partitions of interval $[a, b]$.

The curve is called rectifiable iff s is finite.

s is finite iff all $x_i(t)$ is of bounded variation.

Proof. "If" direction:

By triangular inequality,

$$\|\mathbf{r}(i) - \mathbf{r}(i-1)\| \le \sum_{j=1}^{n} \left|x_j(i) - x_j(i-1)\right|.$$

$$\sum_{i=1}^{k} \|\mathbf{r}(i) - \mathbf{r}(i-1)\| \le \sum_{i=1}^{k}\sum_{j=1}^{n} \left|x_j(i) - x_j(i-1)\right| = \sum_{j=1}^{n}\sum_{i=1}^{k} \left|x_j(i) - x_j(i-1)\right| < \infty.$$

"Only if" direction:

By triangular inequality, for every $j \in \mathbb{N} \land j \le n$,

$$\left|x_j(i) - x_j(i-1)\right| \le \|\mathbf{r}(i) - \mathbf{r}(i-1)\|.$$

$$\sum_{i=1}^{k} \left|x_j(i) - x_j(i-1)\right| \le \sum_{i=1}^{k} \|\mathbf{r}(i) - \mathbf{r}(i-1)\| < \infty.$$

□

The arc length (function) of the curve from a is defined a function of t that maps $t \ge a$ to the arc length of $\mathbf{r}_{[a,t]}$.

ii For differentiable curve

If $\mathbf{r}$ is differentiable, the arc length from a to b is

$$s = \int_a^b \left\|\frac{\mathrm{d}\mathbf{r}(t)}{\mathrm{d}t}\right\| \mathrm{d}t\,.$$

It's often useful and in some context assumed to parameterize a curve w.r.t. its arc length.

iii Surface of revolution

In $\mathbb{R}^n$, the area of the surface formed by rotating the curve $\mathbf{r}(t), \quad t \in [a, b]$ about a rotational axis $\mathbf{P} + \mathbf{k}u, \quad u \in \mathbb{R}$ with $\|\mathbf{k}\| = 1$ is

$$A = 2\pi \int_{t=a}^{b} \left\|\mathbf{r}(t) - \mathbf{P} - \big(\mathbf{k} \cdot (\mathbf{r}(t) - \mathbf{P})\big)\,\mathbf{k}\right\| \mathrm{d}s(t)\,,$$

where $s(t)$ is the arc length function of $\mathbf{r}(t)$ from a to t.

The area is called rectifiable iff A is finite.

A is finite if all components of $\mathbf{r}$ are all of bounded variation.

III Tangent, normal, binormal, curvature, and torsion

i (Unit) tangent (vector)

The unit tangent vector of a curve $\mathbf{r}$ at t is

$$\mathbf{T} = \hat{\mathbf{r}'} = \frac{\mathbf{r}'}{|\mathbf{r}'|}.$$

ii (Principal/unit) normal (vector)

$$\mathbf{N} = \hat{\mathbf{T}'} = \frac{\mathbf{T}'}{|\mathbf{T}'|}.$$

$$\mathbf{T} \cdot \mathbf{N} = 0.$$

$$\begin{aligned}\mathbf{N} &= \frac{\left\|\mathbf{r}'\right\|^2 \mathbf{r}'' - \left(\mathbf{r}'' \cdot \mathbf{r}'\right) \mathbf{r}'}{\left\|\|\mathbf{r}'\|^2 \mathbf{r}'' - (\mathbf{r}'' \cdot \mathbf{r}')\, \mathbf{r}'\right\|} \\ &= \widehat{\mathbf{r}''_{\perp \mathbf{r}'}} \\ &= \frac{\mathbf{r}'' - (\mathbf{r}'' \cdot \mathbf{T})\mathbf{T}}{\|\mathbf{r}'' - (\mathbf{r}'' \cdot \mathbf{T})\mathbf{T}\|}\end{aligned}$$

In three dimensions,

$$\mathbf{N} = \frac{\left(\mathbf{r}' \times \mathbf{r}''\right) \times \mathbf{r}'}{\|(\mathbf{r}' \times \mathbf{r}'') \times \mathbf{r}'\|}$$

In two dimensions, $\mathbf{N}$ is $\frac{\pi}{2}$ counterclockwise rotated from $\mathbf{T}$ if $x' y'' - x'' y' > 0$, and $\mathbf{N}$ is $\frac{\pi}{2}$ clockwise rotated from $\mathbf{T}$ if $x' y'' - x'' y' < 0$.

Proof.

$$\begin{aligned}T' &= \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\mathbf{r}'}{\|\mathbf{r}'\|}\right) \\ &= \frac{\mathbf{r}''\left\|\mathbf{r}'\right\| - \mathbf{r}' \frac{\mathbf{r}' \cdot \mathbf{r}''}{\|\mathbf{r}'\|}}{\|\mathbf{r}'\|^2} \\ &= \frac{\left\|\mathbf{r}'\right\|^2 \mathbf{r}'' - \left(\mathbf{r}'' \cdot \mathbf{r}'\right) \mathbf{r}'}{\|\mathbf{r}'\|^3} \\ &= \frac{\mathbf{r}''_{\perp \mathbf{r}'}}{\|\mathbf{r}'\|} \\ &= \frac{\mathbf{r}'' - \left(\mathbf{r}'' \cdot \mathbf{T}\right) \mathbf{T}}{\|\mathbf{r}'\|}\end{aligned}$$

□

iii (Unit) binormal (vector)

In three dimensions,

$$\begin{aligned}\mathbf{B} &\coloneqq \mathbf{T} \times \mathbf{N} \\ &= \frac{\mathbf{r}' \times \mathbf{r}''}{\|\mathbf{r}' \times \mathbf{r}''\|}\end{aligned}$$

iv Frenet-Serret frame

$\mathbf{T}, \mathbf{N}, \mathbf{B}$ are collectively called Frenet-Serret basis or TNB basis, which is a local orthogonal basis. This basis with origin being the point $\mathbf{r}(t)$ is called Frenet-Serret frame.

v Curvature (曲率)

The curvature κ and curvature vector $\mathbf{K}$ of a curve $\mathbf{r}$ at t are defined, with unit tagent vector $\mathbf{T}$ and arc length s, as

$$\begin{aligned}\mathbf{K} &:= \frac{\mathrm{d}\mathbf{T}}{\mathrm{d}s} \\ &= \frac{\mathrm{d}\mathbf{T}}{\mathrm{d}t} \bigg/ \frac{\mathrm{d}s}{\mathrm{d}t} \\ &= \frac{\mathbf{T}'}{\|\mathbf{r}'\|} \\ &= \frac{\left\|\mathbf{r}'\right\|^2 \mathbf{r}'' - \left(\mathbf{r}'' \cdot \mathbf{r}'\right) \mathbf{r}'}{\|\mathbf{r}'\|^4} \\ &= \frac{\mathbf{r}''_{\perp \mathbf{r}'}}{\|\mathbf{r}'\|^2} \\ &= \frac{\mathbf{r}'' - \left(\mathbf{r}'' \cdot \mathbf{T}\right) \mathbf{T}}{\|\mathbf{r}'\|^2}\end{aligned}$$

$$\kappa := \|\mathbf{K}\|$$

$$\mathbf{K} = \kappa \mathbf{N}$$

In two dimensions, let $\mathbf{r}(t) = (x(t), y(t))$,

$$\kappa = \frac{\left\|x' y'' - x'' y'\right\|}{\left(x'^2 + y'^2\right)^{3/2}},$$

and signed curvature k is defined as

$$k = \frac{x' y'' - x'' y'}{\left(x'^2 + y'^2\right)^{3/2}}.$$

In three dimensions,

$$\mathbf{K} = \frac{\left(\mathbf{r}' \times \mathbf{r}''\right) \times \mathbf{r}'}{\|\mathbf{r}'\|^4}$$

$$\kappa = \frac{\left\|\mathbf{r}' \times \mathbf{r}''\right\|}{\|\mathbf{r}'\|^3}.$$

vi Torsion

The torsion τ of a curve $\mathbf{r}$ at t is defined, with unit normal vector $\mathbf{N}$ and unit binormal vector $\mathbf{B}$, as

$$\begin{aligned}\tau &:= -\frac{\mathrm{d}\mathbf{B}}{\mathrm{d}s} \cdot \mathbf{N} \\ &= \frac{\det\left(\mathbf{r}', \mathbf{r}'', \mathbf{r}'''\right)}{\|\mathbf{r}' \times \mathbf{r}''\|^2}\end{aligned}$$

Proof.

$$\frac{\mathrm{d}\mathbf{B}}{\mathrm{d}t} = \frac{\left(\mathbf{r}' \times \mathbf{r}''\right)'}{\|\mathbf{r}' \times \mathbf{r}''\|} - \mathbf{r}' \times \mathbf{r}'' \frac{\left(\|\mathbf{r}' \times \mathbf{r}''\|^2\right)'}{\|\mathbf{r}' \times \mathbf{r}''\|^3}$$

Since the second term is orthogonal to $\mathbf{N}$,

$$\begin{aligned}
-\frac{\mathrm{d}\mathbf{B}}{\mathrm{d}s} \cdot \mathbf{N} &= -\frac{\mathbf{r}' \times \mathbf{r}'''}{\|\mathbf{r}' \times \mathbf{r}''\| \, \|\mathbf{r}'\|} \cdot \frac{\left(\mathbf{r}' \times \mathbf{r}''\right) \times \mathbf{r}'}{\|(\mathbf{r}' \times \mathbf{r}'') \times \mathbf{r}'\|} \\
&= \frac{\left(\mathbf{r}''' \times \mathbf{r}'\right) \cdot \left(\left(\mathbf{r}' \times \mathbf{r}''\right) \times \mathbf{r}'\right)}{\|\mathbf{r}' \times \mathbf{r}''\|^2 \, \|\mathbf{r}'\|^2} \\
&= \frac{\det\left(\mathbf{r}'', \mathbf{r}''', \mathbf{r}'\right)}{\|\mathbf{r}' \times \mathbf{r}''\|^2} \\
&= \frac{\det\left(\mathbf{r}', \mathbf{r}'', \mathbf{r}'''\right)}{\|\mathbf{r}' \times \mathbf{r}''\|^2}
\end{aligned}$$

□

vii Frenet-Serret formulas or Frenet–Serret theorem

$$\begin{aligned}
\frac{\mathrm{d}T}{\mathrm{d}s} &= \kappa\mathbf{N}, \\
\frac{\mathrm{d}N}{\mathrm{d}s} &= \kappa\mathbf{T} + \tau\mathbf{B}, \\
\frac{\mathrm{d}B}{\mathrm{d}s} &= -\tau\mathbf{N},
\end{aligned}$$

$$\begin{bmatrix} \frac{\mathrm{d}\mathbf{T}}{\mathrm{d}s} \\ \frac{\mathrm{d}\mathbf{N}}{\mathrm{d}s} \\ \frac{\mathrm{d}\mathbf{B}}{\mathrm{d}s} \end{bmatrix} = \begin{bmatrix} 0 & \kappa & 0 \\ -\kappa & 0 & \tau \\ 0 & -\tau & 0 \end{bmatrix} \begin{bmatrix} \mathbf{T} \\ \mathbf{N} \\ \mathbf{B} \end{bmatrix}$$

Proof. PLACEHOLDER □

IV Polar curve

i Definition

Plane curve or polar equation is a plane curve of the form $r = f(\theta)$, also denoted as $r(\theta)$.

ii Radical scaling

Replacing r with kr scales the curve by k.

iii Rotation

Replacing θ with $(\theta - \varphi)$ rotates the curve counterclockwise by φ.

iv Reflection and symmetry

Replacing θ with $-\theta$ reflects the curve through the polar axis.

If a polar curve is unchanged when θ is replaced with $-\theta$, then it is symmetric about the polar axis.

Replacing θ with $(\pi - \theta)$ reflects the curve through $\theta = \frac{\pi}{2}$.

If a polar curve is unchanged when θ is replaced with $(\pi - \theta)$, then it is symmetric about $\theta = \frac{\pi}{2}$.

Replacing either θ with $(\pi + \theta)$ or r with $-r$ reflects the curve through the pole.

If a polar curve is unchanged when θ is replaced with $(\pi + \theta)$ or when r is replaced with $-r$, then it is symmetric about the pole.

A polar equation of the form $r = f(\cos(\theta - \varphi))$ is symmetric about the line $\theta = \varphi$.

v Arc length

The arc length of differentiable polar curve $r(\theta)$ from a to b is

$$s = \int_a^b \sqrt{(r(\theta))^2 + (r'(\theta))^2}\,\mathrm{d}\theta\,.$$

Proof.

$$x(\theta) = r\cos(\theta)$$

$$y(\theta) = r\sin(\theta)$$

$$x'(\theta) = -r\sin(\theta) + r'\cos(\theta)$$

$$y'(\theta) = r\cos(\theta) + r'\sin(\theta)$$

$$(x'(\theta))^2 + (y'(\theta))^2 = (r(\theta))^2 + \left(r'(\theta)\right)^2$$

□

V Tangent space

i Tagent space (切空間)

Suppose that M is an n-dimensional differentiable manifold and $x \in M$. Pick a coordinate chart $\varphi\colon U \to \mathbb{R}^n$, where U is an open subset of M containing x. Suppose further that two curves $\gamma_1, \gamma_2\colon (-1,1) \to M$ with $\gamma_1(0) = \gamma_2(0) = x$ are given such that both $\varphi \circ \gamma_1, \varphi \circ \gamma_2\colon (-1,1) \to \mathbb{R}^n$ are differentiable. We call these differentiable curves initialized at x. Then γ_1 and γ_2 are said to be equivalent at 0 if and only if the derivatives of $\varphi \circ \gamma_1$ and $\varphi \circ \gamma_2$ at 0 coincide. This defines an equivalence relation on the set of all differentiable curves initialized at x, and equivalence classes of such curves are called tangent vectors (切向量) of M at x, sometimes denoted as γ' for some curve γ in the equivalence class. The tangent space of M at (aka near) x, denoted by T_xM, is then defined as the set of all tangent vectors at x, which does not depend on the choice of coordinate chart φ.

ii Dimension of tangent space

The dimension of $T_x M$ is the same as M. PLACEHOLDER?

Proof. PLACEHOLDER □

iii Tagent bundle (切叢)

A tangent bundle TM of a smooth manifold M is defined to be

$$TM = \{(x, v) : x \in M \wedge v \in T_x M\},$$

where $T_x M$ is the tangent space at x.

iv Normal space (法空間)

Suppose that M is a differentiable n-dimensional manifold in a vector space V equipped with a bilinear form B and that $x \in M$. The normal space of M at x, denoted by $N_x M$, is defined as the orthogonal complement of the tangent space of M at x. A vector $v \in N_x M$ such that $B(v, v) \neq 0$ is called a normal vector (法向量) of M at x.

v Dimension of normal space

Suppose further that V is a k-dimensional manifold and M is an topological embedding, the dimension of $N_x M$ is $k - m$. PLACEHOLDER?

Proof. PLACEHOLDER □

vi Direction space

The direction space of an affine subspace is the tangent space of it at any point on it. Note that the tangent spaces of an affine subspace at any two points on it are the same.

vii Metric tensor

Let M be a smooth manifold of dimension n. A metric tensor or tensor g_p at a point p is a function $g_p : (T_p M)^2 \to \mathbb{R}$ that is

- Bilinear:

$$g_p(aU_p + bV_p, Y_p) = ag_p(U_p, Y_p) + bg_p(V_p, Y_p)$$

$$g_p(Y_p, aU_p + bV_p) = ag_p(Y_p, U_p) + bg_p(Y_p, V_p)$$

- Symmetric:

$$g_p(X_p, Y_p) = g_p(Y_p, X_p).$$

- Nondegenerate: For any $X_p \neq 0$ in $T_p M$, there exists $Y_p \in T_p M$ such that

$$g_p(X_p, Y_p) \neq 0.$$

viii Signature of metric tensor

The signature of a metric tensor is the numbers (counted with multiplicity) (v, p, r) of the positive, negative, and zero eigenvalues of it, also denoted as v plus signs, p minus signs, and r zeros, e.g. $(1, 3, 0)$ as $(+, -, -, -)$.

VI Differentiable map between differentiable manifolds

Let $\varphi: M \to N$ be a differentiable map between differentiable manifolds M, N.

i Differential, derivative, or total derivative

Given $x \in M$, the differential, derivative, or total derivative of φ at x, denoted as $\mathrm{d}\varphi_x$ or $D_x\varphi$, is a linear map

$$\mathrm{d}\varphi_x : T_xM \to T_{\varphi(x)}N,$$

where T_xM is the tangent space of M at x and $T_{\varphi(x)}N$ is the tangent space of N at $\varphi(x)$, that maps a tangent vector $v \in T_xM$, which is the equivalence class of curves including a curve γ, to a tangent vector $w \in T_{\varphi(x)}N$ that is the equivalence class of curves including the curve $\varphi \circ \gamma$, called the pushforward of v by φ.

ii Rank

$$\operatorname{rank}\left(\mathrm{d}\varphi_x\right) = \dim\left(\mathrm{d}\varphi_x\left(T_xM\right)\right).$$

iii Regular or smooth

$\mathrm{d}\varphi_x$ is injective, also called regular or smooth, iff

$$\operatorname{rank}\left(\mathrm{d}\varphi_x\right) = \dim\left(T_xM\right).$$

iv Immersion

Let $\varphi: M \to N$ be a differentiable map between differentiable manifolds M, N. It's called an immersion if $\mathrm{d}\varphi_x$ is an injective function for all $x \in M$.

VII Parametric manifold

i Parametric manifold

An m-dimensional (parameterized or parametric) (topological) manifold in a topological space X is (the image or graph of) a continuous map $\gamma(u_1, \ldots, u_m): D \subseteq \mathbb{R}^m \to X$ whose domain is an open set and range is an m-dimensional manifold, where $u_1, \ldots, u_m$ are called parameters. $\gamma(u_1, \ldots, u_m)$ is also called the parametrization of the curve.

For $X = \mathbb{R}^n$, the equations of each component being equal to a function of $u_1, \ldots, u_m$ are called parametric equations.

A uniary function obtained by fixing $m - 1$ of $u_1, \ldots, u_m$ is called grid curve.

ii Regular or smooth

A parametric manifold whose image is differentiable is called regular or smooth at a point if its differential at the point is an injective function.

For differentiable two-dimensional manifold S in $\mathbb{R}^3$ parameterized as $\mathbf{r}(s,t)$, that is,

$$\frac{\partial \mathbf{r}}{\partial s} \times \frac{\partial \mathbf{r}}{\partial t} \neq \mathbf{0}.$$

A parametric manifold whose image is differentiable is called regular or smooth if it's regular at all points.

iii Fundamental tangent vector

For an m-dimensional differentiable parametric manifold $\mathbf{r}(\mathbf{x})$, each $\dfrac{\partial \mathbf{r}}{\partial x_i}$ is a fundamental tangent vector.

VIII Orientation of hypersurface

For a hypersurface S, if it is possible to choose a unit normal vector for each point on it such that the unit normal vectors as a function of the points is a continuous vector field over S, then S is orientable, and an orientation is a choice of such unit normal vectors.

8 Multivariable Calculus

I Notation and convention

- $\mathbf{0}$ or 0 refers to the zero element of a space V, that is,

$$\forall \mathbf{v} \in V: \ \mathbf{v} + \mathbf{0} = \mathbf{v}.$$

- Independent variable vector: $\mathbf{x} = (x_1, x_2, \dots, x_n)$. For two dimensions also $\mathbf{x} = (x, y)$. For three dimensions also $\mathbf{x} = (x, y, z)$.
- Basis: $\mathbf{e}_i$ is the unit vector in the direction of x_i. For two or three dimensions also $\mathbf{i} = \mathbf{e}_i$ and $\mathbf{j} = \mathbf{e}_2$. For three dimensions also $\mathbf{k} = \mathbf{e}_3$.
- Unit normal vector: $\hat{\mathbf{n}}$ or $\mathbf{n}$
- Vector field: A function, often denoted with bold uppercase Latin letter, whose domain is a subset of $\mathbb{R}^n$ and codomain is $\mathbb{R}^m$.
- Vector component: For a type $(r, 1)$ tensor in multidimensional array representation, its ith component is often denoted as the same character with subscript i, e.g., F_i or $\mathbf{F}_i$ for $\mathbf{F}$.
- Scalar field: A function, often denoted with non-bold Latin or Greek letter, whose domain is a subset of $\mathbb{R}^n$ and codomain is $\mathbb{R}$.
- kth total derivative function operator: D^k

- Dot product (點積) operator: $\cdot$
- Cross product (叉積) operator: $\times$
- Gradient (梯度) operator: ∇ or grad
- Jacobian matrix (雅可比矩陣): ∇, $\mathbf{J}(\mathbf{F})$, $J(\mathbf{F})$, $\mathbf{J}_{\mathbf{F}}$, or $J_{\mathbf{F}}$ for operand vector field $\mathbf{F}$.
- Gram matrix, Gramian matrix, or Gramian: $\mathbf{G}(\mathbf{F})$, $G(\mathbf{F})$, $\mathbf{G}_{\mathbf{F}}$, or $G_{\mathbf{F}}$ for operand vector field $\mathbf{F}$.
- Divergence (散度) operator: $\nabla\cdot$ or div
- Curl (旋度) operator (three dimension): $\nabla\times$ or curl
- Directional derivative (方向導數) operator: $(\mathbf{F}\cdot\nabla)\mathbf{G}$, $D_{\mathbf{F}}\mathbf{G}$, $\dfrac{\partial\mathbf{G}}{\partial\mathbf{F}}$, or $\nabla_{\mathbf{F}}\mathbf{G}$ for directional derivative of tensor field $\mathbf{G}$ along/in the direction of a vector field $\mathbf{F}:\mathbb{R}^n\to\mathbb{R}^n$. For constant vector $\mathbf{F}$, $D_{\mathbf{F}}\mathbf{G}$ notation is typically used and $\mathbf{F}$ is typically a unit vector.
- Hessian (黑塞) (matrix): $D^2 f$, $\nabla\nabla$, $\nabla^2 f$, $\mathbf{H}(f)$, $H(f)$, $\mathbf{H}_f$, H_f, or $\nabla\otimes\nabla f$ for operand scalar field f.
- Laplace operator (拉普拉斯算子) or Laplacian: $\nabla\cdot\nabla$, ∇^2, or Δ
- Vector and covector convention: We use column vector convention of vectors and row vector convention of covectors here, and tensor convention follows. Note that a derivative function of a scalar field is a covector. We use the convention that the notation of partial derivative w.r.t. a column vector v means the column vector of the partial derivatives w.r.t. each component of v.
- Line/path/general integral operator: $\displaystyle\int$
- Double/surface integral operator: $\displaystyle\iint$
- Triple/volume integration operator: $\displaystyle\iiint$
- Closed line integral operator: $\displaystyle\oint$
- Closed surface integral operator: $\displaystyle\oiint$
- Arc length element: $\mathrm{d}\ell$
- Vector arc length element: $\mathrm{d}\boldsymbol{\ell}$, which is defined as $\mathbf{n}\,\mathrm{d}\ell$.
- Surface element: $\mathrm{d}S$
- Vector surface element: $\mathrm{d}\mathbf{S}$, which is defined as $\mathbf{n}\,\mathrm{d}S$ and is only defined for orientable surface equipped with an orientation. For a vector field $\mathbf{F}$, $\displaystyle\iint_S \mathbf{F}\cdot\mathrm{d}\mathbf{S}$ is called flux of f across S.
- Positive orientation: The boundary curve ℓ of a compact surface S, $\mathrm{d}\boldsymbol{\ell}$ transverses counterclockwise around S unless otherwise specified.

- Upward/Positive orientation: In $\mathbb{R}^2$, a surface is oriented upward unless otherwise specified.
- Outward/Positive orientation: The boundary region S of a compact region is oriented outward unless otherwise specified.

II Nabla operators

i Gradient and Jacobian matrix

The gradient of a type $(r, 1)$ tensor field f is a type $(r+1, 1)$ tensor field

$$\nabla f = \frac{\partial f}{\partial(x_1, x_2, \dots, x_n)} = \begin{bmatrix} \left(\frac{\partial f}{\partial x_1}\right)^\top & \left(\frac{\partial f}{\partial x_2}\right)^\top & \cdots & \left(\frac{\partial f}{\partial x_n}\right)^\top \end{bmatrix},$$

that is, the transpose of the first-order derivative of f. Some sources define gradient as the transpose of the definition here.

The gradient of a scalar field is a vector field. The gradient of a vector field is a bilinear form field (i.e., also a matrix) and is also called the Jacobian matrix of it. Some sources use the term gradient only for scalar fields.

The Jacobian matrix of a vector field $\mathbf{F}: U \subseteq \mathbb{R}^n \to \mathbb{R}^n$ is a $m \times n$ matrix

$$J(\mathbf{F}) = \begin{bmatrix} \frac{\partial F_1}{\partial x_1} & \cdots & \frac{\partial F_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial F_m}{\partial x_1} & \cdots & \frac{\partial F_m}{\partial x_n} \end{bmatrix},$$

that is,

$$(J(\mathbf{F}))_{ij} = \frac{\partial F_i}{\partial x_j}.$$

The determinant of Jacobian matrix is called Jacobian determinant or Jacobian.

The notation

$$\frac{\partial(F_1, \dots, F_m)}{\partial(x_1, \dots, x_n)}$$

denotes the Jacobian matrix or sometimes Jacobian of $\mathbf{F}$.

ii Directional derivative (方向導數)

For tensor fields $\mathbf{G}: U \subseteq \mathbb{R}^n \to X$ and $\mathbf{F}: V \subseteq \mathbb{R}^n \to \mathbb{R}^n$, the directional derivative of $\mathbf{G}$ in the direction of $\mathbf{F}$ is a function $((\mathbf{F} \cdot \nabla)\mathbf{G}): W \subseteq \mathbb{R}^n \to X$ defined as

$$(\mathbf{F} \cdot \nabla)\mathbf{G} \coloneqq \lim_{h \to 0} \frac{\mathbf{G}(\mathbf{x} + h\mathbf{F}(\mathbf{x})) - \mathbf{G}(\mathbf{x})}{h},$$

where W is a subset of $U \cap V$ where the definition is defined.

For points where $\mathbf{G}$ is differentiable,

$$(\mathbf{F} \cdot \nabla)\mathbf{G} = \sum_{i=1}^{n} F_i \frac{\partial \mathbf{G}}{\partial x_i} = \mathbf{F}(D\mathbf{A}) = \mathbf{F} \cdot (\nabla \mathbf{G}).$$

iii Gram matrix, Gramian matrix, or Gramian

The Gram matrix, Gramian matrix, or Gramian $G(\mathbf{F})$ of a vector field **F** is defined as the transpose of the Jacobian matrix of **F** right multiplied by the Jacobian matrix of **F**

$$G(\mathbf{F}) = J(\mathbf{F})^\top J(\mathbf{F}),$$

that is,

$$G_{ij} = \frac{\partial \mathbf{F}}{\partial x_i} \cdot \frac{\partial \mathbf{F}}{\partial x_j}.$$

The determinant of Gram matrix is called Gram determinant or Gramian.

iv Divergence (散度)

For a type $(r, 1)$ tensor field **F**,

$$\nabla \cdot \mathbf{F} = \sum_{i=1}^{n} \frac{\partial F_i}{\partial x_i}.$$

v Curl (旋度) (three dimension)

For a three-dimensional vector field $\mathbf{F}(x, y, z)$ defined on a subset of $\mathbb{R}^3$,

$$\begin{aligned}\nabla \times \mathbf{F} &= \begin{bmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{bmatrix} \\ &= \left(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z}, \frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x}, \frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right)\end{aligned}$$

vi Curl (two dimension)

For a two-dimensional vector field $\mathbf{F}(x, y)$ defined on a subset of $\mathbb{R}^2$, $\dfrac{\partial F_2}{\partial x} - \dfrac{\partial F_1}{\partial y}$ is called curl, and F_1, F_2 are often denoted P, Q respectively.

vii Hessian (黑塞) (matrix)

The Hessian matrix $H(f)$ of a scalar field f is its second derivative, which is a tensor field of type $(0, 2)$, represented as a matrix, that is,

$$(H(f))_{ij} = \frac{\partial^2 f}{\partial x_i \partial x_j},$$

that is,

$$H(f) = J(\nabla f).$$

viii Laplace operator (拉普拉斯算子)

The Laplace operator applied to a type $(r, 1)$ tensor f is

$$\nabla^2 f = \nabla \cdot (\nabla f).$$

For one-dimensional f,

$$\nabla^2 f = \text{tr}(H(f)) = \sum_{i=1}^{n} \frac{\partial^2 f}{\partial {x_i}^2}.$$

For m-dimensional f, let $f = \mathbf{F} = (F_1, \dots, F_m)$,

$$\begin{aligned}\nabla^2 \mathbf{F} &= \nabla \cdot \begin{bmatrix} \frac{\partial F_1}{\partial x_1} & \cdots & \frac{\partial F_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial F_m}{\partial x_1} & \cdots & \frac{\partial F_m}{\partial x_n} \end{bmatrix} \\ &= \left(\sum_{i=1}^{n} \frac{\partial^2 F_1}{\partial {x_i}^2}, \dots, \sum_{i=1}^{n} \frac{\partial^2 F_m}{\partial {x_i}^2} \right) \\ &= \left(\nabla^2 F_1, \dots, \nabla^2 F_n \right)\end{aligned}$$

ix Poisson's equation (卜瓦松 or 帕松 or 泊松方程)

An equation in the form

$$\nabla^2 f = g(\mathbf{x})$$

for scalar fields f, g.

x Laplace's equation (拉普拉斯方程)

An equation in the form

$$\nabla^2 f = 0$$

for scalar field f.

A twice differentiable scalar field f is called a harmonic function if f satisfies Laplace's equation.

III Multi-index notation (多重指標記號)

i Definition

Suppose there are n variables $x_1, x_2, \dots, x_n$, then a multi-index α is a vector of n nonnegative integers $\alpha_1, \alpha_2, \dots, \alpha_n$:

$$\alpha = (\alpha_1, \alpha_2, \dots, \alpha_n).$$

ii Norm or length

The space is equipped with norm or length $|\alpha|$ or $\|\alpha\|$;

$$|\alpha| = \alpha_1 + \alpha_2 + \dots \alpha_n,$$

i.e., one-norm.

iii Factorial

The factorial $\alpha!$ of it is defined as:

$$\alpha! = \alpha_1! \cdot \alpha_2! \cdot \dots \alpha_n!.$$

iv Exponent

If $\mathbf{x} = (x_1, x_2, \dots, x_n)$, $\mathbf{x}^\alpha$ is defined as:

$$\mathbf{x}^\alpha = x_1^{\alpha_1} \cdot x_2^{\alpha_2} \dots x_n^{\alpha_n}.$$

v Order

For two multi-indices $\alpha = (\alpha_1, \alpha_2, \dots, \alpha_n)$ and $\beta = (\beta_1, \beta_2, \dots, \beta_n)$, $\alpha \leq \beta$ if $\alpha_i \leq \beta_i$ for all $i \in \mathbb{Z} \cap [1, n]$.

IV Higher-order derivatives

i Higher-order mixed partial derivatives

For a tensor field f defined on a subset of $\mathbb{R}^n$ and a multi-index $\alpha = (\alpha_1, \alpha_2, \dots, \alpha_n)$, $D^\alpha f$ or $\partial^\alpha f$ is defined as

$$D^\alpha f = \frac{\partial^{|\alpha|} f}{\partial {x_1}^{\alpha_1} \partial {x_2}^{\alpha_2} \dots \partial {x_n}^{\alpha_n}}.$$

ii Higher-order derivative

The kth order derivative of $\mathbf{F} : \mathbb{R}^n \to \mathbb{R}^m$, denoted as $\mathbf{F}^{(k)}(\mathbf{x})$ or $D^k\mathbf{F}(\mathbf{x})$, is a $(\mathbb{R}^n)^k \to \mathbb{R}$ function, where $(\mathbb{R}^n)^k$ is a Cartesian product of k copies of $\mathbb{R}^n$ vector, that is,

$$D^k\mathbf{F}(\mathbf{x}) = \sum_{|\alpha|=k} \left(D^\alpha \mathbf{F}(\mathbf{a}) \right).$$

In particular, the first-order derivative of $\mathbf{F}$ is the gradient of it.

iii Taylor series or Taylor expansion

Let $f : D \subseteq \mathbb{R}^m \to \mathbb{R}$ be a function and $I \subseteq D$ be an open ball centered at a such that $f \in C^\infty(I)$, then the Taylor series of f at a (or about a, near a, or centered at a) is

$$\sum_{\alpha \in {\mathbb{N}_0}^m} \frac{D^\alpha f(a)}{\alpha!}(x-a)^\alpha,$$

of which convergence is not guaranteed, and convergence to f is not guaranteed even if it's convergent.

Taylor series of at 0 is called Maclaurin series or Maclaurin expansion.

iv Taylor polynomial

Let $f : D \subseteq \mathbb{R}^m \to \mathbb{R}$ be a function and $I \subseteq D$ be an open ball centered at a where f is k times differentiable, then the k-th degree (or order) Taylor polynomial (or approximation) of f at a (or about a, near a, or centered at a) is

$$T_n(x) = \sum_{\alpha \in {\mathbb{N}_0}^m \wedge |\alpha| \leq k} \frac{D^\alpha f(a)}{\alpha!}(x-a)^\alpha,$$

in which the first degree one is also called linear approximation, or linearization, or in three dimensions, tangent plane (approximation), and

$$R_n(x) = f(x) - T_n(x)$$

is called remainder.

Taylor polynomial of at 0 is called Maclaurin polynomial.

V Graph

i Level curve, level slice, or contour line

For a real-valued function f of two variables, the level curve of f at level $k \in \mathbb{R}$ is defined as the set of **x** such that

$$f(\mathbf{x}) = k.$$

ii Contour map or contour plot

Contour map or contour plot is the graphical representation of the level curves of a function of two variable, typical represented with a collection of level curves labelled with corresponding levels that are often of same difference relative to their adjacent curves, or with color or height (z-axis) showing function value at all points.

iii Level surface

For a real-valued function f of three variables, the level surface of f at level $k \in \mathbb{R}$ is defined as the set of **x** such that

$$f(\mathbf{x}) = k.$$

iv Gradient perpendicular to tangent of level

For function $F(\mathbf{x}) : D \subseteq \mathbb{R}^n \to \mathbb{R}$, suppose $\mathbf{r}(t)$ is any parameterized curve lies on $F(\mathbf{x}) = k$, then

$$\nabla F \cdot \mathbf{r}' = 0.$$

v Tangent flat

For function $F(\mathbf{x}) : D \subseteq \mathbb{R}^n \to \mathbb{R}$, for each point $\mathbf{x}_0 \in D$, the tangent flat of F at it is given by

$$\nabla F \cdot (\mathbf{x} - \mathbf{x}_0) = 0.$$

vi Normal line

For function $F(\mathbf{x}) : D \subseteq \mathbb{R}^n \to \mathbb{R}$, for each point $\mathbf{x}_0 \in D$, the normal line of F at it is given by

$$\mathbf{x}(t) = t\nabla F(\mathbf{x}_0) + \mathbf{x}_0, \quad t \in \mathbb{R}.$$

VI Line integral (線積分) or Path integral (路徑積分)

i Definition

For a parameterized curve $\gamma(t) : [a, b] \to \mathbb{R}^n$ that is piecewise C^1, the line integral of a function f defined on the image of γ over γ is defined as the Riemann–Stieltjes integral:

$$\int_\gamma f(\mathbf{x})\,\mathrm{d}(x) \coloneqq \int_{t=a}^{b} f(\gamma(t))\,\mathrm{d}\gamma(t)\,,$$

In which γ is called the integral path.

The integral does not depend on the parametrization of the curve as long as the curve starts at $\gamma(a)$ and ends at $\gamma(b)$ and is piecewise C^1.

ii Conservative field (保守場) or gradient field

An n-dimensional vector field $\mathbf{v}$ defined on an open subset U of $\mathbb{R}^n$ is said to be a conservative field or gradient field on U if there exists a C^1 scalar field φ on U such that

$$\mathbf{v} = \nabla\varphi$$

on U. φ is called the (scalar) potential function.

iii Path independence

Line integrals of an n-dimensional vector field $\mathbf{v}$ defined on an open subset U of $\mathbb{R}^n$ is said to be path-independent or independent of path on U if for any two piecewise C^1 paths $C, D \subseteq U$ with same initial and terminal points,

$$\int_C \mathbf{v} \cdot \mathrm{d}\mathbf{x} = \int_D \mathbf{v} \cdot \mathrm{d}\mathbf{x}$$

VII Multiple integral

i Integral region

The integral region of n-tuple integral is a measurable subset of $\mathbb{R}^n$, often called plane region when $n = 2$ and solid region when $n = 3$.

ii Lebesgue integral

Defined via product measure space. The $\mathrm{d}\mu$ where μ is Lebesgue measure is often denoted as $\mathrm{d}A$ or sometimes $\mathrm{d}S$ for two dimensions and $\mathrm{d}V$ for three dimensions or sometimes higher dimensions.

iii Riemann integral

For rectangle

$$R = \prod_{i=1}^{n} [x_{i0}, x_{i1}],$$

and function f bounded on R,

$$\int_R f(\mathbf{x})\,\mathrm{d}V := \lim_{(m_1,m_2,\ldots,m_n)\to\infty} \left(\sum_{j_i=1}^{m_i}\right)_{i=1}^{n} f\left(\mathbf{x}_{(j_i)_{i=1}^n}*\right)\Delta V,$$

called multiple Riemann sum, where

$$\Delta x_i = \frac{\left(x_{i1} - x_{i0}\right)}{m_i},$$

$$\Delta V = \prod_{i=1}^{n} \Delta x_i,$$

and $\mathbf{x}_{(j_i)_{i=1}^n}*$ is an arbitrary chosen sample point in

$$\prod_{i=1}^{n}\left[x_{i0} + \left(j_i - 1\right)\Delta x_i, x_{i0} + j_i \Delta x_i\right].$$

For arbitrary subset $E \subseteq R$ and function g bounded on E, let h be the trivial extension of g_E to R,

$$\int_E g(\mathbf{x})\,\mathrm{d}V := \int_R h(\mathbf{x})\,\mathrm{d}V.$$

iv Iterated integral

An integral whose integrand is an integral.

VIII Type of region

i Simple region

A x_i simple region V in $\mathbb{R}^n$ is a region that can be written in the form

$$V = \left\{(x_1,\ldots,x_n):\ (x^n_{\substack{j=1\\ j\neq i}}) \in D \wedge \varphi_1(x^n_{\substack{j=1\\ j\neq i}}) \le x_i \le \varphi_2(x^n_{\substack{j=1\\ j\neq i}})\right\}.$$

where φ_1, φ_2 are continuous functions.

ii Type of region in two dimensions

A Type I region is a subset of $\mathbb{R}^2$ of the form

$$D = \{(x,y) \mid a \le x \le b \wedge g_1(x) \le y \le g_2(x)\}$$

where g_1, g_2 are continuous functions defined on $[a,b]$.

A Type II region is a subset of $\mathbb{R}^2$ of the form

$$D = \{(x,y) \mid a \le y \le b \wedge g_1(y) \le x \le g_2(y)\}$$

where g_1, g_2 are continuous functions defined on $[a,b]$.

iii Type of region in three dimensions

A Type I region is a subset of $\mathbb{R}^3$ of the form

$$D = \{(x, y, z) \mid (x, y) \in E \wedge g_1(x, y) \le z \le g_2(x, y)\}$$

where E is a plane g_1, g_2 are continuous functions defined on E.

A Type II region is a subset of $\mathbb{R}^3$ of the form

$$D = \{(x, y, z) \mid (y, z) \in E \wedge g_1(y, z) \le x \le g_2(y, z)\}$$

where E is a plane g_1, g_2 are continuous functions defined on E.

A Type III region is a subset of $\mathbb{R}^3$ of the form

$$D = \{(x, y, z) \mid (x, z) \in E \wedge g_1(x, z) \le y \le g_2(x, z)\}$$

where E is a plane g_1, g_2 are continuous functions defined on E.

IX Manifold element in terms of Hausdorff measure

i Differentiable parametric manifold

An m-dimensional differentiable manifold S in $\mathbb{R}^n$ with $1 \le m < n$ parameterized as $\mathbf{r}(t_1, t_2, \ldots, t_m)$ on domain D has m-dimensional Hausdorff measure

$$\int_D \sqrt{\det(G)}\, \mathrm{d}t_1\, \mathrm{d}t_2 \ldots \mathrm{d}t_m$$

and manifold element

$$\mathrm{d}S = \sqrt{\det(G)}\, \mathrm{d}t_1\, \mathrm{d}t_2 \ldots \mathrm{d}t_m\,,$$

where $\det(G)$ is the Gram determinant of $\mathbf{r}$.

ii Differentiable parametric surface

A differentiable surface S in $\mathbb{R}^3$ parameterized as $\mathbf{r}(s, t)$ on domain D has two-dimensional Hausdorff measure

$$\int_D \left|\frac{\partial \mathbf{r}}{\partial s} \times \frac{\partial \mathbf{r}}{\partial t}\right| \mathrm{d}s\, \mathrm{d}t$$

and surface element

$$\mathrm{d}S = \left|\frac{\partial \mathbf{r}}{\partial s} \times \frac{\partial \mathbf{r}}{\partial t}\right| \mathrm{d}s\, \mathrm{d}t\,.$$

If S is orientable and equipped with an orientation, the vector surface element is

$$\mathrm{d}\mathbf{S} = \pm\frac{\partial \mathbf{r}}{\partial s} \times \frac{\partial \mathbf{r}}{\partial t}\, \mathrm{d}s\, \mathrm{d}t\,,$$

where $\pm$ is such that $\pm\,\mathrm{UnitVector}\left(\frac{\partial \mathbf{r}}{\partial s} \times \frac{\partial \mathbf{r}}{\partial t}\right)$ is in the orientation.

iii Hypersurface as graph

An $(n-1)$-dimensional differentiable hypersurface S in $\mathbb{R}^n$ parameterized w.r.t. $n-1$ of the Cartesian coordinates, say, $x_1, x_2, \dots, x_{n-1}$, as $x_n = f(x_1, x_2, \dots, x_{n-1})$, $(x_1, x_2, \dots, x_{n-1}) \in D$, where $f \in W^{1,1}(\mathbb{R})$, has $(n-1)$-dimensional Hausdorff measure

$$\int_D \sqrt{1 + |\nabla f|^2}\, \mathrm{d}\mu(x_1, x_2, \dots, x_{n-1})$$

and Hausdorff measure element

$$\mathrm{d}S = \sqrt{1 + |\nabla f|^2}\, \mathrm{d}\mu(x_1, x_2, \dots, x_{n-1})$$

9 Non-Cartesian Coordinate Systems

I General

i Scale factors

For a coordinate system, the scale factor h_a of each coordinate a is defined as

$$h_a = \left\| \frac{\partial \mathbf{r}}{\partial a} \right\|$$

And the scale factors are denoted with a vector of the same dimension as the space where each component is the scaling factor of that coordinate.

ii Local basis

For a coordinate system, the local basis is a position-dependent basis where each unit vector $\hat{\mathbf{e}}_a$ of each coordinate a is defined as

$$\hat{\mathbf{e}}_a = \frac{1}{h_a} \frac{\partial \mathbf{r}}{\partial a}$$

iii Orthogonal coordinate system

An orthogonal coordinate system is a coordinate system whose local basis is orthogonal.

iv Line element

For a coordinate system $(a_1, a_2, \dots, a_n)$,

$$\mathrm{d}\mathbf{r} = \sum_{i=1}^{n} h_{a_i} \hat{\mathbf{e}}_{a_i}\, \mathrm{d}a_i\,.$$

v Lebesgue measure element

For an orthogonal curvilinear coordinate system $(a_1, a_2, \dots, a_n)$,

$$\mathrm{d}\mu = \prod_{i=1}^{n} h_{a_i} \hat{\mathbf{e}}\, \mathrm{d}a_i\,.$$

vi Hypersurface elements

For an orthogonal curvilinear coordinate system $(a_1, a_2, \dots, a_n)$, the hypersurface element when a_i is constants is

$$\mathrm{d}\mathbf{S} = \pm \hat{\mathbf{e}}_{a_i} \prod_{\substack{j=1 \\ j \neq i}}^{n} h_{a_j} \prod_{\substack{j=1 \\ j \neq i}}^{n} \mathrm{d}a_j \,,$$

where $\pm$ is $+$ for right-handed and $-$ for left-handed.

II Polar coordinate system (極座標系)

i Definition

A polar coordinate system is a two-dimensional coordinate system that specifies point positions by a point called pole and a polar axis (a reference ray) that starts from the pole. The two polar coordinates (r, θ) are:

- the radial distance $r \in \mathbb{R}_{\geq 0}$ defined as the distance of the point from the origin; and
- the polar angle, azimuth, or azimuthal angle θ defined as any signed angle of the ray from the pole to the point from the polar axis with counterclockwise being positive and clockwise being negative.

For $r < 0$, (r, θ) is defined as $(-r, \theta + \pi)$, which is not canonical.

ii Conversion from and to Cartesian coordinate system

If the pole and polar axis of a polar coordinate system is the origin and the positive direction of the x axis of a Cartesian coordinate system, then:

To:

$$x = r \cos \theta,$$

$$y = r \sin \theta;$$

From:

$$r = \sqrt{x^2 + y^2},$$

$$\theta = \operatorname{arctan2}(y, x) + 2k\pi, \quad k \in \mathbb{Z}.$$

iii Polar rectangle

A polar rectangle is a region in $\mathbb{R}^2$ of the form

$$R = \{(r, \theta) \mid a \leq r \leq b \wedge \alpha \leq \theta \leq \beta\}$$

where $0 \leq a \leq b$ and $0 \leq \beta - \alpha \leq 2\pi$.

iv Local orthogonal basis

Radial unit vector $\hat{\mathbf{e}}_r$ is $(\cos \theta, \sin \theta)$ (Cartesian). Angular (transverse) unit vector $\hat{\mathbf{e}}_\theta$ is $(-\sin \theta, \cos \theta)$ (Cartesian).

v Scaling factors

$$(h_r, h_\theta) = (1, r)$$

vi Line element

$$\mathrm{d}\mathbf{r} = \mathrm{d}r\,\hat{\mathbf{e}}_r + r\,\mathrm{d}\theta\,\hat{\mathbf{e}}_\theta$$

Proof.

$$\begin{aligned}\mathrm{d}\mathbf{r} &= \mathrm{d}(r\hat{\mathbf{e}}_r)\\ &= \mathrm{d}r\,\hat{\mathbf{e}}_r + r\,\mathrm{d}\hat{\mathbf{e}}_r\\ &= \mathrm{d}r\,\hat{\mathbf{e}}_r + r\,\mathrm{d}\theta\,\hat{\mathbf{e}}_\theta\end{aligned}$$

□

vii Area element

$$\mathrm{d}A = r\,\mathrm{d}r\,\mathrm{d}\theta$$

Proof.

$$T:\ (r,\theta) \mapsto (x,y) = (r\cos\theta, r\sin\theta).$$

$$J_T = \begin{pmatrix}\cos\theta & -r\sin\theta\\ \sin\theta & r\cos\theta\end{pmatrix}$$

$$\left|\det(J_T)\right| = r$$

□

Riemann integral:

Let

$$R = \{(r,\theta) \mid a \le r \le b \wedge \alpha \le \theta \le \beta\}, \quad 0 \le a \le b \wedge 0 \le \beta - \alpha \le 2\pi.$$

Then,

$$\iint_R f(x,y)\,\mathrm{d}A = \int_\alpha^\beta \int_a^b f(r\cos\theta, r\sin\theta) r\,\mathrm{d}r\,\mathrm{d}\theta\,.$$

Proof. For each

$$R_{ij} = [r_{i-1}, r_i] \times [\theta_{i-1}, \theta_i],$$

let

$$\begin{aligned}r_i^* &= \frac{1}{2}(r_{i-1} + r_i)\\ \theta_i^* &= \frac{1}{2}(\theta_{i-1} + \theta_i)\\ \Delta A_i &= r_i^* \Delta r \Delta\theta\end{aligned}$$

$$\sum_{i=1}^m \sum_{j=1}^n f(r_i^*\cos\theta_j^*, r_i^*\sin\theta_j^*)\Delta A_i = \sum_{i=1}^m \sum_{j=1}^n f(r_i^*\cos\theta_j^*, r_i^*\sin\theta_j^*) r_i^* \Delta r \Delta\theta$$

□

III Cylindrical coordinate system (圓柱座標系)

i ISO 31-11 Definition

A cylindrical coordinate system is a three-dimensional coordinate system that specifies point positions by a main axis (a chosen directed line) and an auxiliary axis (a reference ray) with the intersection of the two axes called the origin and the plane passing through the origin and perpendicular to the main axis equipped with polar coordinate system with the polar axis being the auxiliary axis called the reference plane. The three cylindrical coordinates (r, θ, z) are:

- the radial distance $r \in \mathbb{R}_{\geq 0}$ defined as the perpendicular distance of the point from the main axis;
- the azimuth or azimuthal angle $\theta \in (-\pi, \pi]$ defined as the polar angle of the projection of the point on the reference plane and undefined on the main axis; and
- the axial coordinate or height $z \in \mathbb{R}_{\geq 0}$ defined as the signed distance of the point from the reference plane where the sign is defined to be positive if the direction is the same as the main axis and negative if the direction is the negation of the direction of the main axis.

ii Conversion from and to Cartesian coordinate system

If the main axis and the auxiliary axis of a cylindrical coordinate system is the z axis and the positive direction of the x axis of a Cartesian coordinate system, then:

To:

$$x = r\cos\theta,$$

$$y = r\sin\theta,$$

$$z = z;$$

From:

$$r = \sqrt{x^2 + y^2},$$

$$\theta = \begin{cases} \operatorname{sgn}(y)\arccos\left(\dfrac{x}{\sqrt{x^2+y^2}}\right), & y \neq 0 \lor x > 0 \\ \pi, & y = 0 \land x < 0 \end{cases},$$

$$z = z.$$

iii Local orthogonal basis

Radial unit vector $\hat{\mathbf{e}}_r$ is $(\cos\theta, \sin\theta, 0)$ (Cartesian). Azimuthal unit vector $\hat{\mathbf{e}}_\theta$ is $(-\sin\theta, \cos\theta, 0)$ (Cartesian). Axial unit vector $\hat{\mathbf{e}}_z$ is $(0, 0, 1)$ (Cartesian).

iv Scaling factors

$$(h_r, h_\theta, h_z) = (1, r, 1)$$

v Line element

$$\mathrm{d}\mathbf{r} = \mathrm{d}r\,\hat{\mathbf{e}}_r + r\,\mathrm{d}\theta\,\hat{\mathbf{e}}_\theta + \mathrm{d}z\,\hat{\mathbf{e}}_z$$

vi Volume element

$$\mathrm{d}V = r\,\mathrm{d}r\,\mathrm{d}\theta\,\mathrm{d}z$$

Proof.

$$T:\ (r, \theta, z) \mapsto (x, y, z) = (r\cos\theta, r\sin\theta, z).$$

$$J_T = \begin{pmatrix} \cos\theta & -r\sin\theta & 0 \\ \sin\theta & r\cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\left|\det(J_T)\right| = r$$

□

vii Surface element

$r = \text{constant}$:

$$\mathrm{d}\mathbf{S} = \hat{\mathbf{e}}_r r\,\mathrm{d}\theta\,\mathrm{d}z$$

$\theta = \text{constant}$:

$$\mathrm{d}\mathbf{S} = \hat{\mathbf{e}}_\theta\,\mathrm{d}r\,\mathrm{d}z$$

$z = \text{constant}$:

$$\mathrm{d}\mathbf{S} = \hat{\mathbf{e}}_z r\,\mathrm{d}r\,\mathrm{d}\theta$$

IV Spherical coordinate system (球座標系)

i Physics convention, ISO 80000-2, or inclination-azimuth convention definition

A spherical coordinate system is a three-dimensional coordinate system that specifies point positions by a polar axis (a chosen directed line) and an auxiliary axis (a reference ray) with the intersection of the two axes called the origin and the plane passing through the origin and perpendicular to the polar axis equipped with polar coordinate system with the polar axis being the auxiliary axis called the reference plane. The three spherical coordinates (r, θ, φ) are:

- the radial distance or radius $r \in \mathbb{R}_{\geq 0}$ defined as the distance of the point from the origin;
- the polar angle, zenith angle, or inclination angle $\theta \in [0, \pi]$ defined as the angle between the ray from the origin to the point and the positive half of the polar axis and undefined at the origin; and
- the azimuth or azimuthal angle $\varphi \in (-\pi, \pi]$ defined as the polar angle of the projection of the point on the reference plane and undefined on the polar axis (of the coordinate system instead of the reference plane).

ii Mathematics convention or azimuth-inclination convention definition

A spherical coordinate system is a three-dimensional coordinate system that specifies point positions by a polar axis (a chosen directed line) and an auxiliary axis (a reference ray) with the intersection of the two axes called the origin and the plane passing through the origin and perpendicular to the polar axis equipped with polar coordinate system with the polar axis being the auxiliary axis called the reference plane. The three spherical coordinates (ρ, θ, φ) are:

- the radial distance or radius $\rho \in \mathbb{R}_{\geq 0}$ defined as the distance of the point from the origin;
- the azimuth or azimuthal angle $\theta \in (-\pi, \pi]$ defined as the polar angle of the projection of the point on the reference plane and undefined on the polar axis (of the coordinate system instead of the reference plane); and
- the polar angle, zenith angle, or inclination angle $\varphi \in [0, \pi]$ defined as the angle between the ray from the origin to the point and the positive half of the polar axis and undefined at the origin.

iii Conversion from and to Cartesian coordinate system (physics convention)

If the polar axis and the auxiliary axis of a spherical coordinate system is the z axis and the positive direction of the x axis of a Cartesian coordinate system, then:

To:

$$x = r \sin\theta \cos\phi,$$

$$y = r \sin\theta \sin\phi,$$

$$z = r \cos\theta;$$

From:

$$r = \sqrt{x^2 + y^2 + z^2},$$

$$\theta = \arccos\left(\frac{z}{r}\right),$$

$$\varphi = \operatorname{atan2}(y, x) = \begin{cases} \operatorname{sgn}(y) \arccos\left(\dfrac{x}{\sqrt{x^2 + y^2}}\right), & y \neq 0 \lor x > 0 \\ \pi, & y = 0 \land x < 0 \end{cases}.$$

iv Conversion from and to Cartesian coordinate system (mathematics convention)

If the polar axis and the auxiliary axis of a spherical coordinate system is the z axis and the positive direction of the x axis of a Cartesian coordinate system, then:

To:

$$x = \rho \sin\varphi \cos\theta,$$

$$y = \rho \sin\varphi \sin\theta,$$

$$z = \rho \cos\varphi;$$

From:

$$\rho = \sqrt{x^2 + y^2 + z^2},$$

$$\theta = \operatorname{atan2}(y, x) = \begin{cases} \operatorname{sgn}(y) \arccos\left(\dfrac{x}{\sqrt{x^2 + y^2}}\right), & y \neq 0 \lor x > 0 \\ \pi, & y = 0 \land x < 0 \end{cases},$$

$$\varphi = \arccos\left(\frac{z}{r}\right).$$

v Conversion from and to cylindrical coordinate system (physics convention)

If the polar axis and the auxiliary axis of a spherical coordinate system is the main axis and the auxiliary axis of a cylindrical coordinate system, then:

To:

$$r = r \sin\theta,$$

$$\theta = \varphi,$$

$$z = r \cos\theta;$$

From:

$$r = \sqrt{r^2 + z^2},$$
$$\theta = \arctan\left(\frac{r}{z}\right),$$
$$\varphi = \theta.$$

vi Conversion from and to cylindrical coordinate system (mathematics convention)

If the polar axis and the auxiliary axis of a spherical coordinate system is the main axis and the auxiliary axis of a cylindrical coordinate system, then:

To:

$$r = \rho \sin\varphi,$$

$$\theta = \theta,$$

$$z = \rho \cos\varphi;$$

From:

$$\rho = \sqrt{r^2 + z^2},$$

$$\theta = \theta,$$

$$\varphi = \arctan\left(\frac{r}{z}\right).$$

vii Local orthogonal basis (physics convention)

Radial unit vector $\hat{\mathbf{e}}_r$ is $(\sin\theta\cos\varphi, \sin\theta\sin\varphi, \cos\theta)$ (Cartesian). Zenith unit vector $\hat{\mathbf{e}}_\theta$ is $(\cos\theta\cos\varphi, \cos\theta\sin\varphi, -\sin\theta)$ (Cartesian). Azimuthal unit vector $\hat{\mathbf{e}}_\varphi$ is $(-\sin\varphi, \cos\varphi, 0)$ (Cartesian).

viii Local orthogonal basis (mathematics convention)

Radial unit vector $\hat{\mathbf{e}}_\rho$ is $(\sin\varphi\cos\theta, \sin\varphi\sin\theta, \cos\varphi)$ (Cartesian). Azimuthal unit vector $\hat{\mathbf{e}}_\theta$ is $(-\sin\theta, \cos\theta, 0)$ (Cartesian). Zenith unit vector $\hat{\mathbf{e}}_\varphi$ is $(\cos\varphi\cos\theta, \cos\varphi\sin\theta, -\sin\varphi)$ (Cartesian).

ix Scale factor (physics convention)

$$(h_r, h_\theta, h_\varphi) = (1, r, r\sin\theta)$$

x Scale factor (mathematics convention)

$$(h_r, h_\theta, h_\varphi) = (1, \rho\sin\varphi, \rho)$$

xi Line element (physics convention)

$$\mathrm{d}\mathbf{r} = \mathrm{d}r\,\hat{\mathbf{e}}_r + r\,\mathrm{d}\theta\,\hat{\mathbf{e}}_\theta + r\sin\theta\,\mathrm{d}\varphi\,\hat{\mathbf{e}}_\varphi$$

xii Line element (mathematics convention)

$$\mathrm{d}\mathbf{r} = \mathrm{d}\rho\,\hat{\mathbf{e}}_\rho + \rho\sin\varphi\,\mathrm{d}\theta\,\hat{\mathbf{e}}_\theta + \rho\,\mathrm{d}\varphi\,\hat{\mathbf{e}}_\varphi$$

xiii Volume element (physics convention)

$$\mathrm{d}V = r^2\sin\theta\,\mathrm{d}r\,\mathrm{d}\theta\,\mathrm{d}\varphi$$

xiv Volume element (mathematics convention)

$$\mathrm{d}V = \rho^2\sin\varphi\,\mathrm{d}r\,\mathrm{d}\theta\,\mathrm{d}\varphi$$

xv Surface elements (physics convention)

$r =$ constant:

$$\mathrm{d}\mathbf{S} = \hat{\mathbf{e}}_r r^2\sin\theta\,\mathrm{d}\theta\,\mathrm{d}\varphi$$

$\theta =$ constant:

$$\mathrm{d}\mathbf{S} = \hat{\mathbf{e}}_\theta r\sin\theta\,\mathrm{d}r\,\mathrm{d}\varphi$$

$\varphi =$ constant:

$$\mathrm{d}\mathbf{S} = \hat{\mathbf{e}}_\varphi r\,\mathrm{d}r\,\mathrm{d}\theta$$

xvi Surface elements (mathematics convention)

$\rho =$ constant:

$$\mathrm{d}\mathbf{S} = \hat{\mathbf{e}}_\rho \rho^2\sin\varphi\,\mathrm{d}\theta\,\mathrm{d}\varphi$$

$\theta =$ constant:

$$\mathrm{d}\mathbf{S} = \hat{\mathbf{e}}_\theta \rho\,\mathrm{d}\rho\,\mathrm{d}\varphi$$

$\varphi =$ constant:

$$\mathrm{d}\mathbf{S} = \hat{\mathbf{e}}_\varphi \rho\sin\varphi\,\mathrm{d}\rho\,\mathrm{d}\theta$$

10 Limit Theorems and Methods

I Uniqueness and Existence

i For real sequence

- A limit inferior must exist uniquely in $[-\infty, \infty)$.
- A limit superior must exist uniquely in $(-\infty, \infty]$.
- If a limit exists, it is unique, and the limit inferior and limit superior are it.

ii For real series

If a limit exists, it is unique.

iii For real-valued net

- A limit inferior must exist uniquely in $[-\infty, \infty)$, and all cluster points are greater than or equal to it.
- A limit superior must exist uniquely in $(-\infty, \infty]$, and all cluster points are less than or equal to it.
- If a limit exists, it is unique, and the limit inferior, limit superior, and only cluster point are it.

iv For real-domain function at a point

- If an one-sided limit exists, it is unique.
- If a limit exists, it is unique, and the left-hand limit and right-hand limit are it.

v For real-valued function from topological space

- A limit inferior must exist uniquely in $[-\infty, \infty)$.
- A limit superior must exist uniquely in $(-\infty, \infty]$.
- If a limit exists, it is unique, and the limit inferior and limit superior are it.

vi For net of functions

- If a pointwise limit function exists, it is unique.
- If a uniform limit function exists, it is unique, and the pointwise limit function is it.

II Limit Laws

If the limit (including one-sided ones for functions with real domains) of sequences, series, nets, or functions with real domains, hereinafter referred to as a function, in one side exists in $\mathbb{R} \cup \{-\infty, \infty\}$, then the limit in the other side exists in $\mathbb{R} \cup \{-\infty, \infty\}$ and follows the following law.

i Linearity, or sum law, difference law, and constant multiple law

The limit of a sum of constant multiples of functions is the sum of the constant multiples of the limits of the functions.

ii Product law

The limit of a product of functions is the product of the limits of the functions.

iii Quotient law

The limit of a quotient of functions is the quotient of the limits of the functions, provided that the limit of the denominator is not 0, in which the reciprocals of ∞ and $-\infty$ are 0.

Thus, if lim $f = \pm\infty$ and lim$(fg) \in \mathbb{R}$, then lim $g = 0$.

iv Power law

The limit of the nth power of a function, in which n is a positive integer, is the nth power of the limit of the function.

v Root law

The limit of the nth root of a function, in which n is a positive integer, is the nth root of the limit of the function.

III Limit inferior and limit superior laws

i Addition law

The limit inferior of a sum of functions is greater than or equal to the sum of the limit inferiors of the functions.

The limit superior of a sum of functions is less than or equal to the sum of the limit superiors of the functions.

ii Constant multiple law

The limit inferior of a constant multiple of a function is the constant multiple of the limit inferior of the function.

The limit superior of a constant multiple of a function is the constant multiple of the limit superior of the function.

iii Multiplication law

The limit inferior of a product of nonnegative real-valued functions is the product of the limit inferiors of the functions.

The limit superior of a product of nonnegative real-valued functions is the product of the limit superiors of the functions.

IV Preservation of equal to

i For real sequence

Given real sequences $\langle a_n \rangle$ and $\langle b_n \rangle$ which for all k that is greater than or equal to a specific $j \in \mathbb{N}$, $a_k = b_k$. If $\lim\limits_{n\to\infty} a_n = L \in \mathbb{R} \cup \{-\infty, \infty\}$, then

$$\lim_{n\to\infty} b_n = L.$$

ii For real series

Given real sequences $\langle a_n \rangle$ and $\langle b_n \rangle$ which for all k that is greater than or equal to a specific $j \in \mathbb{N}$:

$$\sum_{i=1}^{k} a_i = \sum_{i=1}^{k} b_i.$$

If $\sum\limits_{i=1}^{\infty} a_i = L \in \mathbb{R} \cup \{-\infty, \infty\}$, then

$$\sum_{i=1}^{\infty} b_i = L.$$

iii For real-valued net

Let A be a directed set. Given real-valued nets $\langle x_a \rangle_{a\in A}$ and $\langle y_a \rangle_{a\in A}$ which for all $k \in A$ such that there exists $j \in A$ with $j \le k$, $x_k = y_k$. If $\lim\limits_{a} x_a$ exists, then:

$$\lim_{a} x_a = \lim_{a} y_a.$$

iv For real function

Let I be an open interval and $a \in I$, and $f(x)$ and $g(x)$ be functions defined on $I \setminus \{a\}$ which for all $x \in I \wedge x \neq a$, $f(x) = g(x)$. If $\lim\limits_{x\to a} f(x) = L \in \mathbb{R} \cup \{-\infty, \infty\}$, then

$$\lim_{x\to a} g(x) = L.$$

Let $f(x)$ and $g(x)$ be functions defined on (a, b) with $a < b$ which for all $x \in I$, $f(x) = g(x)$. If $\lim\limits_{x\to a^+} f(x) = L \in \mathbb{R} \cup \{-\infty, \infty\}$, then

$$\lim_{x\to a^+} g(x) = L.$$

Let $f(x)$ and $g(x)$ be functions defined on (b, a) with $a > b$ which for all $x \in I$, $f(x) = g(x)$. If $\lim\limits_{x\to a^-} f(x) = L \in \mathbb{R} \cup \{-\infty, \infty\}$, then

$$\lim_{x\to a^-} g(x) = L.$$

Let $f(x)$ and $g(x)$ be functions defined on (a, ∞) which for all $x \in I$, $f(x) = g(x)$. If $\lim\limits_{x\to\infty} f(x) = L \in \mathbb{R} \cup \{-\infty, \infty\}$, then

$$\lim_{x\to\infty} g(x) = L.$$

Let $f(x)$ and $g(x)$ be functions defined on $(-\infty, a)$ which for all $x \in I$, $f(x) = g(x)$. If $\lim_{x\to-\infty} f(x) = L \in \mathbb{R} \cup \{-\infty, \infty\}$, then

$$\lim_{x\to-\infty} g(x) = L.$$

V Preservation of less than or equal to

i For real sequence

Given real sequences $\langle a_n \rangle$ and $\langle b_n \rangle$ which for all k that is greater than or equal to a specific $j \in \mathbb{N}$, $a_k \leq b_k$. If both $\lim_{n\to\infty} a_n$ and $\lim_{n\to\infty} b_n$ are in $\mathbb{R} \cup \{-\infty, \infty\}$, then

$$\lim_{n\to\infty} a_n \leq \lim_{n\to\infty} b_n.$$

If $\lim_{n\to\infty} a_n = \infty$, then $\lim_{n\to\infty} b_n = \infty$; if $\lim_{n\to\infty} b_n = -\infty$, then $\lim_{n\to\infty} a_n = -\infty$.

ii For real series

Given real sequences $\langle a_n \rangle$ and $\langle b_n \rangle$ which for all k that is greater than or equal to a specific $j \in \mathbb{N}$:

$$\sum_{i=1}^{k} a_i \leq \sum_{i=1}^{k} b_i.$$

If both $\sum_{i=1}^{\infty} a_i$ and $\sum_{i=1}^{\infty} b_i$ are in $\mathbb{R} \cup \{-\infty, \infty\}$, then

$$\sum_{i=1}^{\infty} a_i \leq \sum_{i=1}^{\infty} b_i.$$

If $\sum_{i=1}^{\infty} a_i = \infty$, then $\sum_{i=1}^{\infty} b_i = \infty$; if $\sum_{i=1}^{\infty} b_i = -\infty$, then $\sum_{i=1}^{\infty} a_i = -\infty$.

iii For real-valued net

Let A be a directed set. Given real-valued nets $\langle x_a \rangle_{a\in A}$ and $\langle y_a \rangle_{a\in A}$ which for all $k \in A$ such that there exists $j \in A$ with $j \leq k$, $x_k \leq y_k$. If both $\lim_a x_a$ and $\lim_a y_a$ exist, then:

$$\lim_a x_a \leq \lim_a y_a.$$

iv For real function

Let I be an open interval and $a \in I$, and $f(x)$ and $g(x)$ be functions defined on $I \setminus \{a\}$ which for all $x \in I \wedge x \neq a$, $f(x) \leq g(x)$. If both $\lim_{x\to a} f(x)$ and $\lim_{x\to a} g(x)$ are in $\mathbb{R} \setminus \{-\infty, \infty\}$, then

$$\lim_{x\to a} f(x) \leq \lim_{x\to a} g(x).$$

If $\lim_{x\to a} f(x) = \infty$, then $\lim_{x\to a} g(x) = \infty$; if $\lim_{x\to a} g(x) = -\infty$, then $\lim_{x\to a} f(x) = -\infty$.

Let $f(x)$ and $g(x)$ be functions defined on (a, b) with $a < b$ which for all $x \in I$, $f(x) \leq g(x)$. If both $\lim_{x \to a^+} f(x)$ and $\lim_{x \to a^+} g(x)$ are in $\mathbb{R} \cup \{-\infty, \infty\}$, then

$$\lim_{x \to a^+} f(x) \leq \lim_{x \to a^+} g(x).$$

If $\lim_{x \to a^+} f(x) = \infty$, then $\lim_{x \to a^+} g(x) = \infty$; if $\lim_{x \to a^+} g(x) = -\infty$, then $\lim_{x \to a^+} f(x) = -\infty$.

Let $f(x)$ and $g(x)$ be functions defined on (b, a) with $a > b$ which for all $x \in I$, $f(x) \leq g(x)$. If both $\lim_{x \to a^-} f(x)$ and $\lim_{x \to a^-} g(x)$ are in $\mathbb{R} \cup \{-\infty, \infty\}$, then

$$\lim_{x \to a^-} f(x) \leq \lim_{x \to a^-} g(x).$$

If $\lim_{x \to a^-} f(x) = \infty$, then $\lim_{x \to a^-} g(x) = \infty$; if $\lim_{x \to a^-} g(x) = -\infty$, then $\lim_{x \to a^-} f(x) = -\infty$.

Let $f(x)$ and $g(x)$ be functions defined on (a, ∞) which for all $x \in I$, $f(x) \leq g(x)$. If both $\lim_{x \to \infty} f(x)$ and $\lim_{x \to \infty} g(x)$ are in $\mathbb{R} \cup \{-\infty, \infty\}$, then

$$\lim_{x \to \infty} f(x) \leq \lim_{x \to \infty} g(x).$$

If $\lim_{x \to \infty} f(x) = \infty$, then $\lim_{x \to \infty} g(x) = \infty$; if $\lim_{x \to \infty} g(x) = -\infty$, then $\lim_{x \to \infty} f(x) = -\infty$.

Let $f(x)$ and $g(x)$ be functions defined on $(-\infty, a)$ which for all $x \in I$, $f(x) \leq g(x)$. If both $\lim_{x \to -\infty} f(x)$ and $\lim_{x \to -\infty} g(x)$ are in $\mathbb{R} \cup \{-\infty, \infty\}$, then

$$\lim_{x \to -\infty} f(x) \leq \lim_{x \to -\infty} g(x).$$

If $\lim_{x \to -\infty} f(x) = \infty$, then $\lim_{x \to -\infty} g(x) = \infty$; if $\lim_{x \to -\infty} g(x) = -\infty$, then $\lim_{x \to -\infty} f(x) = -\infty$.

VI Squeeze (夾擠) theorem or Sandwich (三明治) theorem

i For real sequence

Given real sequences $\langle a_n \rangle$, $\langle b_n \rangle$, and $\langle c_n \rangle$ which for all k that is greater than or equal to a specific $j \in \mathbb{N}$:

$$a_k \leq c_k \leq b_k.$$

If

$$\lim_{n \to \infty} a_n = \lim_{n \to \infty} b_n = L \in \mathbb{R} \cup \{-\infty, \infty\},$$

then

$$\lim_{n \to \infty} c_n = L.$$

ii For real series

Given real sequences $\langle a_n \rangle$, $\langle b_n \rangle$, and $\langle c_n \rangle$ which for all k that is greater than or equal to a specific $j \in \mathbb{N}$:

$$\sum_{i=1}^{k} a_i \leq \sum_{i=1}^{k} c_i \leq \sum_{i=1}^{k} b_i.$$

If

$$\sum_{i=1}^{\infty} a_i = \sum_{i=1}^{\infty} b_i = L \in \mathbb{R} \cup \{-\infty, \infty\},$$

then:

$$\sum_{i=1}^{\infty} c_i = L.$$

iii For real-valued net

Let A be a directed set. Given real-valued nets $\langle x_a \rangle_{a \in A}$, $\langle y_a \rangle_{a \in A}$, and $\langle z_a \rangle_{a \in A}$ which for all $k \in A$ such that there exists $j \in A$ with $j \leq k$:

$$x_k \leq z_k \leq y_k.$$

If

$$\lim_a x_a = \lim_a y_a = L,$$

then:

$$\lim_a z_a = L.$$

Proof. Start from the limits of x_a and y_a, by definition of limit:

For every $\varepsilon \in \mathbb{R}_{>0}$, there exists some $a_1 \in A$ such that for every $b \in A$ with $b \geq a_1$, the point $L - \varepsilon \leq x_b \leq L + \varepsilon$.

For every $\varepsilon \in \mathbb{R}_{>0}$, there exists some $a_2 \in A$ such that for every $b \in A$ with $b \geq a_2$, the point $L - \varepsilon \leq y_b \leq L + \varepsilon$.

Choose $a \in A$ such that $a_1 \leq a$ and $a_2 \leq a$.

so,

$$L - \varepsilon \leq x_a \leq z_a \leq y_a \leq L + \varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, by definition of limit:

$$\lim_a z_a = L.$$

□

iv For real function

Let I be an open interval and $a \in I$, and $f(x)$, $g(x)$, and $h(x)$ be functions defined on $I \setminus \{a\}$ which for all $x \in I \land x \neq a$, $f(x) \leq h(x) \leq g(x)$. If

$$\lim_{x \to a} f(x) = \lim_{x \to a} g(x) = L \in \mathbb{R} \setminus \{-\infty, \infty\},$$

then

$$\lim_{x \to a} h(x) = L.$$

Let $f(x)$, $g(x)$, and $h(x)$ be functions defined on (a, b) with $a < b$ which for all $x \in I$, $f(x) \le h(x) \le g(x)$. If

$$\lim_{x\to a^+} f(x) = \lim_{x\to a^+} g(x) = L \in \mathbb{R} \setminus \{-\infty, \infty\},$$

then

$$\lim_{x\to a^+} h(x) = L.$$

Let $f(x)$, $g(x)$, and $h(x)$ be functions defined on (b, a) with $a > b$ which for all $x \in I$, $f(x) \le h(x) \le g(x)$. If

$$\lim_{x\to a^-} f(x) = \lim_{x\to a^-} g(x) = L \in \mathbb{R} \setminus \{-\infty, \infty\},$$

then

$$\lim_{x\to a^-} h(x) = L.$$

Let $f(x)$, $g(x)$, and $h(x)$ be functions defined on (a, ∞) which for all $x \in I$, $f(x) \le h(x) \le g(x)$. If

$$\lim_{x\to \infty} f(x) = \lim_{x\to \infty} g(x) = L \in \mathbb{R} \setminus \{-\infty, \infty\},$$

then

$$\lim_{x\to \infty} h(x) = L.$$

Let $f(x)$, $g(x)$, and $h(x)$ be functions defined on $(-\infty, a)$ which for all $x \in I$, $f(x) \le h(x) \le g(x)$. If

$$\lim_{x\to -\infty} f(x) = \lim_{x\to -\infty} g(x) = L \in \mathbb{R} \setminus \{-\infty, \infty\},$$

then

$$\lim_{x\to -\infty} h(x) = L.$$

Proof. Take the first statement for example. The other statements can be proven similarly.

Start from the limits of f and g, by definition of limit:

$$\forall \varepsilon > 0, \exists \delta_1 > 0 \text{ s.t. } 0 < |x-a| < \delta_1 \implies |f(x) - L| < \varepsilon,$$

and

$$\forall \varepsilon > 0, \exists \delta_2 > 0 \text{ s.t. } 0 < |x-a| < \delta_2 \implies |g(x) - L| < \varepsilon,$$

Choose:

$$\delta \coloneqq \min\left(\delta_1, \delta_2\right),$$

so,

$$L - \varepsilon < f(x) \le h(x) \le g(x) < L + \varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, by definition of limit:

$$\lim_{x\to a} h(x) = L.$$

$\square$

VII Limit of real-valued net theorems

i Continuity principle

If real-valued net $\langle x_a \rangle_{a \in A}$ with $\lim_a x_a = L \in \mathbb{R}$, then for any real-valued function f continuous at L,

$$\lim_a f(x_a) = f(L).$$

If real-valued net $\langle x_a \rangle_{a \in A}$ with $\lim_a x_a = \pm\infty$, then for any real-valued function f such that $\lim_{x \to \pm\infty} f(x) = L \in \mathbb{R} \cup \{-\infty, \infty\}$,

$$\lim_a f(x_a) = L.$$

ii Monotone Convergence Theorem (MCT) (單調收斂定理)

For any nonempty, non-decreasing net of real numbers $\langle a_n \rangle_{n \in A}$, it is bounded-above iff it converges to a real number, and its limit is its supremum.

Proof. $\Rightarrow$ direction:

Let $\{a_n\}$ be the set of values of $\langle a_n \rangle_{n \in A}$. Assume that $\{a_n\}$ is nonempty and bounded-above by $c = \sup_n a_n$.

We have

$$\forall \varepsilon > 0 : \exists M \in A \text{ s.t. } c \geq a_M > c - \varepsilon,$$

since otherwise $c - \varepsilon$ is a strictly smaller upper bound of $\langle a_n \rangle$, contradicting the definition of the supremum.

Then since $\langle a_n \rangle$ is non-decreasing, and c is an upper bound:

$$\forall \varepsilon > 0 : \exists M \in A \text{ s.t. } \forall n \geq M : |c - a_n| = c - a_n \leq c - a_M = |c - a_M| < \varepsilon.$$

$\Leftarrow$ direction:

Assume that $\lim_n a_n = L \in \mathbb{R}$. Assume for contradiction that for some $M \in A$,

$$a_M > L.$$

Let

$$\varepsilon = \frac{a_M - L}{2} > 0.$$

Since

$$\forall n \geq M : a_n \geq a_M > L,$$

we have

$$\forall n \geq M : |a_n - L| = (a_n - a_M) + (a_M - L) \geq a_M - L > \varepsilon.$$

This contradicts that

$$\lim_n a_n = L.$$

$\square$

For any nonempty, non-increasing net of real numbers $\langle a_n \rangle_{n\in A}$, it is bounded-below iff it converges to a real number, and its limit is its infimum.

The specialization of this theorem on sequences is also called monotonic sequence theorem.

VIII Limit of real sequence theorems

i Bolzano–Weierstrass Theorem (波爾查諾-魏爾斯特拉斯定理)

Every bounded sequence in an Euclidean space $\mathbb{R}^n$ has a convergent subsequence.

Proof. For $\mathbb{R}$, take any bounded sequence (x_n) in $\mathbb{R}$. Then there exist real numbers m, M such that

$$m \le x_n \le M, \quad \forall n.$$

So the sequence lies in the closed interval $[m, M]$.

For $\varepsilon = 1$. Divide $[m, M]$ into disjoint subintervals with length $\dfrac{M-m}{2^\varepsilon}$:

$$\left[m, \frac{m+M}{2}\right], \left[\frac{m+M}{2}, M\right].$$

Since (x_n) is infinite, by the pigeonhole principle, one of the subintervals must contain infinitely many terms of (x_n). Denote it by I_1 and pick the subsequence $\left(x_n^{(1)}\right)$ of (x_n) that is entirely in I_1.

Repeat it for $\varepsilon = 2, 3, \ldots$, by dividing $I_{\varepsilon-1}$ into two halves, we can construct a nested sequence of subsequences

$$\left(x_n^{(1)}\right) \supseteq \left(x_n^{(2)}\right) \supseteq \ldots$$

where $\left(x_n^{(k)}\right)$ lies in a closed interval of length $\dfrac{M-m}{2^k}$, I_k.

Pick the diagonal sequence $y_k = x_k^{(k)}$. Then

$$y_k \in \left(x_n^{(k)}\right) \subseteq I_k,$$

and the sequence (y_k) is a convergent subsequence of (x_n) because

$$\forall N \in \mathbb{N}: \ m, n \ge N \implies |y_m - y_n| \le \frac{M-m}{2^N}.$$

To generalized to $\mathbb{R}^n$, consider each coordinate sequence in $\mathbb{R}$, it has a convergent subsequence. Diagonalization gives a subsequence converging in all coordinates. □

ii Abel transformation or summation by parts

Let $\langle f_k \rangle$ and $\langle g_k \rangle$ be two sequences.

$$\sum_{k=m}^{n} f_k \left(g_{k+1} - g_k\right) = \left(f_{n+1}g_{n+1} - f_m g_m\right) - \sum_{k=m}^{n} g_{k+1} \left(f_{k+1} - f_k\right).$$

$$\sum_{k=m}^{n} f_k \left(g_k - g_{k-1}\right) = \left(f_n g_n - f_{m-1}g_{m-1}\right) - \sum_{k=m}^{n} g_{k-1} \left(f_k - f_{k-1}\right).$$

$$\sum_{k=m}^{n} \left(f_k \left(g_{k+1} - g_k\right) + g_k \left(f_k - f_{k-1}\right)\right) = f_n g_{n+1} - f_{m-1} g_m.$$

IX Limit of real function theorems

i Limits involving quotient functions

Let a and b be real numbers, set

$$A = \left\{ f : U \subseteq \mathbb{R} \to \mathbb{R} \middle| f(x) = x \vee \ln(f(x)) \in A \vee e^{f(x)} \in A \right\},$$

and function $f \in A$. Then:

$$\lim_{x\to\infty} \frac{(f(x))^a}{(f(x))^b} = \infty, \quad a > b$$

$$\lim_{x\to\infty} \frac{af(x)}{bf(x)} = \frac{a}{b}, \quad b \neq 0$$

$$\lim_{x\to\infty} \frac{n^{af(x)}}{bf(x)} = \infty, \quad a, b > 0 \wedge n > 1$$

$$\lim_{x\to\infty} \frac{n^{af(x)}}{bf(x)} = 0, \quad a, b > 0 \wedge 0 \leq n < 1$$

$$\lim_{x\to\infty} \frac{af(x)}{b\log_n f(x)} = \infty, \quad a, b > 0 \wedge n > 1$$

ii BV function has limit

If function $f : [a, b) \to \mathbb{R}$ with $b \in (a, \infty]$ is of bounded variation on $[a, b)$, then $\lim\limits_{x\to b^-} f(x)$ exists in $\mathbb{R}$.

Proof. PLACEHOLDER □

X Limit of function between normed spaces theorem

i Different paths criterion for nonexistence of a limit

Let V, W be two normed spaces. Let I be an open subset of V containing the point a. Let f be a function with codomain W defined on I, except possibly at a itself. Let $C(t)$, $D(t)$ be two paths in $I \cup \{a\}$ over $[0, \tau]$ for some positive real number τ such that $C(0) = D(0) = a$. If

$$\lim_{t\to 0} f(C(t)) \neq \lim_{t\to 0} f(D(t))$$

or either $\lim\limits_{t\to 0} f(C(t))$ or $\lim\limits_{t\to 0} f(D(t))$ doesn't exist, then

$$\lim_{x\to a} f(x)$$

doesn't exist.

ii Polar coordinate method

In $\mathbb{R}^2$, for any normed vector-valued function $f(x, y)$, if there exists a constant L in the codomain of f such that

$$\lim_{r\to 0} f(r\cos\theta, r\sin\theta) = L$$

then

$$\lim_{(x,y)\to(0,0)} f(x, y) = L.$$

iii Cylindrical coordinate method

In $\mathbb{R}^3$, for any normed vector-valued function $f(x, y, z)$, if there exists a constant L in the codomain of f such that

$$\lim_{(\rho,z)\to(0,0)} f(\rho\cos\varphi, \rho\sin\varphi, z) = L$$

then

$$\lim_{(x,y,z)\to(0,0,0)} f(x, y, z) = L.$$

iv Spherical coordinate method

In $\mathbb{R}^3$, for any normed vector-valued function $f(x, y, z)$, if there exists a constant L in the codomain of f such that

$$\lim_{r\to 0} f(r\sin\theta\cos\phi, r\sin\theta\sin\phi, r\cos\theta) = L$$

then

$$\lim_{(x,y,z)\to(0,0,0)} f(x, y, z) = L.$$

v Generalized polar coodinate method

Suppose there is a coordinate of n real number components that can express each point in $\mathbb{R}^n$ uniquely, for any fixed point $\mathbf{a} = (a_1, a_2, \dots, a_n) \in \mathbb{R}^n$, if the coordinate is such that, for any point $\mathbf{x} = (x_1, x_2, \dots, x_n)$ in a neighborhood of $\mathbf{a}$, for any real numbers $y_1, y_2, \dots, y_m$ with $m \leq n$ such that $\mathbf{y} = (x_1 + y_1, x_2 + y_2, \dots, x_m + y_m, x_{m+1}, x_{m+2}, \dots, x_n)$ is a point in $\mathbb{R}^n$,

$$\|\mathbf{x} - \mathbf{a}\| = \left\|\mathbf{y} - \mathbf{a}\right\|,$$

then for any normed vector-valued function f defined on some neighborhood of $\mathbf{a}$, if there exists a constant L in the codomain of f such that

$$\lim_{(x_{m+1},x_{m+2},\dots,x_n)\to(a_{m+1},a_{m+2},\dots,a_n} f(\mathbf{x}) = L$$

then,

$$\lim_{\mathbf{x}\to\mathbf{a}} f(\mathbf{x}) = L.$$

vi Homogenous decomposition

For a neighborhood D of a point $\mathbf{a} \in \mathbb{R}^n$ and functions $P, Q : D \subseteq \mathbb{R}^n \to \mathbb{R}$ that are analytic at some neighborhood of $\mathbf{a}$,

- If $Q(\mathbf{a}) \neq 0$,
$$\lim_{\mathbf{x}\to\mathbf{a}} \frac{P(\mathbf{x})}{Q(\mathbf{x})} = \frac{P(\mathbf{a})}{Q(\mathbf{a})}.$$
- If $Q(\mathbf{a}) = 0$ and $P(\mathbf{a}) \neq 0$, the limit doesn't exists.

- If $P(\mathbf{a}) = Q(\mathbf{a}) = 0$, use homogenous decomposition. Let

$$P_i(\mathbf{x}) = \frac{P^{(i)}(\mathbf{a})}{i!}\mathbf{x}^i,$$

$$Q_i(\mathbf{x}) = \frac{Q^{(i)}(\mathbf{a})}{i!}\mathbf{x}^i,$$

  which are homogeneous. Suppose that k is the smallest integer such that P_k is nonzero, and m is the smallest integer such that Q_m is nonzero.

$$\lim_{\mathbf{x}\to\mathbf{a}} \frac{P(\mathbf{x})}{Q(\mathbf{x})} = \lim_{\mathbf{x}\to\mathbf{a}} \frac{P_k(\mathbf{x}-\mathbf{a})}{Q_m(\mathbf{x}-\mathbf{a})}$$
$$= \lim_{\mathbf{x}\to\mathbf{0}} \|\mathbf{x}\|^{k-m} \frac{P_k\left(\hat{\mathbf{x}}\right)}{Q_m\left(\hat{\mathbf{x}}\right)}$$

  - If $k - m > 0$, the limit is 0.
  - If $k - m < 0$, the limit doesn't exists.
  - If $k = m$, if

$$\frac{P_k\left(\hat{\mathbf{x}}\right)}{Q_m\left(\hat{\mathbf{x}}\right)}$$

    is a constant $L \in \mathbb{R}^m$, then the limit is L; otherwise the limit doesn't exist.

XI $\left(1 + \frac{a_n}{n}\right)^n$

For a sequence $(a_n)_{n=1}^{\infty}$, if $\lim_{n\to\infty} a_n = a \in \mathbb{R}$, then

$$\lim_{n\to\infty}\left(1 + \frac{a_n}{n}\right)^n = e^a.$$

Proof. Define

$$x_n = \frac{a_n}{n}$$
$$\lim_{n\to\infty} x_n = 0$$
$$\lim_{x\to 0}\frac{\ln(1+x)}{x} = 1$$
$$\lim_{n\to\infty} n\ln\left(1 + x_n\right) = \lim_{n\to\infty} a_n = a$$
$$\lim_{n\to\infty}\left(1 + \frac{a_n}{n}\right)^n = \lim_{n\to\infty} \exp\left(n\ln\left(1 + x_n\right)\right) = e^a$$

□

11 Continuity Theorems

I Basic laws

i Arithmetic Laws

If f and g are continuous at a and c is a constant, then the following functions are also continuous at a:

$$f + g; \quad f - g; \quad cf; \quad fg;$$

$$\frac{f}{g} \text{ if } g(a) \neq 0.$$

ii Composte Laws

If g is continuous at a and f is continuous at $g(a)$, then the composite function $f \circ g$ is continuous at a.

II Intermediate Value Theorem (IVT) (中間值定理)

i For real functions

Let $f : I \subseteq \mathbb{R} \to \mathbb{R}$ be a function, then for any interval $[a, b] \subseteq I$ such that f is continuous on $[a, b]$ and $f(a) \neq f(b)$, N is strictly between $f(a)$ and $f(b)$ implies there exists $c \in (a, b)$ such that $f(c) = N$.

Proof. Without loss of generality, assume $f(a) < N < f(b)$.

Consider the set

$$S = \{x \in [a, b] \mid f(x) \leq N\}.$$

Since $f(a) < N$, $a \in S$; since $f(b) > N$, $b \notin S$. So S is nonempty and bounded-above by b.

Let $c = \sup S$. We claim that $f(c) = N$. Argue by contradiction.

Assume $f(c) < N$. By continuity, there exists $\delta > 0$ such that for all $x \in (c, c + \delta) \cap [a, b]$, $|f(x) - f(c)| < N - f(c)$. So $f(x) < N$ for some $x > c$. Contradicting that c is the least upper bound.

Assume $f(c) > N$. By continuity, there exists $\delta > 0$ such that for all $x \in (c - \delta, c) \cap [a, b]$, $|f(x) - f(c)| < f(c) - N$. So $f(x) > N$ for some $x < c$. Contradicting that c is the least upper bound.

Therefore, $f(c) = N$. □

ii For functions between topological spaces

Let X and Y be topological spaces and function $f : I \subseteq X \to Y$ be continuous on a subset J of I, then for any connected subset K of J, $f(K)$ is a connected subset of Y.

Let X and Y be topological spaces and function $f : I \subseteq X \to Y$ be continuous on a subset J of I, then for any path-connected subset K of J, $f(K)$ is a path-connected subset of Y.

III Monotone function have at most countably many discontinuities

Suppose a real-valued function f is monotone on an interval (a, b), then it has at most countably many discontinuities on (a, b).

Proof. Suppose f is non-decreasing on (a, b). For any $x \in (a, b)$, $f(x^-)$ and $f(x^+)$ both exist and $f(x^-) \leq f(x) \leq f(x^+)$. For any discontinuity x, assign a rational index $r_x \in (f(x^-), f(x^+))$. This is always possible since $(f(x^-), f(x^+))$ is nonempty and $\mathbb{Q}$ is dense in $\mathbb{R}$. The cardinality of the set of cardinality is the same as that of the set of the indices, which is at most $|\mathbb{Q}| = \aleph_0$. □

IV Uniform continuity implies continuity

i For functions between metric spaces

PLACEHOLDER

ii For functions between topological vector spaces

PLACEHOLDER

V Absolute continuity implies uniform continuity for functions from an interval to a metric space

PLACEHOLDER

VI Lipschitz continuity theorems

i Lipschitz continuity implies absolute continuity for functions from an interval to a metric space

PLACEHOLDER

ii Local Lipschitz continuity on compact set implies global Lipschitz continuity on it

PLACEHOLDER

iii Banach fixed-point theorem (巴拿赫不動點定理), contraction mapping theorem (壓縮映射定理), contraction principle, or Banach–Caccioppoli theorem

For any contraction mapping T over a non-empty complete metric space X with best contraction constant K, there must exist a unique fixed-point x^* of T in X, and for any $x \in X$, for any sequence $\langle x_n \rangle_{n \in \mathbb{N}_0}$ defined as

$$\begin{cases} x_0 = x \\ x_n = T\left(x_{n-1}\right), n \in \mathbb{N} \end{cases},$$

$$\lim_{n \to \infty} x_n = x^*$$

and

$$d(x_n, x^*) \leq \frac{K^n}{1-K} d(x_1, x_0).$$

12 Derivative Theorems and Methods

I Differentiation Theorems and Methods

i Linearity, or sum rule, difference rule, and constant multiple rule

If functions f and g are differentiable on I and $a, b \in \mathbb{R}$, then $af(x) + bg(x)$ is also differentiable on I and its derivative is given by

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(af(x) + bg(x)\right) = af'(x) + bg'(x).$$

Proof.

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}x}\left(af(x) + bg(x)\right) &= \lim_{h\to 0}\frac{af(x+h) + bg(x+h) - af(x) - bg(x)}{h}\\ &= af'(x) + bg'(x)\end{aligned}$$

□

ii Product rule (乘法定則)

If functions f and g are differentiable on I, then the product fg is also differentiable on I and its derivative is given by

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(f(x)g(x)\right) = f'(x)g(x) + f(x)g'(x).$$

Proof.

$$\begin{aligned}\frac{\mathrm{d}f(x)g(x)}{\mathrm{d}x} &= \lim_{h\to 0}\frac{f(x+h)g(x+h) - f(x)g(x)}{h}\\ &= \lim_{h\to 0}\frac{f(x+h)(g(x+h) - g(x)) + g(x)(f(x+h) - f(x))}{h}\\ &= f(x)g'(x) + f'(x)g(x).\end{aligned}$$

□

iii General Leibniz rule

If functions f and g are n-times differentiable on I, then the product fg is also n-times differentiable on I and its derivative is given by

$$(fg)^{(n)} = \sum_{k=0}^{n}\binom{n}{k} f^{(n-k)} g^{(k)}.$$

iv Product rule for dot product

If functions f and g are differentiable on $I \subseteq \mathbb{R}^n$, then the dot product $f \cdot g$ is also differentiable on I and its derivative is given by

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(f(x) \cdot g(x)\right) = f'(x) \cdot g(x) + f(x) \cdot g'(x).$$

Proof.

$$
\begin{aligned}
\frac{\mathrm{d} f(x)\cdot g(x)}{\mathrm{d}x} &= \lim_{h\to 0}\frac{f(x+h)\cdot g(x+h) - f(x)\cdot g(x)}{h} \\
&= \lim_{h\to 0}\frac{f(x+h)\cdot (g(x+h)-g(x)) + g(x)\cdot (f(x+h)-f(x))}{h} \\
&= f(x)\cdot g'(x) + f'(x)\cdot g(x).
\end{aligned}
$$

□

v Product rule for cross product

If functions f and g are differentiable on $I \subseteq \mathbb{R}^3$, then the cross product $f \times g$ is also differentiable on I and its derivative is given by

$$\frac{\mathrm{d}}{\mathrm{d}x}(f(x)\times g(x)) = f'(x)\times g(x) + f(x)\times g'(x).$$

Proof.

$$
\begin{aligned}
\frac{\mathrm{d} f(x)\times g(x)}{\mathrm{d}x} &= \lim_{h\to 0}\frac{f(x+h)\times g(x+h) - f(x)\times g(x)}{h} \\
&= \lim_{h\to 0}\frac{f(x+h)\times (g(x+h)-g(x)) + g(x)\times (f(x+h)-f(x))}{h} \\
&= f(x)\times g'(x) + f'(x)\times g(x).
\end{aligned}
$$

□

vi Quotient rule (除法定則)

If functions f and g are differentiable on I and $g \neq 0$ on I, then the quotient $\frac{f}{g}$ is also differentiable on I and its derivative is given by

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{f(x)}{g(x)}\right) = \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2}.$$

Proof.

$$
\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}x}\frac{f(x)}{g(x)} &= \lim_{h\to 0}\frac{\frac{f(x+h)}{g(x+h)} - \frac{f(x)}{g(x)}}{h} \\
&= \lim_{h\to 0}\frac{f(x+h)g(x) - f(x)g(x+h)}{h\,g(x+h)g(x)} \\
&= \lim_{h\to 0}\frac{(f(x+h)-f(x))g(x) - f(x)(g(x+h)-g(x))}{h\,g(x+h)g(x)} \\
&= \lim_{h\to 0}\frac{f(x+h)-f(x)}{h\,g(x+h)} - f(x)\lim_{h\to 0}\frac{g(x+h)-g(x)}{h\,g(x+h)g(x)} \\
&= \frac{f'(x)g(x) - f(x)g'(x)}{g(x)^2}
\end{aligned}
$$

□

vii Chain rule (連鎖律) of real functions

Let $f : X \subseteq \mathbb{R} \to \mathbb{R}$ and $g : Y \subseteq \mathbb{R} \to \mathbb{R}$ be functions, g be differentiable on $I \subseteq \mathbb{R}$, and f be differentiable on $g(I)$. Then the composite $f \circ g$ is differentiable on I and its derivative on I is given by

$$\frac{\mathrm{d}}{\mathrm{d}x}\left((f\circ g)(x)\right) = f'\left(g(x)\right)g'(x).$$

Proof.

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}x}\left((f\circ g)(x)\right) &= \lim_{h\to 0}\frac{f\left(g(x+h)\right)-f\left(g(x)\right)}{h}\\ &= \lim_{h\to 0}\frac{f\left(g(x+h)\right)-f\left(g(x)\right)}{g(x+h)-g(x)}\cdot\frac{g(x+h)-g(x)}{h}\\ &= f'\left(g(x)\right)g'(x)\end{aligned}$$

□

viii Multivariable chain rule

Suppose

$$F\left(t,\mathbf{s}_1,\mathbf{s}_2,\ldots,\mathbf{s}_n\right) = f\left(x_1(t,\mathbf{s}_1),x_2(t,\mathbf{s}_2),\ldots,x_n(t,\mathbf{s}_n)\right),$$

then

$$\frac{\partial F}{\partial t} = \sum_{i=1}^{n}\frac{\partial f}{\partial x_i}\frac{\partial x_i}{\partial t}.$$

ix Faà di Bruno's formula

$$\begin{aligned}&\frac{\mathrm{d}^n}{\mathrm{d}x}f\left(g(x)\right)\\ &= \sum_{\sum_{i=1}^{n} i m_i = n,\quad m_i\in\mathbb{N}_0}\frac{n!}{\prod_{j=1}^{n} m_j!}\cdot\\ &f^{\left(\sum_{j=1}^{n} m_j\right)}\left(g(x)\right)\cdot\\ &\prod_{j=1}^{n}\left(\frac{g^{(j)}(x)}{j!}\right)^{m_j}.\end{aligned}$$

x Chain rule of real vector functions

Let $f : X\subseteq\mathbb{R}^n\to\mathbb{R}^p$ and $g : Y\subseteq\mathbb{R}^m\to\mathbb{R}^n$ be functions, g be differentiable on $\Omega\subseteq\mathbb{R}^m$, and f be differentiable on $g(\Omega)$. Then the composite $f\circ g$ is differentiable on Ω and its derivative on Ω is given by

$$\nabla\left(f\circ g\right)(\mathbf{x}) = (\nabla f)(g(\mathbf{x}))\cdot(\nabla g)(\mathbf{x}).$$

xi Exponential Bell polynomials

Partially or incomplete exponential Bell polynomials $B_{n,k}\left(x_1,x_2,\ldots,x_{n-k+1}\right)$ is defined as

$$B_{n,k}\left(x_1,x_2,\ldots,x_{n-k+1}\right) = n!\sum_{\langle j_i\rangle\in C}\prod_{i=1}^{n-k+1}\frac{x_i^{\,j_i}}{(i!)^{j_i}j_i!},$$

where

$$C = \left\{\langle j_i\rangle_{i=1}^{n-k+1}\,\middle|\,\sum_{i=1}^{n-k+1} = k\wedge\sum_{i=1}^{n-k+1} i j_i = n\right\}.$$

Complete exponential Bell polynomial $B_n\left(x_1, x_2, \ldots, x_n\right)$ is defined as

$$B_n\left(x_1, x_2, \ldots, x_n\right) = n! \sum_{k=0}^{n} B_{n,k}\left(x_1, x_2, \ldots, x_{n-k+1}\right).$$

xii Derivative of inverse functions

$$\left(f^{-1}\right)'(x) = \frac{1}{f'\left(f^{-1}(x)\right)}.$$

Proof.

$$f\left(f^{-1}(x)\right) = x.$$

Differentiate both sides:

$$f'\left(f^{-1}(x)\right)\left(f^{-1}\right)'(x) = 1.$$

□

$$\left(f^{-1}\right)''(x) = \frac{f''\left(f^{-1}(x)\right)}{\left(f'\left(f^{-1}(x)\right)\right)^3}.$$

Proof.

$$f'\left(f^{-1}(x)\right)\left(f^{-1}\right)'(x) = 1.$$

Differentiate both sides:

$$f''\left(f^{-1}(x)\right)\left(\left(f^{-1}\right)'(x)\right)^2 + f'\left(f^{-1}(x)\right)\left(f^{-1}\right)''(x) = 0.$$

$$\left(f^{-1}\right)''(x) = \frac{f''\left(f^{-1}(x)\right)\left(\left(f^{-1}\right)'(x)\right)^2}{f'\left(f^{-1}(x)\right)} = \frac{f''\left(f^{-1}(x)\right)}{\left(f'\left(f^{-1}(x)\right)\right)^3}.$$

□

$$\frac{\mathrm{d}^n f^{-1}(x)}{\mathrm{d}x^n} = \frac{(-1)^{n-1}}{\left(f'\left(f^{-1}(x)\right)\right)^{2n-1}} B_{n-1} \sum_{i=2}^{n} f^{(i)}\left(f^{-1}(x)\right), \quad n \in \mathbb{N},$$

where B_n is complete exponential Bell polynomial.

xiii Jacobian of inverse

Let $\mathbf{T}: I \subseteq \mathbb{R}^n \to \mathbb{R}^n$ be injective and differentiable on $D \subseteq I$, and $\mathbf{T}^{-1}$ be differentiable on $\mathbf{T}(D)$. Then,

$$J(\mathbf{T})J(\mathbf{T}^{-1}) = \mathbf{I}$$

where $\mathbf{I}$ is the $n \times n$ unit matrix.

Proof. PLACEHOLDER □

xiv Implicit function theorem (隱函數定理) of two real variables

Given a function $f(x, y)$ is a function that is continuously differentiable on a neighborhood of a point (x_0, y_0), $f(x_0, y_0) = 0$, and $\dfrac{\partial f}{\partial y}\left(x_0, y_0\right) \neq 0$ on a neighborhood of (x_0, y_0), then there exists a unique differentiable function g such that $y_0 = g(x_0)$ and $f(x, g(x)) = 0$ on a neighborhood of x_0. Moreover, by differentiating both sides of $f(x, g(x)) = 0$, called implicit differentiation, we obtain

$$g'(x) = -\frac{\frac{\partial f}{\partial x}(x, g(x))}{\frac{\partial f}{\partial y}(x, \varphi(x))}.$$

xv Implicit function theorem of multiple real variables

Below, we will denote $(\mathbf{x}, \mathbf{y})$ as an element of $\mathbb{R}^{(n+m)}$ for some $\mathbf{x} \in \mathbb{R}^n$ and $\mathbf{y} \in \mathbb{R}^m$, and denote the left n columns of $(Df)(\mathbf{a}, \mathbf{b})$ as $J_{f,\mathbf{x}}$, and the right m columns of $(Df)(\mathbf{a}, \mathbf{b})$ as $J_{f,\mathbf{y}}$. Given a function $f : D \subseteq \mathbb{R}^{(n+m)} \to \mathbb{R}^m$ that is continuously differentiable on a neighborhood of a point $(\mathbf{a}, \mathbf{b})$, $f(\mathbf{a}, \mathbf{b}) = \mathbf{0}$, and $J_{f,\mathbf{y}}(\mathbf{a}, \mathbf{b})$ is invertible, then there exists a unique function g such that $\mathbf{b} = g(\mathbf{a})$ and $f(\mathbf{x}, g(\mathbf{x})) = \mathbf{0}$ on a neighborhood of $\mathbf{a}$. Moreover, by differentiating both sides of $f(\mathbf{x}, g(\mathbf{x})) = \mathbf{0}$, called implicit differentiation, we obtain

$$(Dg)(\mathbf{x}) = -\left[J_{f,\mathbf{y}}(\mathbf{x}, g(\mathbf{x}))\right]^{-1}\left[J_{f,\mathbf{x}}(\mathbf{x}, g(\mathbf{x}))\right],$$

or for each component x_i of $\mathbf{x}$,

$$\frac{\partial g}{\partial x_i} = -\left[J_{f,\mathbf{y}}(\mathbf{x}, g(\mathbf{x}))\right]^{-1}\frac{\partial f}{\partial x_i}.$$

xvi Derivative of natural logarithm of absolute value of function

If functions f is differentiable and nonzero on I, then $\ln|f(x)|$ is also differentiable on I and its derivative is given by

$$\frac{\mathrm{d}\ln|f(x)|}{\mathrm{d}x} = \frac{f'(x)}{f(x)}.$$

xvii Logarithmic differentiation

Logarithmic differentiation is a method for finding derivatives by taking the natural logarithm of (the absolute values of) both sides of an equation first, and then differentiating.

It’s especially useful when the function involves products or quotients of many factors, or variables in both the base and the exponent.

xviii Derivative of variable base, constant exponent funcion

If $n \in \mathbb{R}$ and $f(x)$ is differentiable on I, then $(f(x))^n$ is also differentiable on I and its derivative is given by

$$\frac{\mathrm{d}}{\mathrm{d}x}(f(x))^n = n(f(x))^{n-1} f'(x).$$

xix Derivative of constant base, variable exponent funcion

If $n \in \mathbb{R}$ and $f(x)$ is differentiable on I, then $n^{f(x)}$ is also differentiable on I and its derivative is given by

$$\frac{\mathrm{d}}{\mathrm{d}x} n^{f(x)} = n^{f(x)} \left(\ln n\right) f'(x).$$

xx Derivative of variable base, variable exponent funcion

$$\frac{\mathrm{d}}{\mathrm{d}x} (f(x))^{g(x)} = (f(x))^{g(x)} \left(g'(x) \ln(f(x)) + g(x) \frac{f'(x)}{f(x)} \right).$$

Proof.

$$y := (f(x))^{g(x)}$$

Take natural logarithm of both sides:

$$\ln y = g(x) \ln(f(x)).$$

Take differentiation of both sides:

$$\frac{y'}{y} = g'(x) \ln(f(x)) + g(x) \frac{f'(x)}{f(x)}.$$

$$y' = (f(x))^{g(x)} \left(g'(x) \ln(f(x)) + g(x) \frac{f'(x)}{f(x)} \right).$$

□

II Existence

i Monotone function is differentiable a.e.

If a real function f is monotone on an interval I, it is differentiable a.e. on I.

Proof. PLACEHOLDER □

ii Existence of total derivative

For a function $f : D \subseteq \mathbb{R}^n \to \mathbb{R}^m$, f' exists at a point $a \in D$ iff

$$\lim_{h \to 0} \frac{f(a+h) - L(a+h)}{\|h\|} = 0,$$

where L is the linearization of f at a, that is,

$$L(x) = f(a) + \nabla f(a)(x - a),$$

and the derivative of f at a is $\nabla f(a)$.

Proof. $f'(a)$ exists iff there exists a bounded linear operator $A : \mathbb{R}^n \to \mathbb{R}^m$ such that

$$\lim_{h \to 0} \frac{f(a+h) - f(a) - A(h)}{\|h\|} = 0.$$

Suppose the limit is 0. For each x_i, let the unit vector along its positive direction be e_i, along the line $x_i t, \quad t \in \mathbb{R}$,

$$\lim_{t \to 0} \frac{f(a + e_i t) - f(a) - A(e_i t)}{t} = \frac{\partial f}{\partial x_i}(a) - A(e_i)$$

Thus $A_i = \dfrac{\partial f}{\partial x_i}(a)$, and

$$A = \nabla f(a)$$

And

$$\frac{f(a+h) - f(a) - A(h)}{\|h\|} = \frac{f(a+h) - f(a) + \nabla f(a)(h)}{\|h\|} = \frac{f(a+h) - L(a+h)}{\|h\|}$$

□

III Graph-related

i Differentiability implies continuity

Let V and W be normed vector spaces, U be an open subset of V, and $f : U \subseteq V \to W$ be a function. For a point $a \in U$, f is differentiable at a implies it is continuous at a; for an open subset $I \subseteq U$, f is differentiable on I implies it is continuous on I.

Proof. f is differentiable at a iff there exists bounded linear operator $A : V \to W$ such that

$$\lim_{\|h\|_V \to 0} \frac{\|f(a+h) - f(a) - A(h)\|_W}{\|h\|_V} = 0,$$

that is,

$$\forall \varepsilon > 0 : \exists \delta > 0 \text{ s.t. } \|h\|_V < \delta \implies \|f(a+h) - f(a) - A(h)\|_W \le \varepsilon \|h\|_V.$$

By the triangular identity,

$$\|f(a+h) - f(a)\|_W \le \|A(h)\|_W + \|f(a+h) - f(a) - A(h)\|_W.$$

By the definition of bounded linear operator, there exists $C \in \mathbb{R}_{>0}$ such that

$$\|A(h)\|_W \le C\|h\|_V.$$

$$\|f(a+h) - f(a)\|_W \le (C + \varepsilon)\|h\|_V.$$

Thus,

$$\forall \varepsilon > 0 : \exists \delta' = \frac{\varepsilon}{C + \varepsilon} \text{ s.t. } \|h\|_V < \delta' \implies \|f(a+h) - f(a)\|_W \le (C + \varepsilon)\|h\|_V < (C + \varepsilon)\delta' = \varepsilon.$$

□

ii Extreme Value Theorem (EVT) (極值定理)

Let function $f : I \subseteq \mathbb{R} \to \mathbb{R}$ be continuous on an interval $[a, b] \subseteq I$, then

$$\exists c, d \in [a, b] \text{ s.t. } \forall x \in [a, b] : f(c) \ge f(x) \ge f(d).$$

Proof. We first claim that f is bounded on $[a,b]$. Argue by contradiction. Suppose f is not bounded-above.

For any $n \in \mathbb{N}$, there exists $x_n \in [a,b]$ such that $f\left(x_n\right) > n$. This defines a sequence $\langle x_n \rangle$.

By the Bolzano-Weierstrass Theorem and that $[a,b]$ is bounded, it has a convergent subsequence x_{n_k} that converges to some limit L. Since $[a,b]$ is closed, $L \in [a,b]$.

Since f is continuous on $[a,b]$, the sequence $f\left(x_{n_k}\right)$ converges to $f(L)$, which is finite.

This contradicts $f\left(x_n\right) > n$ for every $n \in \mathbb{N}$. So the set

$$S = \{f(x) \mid x \in [a,b]\}.$$

must be bounded-above.

By considering $-f$, we obtain that S is also bounded-below.

By the Completeness of the Real Numbers, every non-empty set bounded-above has a least upper bound; every non-empty set bounded-below has a greatest lower bound.

Let

$$M = \sup S, \quad m = \inf S.$$

Since M is the least upper bound of S, for any $n \in \mathbb{N}$, $M - \frac{1}{n}$ is not an upper bound of S. This means that for each $n \in \mathbb{N}$, there exists a point $y_n \in [a,b]$ such that:

$$M - \frac{1}{n} < f(y_n) \leq M.$$

This defines another sequence $\langle y_n \rangle$ within $[a,b]$.

By the Bolzano-Weierstrass Theorem and that $[a,b]$ is bounded, it has a convergent subsequence y_{n_k} that converges to some limit c. Since $[a,b]$ is closed, $c \in [a,b]$.

$$\lim_{n_k \to \infty} \left(M - \frac{1}{n_k}\right) = M.$$

By the Squeeze Theorem:

$$\lim_{n_k \to \infty} f\left(y_{n_k}\right) = M.$$

Since f is continuous on $[a,b]$, the sequence $f\left(y_{n_k}\right)$ converges to $f(c)$.

Similarly, we can prove that there exists $d \in [a,b]$ such that $f(d) = m$. □

iii Extreme Value Theorem (EVT) for compact set

Let X be a topological space, and function $f : I \subset X \to \mathbb{R}$ be continuous on a compact set $D \subseteq I$, then

$$\exists a, b \in D \text{ s.t. } \forall x \in D: \ f(a) \geq f(x) \geq f(b).$$

iv Fermat's Theorem (費馬引理) or Interior Extremum Theorem (內部極值定理)

Let X be a LCTVS and function $f: D \subseteq X \to \mathbb{R}$ be differentiable on an open set $U \subseteq D$ and $c \in U$ be a local extreme of f, then

$$f'(c) = 0,$$

where $f'(c) \in L(X, \mathbb{R})$ and 0 is the zero element of $L(X, \mathbb{R})$.

Proof. Let $h \in X$ be arbitrary.

Because U is open, there exists $\varepsilon > 0$ such that $c + th \in U$ for $|t| < \varepsilon$. Define

$$g(t) = f(c + th), \quad |t| < \varepsilon.$$

Since f is differentiable on U, g is differentiable and

$$g'(0) = f'(c)(h).$$

Without loss of generality, suppose c is a local maximum. Then, there exists $\delta > 0$ such that

$$g(t) \leq g(0)$$

for all $|t| < \delta$.

Thus for all $\delta > t > 0$,

$$\frac{g(t) - g(0)}{t} \leq 0$$

and for all $-\delta < t < 0$,

$$\frac{g(t) - g(0)}{t} \geq 0.$$

Therefore

$$g'(0) = 0.$$

Since

$$g'(0) = f'(c)(h) = 0$$

for all $h \in X$

$$f'(c) = 0.$$

□

In terms of critical point, Fermat's Theorem can be rephrased as, a local extreme is a critical point.

v Rolle's Theorem (羅爾定理)

Let function $f: I \subseteq \mathbb{R} \to \mathbb{R}$ be continuous on an interval $[a, b] \subseteq I$ and differentiable on (a, b), and $f(a) = f(b)$, then

$$\exists c \in (a, b) \text{ s.t. } f'(c) = 0.$$

Proof. By the Extreme Value Theorem, f must have maximum and minimum on $[a, b]$.

Let

$$M = \max_{x\in[a,b]} f(x), \quad m = \min_{x\in[a,b]} f(x).$$

If $M = m$, then f is constant on $[a, b]$, then $f'(x) = 0$ for any $c \in (a, b)$.

Otherwise, $M > m$. Because $f(a) = f(b)$.

We claim that there must be $c \in (a, b)$ such that $M = f(c)$ or $m = f(c)$. Argue by contradiction. If the claim is false, then $M = m = f(a) = f(b)$, contradicting $M > m$.

By Fermat's Theorem,

$$f'(c) = 0.$$

□

vi Mean Value Theorem (MVT) (均值定理) (for Derivatives)

Let function $f : I \subseteq \mathbb{R} \to \mathbb{R}$ be continuous on an interval $[a, b] \subseteq I$ and differentiable on (a, b) with $a < b$, then

$$\exists c \in (a, b) \text{ s.t. } f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Proof. Let

$$\phi(x) = f(x) - \frac{f(b) - f(a)}{b - a}(x - a).$$
$$\phi(a) = \phi(b) = f(a).$$

By Rolle's Theorem, there exists $c \in (a, b)$ such that $\phi'(c) = 0$.

$$\phi'(c) = f'(c) - \frac{f(b) - f(a)}{b - a}.$$

□

vii Cauchy's Mean Value Theorem (CMVT) (柯西均值定理)

Let functions $f : I \subseteq \mathbb{R} \to \mathbb{R}$ and $g : J \subseteq \mathbb{R} \to \mathbb{R}$ be continuous on an interval $[a, b] \subseteq I \cap J$ and differentiable on (a, b) with $a < b$, then

$$\exists c \in (a, b) \text{ s.t. } (f(b) - f(a))\, g'(c) = (g(b) - g(a))\, f'(c).$$

Proof. If $f(b) - f(a) = g(b) - g(a) = 0$, it holds. For the other cases, without loss of generality, assume $g(b) - g(a) \neq 0$.

Define a function $\phi(x)$

$$\phi(x) = f(x) - \frac{f(b) - f(a)}{g(b) - g(a)} g(x).$$
$$\phi(b) - \phi(a) = (f(b) - f(a)) - \frac{f(b) - f(a)}{g(b) - g(a)} (g(b) - g(a)) = 0.$$

By Rolle's Theorem, there exists $c \in (a, b)$ such that $\phi'(c) = 0$.

$$\phi'(x) = f'(x) - \frac{f(b) - f(a)}{g(b) - g(a)} g'(x).$$

□

viii Generalized Racetrack principle

If $f'(x) > g'(x)$ for all $x \in (a, b)$ with $b \in \mathbb{R}_{>a} \cup \{\infty\}$, and $f(a) = g(a)$, then $f(x) > g(x)$ for all $x \in (a, b)$.

If $f'(x) > g'(x)$ for all $x \in (a, b)$ with $b \in \mathbb{R}_{>a} \cup \{\infty\}$, f is continuous on $(a, b]$, and $f(a) = g(a)$, then $f(x) > g(x)$ for all $x \in (a, b]$.

If $f'(x) \geq g'(x)$ for all $x \in (a, b)$ with $b \in \mathbb{R}_{>a} \cup \{\infty\}$, and $f(a) = g(a)$, then $f(x) \geq g(x)$ for all $x \in (a, b]$.

If $f'(x) \geq g'(x)$ for all $x \in (a, b)$ with $b \in \mathbb{R}_{>a} \cup \{\infty\}$, f is continuous on $(a, b]$, and $f(a) = g(a)$, then $f(x) \geq g(x)$ for all $x \in (a, b]$.

Proof. For the first and second statements, define a function

$$h(x) = f(x) - g(x).$$

$$h'(x) = f'(x) - g'(x) > 0, \quad \forall x \in (a, b).$$

$$h(a) = f(a) - g(a) = 0.$$

By MVT, for all $x \in (a, b)$ for the first statement and $x \in (a, b]$ for the second statement, there exists $c \in (a, x)$ such that

$$\frac{h(x) - h(a)}{x - a} = h'(c) > 0.$$

$$f(x) - g(x) = h(x) = (x - a)h'(c) > 0.$$

The other statements can be proven similarly. □

ix Increasing/Decreasing Test (I/D Test) Theorem

Let function $f : I \subseteq \mathbb{R} \to \mathbb{R}$ be differentiable on an open interval $(a, b) \subseteq I$.

- if $f'(x) > 0$ for any $x \in (a, b)$, then f is strictly increasing on (a, b).
- If $f'(x) > 0$ for any $x \in (a, b)$ and f is continuous on $[a, b)$, then f is strictly increasing on $[a, b)$.
- If $f'(x) > 0$ for any $x \in (a, b)$ and f is continuous on $(a, b]$, then f is strictly increasing on $(a, b]$.
- if $f'(x) < 0$ for any $x \in (a, b)$, then f is strictly decreasing on (a, b).
- If $f'(x) < 0$ for any $x \in (a, b)$ and f is continuous on $[a, b)$, then f is strictly decreasing on $[a, b)$.
- If $f'(x) < 0$ for any $x \in (a, b)$ and f is continuous on $(a, b]$, then f is strictly decreasing on $(a, b]$.
- $f'(x) \geq 0$ for any $x \in (a, b)$ iff f is non-decreasing on (a, b).
- If $f'(x) \geq 0$ for any $x \in (a, b)$ and f is continuous on $[a, b)$, then f is non-decreasing on $[a, b)$.

- If $f'(x) \geq 0$ for any $x \in (a,b)$ and f is continuous on $(a,b]$, then f is non-decreasing on $(a,b]$.
- $f'(x) \leq 0$ for any $x \in (a,b)$ iff f is non-increasing on (a,b).
- If $f'(x) \leq 0$ for any $x \in (a,b)$ and f is continuous on $[a,b)$, then f is non-increasing on $[a,b)$.
- If $f'(x) \leq 0$ for any $x \in (a,b)$ and f is continuous on $(a,b]$, then f is non-increasing on $(a,b]$.

Proof. For the first statement, we want to show that $\forall c,d \in (a,b) : c < d \implies f(c) < f(d)$ if $f'(x) > 0$ for any $x \in (a,b)$.

Take any two points $c,d \in (a,b)$ with $c < d$. By MVT, there exists $g \in (c,d)$ such that

$$f'(g) = \frac{f(d) - f(c)}{d - c}.$$

Because $f'(g) > 0$ and $d - c > 0$, we get $f(d) - f(c) > 0$.

The second and third statements are obviously true as the first holds.

The "if" directions of the other statements can be proven similarly.

For the "only if" direction of the seventh statement, we want to show that $f'(x) \geq 0$ if $\forall c,d \in (a,b) : c < d \implies f(c) \leq f(d)$ and f is differentiable on (a,b).

Fix any $x \in (a,b)$. For any $h > 0$ with $(x+h) \in (a,b)$, we have

$$f(x) \leq f(x+h).$$

For any $h < 0$ with $(x+h) \in (a,b)$, we have

$$f(x) \geq f(x+h).$$

Thus for any h with $(x+h) \in (a,b)$, we have

$$\frac{f(x+h) - f(x)}{h} \geq 0.$$

The "only if" direction of the tenth statement can be proven similarly. □

x First Derivative Test for Local Extrema Theorem

Let $f : I \subseteq \mathbb{R} \to \mathbb{R}$ be a function, c be a critical point of f, and f' be continuous at c.

- If exist $a < c < b$ such that $f'(c) > 0$ for any $x \in (a,c)$ and $f'(c) < 0$ for any $x \in (c,b)$, then f has a strict local maximum at c.
- If exist $a < c < b$ such that $f'(c) \geq 0$ for any $x \in (a,c)$ and $f'(c) \leq 0$ for any $x \in (c,b)$, then f has a local maximum at c.
- If exist $a < c < b$ such that $f'(c) < 0$ for any $x \in (a,c)$ and $f'(c) > 0$ for any $x \in (c,b)$, then f has a strict local minimum at c.
- If exist $a < c < b$ such that $f'(c) \leq 0$ for any $x \in (a,c)$ and $f'(c) \geq 0$ for any $x \in (c,b)$, then f has a local minimum at c.

- If exist $a < c < b$ such that $f'(c) > 0$ for any $x \in (a, b)$, then f has no local exteme at c.
- If exist $a < c < b$ such that $f'(c) < 0$ for any $x \in (a, b)$, then f has no local ext eme at c.

Proof. By Increasing/Decreasing Test Theorem, it is obvious. □

xi Second derivative test of one variable

Let $f : I \subseteq \mathbb{R} \to \mathbb{R}$ be a function, c be a critical point of f, and f be twice differentiable at c.

- If $f'' < 0$, then f has a strict local maximum at c.
- If $f'' > 0$, then f has a strict local minimum at c.

xii Second (partial) derivative test of two variables

Let $f(x, y) : I \subseteq \mathbb{R}^2 \to \mathbb{R}$ be a function, c be a critical point of f, f be twice differentiable at c, and

$$D = \begin{vmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{vmatrix}.$$

- If $D(c) > 0$ and $f_{xx}(c) < 0$, then f has a strict local maximum at c.
- If $D(c) > 0$ and $f_{xx}(c) > 0$, then f has a strict local minimum at c.
- If $D(c) < 0$, then f has a saddle point at c.

xiii Second derivative test of multiple derivatives

Let $f : I \subseteq \mathbb{R}^n \to \mathbb{R}$ be a function, c be a critical point of f, and f be twice differentiable at c.

- If $H(f)$ is negative definite, then f has a strict local maximum at c.
- If $H(f)$ is postive definite, then f has a strict local minimum at c.
- If $H(f)$ is negative semidefinite, then f has a local maximum at c.
- If $H(f)$ is postive semidefinite, then f has a local minimum at c.
- If $H(f)$ is indefinite, then f has a saddle point at c.

xiv First Derivative Test for Concavity Theorem

Let $f : I \subseteq \mathbb{R} \to \mathbb{R}$ be a function and $(a, b) \subseteq I$ be an open interval.

- If f' is strictly increasing on (a, b), often called "f lies strictly below all its tagent lines on (a, b)", then f is strictly concave upward on (a, b).
- If f' is strictly increasing on (a, b) and f is continuous on $[a, b)$, then f is strictly concave upward on $[a, b)$.
- If f' is strictly increasing on (a, b) and f is continuous on $(a, b]$, then f is strictly concave upward on $(a, b]$.

- If f' is strictly decreasing on I, often called "f lies strictly above all its tagent lines on (a,b)", then f is strictly concave downward on (a,b).
- If f' is strictly decreasing on (a,b) and f is continuous on $[a,b)$, then f is strictly concave downward on $[a,b)$.
- If f' is strictly decreasing on (a,b) and f is continuous on $(a,b]$, then f is strictly concave downward on $(a,b]$.
- f' is non-decreasing on (a,b), often called "f lies weakly below all its tagent lines on (a,b)", iff f is weakly concave upward on (a,b).
- If f' is non-decreasing on (a,b) and f is continuous on $[a,b)$, then f is weakly concave upward on $[a,b)$.
- If f' is non-decreasing on (a,b) and f is continuous on $(a,b]$, then f is weakly concave upward on $(a,b]$.
- f' is non-increasing on (a,b), often called "f lies weakly above all its tagent lines on (a,b)", iff f is weakly concave downward on (a,b).
- If f' is non-increasing on (a,b) and f is continuous on $[a,b)$, then f is weakly concave downward on $[a,b)$.
- If f' is non-increasing on (a,b) and f is continuous on $(a,b]$, then f is weakly concave downward on $(a,b]$.

Proof. For the first statement, we want to show that for any $c,d \in (a,b)$ and $\alpha \in (0,1)$, $f((1-\alpha)c+\alpha d) < (1-\alpha)f(c)+\alpha f(d)$ if for any $a \le A < B \le b$, $f'(A) < f'(B)$.

Fix any $a \le c < x < d \le b$. By MVT, there exists $A \in (c,x)$ such that

$$f'(A) = \frac{f(x)-f(c)}{x-c}.$$

By MVT, there exists $B \in (b,c)$ such that

$$f'(B) = \frac{f(d)-f(x)}{d-x}.$$

Since $A < B$, $f'(A) \le f'(B)$.

$$\frac{f(x)-f(c)}{x-c} \le \frac{f(d)-f(x)}{d-x}.$$

$$(f(x)-f(c))(d-x) \le (f(d)-f(x))(x-c).$$

$$(d-c)f(x) \le (d-x)f(c)+(x-c)f(d).$$

$$f(x) \le \frac{(d-x)}{(d-c)}f(c) + \frac{(x-c)}{(d-c)}f(d).$$

Set

$$\alpha = \frac{(x-c)}{(d-c)} \in (0,1).$$

Then

$$x = (1-\alpha)c + \alpha d.$$

Thus the inequality becomes

$$f((1-\alpha)c + \alpha d) \leq (1-\alpha)f(c) + \alpha f(d).$$

For the second statement, argue by contradiction. Assume that there exists $d \in (a, b)$ and $\alpha \in (0, 1)$ such that

$$f((1-\alpha)a + \alpha d) \geq (1-\alpha)f(a) + \alpha f(d).$$

Let $x = (1-\alpha)a + \alpha d$.

$$f(x) - f(a) \geq \alpha(f(d) - f(a)).$$

By MVT, there exists $B \in (a, x)$ such that

$$f(x) - f(a) = f'(B)(x-a) = f'(B)\alpha(d-a).$$

By MVT, there exists $c \in (x, d)$ such that

$$f(d) - f(x) = f'(C)(d-x) = f'(C)(1-\alpha)(d-a).$$

$$f(d) - f(a) = (f'(B)\alpha + f'(C)(1-\alpha))(d-a).$$

$$f(x) - f(a) = f'(B)\alpha(d-a) \geq \alpha(f'(B)\alpha + f'(C)(1-\alpha))(d-a).$$

$$f'(B) \geq f'(B)\alpha + f'(C)(1-\alpha).$$

$$f'(B) \geq f'(C).$$

This contradicts the condition that $B < C \implies f'(B) < f'(C)$ for any $a < B < C < b$.

The third statement can be proven similarly.

The "if" directions of the other statements can be proven similarly.

For the "only if" direction of the seventh statement, we want to show that for any $c, d \in (a, b)$ and $\alpha \in (0, 1)$, $f((1-\alpha)c + \alpha d) \leq (1-\alpha)f(c) + \alpha f(d)$ if $\forall c, d \in (a, b):\ c < d \implies f'(c) \leq f'(d)$.

Fix any $c, d \in (a, b)$ with $c < d$.

By MVT, there exists $g \in (c, ((1-\alpha)c + \alpha d))$ such that

$$f((1-\alpha)c + \alpha d) - f(c) = f'(g)((1-\alpha)c + \alpha d - c) = f'(g)\alpha(d-c).$$

By MVT, there exists $h \in (((1-\alpha)c + \alpha d)), d)$ such that

$$f(d) - f((1-\alpha)c + \alpha d) = f'(h)(d - (1-\alpha)c - \alpha d) = f'(h)(1-\alpha)(d-c).$$

Thus,

$$f((1-\alpha)c + \alpha d) = f(c) + f'(g)\alpha(d-c) = f(d) - f'(h)(1-\alpha)(d-c).$$

$$f'(g)\alpha + f'(h)(1-\alpha) = \frac{f(d) - f(c)}{d-c}.$$

Because $f'(g) \leq f'(h)$, we have

$$f'(g)(d-c) \leq f(d) - f(c).$$

Substitute:

$$f((1-\alpha)c + \alpha d) \leq f(c) + \alpha(f(d) - f(c)) = (1-\alpha)f(c) + \alpha f(d).$$

The "only if" direction of the tenth statement can be proven similarly. □

xv Concavity Test Theorem or Second Derivative Test for Concavity Theorem

Let function $f: I \subseteq \mathbb{R} \to \mathbb{R}$ be twice differentiable on an open interval $(a, b) \subseteq I$.

- If $f'' > 0$ for any $x \in (a, b)$, then the graph of f is strictly concave upward on (a, b).
- If $f'' > 0$ for any $x \in (a, b)$ and f' is continuous on $[a, b)$, then the graph of f is strictly concave upward on $[a, b)$.
- If $f'' > 0$ for any $x \in (a, b)$ and f' is continuous on $(a, b]$, then the graph of f is strictly concave upward on $(a, b]$.
- If $f'' < 0$ for any $x \in (a, b)$, then the graph of f is strictly concave downward on (a, b).
- If $f'' < 0$ for any $x \in (a, b)$ and f' is continuous on $[a, b)$, then the graph of f is strictly concave downward on $[a, b)$.
- If $f'' < 0$ for any $x \in (a, b)$ and f' is continuous on $(a, b]$, then the graph of f is strictly concave downward on $(a, b]$.
- $f'' \geq 0$ for any $x \in (a, b)$ iff the graph of f is weakly concave upward on (a, b).
- If $f'' \geq 0$ for any $x \in (a, b)$ and f' is continuous on $[a, b)$, then the graph of f is weakly concave upward on $[a, b)$.
- If $f'' \geq 0$ for any $x \in (a, b)$ and f' is continuous on $(a, b]$, then the graph of f is weakly concave upward on $(a, b]$.
- $f'' \leq 0$ for any $x \in (a, b)$ iff the graph of f is weakly concave downward on (a, b).
- If $f'' \leq 0$ for any $x \in (a, b)$ and f' is continuous on $[a, b)$, then the graph of f is weakly concave downward on $[a, b)$.
- If $f'' \leq 0$ for any $x \in (a, b)$ and f' is continuous on $(a, b]$, then the graph of f is weakly concave downward on $(a, b]$.

Proof. By Increasing/Decreasing Test Theorem and First Derivative Test for Concavity Theorem, it is obvious. □

xvi Inflection Point Theorem

Let $f: I \subseteq \mathbb{R} \to \mathbb{R}$ be a function. If $(c, f(c))$ is an inflection point of f and f is differentiable at c, then c is a critical point of f'.

Proof. By the First Derivative Test for Concavity Theorem, it is obvious. □

xvii Extremum of direction derivatives

For each point $\mathbf{x}$,

$$\max_{\|\mathbf{u}\|=1} D_{\mathbf{u}}\mathbf{A}(\mathbf{x}) = \|\nabla \mathbf{A}(\mathbf{x})\|,$$

$$\arg_{\mathbf{u}} \max_{\|\mathbf{u}\|=1} D_{\mathbf{u}}\mathbf{A}(\mathbf{x}) = \frac{\nabla \mathbf{A}(\mathbf{x})}{\|\nabla \mathbf{A}(\mathbf{x})\|},$$

$$\min_{\|\mathbf{u}\|=1} D_{\mathbf{u}}\mathbf{A}(\mathbf{x}) = -\|\nabla\mathbf{A}(\mathbf{x})\|,$$

$$\arg_{\mathbf{u}}\min_{\|\mathbf{u}\|=1} D_{\mathbf{u}}\mathbf{A}(\mathbf{x}) = -\frac{\nabla\mathbf{A}(\mathbf{x})}{\|\nabla\mathbf{A}(\mathbf{x})\|}.$$

IV Indeterminate forms and L'Hôpital's rule (羅必達法則) or Bernoulli's rule

i (General form of) L'Hôpital's rule or Bernoulli's rule and indeterminate quotients

Let I be an open interval and $a \in I$, and $f(x)$ and $g(x)$ be functions differentiable on $I \setminus \{a\}$. If

- $\lim_{x\to a} f(x) = \lim_{x\to a} g(x) = 0$, where $\lim_{x\to a}\frac{f(x)}{g(x)}$ is called indeterminate form of type $\frac{0}{0}$, or $\left|\lim_{x\to a} f(x)\right| = \left|\lim_{x\to a} g(x)\right| = \infty$, where $\lim_{x\to a}\frac{f(x)}{g(x)}$ is called indeterminate form of type $\frac{\infty}{\infty}$, and
- $\lim_{x\to a}\frac{f'(x)}{g'(x)} \in \mathbb{R}\cup\{-\infty,\infty\}$,

then:

$$\lim_{x\to a}\frac{f(x)}{g(x)} = \lim_{x\to a}\frac{f'(x)}{g'(x)}.$$

Let $f(x)$ and $g(x)$ be functions differentiable on (a,b) with $a < b$. If

- $\lim_{x\to a^+} f(x) = \lim_{x\to a^+} g(x) = 0$, where $\lim_{x\to a^+}\frac{f(x)}{g(x)}$ is called indeterminate form of type $\frac{0}{0}$, or $\left|\lim_{x\to a^+} f(x)\right| = \left|\lim_{x\to a^+} g(x)\right| = \infty$, where $\lim_{x\to a^+}\frac{f(x)}{g(x)}$ is called indeterminate form of type $\frac{\infty}{\infty}$, and
- $\lim_{x\to a^+}\frac{f'(x)}{g'(x)} \in \mathbb{R}\cup\{-\infty,\infty\}$,

then:

$$\lim_{x\to a^+}\frac{f(x)}{g(x)} = \lim_{x\to a^+}\frac{f'(x)}{g'(x)}.$$

Let $f(x)$ and $g(x)$ be functions differentiable on (b,a) with $a > b$. If

- $\lim_{x\to a^-} f(x) = \lim_{x\to a^-} g(x) = 0$, where $\lim_{x\to a^-}\frac{f(x)}{g(x)}$ is called indeterminate form of type $\frac{0}{0}$, or $\left|\lim_{x\to a^-} f(x)\right| = \left|\lim_{x\to a^-} g(x)\right| = \infty$, where $\lim_{x\to a^-}\frac{f(x)}{g(x)}$ is called indeterminate form of type $\frac{\infty}{\infty}$, and
- $\lim_{x\to a^-}\frac{f'(x)}{g'(x)} \in \mathbb{R}\cup\{-\infty,\infty\}$,

then:

$$\lim_{x\to a^-}\frac{f(x)}{g(x)} = \lim_{x\to a^-}\frac{f'(x)}{g'(x)}.$$

Let $f(x)$ and $g(x)$ be functions differentiable on (a,∞). If

- $\lim_{x\to\infty} f(x) = \lim_{x\to\infty} g(x) = 0$, where $\lim_{x\to\infty}\frac{f(x)}{g(x)}$ is called indeterminate form of type $\frac{0}{0}$, or $\left|\lim_{x\to\infty} f(x)\right| = \left|\lim_{x\to\infty} g(x)\right| = \infty$, where $\lim_{x\to\infty}\frac{f(x)}{g(x)}$ is called indeterminate form of type $\frac{\infty}{\infty}$, and

- $\lim_{x\to\infty} \dfrac{f'(x)}{g'(x)} \in \mathbb{R} \cup \{-\infty, \infty\}$,

then:

$$\lim_{x\to\infty} \frac{f(x)}{g(x)} = \lim_{x\to\infty} \frac{f'(x)}{g'(x)}.$$

Let $f(x)$ and $g(x)$ be functions differentiable on $(-\infty, a)$. If

- $\lim_{x\to-\infty} f(x) = \lim_{x\to-\infty} g(x) = 0$, where $\lim_{x\to-\infty} \dfrac{f(x)}{g(x)}$ is called indeterminate form of type $\dfrac{0}{0}$, or $\left|\lim_{x\to-\infty} f(x)\right| = \left|\lim_{x\to-\infty} g(x)\right| = \infty$, where $\lim_{x\to-\infty} \dfrac{f(x)}{g(x)}$ is called indeterminate form of type $\dfrac{\infty}{\infty}$, and
- $\lim_{x\to-\infty} \dfrac{f'(x)}{g'(x)} \in \mathbb{R} \cup \{-\infty, \infty\}$,

then:

$$\lim_{x\to-\infty} \frac{f(x)}{g(x)} = \lim_{x\to-\infty} \frac{f'(x)}{g'(x)}.$$

Proof. Let $f(x)$ and $g(x)$ be functions differentiable on (a, b) with $a < b$. We want to show that if $\lim_{x\to a^+} f(x) = \lim_{x\to a^+} g(x) = 0$ and $\lim_{x\to a^+} \dfrac{f'(x)}{g'(x)} \in \mathbb{R} \cup \{-\infty, \infty\}$, then

$$\lim_{x\to a^+} \frac{f(x)}{g(x)} = \lim_{x\to a^+} \frac{f'(x)}{g'(x)}.$$

Because the existence of the limits, we know that there exists interval (a, c) such that $g(x)$ and $g'(x)$ both $\neq 0$ on (a, c). Let

$$F(x) = \begin{cases} f(x), & x \in (a, c) \\ 0, & x = a \end{cases}$$

and

$$G(x) = \begin{cases} g(x), & x \in (a, c) \\ 0, & x = a \end{cases}$$

By CMVT, for any $x \in (a, c)$, there exists $c \in (a, x)$ such that

$$(F(x) - F(a))\, G'(c) = (G(x) - G(a))\, F'(c).$$

Since $F(a) = G(a) = 0$,

$$F(x)G'(c) = G(x)F'(c).$$

Thus,

$$\lim_{x\to a^+} \frac{f(x)}{g(x)} = \lim_{x\to a^+} \frac{F(x)}{G(x)} = \lim_{x\to a^+} \frac{F'(x)}{G'(x)} = \lim_{x\to a^+} \frac{f'(x)}{g'(x)}.$$

Let $f(x)$ and $g(x)$ be functions differentiable on (a, b) with $a < b$. We want to show that if $\lim_{x\to a^+} f(x) = \lim_{x\to a^+} g(x) = \infty$ and $\lim_{x\to a^+} \dfrac{f'(x)}{g'(x)} \in \mathbb{R} \cup \{-\infty, \infty\}$, then

$$\lim_{x\to a^+} \frac{f(x)}{g(x)} = \lim_{x\to a^+} \frac{f'(x)}{g'(x)}.$$

Because $\lim_{x\to a^+} f(x) = \lim_{x\to a^+} g(x) = \infty$, by quotient law, we know that

$$\lim_{x\to a^+} \frac{1}{f(x)} = \lim_{x\to a^+} \frac{1}{g(x)} = 0.$$

Because the existence of the limits, there exists interval (a, c) such that $\frac{1}{f(x)}$ and $\frac{1}{g(x)}$ both $\neq 0$ on (a, c). Let

$$F(x) = \begin{cases} \frac{1}{f(x)}, & x \in (a, b) \\ 0, & x = a \end{cases}$$

and

$$G(x) = \begin{cases} \frac{1}{g(x)}, & x \in (a, b) \\ 0, & x = a \end{cases}$$

$$F'(x) = \frac{-f'(x)}{(f(x))^2}.$$

$$G'(x) = \frac{-g'(x)}{(g(x))^2}.$$

By CMVT, for any $x \in (a, c)$, there exists $c \in (a, x)$ such that

$$(F(x) - F(a))\, G'(c) = (G(x) - G(a))\, F'(c).$$

Since $F(a) = G(a) = 0$,

$$F(x)G'(c) = G(x)F'(c).$$

Thus,

$$\begin{aligned} \lim_{x\to a^+} \frac{f(x)}{g(x)} = \lim_{x\to a^+} \frac{G(x)}{F(x)} &= \lim_{x\to a^+} \frac{G'(x)}{F'(x)} \\ &= \lim_{x\to a^+} \frac{g'(x)}{f'(x)} \lim_{x\to a^+} \left(\frac{f(x)}{g(x)}\right)^2 \\ \lim_{x\to a^+} \frac{f(x)}{g(x)} &= \lim_{x\to a^+} \frac{f'(x)}{g'(x)}. \end{aligned}$$

All other statements can be proven either similarly or by applying other statements. □

ii Indeterminate products

Let I be an open interval and $a \in I$, and $f(x)$ and $g(x)$ be functions differentiable on $I \setminus \{a\}$. If $\lim_{x\to a} f(x) = 0$ and $\left|\lim_{x\to a} g(x)\right| = \infty$, where $\lim_{x\to a} f(x)g(x)$ is called indeterminate form of type $0 \cdot \infty$, we may find $\lim_{x\to a} f(x)g(x)$ by writing the product $f(x)g(x)$ as a quotient:

$$fg = \frac{f}{1/g}, \quad \text{or } fg = \frac{g}{1/f}$$

and use general form of L'Hôpital's rule.

Let $f(x)$ and $g(x)$ be functions differentiable on (a,b) with $a<b$. If $\lim_{x\to a^+} f(x)=0$ and $\left|\lim_{x\to a^+} g(x)\right| = \infty$, where $\lim_{x\to a^+} f(x)g(x)$ is called indeterminate form of type $0\cdot\infty$, we may find $\lim_{x\to a^+} f(x)g(x)$ by writing the product $f(x)g(x)$ as a quotient:

$$fg=\frac{f}{1/g},\quad \text{or } fg=\frac{g}{1/f}$$

and use general form of L'Hôpital's rule.

Let $f(x)$ and $g(x)$ be functions differentiable on (b,a) with $a>b$. If $\lim_{x\to a^-} f(x)=0$ and $\left|\lim_{x\to a^-} g(x)\right| = \infty$, where $\lim_{x\to a^-} f(x)g(x)$ is called indeterminate form of type $0\cdot\infty$, we may find $\lim_{x\to a^-} f(x)g(x)$ by writing the product $f(x)g(x)$ as a quotient:

$$fg=\frac{f}{1/g},\quad \text{or } fg=\frac{g}{1/f}$$

and use general form of L'Hôpital's rule.

Let $f(x)$ and $g(x)$ be functions differentiable on (a,∞). If $\lim_{x\to\infty} f(x)=0$ and $\left|\lim_{x\to\infty} g(x)\right| = \infty$, where $\lim_{x\to\infty} f(x)g(x)$ is called indeterminate form of type $0\cdot\infty$, we may find $\lim_{x\to\infty} f(x)g(x)$ by writing the product $f(x)g(x)$ as a quotient:

$$fg=\frac{f}{1/g},\quad \text{or } fg=\frac{g}{1/f}$$

and use general form of L'Hôpital's rule.

Let $f(x)$ and $g(x)$ be functions differentiable on $(-\infty,a)$. If $\lim_{x\to-\infty} f(x)=0$ and $\left|\lim_{x\to-\infty} g(x)\right| = \infty$, where $\lim_{x\to-\infty} f(x)g(x)$ is called indeterminate form of type $0\cdot\infty$, we may find $\lim_{x\to-\infty} f(x)g(x)$ by writing the product $f(x)g(x)$ as a quotient:

$$fg=\frac{f}{1/g},\quad \text{or } fg=\frac{g}{1/f}$$

and use general form of L'Hôpital's rule.

iii Indeterminate differences

Let I be an open interval and $a\in I$, and $f(x)$ and $g(x)$ be functions differentiable on $I\setminus\{a\}$. If $\lim_{x\to a} f(x)=\lim_{x\to a} g(x)\in\{\infty,-\infty\}$, where $\lim_{x\to a}(f(x)-g(x))$ is called indeterminate form of type $\infty-\infty$, we may find $\lim_{x\to a}(f(x)-g(x))$ by writing the difference $f(x)-g(x)$ as a quotient and use general form of L'Hôpital's rule.

Let $f(x)$ and $g(x)$ be functions differentiable on (a,b) with $a<b$. If $\lim_{x\to a^+} f(x)=\lim_{x\to a^+} g(x)\in\{\infty,-\infty\}$, where $\lim_{x\to a^+}(f(x)-g(x))$ is called indeterminate form of type $\infty-\infty$, we may find $\lim_{x\to a^+}(f(x)-g(x))$ by writing the difference $f(x)-g(x)$ as a quotient and use general form of L'Hôpital's rule.

Let $f(x)$ and $g(x)$ be functions differentiable on (b,a) with $a>b$. If $\lim_{x\to a^-} f(x)=\lim_{x\to a^-} g(x)\in\{\infty,-\infty\}$, where $\lim_{x\to a^-}(f(x)-g(x))$ is called indeterminate form of type $\infty-\infty$, we may find

$\lim_{x\to a^-} (f(x) - g(x))$ by writing the difference $f(x) - g(x)$ as a quotient and use general form of L'Hôpital's rule.

Let $f(x)$ and $g(x)$ be functions differentiable on (a, ∞). If $\lim_{x\to\infty} f(x) = \lim_{x\to\infty} g(x) \in \{\infty, -\infty\}$, where $\lim_{x\to\infty} (f(x) - g(x))$ is called indeterminate form of type $\infty - \infty$, we may find $\lim_{x\to\infty} (f(x) - g(x))$ by writing the difference $f(x) - g(x)$ as a quotient and use general form of L'Hôpital's rule.

Let $f(x)$ and $g(x)$ be functions differentiable on $(-\infty, a)$. If $\lim_{x\to-\infty} f(x) = \lim_{x\to-\infty} g(x) \in \{\infty, -\infty\}$, where $\lim_{x\to-\infty} (f(x) - g(x))$ is called indeterminate form of type $\infty - \infty$, we may find $\lim_{x\to-\infty} (f(x) - g(x))$ by writing the difference $f(x) - g(x)$ as a quotient and use general form of L'Hôpital's rule.

iv Indeterminate powers/exponentials

Let I be an open interval and $a \in I$, and $f(x)$ and $g(x)$ be functions differentiable on $I \setminus \{a\}$. If $\lim_{x\to a} f(x) = \lim_{x\to a} g(x) = 0$, where $\lim_{x\to a} f(x)^{g(x)}$ is called indeterminate form of type 0^0, we may find $\lim_{x\to a} f(x)^{g(x)}$ by taking natural logarithm:

$$g(x)\ln(f(x)),$$

which is in indeterminate form of $0 \cdot \infty$ because $\lim_{x\to a} \ln(f(x)) = -\infty$.

Let I be an open interval and $a \in I$, and $f(x)$ and $g(x)$ be functions differentiable on $I \setminus \{a\}$. If $\lim_{x\to a} f(x) = \infty$ and $\lim_{x\to a} g(x) = 0$, where $\lim_{x\to a} f(x)^{g(x)}$ is called indeterminate form of type ∞^0, we may find $\lim_{x\to a} f(x)^{g(x)}$ by taking natural logarithm:

$$g(x)\ln(f(x)),$$

which is in indeterminate form of $0 \cdot \infty$ because $\lim_{x\to a} \ln(f(x)) = \infty$.

Let I be an open interval and $a \in I$, and $f(x)$ and $g(x)$ be functions differentiable on $I \setminus \{a\}$. If $\lim_{x\to a} f(x) = 1$ and $\lim_{x\to a} g(x) = \infty$, where $\lim_{x\to a} f(x)^{g(x)}$ is called indeterminate form of type 1^∞, we may find $\lim_{x\to a} f(x)^{g(x)}$ by taking natural logarithm:

$$g(x)\ln(f(x)),$$

which is in indeterminate form of $0 \cdot \infty$ because $\lim_{x\to a} \ln(f(x)) = 0$.

Let $f(x)$ and $g(x)$ be functions differentiable on (a, b) with $a < b$. If $\lim_{x\to a^+} f(x) = \lim_{x\to a^+} g(x) = 0$, where $\lim_{x\to a^+} f(x)^{g(x)}$ is called indeterminate form of type 0^0, we may find $\lim_{x\to a^+} f(x)^{g(x)}$ by taking natural logarithm:

$$g(x)\ln(f(x)),$$

which is in indeterminate form of $0 \cdot \infty$ because $\lim_{x\to a^+} \ln(f(x)) = -\infty$.

Let $f(x)$ and $g(x)$ be functions differentiable on (a, b) with $a < b$. If $\lim_{x\to a^+} f(x) = \infty$ and $\lim_{x\to a^+} g(x) = 0$, where $\lim_{x\to a^+} f(x)^{g(x)}$ is called indeterminate form of type ∞^0, we may find $\lim_{x\to a^+} f(x)^{g(x)}$ by taking natural logarithm:

$$g(x)\ln(f(x)),$$

which is in indeterminate form of $0 \cdot \infty$ because $\lim_{x \to a^+} \ln(f(x)) = \infty$.

Let $f(x)$ and $g(x)$ be functions differentiable on (a, b) with $a < b$. If $\lim_{x \to a^+} f(x) = 1$ and $\lim_{x \to a^+} g(x) = \infty$, where $\lim_{x \to a^+} f(x)^{g(x)}$ is called indeterminate form of type 1^∞, we may find $\lim_{x \to a^+} f(x)^{g(x)}$ by taking natural logarithm:

$$g(x) \ln(f(x)),$$

which is in indeterminate form of $0 \cdot \infty$ because $\lim_{x \to a^+} \ln(f(x)) = 0$.

Let $f(x)$ and $g(x)$ be functions differentiable on (b, a) with $a > b$. If $\lim_{x \to a^-} f(x) = \lim_{x \to a^-} g(x) = 0$, where $\lim_{x \to a^-} f(x)^{g(x)}$ is called indeterminate form of type 0^0, we may find $\lim_{x \to a^-} f(x)^{g(x)}$ by taking natural logarithm:

$$g(x) \ln(f(x)),$$

which is in indeterminate form of $0 \cdot \infty$ because $\lim_{x \to a^-} \ln(f(x)) = -\infty$.

Let $f(x)$ and $g(x)$ be functions differentiable on (b, a) with $a > b$. If $\lim_{x \to a^-} f(x) = \infty$ and $\lim_{x \to a^-} g(x) = 0$, where $\lim_{x \to a^-} f(x)^{g(x)}$ is called indeterminate form of type ∞^0, we may find $\lim_{x \to a^-} f(x)^{g(x)}$ by taking natural logarithm:

$$g(x) \ln(f(x)),$$

which is in indeterminate form of $0 \cdot \infty$ because $\lim_{x \to a^-} \ln(f(x)) = \infty$.

Let $f(x)$ and $g(x)$ be functions differentiable on (b, a) with $a > b$. If $\lim_{x \to a^-} f(x) = 1$ and $\lim_{x \to a^-} g(x) = \infty$, where $\lim_{x \to a^-} f(x)^{g(x)}$ is called indeterminate form of type 1^∞, we may find $\lim_{x \to a^-} f(x)^{g(x)}$ by taking natural logarithm:

$$g(x) \ln(f(x)),$$

which is in indeterminate form of $0 \cdot \infty$ because $\lim_{x \to a^-} \ln(f(x)) = 0$.

Let $f(x)$ and $g(x)$ be functions differentiable on (a, ∞). If $\lim_{x \to \infty} f(x) = \lim_{x \to \infty} g(x) = 0$, where $\lim_{x \to \infty} f(x)^{g(x)}$ is called indeterminate form of type 0^0, we may find $\lim_{x \to \infty} f(x)^{g(x)}$ by taking natural logarithm:

$$g(x) \ln(f(x)),$$

which is in indeterminate form of $0 \cdot \infty$ because $\lim_{x \to \infty} \ln(f(x)) = -\infty$.

Let $f(x)$ and $g(x)$ be functions differentiable on (a, ∞). If $\lim_{x \to \infty} f(x) = \infty$ and $\lim_{x \to \infty} g(x) = 0$, where $\lim_{x \to \infty} f(x)^{g(x)}$ is called indeterminate form of type ∞^0, we may find $\lim_{x \to \infty} f(x)^{g(x)}$ by taking natural logarithm:

$$g(x) \ln(f(x)),$$

which is in indeterminate form of $0 \cdot \infty$ because $\lim_{x \to \infty} \ln(f(x)) = \infty$.

Let $f(x)$ and $g(x)$ be functions differentiable on (a, ∞). If $\lim_{x \to \infty} f(x) = 1$ and $\lim_{x \to \infty} g(x) = \infty$, where $\lim_{x \to \infty} f(x)^{g(x)}$ is called indeterminate form of type 1^∞, we may find $\lim_{x \to \infty} f(x)^{g(x)}$ by taking natural logarithm:

$$g(x) \ln(f(x)),$$

which is in indeterminate form of $0 \cdot \infty$ because $\lim_{x\to\infty} \ln(f(x)) = 0$.

Let $f(x)$ and $g(x)$ be functions differentiable on $(-\infty, a)$. If $\lim_{x\to-\infty} f(x) = \lim_{x\to-\infty} g(x) = 0$, where $\lim_{x\to-\infty} f(x)^{g(x)}$ is called indeterminate form of type 0^0, we may find $\lim_{x\to-\infty} f(x)^{g(x)}$ by taking natural logarithm:

$$g(x)\ln(f(x)),$$

which is in indeterminate form of $0 \cdot \infty$ because $\lim_{x\to-\infty} \ln(f(x)) = -\infty$.

Let $f(x)$ and $g(x)$ be functions differentiable on $(-\infty, a)$. If $\lim_{x\to-\infty} f(x) = \infty$ and $\lim_{x\to-\infty} g(x) = 0$, where $\lim_{x\to-\infty} f(x)^{g(x)}$ is called indeterminate form of type ∞^0, we may find $\lim_{x\to-\infty} f(x)^{g(x)}$ by taking natural logarithm:

$$g(x)\ln(f(x)),$$

which is in indeterminate form of $0 \cdot \infty$ because $\lim_{x\to-\infty} \ln(f(x)) = \infty$.

Let $f(x)$ and $g(x)$ be functions differentiable on $(-\infty, a)$. If $\lim_{x\to-\infty} f(x) = 1$ and $\lim_{x\to-\infty} g(x) = \infty$, where $\lim_{x\to-\infty} f(x)^{g(x)}$ is called indeterminate form of type 1^∞, we may find $\lim_{x\to-\infty} f(x)^{g(x)}$ by taking natural logarithm:

$$g(x)\ln(f(x)),$$

which is in indeterminate form of $0 \cdot \infty$ because $\lim_{x\to-\infty} \ln(f(x)) = 0$.

V Taylor series

i Uniqueness

Let $f : D \subseteq \mathbb{R}^m \to \mathbb{R}$ be a function, let a be any point, $I \subseteq D$ be any open ball centered a, if there exists a series such that:

$$f(x) = \sum_{\alpha\in\mathbb{N}_0{}^m} \frac{c_\alpha}{\alpha!}(x-a)^\alpha,$$

for all $x \in I$, then

$$c_\alpha = D^\alpha f(a).$$

Proof. Fix any multi-index β. Apply D^β to both sides

$$D^\beta f(x) = \sum_{\alpha\in\mathbb{N}_0{}^m} \frac{c_\alpha}{\alpha!} D^\beta (x-a)^\alpha$$

$$D^\beta (x-a)^\alpha = \begin{cases} \dfrac{\alpha!}{(\alpha-\beta)!}(x-a)^{\alpha-\beta}, & \beta \le \alpha \\ 0, & \text{otherwise} \end{cases}.$$

$$D^\beta f(x) = \sum_{\alpha\in\mathbb{N}_0{}^m \wedge \beta\le\alpha} \frac{c_\alpha}{(\alpha-\beta)!}(x-a)^{\alpha-\beta}$$

Evaluate a, only $\alpha = \beta$ term survives. Thus,

$$D^\beta f(a) = c_\beta.$$

□

ii Taylor's theorem on the reals

Let $k \in \mathbb{N}$, and function $f : D \subseteq \mathbb{R} \to \mathbb{R}$ be k times differentiable at a point $a \in D$. Then there exists a real-valued function R_k defined on an open interval containing a such that

$$f(x) = \sum_{n=0}^{k} \frac{f^{(n)}(a)}{n!}(x-a)^n + R_k(x),$$

and that

- Peano form of remainder:
$$\lim_{x \to a} \frac{R_k(x)}{|x-a|^k} = 0.$$
- Lagrange form of remainder: Suppose $f^{(k+1)}$ exists on $[a, x)$ for $x \geq a$ and $(x, a]$ for $x \leq a$, and $f^{(k)}$ is continuous on $[a, x]$ for $x \geq a$ and $[x, a]$ for $x \leq a$. Then, there exists ξ in $[a, x]$ for $x \geq a$ and $[x, a]$ for $x \leq a$:
$$R_k(x) = \frac{f^{(k+1)}(\xi)}{(k+1)!}(x-a)^{k+1}.$$
- Integral form of the remainder: Suppose $f^{(k)}$ is absolutely continuous on $[a, x]$ for $x \geq a$ and $[x, a]$ for $x \leq a$. Then
$$R_k(x) = \frac{1}{k!}(x-a)^{k+1} \int_0^1 (1-t)^k f^{(k+1)} \left(a + t(x-a)\right) \mathrm{d}t\,.$$

Proof. PLACEHOLDER □

iii Taylor's theorem on higher dimensions

Let $k, m \in \mathbb{N}$, and function $f : D \subseteq \mathbb{R}^m \to \mathbb{R}$ be k times differentiable at a point $a \in D$. Then there exists a real-valued function R_k defined on a neighborhood of a such that

$$f(x) = \sum_{\alpha \in {\mathbb{N}_0}^m \wedge |\alpha| \leq k} \frac{D^\alpha f(a)}{\alpha!}(x-a)^\alpha + R_k(x),$$

and that

- Peano form of remainder:
$$\lim_{x \to a} \frac{R_k(x)}{|x-a|^k} = 0.$$
- Lagrange form of remainder: Suppose for all $\alpha \in {\mathbb{N}_0}^m \wedge |\alpha| = k+1$, f^α exists on a neighborhood of a whose closure contains x, and for all $\alpha \in {\mathbb{N}_0}^m \wedge |\alpha| = k$, f^α is continuous on a neighborhood of a containing x. Then, there exists $t \in (0, 1)$ such that for $\xi = a + t(x-a)$:
$$R_k(x) = \sum_{\beta \in {\mathbb{N}_0}^m \wedge |\beta| = k+1} \frac{D^\beta f(\xi)}{\beta!}(x-a)^\beta.$$
- Integral form of the remainder: Suppose for all $\alpha \in {\mathbb{N}_0}^m \wedge |\alpha| = k$, f^α is absolutely continuous on a neighborhood of a containing x. Then
$$R_k(x) = \sum_{\beta \in {\mathbb{N}_0}^m \wedge |\beta| = k+1} \frac{k+1}{\beta!}(x-a)^\beta \int_0^1 (1-t)^k \left(D^\beta f\right)\left(a + t(x-a)\right) \mathrm{d}t\,.$$

Proof. PLACEHOLDER □

iv Taylor's inequality

If $\left|f^{(n+1)}(x)\right| \leq M$ for $|x-a| \leq d$, then the remainder $R_n(x)$ of the Taylor series satisfies the inequality

$$\left|R_n(x)\right| \leq \frac{M}{(n+1)!}|x-a|^{n+1}, \quad |x-a| \leq d.$$

Proof. PLACEHOLDER □

VI Lagrange multiplier (method) (拉格朗日乘數/乘子（法）)

The Lagrange multiplier method is a method for finding the points where extremes occur of a differentiable function under constraints.

i Univariate form

Let $f : \mathbb{R} \to \mathbb{R}$ and $g : \mathbb{R} \to \mathbb{R}$. We want to find the points where extremes of $f(x)$ under the constraint $g(x) = \mathbf{0}$ occur.

First, construct the Lagrangian function $\mathcal{L}(x, \lambda)$:

$$\mathcal{L}(x, \lambda) = f(x) - \lambda \cdot g(x),$$

where $\lambda \in \mathbb{R}$ is the Lagrange multiplier (拉格朗日乘數 or 乘子).

Find the derivative of $\mathcal{L}$ and set it to zero:

$$\frac{\mathrm{d}\mathcal{L}}{\mathrm{d}x} = 0$$

Solve the equation to find x and λ .

Statement: Let A be the set of all solutions for x such that $\frac{\mathrm{d}\mathcal{L}}{\mathrm{d}x} = 0$, and B be the set of all points where extremes of $f(x)$ under the constraint $g(x) = \mathbf{0}$ occur. We claim that $B \subseteq A$.

ii Multivariate form

Let $\mathbf{x} = (x_1, x_2, \ldots, x_n)$ be the independent variable vector and $\mathbf{0}$ be the zero vector. Now we have $f : \mathbb{R}^n \to \mathbb{R}$ and $g : \mathbb{R}^n \to \mathbb{R}^c$. We want to find the points where extremes of $f(\mathbf{x})$ under the constraint $g(\mathbf{x}) = \mathbf{0}$ occur.

First, construct the Lagrangian function $\mathcal{L}(\mathbf{x}, \lambda)$:

$$\mathcal{L}(\mathbf{x}, \lambda) = f(\mathbf{x}) - \lambda \cdot g(\mathbf{x}),$$

where $\lambda \in \mathbb{R}^c$ is the Lagrange multiplier vector.

Find the gradient of $\mathcal{L}$ and set it to zero:

$$\nabla\mathcal{L} = \mathbf{0}$$

Solve the equation to find $\mathbf{x}$ and λ .

Statement: Let A be the set of all solutions for $\mathbf{x}$ such that $\nabla \mathcal{L} = \mathbf{0}$, and B be the set of all points where extremes of $f(\mathbf{x})$ under the constraint $g(\mathbf{x}) = \mathbf{0}$ occur. We claim that $B \subseteq A$.

Proof. Consider $\mathbf{x}^* \in B$. It must satisfy the constraint:

$$g(\mathbf{x}^*) = \mathbf{0}.$$

Any feasible point $\mathbf{x}$ near $\mathbf{x}^*$ must satisfy the constraint. We can represent $\mathbf{x}$ as:

$$\mathbf{x} = \mathbf{x}^* + \delta\mathbf{x},$$

where $\delta\mathbf{x}$ is a differential change tangent to the manifold defined by $g(\mathbf{x})$, that is, it lies in the kernel of $\nabla g(\mathbf{x}^*)$, that is,

$$g(\mathbf{x}^* + \delta\mathbf{x}) = \mathbf{0}.$$

Find the first-order approximation of f at $\mathbf{x}^*$:

$$f(\mathbf{x}^* + \delta\mathbf{x}) \approx f(\mathbf{x}^*) + \nabla f(\mathbf{x}^*) \cdot \delta\mathbf{x} + O(\|\delta\mathbf{x}\|^2)$$

Find the first-order approximation of g at $\mathbf{x}^*$:

$$g(\mathbf{x}^* + \delta\mathbf{x}) \approx g(\mathbf{x}^*) + \nabla g(\mathbf{x}^*) \cdot \delta\mathbf{x} + O(\|\delta\mathbf{x}\|^2)$$

Since $\mathbf{x}^* \in B$, for any feasible $\delta\mathbf{x}$ we must have:

$$\nabla f(\mathbf{x}^*) \cdot \delta\mathbf{x} = 0.$$

Since $g(\mathbf{x}^*) = \mathbf{0}$, we have

$$\nabla g(\mathbf{x}^*) \cdot \delta\mathbf{x} = O(\|\delta\mathbf{x}\|^2).$$

Because $\delta\mathbf{x} \in \ker\left(\nabla g(\mathbf{x}^{)}\right)$, $\nabla f(\mathbf{x}^{)}$ can be expressed as a linear combination of $\nabla g(\mathbf{x}^{)}$. This means that there exists a vector to λ such that:

$$\nabla \mathcal{L} = \nabla \left(f(\mathbf{x}) - \lambda \cdot g(\mathbf{x})\right) = \mathbf{0}$$

□

VII Generalized Schwarz's Theorem, Young's Theorem, or Clairaut's Theorem on equality or symmetry of mixed partial derivatives

For a function $f : \Omega \subseteq \mathbb{R}^n \to \mathbb{R}^o$, for any indexed family $A : \mathbb{N}_{\leq m} \to \mathbb{N}_{\leq n}$, for any permutation σ of $\mathbb{N}_{\leq m}$, define

$$D_{A,\sigma} f = \frac{\partial^m f}{\prod_{k=1}^m \partial x_{A_{\sigma(k)}}}.$$

If for all permutations σ of $\mathbb{N}_{\leq m}$, $D_{A,\sigma} f$ exist on a neighborhood of a fixed point $\mathbf{p} \in \Omega$ and is continuous at $\mathbf{p}$, then all of them are equal at $\mathbf{p}$.

VIII Numerical differentiation (數值微分)

i Newton's Method (牛頓法) or Newton-Raphson Method (牛頓-拉普森法)

Newton's method, also known as Newton-Raphson method, is an iterative technique used to approximate the roots of a real-valued function. Given a function $f(x)$ and an initial guess x_0, Newton's method refines this guess by repeatedly applying the formula:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}, \quad n \in \mathbb{N}$$

where:

- x_n is the current approximation,
- $f(x_n)$ is the value of the function at x_n,
- $f'(x_n)$ is the derivative of $f(x)$ evaluated at x_n.

PLACEHOLDER: Newton's method converges quadratically or faster if and proof PLACEHOLDER: Given convergence, suppose that we want the error to be less than a given value v, we can take x_n such that $\left|x_n - x_{n-1}\right| < v$?

13 Integral Theorems and Methods

I Linearity, or sum rule, difference rule, and constant multiple rule

If functions f and g are integrable over an interval I and $a, b \in \mathbb{R}$, then $af(x) + bg(x)$ is also integrable over I and its integral is given by

$$\int_I af(x) + bg(x) \, \mathrm{d}x = a \int_I f(x) \, \mathrm{d}x + b \int_I g(x) \, \mathrm{d}x \, .$$

II Integrals of Symmetric Functions

Suppose real function f is Riemann integrable on $[-a, a]$.

- If f is an even function, that is, $f(-x) = f(x)$, then $\int_{-a}^{a} f(x) \, \mathrm{d}x = 2 \int_0^a f(x) \, \mathrm{d}x$.
- If f is an of function, that is, $f(-x) = -f(x)$, then $\int_{-a}^{a} f(x) \, \mathrm{d}x = 0$.

III Mean Value Theorem (MVT) for Integrals

If a function $f : D \subseteq \mathbb{R} \to \mathbb{R}$ is continuous on a closed interval $[a, b]$ with $a < b$, then

$$\exists c \in (a, b) \text{ s.t. } f(c) = \frac{1}{b - a} \int_a^b f(x) \, \mathrm{d}x \, .$$

Proof. By EVT, f must have maximum and minimum on $[a, b]$.

Let

$$M = \max_{x\in[a,b]} f(x), \quad m = \min_{x\in[a,b]} f(x).$$

Then for any $x \in [a, b]$:

$$m \le f(x) \le M.$$

Integrate it over $[a, b]$:

$$m(b-a) = \int_a^b m \,\mathrm{d}x \le \int_a^b f(x)\,\mathrm{d}x \le \int_a^b M \,\mathrm{d}x = M(b-a).$$

$$m \le \frac{1}{b-a}\int_a^b f(x)\,\mathrm{d}x \le M.$$

Since f is continuous, by IVT, there exists $c \in [a, b]$ such that

$$f(c) = \frac{1}{b-a}\int_a^b f(x)\,\mathrm{d}x\,.$$

□

IV Fundamental Theorem of Calculus (FTC) (微積分基本定理)

i Fundamental Theorem of Calculus Part First or First Fundamental Theorem of Calculus

Let function $f : D \subseteq \mathbb{R} \to \mathbb{R}$ be continuous on a closed interval $[a, b]$, and F be the function defined, for all $x \in [a, b]$ by

$$F(x) = \int_a^x f(t)\,\mathrm{d}t\,.$$

Then F Lipschitz continuous on $[a, b]$ and differentiable on (a, b) and

$$F'(x) = f(x)$$

for all $x \in (a, b)$.

Proof. Lipschitz continuity: Let $(x, y) \in (a, b)$ with $x < y$. Let

$$M = \max_{t\in[a,b]} |f(t)|\,.$$

Then,

$$F(y) - F(x) = \int_a^y f(t)\,\mathrm{d}t - \int_a^x f(t)\,\mathrm{d}t = \int_x^y f(t)\,\mathrm{d}t\,.$$

$$|F(y) + F(x)| = \left|\int_x^y f(t)\,\mathrm{d}t\right| \le \int_x^y |f(t)|\,\mathrm{d}t \le \int_x^y M\,\mathrm{d}t = M(y-x).$$

Thus F is Lipschitz continuous with constant M.

Differentiability:

$$F'(x) = \lim_{h\to 0} \frac{\int_a^{x+h} f(t)\,\mathrm{d}t - \int_a^x f(t)\,\mathrm{d}t}{h} = \lim_{h\to 0} \frac{\int_x^{x+h} f(t)\,\mathrm{d}t}{h}.$$

By MVT for integrals, there exists $c_h \in [x, x+h]$ (or $[x+h, x]$ for h<0) such that

$$f(c_h) = \frac{\int_x^{x+h} f(t)\,\mathrm{d}t}{h}.$$

By continuity of f,

$$\lim_{h\to 0} c_h = x.$$

Hence,

$$\lim_{h\to 0} \frac{\int_x^{x+h} f(t)\,\mathrm{d}t}{h} = f(x).$$

□

ii Fundamental Theorem of Calculus Part Second, Second Fundamental Theorem of Calculus, or Newton–Leibniz Theorem

Let function $f : I \subseteq \mathbb{R} \to \mathbb{R}$ be Riemann integrable on a closed interval $[a, b]$, and $F : J \subseteq \mathbb{R} \to \mathbb{R}$ be a function continuous on $[a, b]$ and differentiable on (a, b) such that

$$F'(x) = f(x)$$

for all $x \in (a, b)$.

Then

$$\int_a^b f(x)\,\mathrm{d}x = F(b) - F(a).$$

Proof. Let $P(x, n)$ be a partition of $[a, b]$. By MVT, for each subinterval $[x_i, x_{i+1}]$, there exists $c_i \in [x_i, x_{i+1}]$ such that

$$F(x_{i+1}) - F(x_i) = F'(c_i)(x_{i+1} - x_i) = f(c_i)(x_{i+1} - x_i).$$

So for any $n \in \mathbb{N}$:

$$\sum_{i=0}^{n-1} f(c_i)(x_{i+1} - x_i) = \sum_{i=0}^{n-1}(F(x_{i+1}) - F(x_i)) = F(x_n) - F(x_0) = F(b) - F(a).$$

Since f is Riemann integrable

$$\int_a^b f(x)\,\mathrm{d}x = \lim_{n\to\infty} \sum_{i=0}^{n-1} f(c_i)(x_{i+1} - x_i)$$

exists. □

V Integrability

i Lebesgue-Vitali theorem (of characterization of the Riemann integrable functions)

A function is Riemann integrable (i.e. Darboux integrable) over a closed interval I iff it is bounded on I and continuous a.e. in I.

Proof. PLACEHOLDER □

ii Riemann–Stieltjes integrability

The Riemann–Stieltjes integral

$$\int_{x=a}^{b} f(x)\,\mathrm{d}g(x)$$

exists if g is of a bounded variation on $[a,b]$, f is bounded on $[a,b]$, f is continuous a.e. on $[a,b]$, and there doesn't exist $x \in [a,b]$ such that f and g both have irremovable discontinuity at x.

Proof. PLACEHOLDER □

iii PLACEHOLDER

PLACEHOLDER

VI Absolutely continuous antiderivative

If function $F(x)$ is absolutely continuous on an open interval (a,b), then for any $f_1, f_2 \in L^1((a,b))$ such that

$$f(a) + \int_a^x f_1(t)\,\mathrm{d}t = \int_a^x f_2(t)\,\mathrm{d}t = F(x),$$

we have

$$f_1(x) = f_2(x)$$

a.e. on (a,b), and

$$f_1(x) = F'(x)$$

a.e. on (a,b). (Note that F is differentiable a.e. since it's absolutely continuous.)

VII Jacobian matrix of injective function

If $\mathbf{F} : U \subseteq \mathbb{R}^n \to \mathbb{R}^n$ is injective and differentiable, and $\mathbf{F}^{-1} : V \subseteq \mathbb{R}^n \to \mathbb{R}^n$ is differentiable, then for all $\mathbf{x} \in U$,

$$(J(\mathbf{F}))(\mathbf{x})(J(\mathbf{F}^{-1}))(\mathbf{F}(\mathbf{x})) = I_n,$$

and for all $\mathbf{y} \in V$,

$$(J(\mathbf{F}))(\mathbf{F}^{-1}(\mathbf{y}))(J(\mathbf{F}^{-1}))(\mathbf{y}) = I_n.$$

Proof.

$$(\mathbf{F} \circ \mathbf{F}^{-1})(\mathbf{x}) = \mathbf{x}, \quad \mathbf{x} \in V$$

$$(J(\mathbf{F}) \circ \mathbf{F}^{-1} \cdot J(\mathbf{F}^{-1}))(\mathbf{x}) = I_n, \quad \mathbf{x} \in V$$

$$(\mathbf{F}^{-1} \circ \mathbf{F})(\mathbf{x}) = \mathbf{x}, \quad \mathbf{x} \in U$$

$$(J(\mathbf{F}^{-1}) \circ \mathbf{F} \cdot J(\mathbf{F}))(\mathbf{x}) = I_n, \quad \mathbf{x} \in U$$

□

VIII Integration by substitution (代換積分法), integration by change of variables (換元積分法), u-substitution (u-代換), reverse chain rule, substitution theroem (代換定理), change of variables theorem (換元定理), or transformation theorem (變換定理)

i Univariate form

Let $g : I \subseteq \mathbb{R} \to \mathbb{R}$ be injective and differentiable on $[a, b] \subseteq I$, with g' being integrable on $[a, b]$, and $f : K \supseteq g([a, b]) \to \mathbb{R}$ be integrable on $g([a, b])$. Then:

$$\int_a^b f \circ g(x) \cdot g'(x)\,\mathrm{d}x = \int_{g(a)}^{g(b)} f(u)\,\mathrm{d}u,$$

and

$$\int f \circ g(x) \cdot g'(x)\,\mathrm{d}x = \int f(u)\,\mathrm{d}u.$$

Substitutions from RHS to LHS are also called inverse substitution.

Proof. Consider the interval $[a, b]$ partitioned as

$$P = \{a = x_0 < x_1 < \cdots < x_n = b\}.$$

For each subinterval $[x_{i-1}, x_i]$, let $\xi_i \in [x_{i-1}, x_i]$ be a sample point. The Riemann sum for the left-hand side of the integral is:

$$S_P = \sum_{i=1}^{n} f(g(\xi_i))g'(\xi_i)(x_i - x_{i-1}).$$

Since g is injective and differentiable, it is either strictly increasing or strictly decreasing on $[a, b]$. Assume g is strictly increasing (the proof for g strictly decreasing follows similarly).

Under this assumption, g maps $[a, b]$ to $[g(a), g(b)]$ with $g(a) < g(b)$. Let $[g(a), g(b)]$ be partitioned as

$$Q = \{g(a) = u_0 < u_1 < \cdots < u_m = g(b)\},$$

where $u_i = g(x_i)$.

The Riemann sum for the right-hand side of the integral is:

$$T_Q = \sum_{i=1}^{n} f(u_i)(u_i - u_{i-1}).$$

Since $u_i = g(x_i)$ and $g'(\xi_i) \approx \dfrac{g(x_i) - g(x_{i-1})}{x_i - x_{i-1}}$, we rewrite:

$$g'(\xi_i)(x_i - x_{i-1}) \approx g(x_i) - g(x_{i-1}) = u_i - u_{i-1}.$$

Thus, the left-hand Riemann sum S_P becomes:

$$S_P = \sum_{i=1}^{n} f(g(\xi_i))(u_i - u_{i-1}),$$

which matches the structure of the right-hand Riemann sum T_Q if we let $\xi_i = g^{-1}(u_i)$.

As the partition P of $[a, b]$ gets finer, the Riemann sum S_P converges to $\displaystyle\int_a^b f(g(x))g'(x)\,\mathrm{d}x$. Similarly, as the partition Q of $[g(a), g(b)]$ gets finer, the Riemann sum T_Q converges to $\displaystyle\int_{g(a)}^{g(b)} f(u)\,\mathrm{d}u$. □

ii Multivariate form

Let $D \subseteq \mathbb{R}^n$ be an open set, $\mathbf{F} : D \to \mathbb{R}^n$ be injective and differentiable on D and C^1 a.e. on D, $|\det(J(\mathbf{F}))|$ be nonzero on D. Then for any piecewise continuous function $f : \mathbf{F}(D) \to \mathbb{R}$ with compact support on D,

$$\int_D f(\mathbf{F}(\mathbf{x}))\,|\det((J(\mathbf{F}))(\mathbf{x}))|\,\mathrm{d}\mathbf{x} = \int_{\mathbf{F}(D)} f(\mathbf{y})\,\mathrm{d}\mathbf{y}$$

and

$$\int f(\mathbf{F}(\mathbf{x}))\,|\det((J(\mathbf{F}))(\mathbf{x}))|\,\mathrm{d}\mathbf{x} = \int f(\mathbf{y})\,\mathrm{d}\mathbf{y}\,.$$

Substitutions from RHS to LHS are also called inverse substitution.

Proof. PLACEHOLDER □

If also that $\mathbf{F}^{-1}$ is differentiable on $\mathbf{F}(D)$ and C^1 a.e. on $\mathbf{F}(D)$, since

$$(J(\mathbf{F}))(\mathbf{x})(J(\mathbf{F}^{-1}))(\mathbf{F}(\mathbf{x})) = I_n,$$

$$\int_D f(\mathbf{F}(\mathbf{x}))\,\mathrm{d}\mathbf{x} = \int_{\mathbf{F}(D)} f(\mathbf{y})\,\left|\det\left((J(\mathbf{F}^{-1}))(\mathbf{y})\right)\right|\,\mathrm{d}\mathbf{y}$$

and

$$\int f(\mathbf{F}(\mathbf{x}))\,\mathrm{d}\mathbf{x} = \int f(\mathbf{y})\,\left|\det\left((J(\mathbf{F}^{-1}))(\mathbf{y})\right)\right|\,\mathrm{d}\mathbf{y}\,.$$

iii Measure theory form

Let X be a locally compact Hausdorff space equipped with a finite Radon measure μ, and let Y be a σ-compact Hausdorff space with a σ-finite Radon measure ρ. Let $\phi : X \to Y$ be an absolutely continuous function. Then there exists a real-valued Borel measurable function w on X such that, for every function $f : Y \to \mathbb{R}$ in $L^1(Y)$, the function $(f \circ \phi) \cdot w$ is in $L^1(X)$, and for every open subset U of X

$$\int_U (f \circ \phi)(x) \cdot w(x)\,\mathrm{d}\mu(x) = \int_{\phi(U)} f(y)\,\mathrm{d}\rho(y)\,.$$

Substitutions from RHS to LHS are also called inverse substitution.

Furthermore, there exists some real-valued Borel measurable function g such that

$$w(x) = (g \circ \phi)(x).$$

Proof. PLACEHOLDER □

IX Integration by parts (IBP) (分部積分法) or partial integration (部分積分法)

i Theorem

$$\frac{\mathrm{d}}{\mathrm{d}x}\prod_{i=1}^{n} f_i(x) = \sum_{j=1}^{n}\left(\frac{\mathrm{d}f_j(x)}{\mathrm{d}x}\frac{\prod_{\substack{i=1\\ i\neq j}}^{n} f_i(x)}{f_j(x)}\right)$$

For example,

$$\int_\Omega u\,\mathrm{d}v = (uv)\big|_\Omega - \int_\Omega v\,\mathrm{d}u\,.$$

ii Application

Integration by parts is a heuristic rather than a purely mechanical process for solving integrals; given a single function to integrate, the typical strategy is to carefully separate this single function into a product of two functions such that the residual integral from the integration by parts formula is easier to evaluate than the single function.

The DETAIL rule or the LIATE rule is a rule of thumb for integration by parts. It involves choosing as u the function that comes first in the following list:

- L: Logarithmic function
- I: Inverse trigonometric function
- A: Algebraic function
- T: Trigonometric function
- E: Exponential function

iii Notation for two parts

We write the integration of

$$\int f(x)\,\mathrm{d}x$$

by parts as:

1. Let

   $$u = g(x), \quad \mathrm{d}v = h(x)\,\mathrm{d}x$$

   with some functions $g(x)$ and $h(x)$ such that

   $$f(x)\,\mathrm{d}x = u\,\mathrm{d}v\,.$$

2. Compute

   $$\mathrm{d}u = g'(x)\,\mathrm{d}x\,, \quad v = \int h(x)\,\mathrm{d}x$$

   with the constant of integration of $\int h(x)\,\mathrm{d}x$ arbitrarily chosen to be assigned to v.

3. Integral by parts
$$\int f(x)\,\mathrm{d}x = uv - \int v\,\mathrm{d}u\,.$$

4. If $\int v\,\mathrm{d}u$ is simplified as
$$\int v\,\mathrm{d}u = k\int f(x)\,\mathrm{d}x + I(x) + C$$
with some constant $k \notin \{-1, 0\}$, we write
$$\begin{aligned}\int f(x)\,\mathrm{d}x &= uv - \int v\,\mathrm{d}u\\ &= uv - k\int f(x)\,\mathrm{d}x - I(x) + C\\ &= \frac{1}{k+1}\left(uv - I(x)\right) + C.\end{aligned}$$

Take
$$\int e^{ax}\sin(bx)\,\mathrm{d}x$$
for example.

Let:
$$u = \sin(bx), \quad \mathrm{d}v = e^{ax}\,\mathrm{d}x\,.$$
$$\mathrm{d}u = b\cos(bx)\,\mathrm{d}x\,, \quad v = \int e^{ax}\,\mathrm{d}x = \frac{1}{a}e^{ax}.$$
$$\begin{aligned}\int e^{ax}\sin(bx)\,\mathrm{d}x &= \int u\,\mathrm{d}v = uv - \int v\,\mathrm{d}u\\ &= \frac{1}{a}e^{ax}\sin(bx) - \frac{b}{a}\int e^{ax}\cos(bx)\,\mathrm{d}x\,.\end{aligned}$$
$$w = \cos(bx).$$
$$\mathrm{d}w = -b\sin(bx).$$
$$\begin{aligned}\int e^{ax}\cos(bx)\,\mathrm{d}x &= \int w\,\mathrm{d}v = wv - \int v\,\mathrm{d}w\\ &= \frac{1}{a}e^{ax}\cos(bx) + \frac{b}{a}\int e^{ax}\sin(bx)\,\mathrm{d}x\,.\end{aligned}$$
$$\begin{aligned}\int e^{ax}\sin(bx)\,\mathrm{d}x &= \frac{1}{a}e^{ax}\sin(bx) - \frac{b}{a}\left(\frac{1}{a}e^{ax}\cos(bx) + \frac{b}{a}\int e^{ax}\sin(bx)\,\mathrm{d}x\right)\\ &= \frac{1}{a}e^{ax}\sin(bx) - \frac{b}{a^2}e^{ax}\cos(bx) - \frac{b^2}{a^2}\int e^{ax}\sin(bx)\,\mathrm{d}x\\ &= \frac{a^2}{a^2+b^2}\left(\frac{1}{a}e^{ax}\sin(bx) - \frac{b}{a^2}e^{ax}\cos(bx)\right) + C\\ &= \frac{e^{ax}}{a^2+b^2}\left(a\sin(bx) - b\cos(bx)\right) + C.\end{aligned}$$

X Riemann–Stieltjes integral theorems

i Integration by parts (IBP) or partial integration of Riemann–Stieltjes integral

$$\int_{x=a}^{b} f(x)\,\mathrm{d}g(x) = f(b)g(b) - f(a)g(a) - \int_{x=a}^{b} g(x)\,\mathrm{d}f(x)\,.$$

Proof. PLACEHOLDER □

ii Riemann–Stieltjes integral with respect to differential function

If $f(x)$ is bounded on $[a,b]$, $g(x)$ is monotonic on $[a,b]$, and $g'(x)$ exists and is Riemann integrable on $[a,b]$, then the Riemann–Stieltjes integral is related to the Riemann integral by

$$\int_{x=a}^{b} f(x)\,\mathrm{d}g(x) = \int_{a}^{b} f(x)g'(x)\,\mathrm{d}x\,.$$

Proof. PLACEHOLDER □

iii Riemann–Stieltjes integral with respect to step function

If $a < s < b$,

$$g(x) = \begin{cases} 0, & a \le x \le s \\ 1, & b \ge x > s \end{cases},$$

and $f : X \supseteq [a,b] \to \mathbb{R}$ is continuous at s, then

$$\int_{x=a}^{b} f(x)\,\mathrm{d}g(x) = f(s).$$

Proof. PLACEHOLDER □

XI General Method of Integration of Rational Functions

i $\dfrac{A}{x-a}$

$$\int \frac{A}{x-a}\,\mathrm{d}x = A\ln|x-a| + C, \quad a, A \in \mathbb{R}$$

ii $\dfrac{A}{(x-a)^g}$

$$\int \frac{A}{(x-a)^g}\,\mathrm{d}x = -\frac{A}{(g-1)\,(x-a)^{g-1}} + C, \quad a, A \in \mathbb{R} \wedge g \in \mathbb{N}_{>1}$$

iii $\dfrac{Ax+B}{x^2+bx+c}$

$$\int \frac{Ax+B}{x^2+bx+c} = \frac{A}{2}\ln\left|x^2+bx+c\right| + \frac{2B-bA}{\sqrt{4c-b^2}}\arctan\left(\frac{2x+b}{\sqrt{4c-b^2}}\right) + C, \quad b, c, A, B \in \mathbb{R} \wedge b^2 < 4c$$

Proof.

$$\int \frac{Ax+B}{x^2+bx+c} = \int \frac{A}{2}\frac{2x+b}{x^2+bx+c} + \frac{B-\frac{bA}{2}}{\left(x+\frac{b}{2}\right)^2+\left(c-\frac{b^2}{4}\right)}\,\mathrm{d}x$$

$$= \frac{A}{2}\ln\left|x^2+bx+c\right| + \left(B-\frac{bA}{2}\right)\int \frac{1}{u^2+\left(c-\frac{b^2}{4}\right)}\,\mathrm{d}u\,, \quad u = x+\frac{b}{2}$$

$$= \frac{A}{2}\ln\left|x^2+bx+c\right| + \frac{2B-bA}{\sqrt{4c-b^2}}\arctan\left(\frac{2x+b}{\sqrt{4c-b^2}}\right) + C$$

□

iv $\dfrac{1}{\left(x^2+bx+c\right)^h}$

$$\int \frac{1}{\left(x^2+bx+c\right)^h}\,\mathrm{d}x = \frac{1}{\left(4c-b^2\right)(h-1)}\frac{2x+b}{\left(x^2+bx+c\right)^{h-1}} + \frac{4h-6}{\left(4c-b^2\right)(h-1)}\int \frac{1}{\left(x^2+bx+c\right)^{h-1}}\,\mathrm{d}x\,, \quad b,c\in$$

Proof.

$$u = x+\frac{b}{2}, \quad \alpha = \sqrt{c-\frac{b^2}{4}}$$

$$\int \frac{1}{\left(x^2+bx+c\right)^h}\,\mathrm{d}x = \int \frac{1}{\left(u^2+\alpha^2\right)^h}\,\mathrm{d}u$$

$$= \frac{1}{\alpha^2}\int \frac{u^2+\alpha^2-u^2}{\left(u^2+\alpha^2\right)^h}\,\mathrm{d}u$$

$$= \frac{1}{\alpha^2}\int \frac{1}{\left(u^2+\alpha^2\right)^{h-1}}\,\mathrm{d}u - \frac{1}{\alpha^2}\int \frac{u^2}{\left(u^2+\alpha^2\right)^h}\,\mathrm{d}u$$

$$\frac{\mathrm{d}}{\mathrm{d}u}\left(\frac{u}{\left(u^2+\alpha^2\right)^{h-1}}\right) = \frac{\left(u^2+\alpha^2\right)^{h-1} - u(h-1)\left(u^2+\alpha^2\right)^{h-2}(2u)}{\left(u^2+\alpha^2\right)^{2(h-1)}}$$

$$= \frac{1}{\left(u^2+\alpha^2\right)^{h-1}} - 2(h-1)\frac{u^2}{\left(u^2+\alpha^2\right)^h}$$

$$\frac{u^2}{\left(u^2+\alpha^2\right)^h} = \frac{1}{2(h-1)}\left(\frac{1}{\left(u^2+\alpha^2\right)^{h-1}} - \frac{\mathrm{d}}{\mathrm{d}u}\left(\frac{u}{\left(u^2+\alpha^2\right)^{h-1}}\right)\right)$$

$$\int \frac{u^2}{\left(u^2+\alpha^2\right)^h}\,\mathrm{d}u = \frac{1}{2(h-1)}\left(\int \frac{1}{\left(u^2+\alpha^2\right)^{h-1}}\,\mathrm{d}u - \frac{u}{\left(u^2+\alpha^2\right)^{h-1}}\right)$$

$$\int \frac{1}{\left(x^2+bx+c\right)^h}\,\mathrm{d}x = \frac{1}{\alpha^2}\int \frac{1}{\left(u^2+\alpha^2\right)^{h-1}}\,\mathrm{d}u - \frac{1}{\alpha^2}\int \frac{u^2}{\left(u^2+\alpha^2\right)^h}\,\mathrm{d}u$$

$$= \frac{1}{\alpha^2}\int \frac{1}{\left(u^2+\alpha^2\right)^{h-1}}\,\mathrm{d}u - \frac{1}{2\alpha^2(h-1)}\left(\int \frac{1}{\left(u^2+\alpha^2\right)^{h-1}}\,\mathrm{d}u - \frac{u}{\left(u^2+\alpha^2\right)^{h-1}}\right)$$

$$= \frac{2h-3}{2\alpha^2(h-1)}\int \frac{1}{\left(u^2+\alpha^2\right)^{h-1}}\,\mathrm{d}u + \frac{1}{2\alpha^2(h-1)}\frac{u}{\left(u^2+\alpha^2\right)^{h-1}}$$

$$= \frac{4h-6}{\left(4c-b^2\right)(h-1)}\int \frac{1}{\left(x^2+bx+c\right)^{h-1}}\,\mathrm{d}x + \frac{1}{\left(4c-b^2\right)(h-1)}\frac{2x+b}{\left(x^2+bx+c\right)^{h-1}}$$

$\square$

v $\dfrac{Ax+B}{\left(x^2+bx+c\right)^h}$

$$\int \frac{Ax+B}{\left(x^2+bx+c\right)^h}\,\mathrm{d}x = -\frac{A}{2(h-1)\left(x^2+bx+c\right)^{h-1}} + \left(B-\frac{bA}{2}\right)\int \frac{1}{\left(x^2+bx+c\right)^h}\,\mathrm{d}x\,, \quad b,c,A,B \in \mathbb{R} \land b^2 < 4c$$

vi Integration of Rational Functions by Partial Fractions

To compute the integral of any arbitrary rational function $f(x)$,

1. Express it as
$$f(x) = S(x) + \frac{P(x)}{Q(x)}$$
where $S(x), P(x), Q(x)$ are polynomials and $\deg(P) < \deg(Q)$.
2. Find $q, a_j, b_k, c_k \in \mathbb{R}$, $m, n \in \mathbb{N}_0$, $r_j, s_k \in \mathbb{N}$ and ${b_k}^2 - 4c_k < 0$ for all k such that
$$Q(x) = q \prod_{j=1}^{m} \left(x-a_j\right)^{r_j} \prod_{k=1}^{n} \left(x^2+b_kx+c_k\right)^{s_k}.$$
Note that all polynomials can be expressed as a product of linear factors and irreducible quadratic factors.
3. Express $\dfrac{P(x)}{Q(x)}$ as
$$\frac{P(x)}{Q(x)} = p\left(\sum_{j=1}^{m}\sum_{g=1}^{r_j} \frac{A_{j_g}}{\left(x-a_j\right)^g} + \sum_{k=1}^{n}\sum_{h=1}^{s_k} \frac{B_{k_h}x + C_{k_h}}{\left(x^2+b_kx+c_k\right)^h}\right)$$
where $p, A_{j_g}, B_{k_h}, C_{k_h} \in \mathbb{R}$.
4. Integrate $S(x)$ and each term of $\dfrac{A_{j_g}}{\left(x-a_j\right)^g}$ and $\dfrac{B_{k_h}x + C_{k_h}}{\left(x^2+b_kx+c_k\right)^h}$ respectively.

XII Substitution for irrational function integrand

i Direct substitution

When computing an integral of
$$R(x, \sqrt[n]{g(x)})$$
with respect to x where R and g are rational functions, the substitution $u = \sqrt[n]{g(x)}$ may be effective.

ii Square root of quadratic polynomial

When computing an integral of
$$R\left(x, \sqrt{ax^2+bx+c}\right)$$
with respect to x where R is a rational function and $a \neq 0$, it may be effective to

1. let $h = \frac{b}{2|a|}$ and $k = -\frac{b^2}{4a^2} + \frac{c}{a}$, rewrite $\sqrt{ax^2+bx+c}$ as

$$\sqrt{ax^2+bx+c} = \sqrt{|ak|}\sqrt{\text{sgn}(a)\left(\frac{x}{\sqrt{|k|}} + \frac{h}{\sqrt{|k|}}\right)^2 + \text{sgn}(k)},$$

2. use substitution $u = \frac{x}{\sqrt{|k|}} + \frac{h}{\sqrt{|k|}}$, which makes the integrand,

$$\sqrt{|k|}R\left(\sqrt{|k|}u - h, \sqrt{|ak|}\sqrt{\text{sgn}(a)u^2 + \text{sgn}(k)}\right),$$

   which is a rational function of u and $\sqrt{\text{sgn}(a)u^2 + \text{sgn}(k)}$, and,

3. depending on sgn(a) and sgn(k),
   - $k = 0$: the integrand is a rational function of u,
   - $a > 0$ and $k > 0$: try the substitution $u = \tan(t)$, $u = \cot(t)$, $u = \sinh(t)$, or $u = \text{csch}(t)$,
   - $a > 0$ and $k < 0$: try the substitution $u = \sec(t)$, $u = \csc(t)$, $u = \cosh(t)$, or $u = \coth(t)$,
   - $a < 0$ and $k > 0$: try the substitution $u = \sin(t)$, $u = \cos(t)$, $u = \tanh(t)$, or $u = \text{sech}(t)$.

iii Euler('s) substitution(s)

When computing an integral of

$$R(x, \sqrt{ax^2+bx+c})$$

with respect to x where R is a rational function, the three Euler('s) substitutions may be effective.

1. Euler's first substitution: For $a \geq 0$, substitute

$$\sqrt{ax^2+bx+c} = \pm x\sqrt{a} + t$$

$$x = \frac{c - t^2}{\pm 2t\sqrt{a} - b}$$

   where either the positive sign or the negative sign can be chosen.

2. Euler's second substitution: For $c \geq 0$, substitute

$$\sqrt{ax^2+bx+c} = xt \pm \sqrt{c}$$

$$x = \frac{\pm 2t\sqrt{c} - b}{a - t^2}$$

   where either the positive sign or the negative sign can be chosen.

3. Euler's third substitution: For $b^2 - 4ac \geq 0$, let

$$ax^2 + bx + c = a(x-\alpha)(x-\beta),$$

   substitute

$$\sqrt{ax^2+bx+c} = (x-\alpha)t$$

$$x = \frac{\alpha\beta - \alpha t^2}{a - t^2}$$

XIII Function of trigonometric functions integral

i Tangent half-angle substitution, Weierstrass substitution, or universal trigonometric substitution

When computing an integral of

$$f(\sin(x), \cos(x))$$

with respect to x, the substitution

$$t = \tan\left(\frac{x}{2}\right)$$

gives

$$\int f(\sin(x), \cos(x)) = \int f\left(\frac{2t}{1+t^2}, \frac{1-t^2}{1+t^2}\right) \frac{2}{1+t^2}\, \mathrm{d}t\,.$$

ii Product of sine and cosine functions

$$\int \sin^m(x) \cos^n(x)\, \mathrm{d}x\,, \quad m, n \in \mathbb{N}.$$

- For odd n, use $\cos^2(x) = 1 - \sin^2(x)$ and then substitute $u = \sin(x), \quad \mathrm{d}u = \cos(x)\, \mathrm{d}x$.
- For odd m, use $\sin^2(x) = 1 - \cos^2(x)$ and then substitute $u = \cos(x), \quad \mathrm{d}u = -\sin(x)\, \mathrm{d}x$.
- For $m = n$, use $\sin(x)\cos(x) = \dfrac{1}{2}\sin(2x)$.
- For $\dfrac{m-n}{2} \in \mathbb{N}$, use $\sin(x)\cos(x) = \dfrac{1}{2}\sin(2x)$ and then substitute $\sin^2(x) = \dfrac{1}{2}(1-\cos(2x))$.
- For $\dfrac{n-m}{2} \in \mathbb{N}$, use $\sin(x)\cos(x) = \dfrac{1}{2}\sin(2x)$ and then substitute $\cos^2(x) = \dfrac{1}{2}(1+\cos(2x))$.

iii Product of tangent and secant functions

$$\int \tan^m(x) \sec^n(x)\, \mathrm{d}x\,, \quad m, n \in \mathbb{N}.$$

- For even n, save a factor of $\sec^2(x)$, use $\sec^2(x) = 1 + \tan^2(x)$ to express the remaining factors in terms of $\tan(x)$, and then substitute $u = \tan(x), \quad \mathrm{d}u = \sec^2(x)\, \mathrm{d}x$.
- For odd m, save a factor of $\tan(x)\sec(x)$, use $\tan^2(x) = \sec^2(x) - 1$ to express the remaining factors in terms of $\sec(x)$, and then substitute $u = \sec(x), \quad \mathrm{d}u = \tan(x)\sec(x)\, \mathrm{d}x$.
- For even m, use $\tan^2(x) = \sec^2(x) - 1$ to express the integrand in terms of $\sec(x)$, and then use reduction formula for secant function.

iv Product of cotangent and cosecant functions

$$\int \cot^m(x) \csc^n(x)\, \mathrm{d}x\,, \quad m, n \in \mathbb{N}.$$

- For even n, save a factor of $\csc^2(x)$, use $\csc^2(x) = 1 + \cot^2(x)$ to express the remaining factors in terms of $\cot(x)$, and then substitute $u = \cot(x), \quad \mathrm{d}u = -\csc^2(x)\, \mathrm{d}x$.
- For odd m, save a factor of $\cot(x)\csc(x)$, use $\cot^2(x) = \csc^2(x) - 1$ to express the remaining factors in terms of $\csc(x)$, and then substitute $u = \csc(x), \quad \mathrm{d}u = -\cot(x)\csc(x)\, \mathrm{d}x$.

- For even m, use $\cot^2(x) = \csc^2(x) - 1$ to express the integrand in terms of $\csc(x)$, and then use reduction formula for cosecant function.

XIV Leibniz integral rule (for differentiation under the integral sign) (（積分符號內取微分的）萊布尼茲積分法則)

i Constant limits

Let $f(x,t)$ be a function of two variables and the following expression exists. Then:

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\int_a^b f(x,t)\,\mathrm{d}t\right) = \int_a^b \frac{\partial}{\partial x} f(x,t)\,\mathrm{d}t\,.$$

ii Variable limits

Let $a(x), b(x)$ be two functions of one variable, $f(x,t)$ be a function of two variables, and the following expression exists. Then:

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\int_{a(x)}^{b(x)} f(x,t)\,\mathrm{d}t\right) = f\,(x, b(x)) \cdot \frac{\mathrm{d}b(x)}{\mathrm{d}x} - f\,(x, a(x)) \cdot \frac{\mathrm{d}a(x)}{\mathrm{d}x} + \int_{a(x)}^{b(x)} \frac{\partial}{\partial x} f(x,t)\,\mathrm{d}t$$

iii Measure theory form for const limits

Let X be an open subset of $\mathbb{R}$, and Ω be a measure space. Suppose $f: X \times \Omega \to \mathbb{R}$ satisfies the following conditions:

1. $f(x,\omega)$ is a Lebesgue-integrable function of ω for each $x \in X$.
2. For almost all $\omega \in \Omega$, the partial derivative $\dfrac{\partial f}{\partial x}$ exists for all $x \in X$.
3. There is an integrable function $\theta: \Omega \to \mathbb{R}$ such that $\left|\dfrac{\partial f}{\partial x}(x,\omega)\right| \le \theta(\omega)$ for all $x \in X$ and almost all $\omega \in \Omega$.

Then, for all $x \in X$,

$$\frac{\mathrm{d}}{\mathrm{d}x}\int_\Omega f(x,\omega)\,\mathrm{d}\omega = \int_\Omega \frac{\partial f}{\partial x}(x,\omega)\,\mathrm{d}\omega\,.$$

XV Lebesgue integral theorems

i Fubini's theorem (富比尼定理)

If real-valued function $f \in L^1(X \times Y)$, then:

$$\int_{X\times Y} f(x,y)\,\mathrm{d}(x,y) = \int_X \int_Y f(x,y)\,\mathrm{d}y\,\mathrm{d}x = \int_Y \int_X f(x,y)\,\mathrm{d}x\,\mathrm{d}y\,.$$

Proof. For nonnegative simple function, it is obvious.

For a nonnegative measurable function f, construct a monotone series of nonnegative simple functions f_n that converges pointwise to f:

$$f_n = \sum_{k=0}^{2^n n-1} \frac{k}{2^n} \cdot \mathbf{1}_{\left\{x \mid \frac{x}{2^n} \le f(x) < \frac{k+1}{2^n}\right\}}(x) + n \cdot 1_{\{x \mid f(x) \ge n\}}(x)$$

with

$$\lim_{n\to\infty} f_n = f.$$

By monotone convergence theorem, the statement holds for all nonnegative measurable functions.

For measurable function, define

$$f^+(x) = \begin{cases} f(x), & \text{if } f(x) > 0 \\ 0, & \text{otherwise} \end{cases}, \quad \text{and}$$

$$f^-(x) = \begin{cases} -f(x), & \text{if } f(x) < 0 \\ 0, & \text{otherwise} \end{cases}.$$

Note that both f^+ and f^- are nonnegative and that

$$f = f^+ - f^-, \quad |f| = f^+ + f^-.$$

Thus, the statement holds for all measurable functions. □

ii Countable additivity

For an indexed family $(A_i)_{i=1}^n$ of piecewise disjoint measurable sets with $n \in \mathbb{N} \cup \{\infty\}$, if $f \in L^1(A_i)$ for all i, then

$$\int_{\bigcup_{i=1}^n A_i} f \,\mathrm{d}\mu = \sum_{i=1}^n \int_{A_i} f \,\mathrm{d}\mu .$$

iii (Lebesgue's) Dominated convergence theorem (DCT) (（勒貝格）控制/受制收斂定理)

Let $\langle f_n \rangle_{n\in I}$ be a net of real or complex-valued measurable functions on a measure space (S, Σ, μ). If f_n is almost everywhere pointwise convergent to a function f, and there is a Lebesgue-integrable function g such that

$$\left| f_n \right| \le g$$

almost everywhere for all $n \in I$.

Then f_n for all $n \in I$ and f are in $L^1(\mu)$ and

$$\lim_n \int_S f_n \,\mathrm{d}\mu = \int_S \lim_n f_n \,\mathrm{d}\mu = \int_S f \,\mathrm{d}\mu ,$$

and

$$\lim_n \int_S \left| f_n - f \right| \mathrm{d}\mu = 0.$$

Proof. PLACEHOLDER □

XVI Fundamental theorem of multivariable calculus (多變數微積分基本定理)

i Gradient theorem or Fundamental theorem for linear integral

Suppose $C:\ \mathbf{r}(t),\quad t\in[a,b]$ is a piecewise C^1, oriented curve. If $\mathbf{F}$ is a tensor field differentiable on a neighborhood of C, then

$$\int_C \nabla\mathbf{F}\,\mathrm{d}\mathbf{r}=\mathbf{F}(\mathbf{r}(b))-\mathbf{F}(\mathbf{r}(a)).$$

Proof.

$$\int_C \nabla\mathbf{F}\,\mathrm{d}\mathbf{r}=\int_a^b(\nabla\mathbf{F})(\mathbf{r}(t))\mathbf{r}'(t)\,\mathrm{d}t$$

Let $\mathbf{F}=(F_1,\dots,F_n)$. For each component $F_i(\mathbf{r}(t))$,

$$\frac{\mathrm{d}}{\mathrm{d}t}F_i(\mathbf{r}(t))=(\nabla F_i)(\mathbf{r}(t))\mathbf{r}_i'(t)$$

Thus

$$(\nabla\mathbf{F})(\mathbf{r}(t))\mathbf{r}'(t)=\frac{\mathrm{d}}{\mathrm{d}t}\mathbf{F}(\mathbf{r}(t))$$

By FTC part 2,

$$\int_a^b\frac{\mathrm{d}}{\mathrm{d}t}\mathbf{F}(\mathbf{r}(t))\,\mathrm{d}t=\mathbf{F}(\mathbf{r}(b))-\mathbf{F}(\mathbf{r}(a))$$

□

ii Divergence theorem, Gauss's theorem, or Ostrogradsky's theorem (高斯散度定理)

Suppose $V\subseteq\mathbb{R}^n$ is a compact region with outward-oriented, piecewise C^1 boundary $\partial V=S$. If $\mathbf{F}$ is a vector field differentiable and piecewise C^1 on a neighborhood of V, then

$$\int_V\nabla\cdot\mathbf{F}\,\mathrm{d}V=\oint_S\mathbf{F}\cdot\mathrm{d}\mathbf{S}.$$

Proof. Simple regions case: Let σ be a permutation on $(x_1,\dots,x_n)$ and

$$V=\left\{(x_1,\dots,x_n):\ \forall i\in\mathbb{Z}\cap[1,n]:\ \varphi_i\left(\left(x_{\sigma(j)}\right)_{j=1}^{i}\right)\le x_{\sigma(i)}\le\psi_i\left(\left(x_{\sigma(j)}\right)_{j=1}^{i}\right)\right\},$$

where φ_i,ψ_i are $(i-1)$-ary C^1 functions.

$$\int_V\nabla\cdot\mathbf{F}\,\mathrm{d}V=\sum_{i=1}^n\int_V\frac{\partial F_{\sigma(i)}}{\partial x_{\sigma(i)}}\,\mathrm{d}V$$

For each term, by FTC part 2,

$$\begin{aligned}\int_V\frac{\partial F_i}{\partial x_i}\,\mathrm{d}V&=\int_{\varphi_1}^{\psi_1}\dots\int_{\varphi_n(x_{\sigma(1)},\dots,x_{\sigma(n-1)})}^{\psi_n(x_{\sigma(1)},\dots,x_{\sigma(n-1)})}\frac{\partial F_{\sigma(i)}}{\partial x_{\sigma(i)}}\,\mathrm{d}x_{\sigma(n)}\dots\mathrm{d}x_{\sigma(1)}\\&=\int_{\varphi_1}^{\psi_1}\dots\int_{\varphi_{i-1}(x_{\sigma(1)},\dots,x_{\sigma(i-2)})}^{\psi_{i-1}(x_{\sigma(1)},\dots,x_{\sigma(i-2)})}\int_{\varphi_{i+1}(x_{\sigma(1)},\dots,x_{\sigma(i)})}^{\psi_{i+1}(x_{\sigma(1)},\dots,x_{\sigma(i)})}\dots\int_{\varphi_n(x_{\sigma(1)},\dots,x_{\sigma(n-1)})}^{\psi_n(x_{\sigma(1)},\dots,x_{\sigma(n-1)})}F_{\sigma(i)}\left(x_1,\dots,x_{i-1},\psi_i(x_{\sigma(1)},\dots,x_{\sigma\end{aligned}$$

$$\begin{aligned}&=\int_{\varphi_1}^{\psi_1}\dots\int_{\varphi_{i-1}(x_{\sigma(1)},\dots,x_{\sigma(i-2)})}^{\psi_{i-1}(x_{\sigma(1)},\dots,x_{\sigma(i-2)})}\int_{\varphi_{i+1}(x_{\sigma(1)},\dots,x_{\sigma(i)})}^{\psi_{i+1}(x_{\sigma(1)},\dots,x_{\sigma(i)})}\dots\int_{\varphi_n(x_{\sigma(1)},\dots,x_{\sigma(n-1)})}^{\psi_n(x_{\sigma(1)},\dots,x_{\sigma(n-1)})}(\mathbf{F}\cdot\mathbf{e}_i)\left(\psi_i(x_{\sigma(1)},\dots,x_{\sigma(i-1)})\right)+(\mathbf{F}\\&=\int_{x_{\sigma(i)}=\psi_i(x_{\sigma(1)},\dots,x_{\sigma(i-1)})}\mathbf{F}\cdot\mathrm{d}\mathbf{S}+\int_{x_i=\varphi_i(x_{\sigma(1)},\dots,x_{\sigma(i-1)})}\mathbf{F}\cdot\mathrm{d}\mathbf{S}\end{aligned}$$

Thus

$$\sum_{i=1}^{n} \int_V \frac{\partial F_i}{\partial x_i} \,\mathrm{d}V = \int_S \mathbf{F} \cdot \mathrm{d}\mathbf{S}$$

This proves for simple regions case.

Finite union of simple regions case: Both sides are additive. Internal faces cancel. So the identity still holds.

Any compact region with piecewise C^1 boundary has finite cover of simple regions. So the identity still holds. □

iii Divergence product rule

For a scalar field f and a vector field **G**,

$$\nabla \cdot (f\mathbf{G}) = \nabla f \cdot \mathbf{G} + f\nabla \cdot \mathbf{G}.$$

iv Green's first identity

Suppose $V \subseteq \mathbb{R}^n$ is a compact region with outward-oriented, piecewise C^1 boundary $\partial V = S$. If a scalar field f is continuous and piecewise C^1 on a neighborhood of S and a scalar field g is C^1 and piecewise C^2 on a neighborhood of S, then

$$\int_V f\nabla^2 g + \nabla f \cdot \nabla g \,\mathrm{d}V = \oint_S f\nabla g \cdot \mathrm{d}\mathbf{S}.$$

Proof. By divergence theorem and divergence product rule,

$$\begin{aligned}
\oint_S (f\nabla g) \cdot \mathrm{d}\mathbf{S} &= \int_V \nabla \cdot (f\nabla g) \,\mathrm{d}V \\
&= \int_V \nabla f \cdot \nabla g + f\nabla \cdot (\nabla g) \,\mathrm{d}V \\
&= \int_V f\nabla^2 g + \nabla f \cdot \nabla g \,\mathrm{d}V
\end{aligned}$$

□

v Green's second identity

Suppose $V \subseteq \mathbb{R}^n$ is a compact region with outward-oriented, piecewise C^1 boundary $\partial V = S$. If two scalar fields f, g are C^1 and piecewise C^2 on a neighborhood of V, and a scalar field ε is continuous and piecewise C^1 on a neighborhood of V, then

$$\int_V f\nabla \cdot (\varepsilon\nabla g) - g\nabla \cdot (\varepsilon\nabla f) \,\mathrm{d}V = \oint_S \varepsilon\,(f\nabla g - g\nabla f) \cdot \mathrm{d}\mathbf{S}.$$

Proof. By divergence theorem and divergence product rule,

$$
\begin{aligned}
\oint_S \varepsilon\,(f\nabla g - g\nabla f)\,\mathrm{d}\mathbf{S} &= \int_V \nabla\cdot(\varepsilon\,(f\nabla g - g\nabla f))\,\mathrm{d}V \\
&= \int_V \nabla\varepsilon\cdot(f\nabla g - g\nabla f) + \varepsilon\nabla\cdot(f\nabla g - g\nabla f)\,\mathrm{d}V \\
&= \int_V f\nabla\varepsilon\cdot\nabla g - g\nabla\varepsilon\cdot\nabla f + \varepsilon\left(\nabla f\cdot\nabla g + f\nabla^2 g - \nabla g\cdot\nabla f - g\nabla^2 f\right)\mathrm{d}V \\
&= \int_V f\nabla\cdot(\varepsilon\nabla g) - f\varepsilon\nabla^2 g - g\nabla\cdot(\varepsilon\nabla f) + g\varepsilon\nabla^2 f + \varepsilon\left(f\nabla^2 g - g\nabla^2 f\right)\mathrm{d}V \\
&= \int_V f\nabla\cdot(\varepsilon\nabla g) - g\nabla\cdot(\varepsilon\nabla f)\,\mathrm{d}V
\end{aligned}
$$

□

For $\varepsilon = 1$,

$$\int_V f\nabla^2 g - g\nabla^2 f\,\mathrm{d}V = \oint_S (f\nabla g - g\nabla f)\cdot\mathrm{d}\mathbf{S}\,.$$

vi Green's theorem (格林定理或綠定理)

Suppose $S \subseteq \mathbb{R}^2$ is a compact surface with positively-oriented, piecewise C^1 boundary $\partial S = \ell$. If two scalar functions $P(x, y)$ and $Q(x, y)$ are continuous and piecewise C^1 on a neighborhood of S, then

$$\iint_S \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\,\mathrm{d}S = \oint_\ell P\,\mathrm{d}x + Q\,\mathrm{d}y\,,$$

that is, let $\mathbf{F} = (Q, -P)$,

$$\iint_S \nabla\cdot\mathbf{F}\,\mathrm{d}S = \oint_\ell \mathbf{F}\cdot\mathrm{d}\boldsymbol{\ell}.$$

Proof.

$$\nabla\cdot\mathbf{F} = \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}$$

$$\mathrm{d}\boldsymbol{\ell} = \mathrm{d}y - \mathrm{d}x$$

By divergence theorem,

$$\iint_S \nabla\cdot\mathbf{F}\,\mathrm{d}S = \oint_\ell \mathbf{F}\cdot\mathrm{d}\boldsymbol{\ell}.$$

□

vii Stokes' theorem (斯托克斯定理) or Kelvin–Stokes theorem

Suppose $S \subseteq \mathbb{R}^3$ is a compact surface with positively-oriented, piecewise C^1 boundary $\partial S = \ell$. If a three-dimensional vector field $\mathbf{F}$ is continuous and piecewise C^1 on a neighborhood of S, then

$$\iint_S (\nabla\times\mathbf{F})\cdot\mathrm{d}\mathbf{S} = \oint_\ell \mathbf{F}\cdot\mathrm{d}\boldsymbol{\ell}.$$

Proof. Parameteriz S and ℓ as T and ω in parameters u, v. Let T be $\mathbf{r}(u, v) = (x(u, v), y(u, v), z(u, v))$, and

$$\mathrm{d}\mathbf{T} = \frac{\partial\mathbf{r}}{\partial u}\times\frac{\partial\mathbf{r}}{\partial v}\,\mathrm{d}u\,\mathrm{d}v\,.$$

$$\iint_S (\nabla \times \mathbf{F})\, \mathrm{d}\mathbf{S} = \iint_T (\nabla \times \mathbf{F})(\mathbf{r}(u,v)) \cdot \left(\frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v} \right) \mathrm{d}u\, \mathrm{d}v$$

Define the pulled-back vector fields in parameter space

$$\mathbf{G}(u,v) = \mathbf{F}(\mathbf{r}(u,v))$$

$$\mathbf{P}(u,v) = \mathbf{F} \cdot \frac{\partial \mathbf{r}}{\partial u}$$

$$\mathbf{Q}(u,v) = \mathbf{F} \cdot \frac{\partial \mathbf{r}}{\partial v}$$

Let $\mathbf{F} = (F_1, F_2, F_3)$.

$$\begin{aligned}
&(\nabla \times \mathbf{F})(\mathbf{r}(u,v)) \cdot \left(\frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v} \right) \\
=&\begin{vmatrix} \frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z} & \frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x} & \frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \\ \frac{\partial x}{\partial u} & \frac{\partial y}{\partial u} & \frac{\partial z}{\partial u} \\ \frac{\partial x}{\partial v} & \frac{\partial y}{\partial v} & \frac{\partial z}{\partial v} \end{vmatrix} \\
=&\left(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z} \right) \left(\frac{\partial y}{\partial u}\frac{\partial z}{\partial v} - \frac{\partial y}{\partial v}\frac{\partial z}{\partial u} \right) - \left(\frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x} \right) \left(\frac{\partial x}{\partial u}\frac{\partial z}{\partial v} - \frac{\partial x}{\partial v}\frac{\partial z}{\partial u} \right) + \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right) \left(\frac{\partial x}{\partial u}\frac{\partial y}{\partial v} - \frac{\partial x}{\partial v}\frac{\partial y}{\partial u} \right) \\
=&\frac{\partial F_3}{\partial y}\frac{\partial y}{\partial u}\frac{\partial z}{\partial v} - \frac{\partial F_3}{\partial y}\frac{\partial y}{\partial v}\frac{\partial z}{\partial u} - \frac{\partial F_2}{\partial z}\frac{\partial y}{\partial u}\frac{\partial z}{\partial v} + \frac{\partial F_2}{\partial z}\frac{\partial y}{\partial v}\frac{\partial z}{\partial u} - \frac{\partial F_1}{\partial z}\frac{\partial x}{\partial u}\frac{\partial z}{\partial v} + \frac{\partial F_1}{\partial z}\frac{\partial x}{\partial v}\frac{\partial z}{\partial u} - \frac{\partial F_3}{\partial x}\frac{\partial x}{\partial u}\frac{\partial z}{\partial v} + \frac{\partial F_3}{\partial x}\frac{\partial x}{\partial v}\frac{\partial z}{\partial u} + \frac{\partial F_2}{\partial x}\frac{\partial x}{\partial u} \\
=&\left(\frac{\partial F_1}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial F_1}{\partial y}\frac{\partial y}{\partial u} + \frac{\partial F_1}{\partial z}\frac{\partial z}{\partial u} \right) \frac{\partial x}{\partial v} - \frac{\partial F_1}{\partial x}\frac{\partial x}{\partial u}\frac{\partial x}{\partial v} - \left(\frac{\partial F_1}{\partial x}\frac{\partial x}{\partial v} + \frac{\partial F_1}{\partial y}\frac{\partial y}{\partial v} + \frac{\partial F_1}{\partial z}\frac{\partial z}{\partial v} \right) \frac{\partial x}{\partial u} + \frac{\partial F_1}{\partial x}\frac{\partial x}{\partial u}\frac{\partial x}{\partial v} + \left(\frac{\partial F_2}{\partial x}\frac{\partial x}{\partial u} + \right. \\
=&\left(\frac{\partial F_1}{\partial u}\frac{\partial x}{\partial v} + \frac{\partial F_2}{\partial u}\frac{\partial y}{\partial v} + \frac{\partial F_3}{\partial u}\frac{\partial z}{\partial v} \right) - \left(\frac{\partial F_1}{\partial v}\frac{\partial x}{\partial u} + \frac{\partial F_2}{\partial v}\frac{\partial y}{\partial u} + \frac{\partial F_3}{\partial v}\frac{\partial z}{\partial u} \right) \\
=&\frac{\partial}{\partial u}\left(\mathbf{F} \cdot \frac{\partial \mathbf{r}}{\partial v} \right) - \frac{\partial}{\partial v}\left(\mathbf{F} \cdot \frac{\partial \mathbf{r}}{\partial u} \right) \\
=&\frac{\partial \mathbf{Q}}{\partial u} - \frac{\partial \mathbf{P}}{\partial v}
\end{aligned}$$

By Green's theorem,

$$\begin{aligned}
\iint_T \frac{\partial \mathbf{Q}}{\partial u} - \frac{\partial \mathbf{P}}{\partial v}\, \mathrm{d}u\, \mathrm{d}v &= \oint_\omega \mathbf{P}\, \mathrm{d}u + \mathbf{Q}\, \mathrm{d}v \\
&= \oint_\omega \mathbf{G}\, \mathrm{d}\boldsymbol{\omega} \\
&= \oint_\ell \mathbf{F}\, \mathrm{d}\boldsymbol{\ell}
\end{aligned}$$

□

viii Generalized Stokes theorem, Stokes–Cartan theorem, or fundamental theorem of multivariable calculus

The generalized Stokes theorem says that the integral of a differential form ω over the boundary $\partial\Omega$ of some orientable manifold Ω is equal to the integral of its exterior derivative $\mathrm{d}\boldsymbol{\omega}$ over the whole of Ω, i.e.,

$$\int_{\partial\Omega} \omega = \int_\Omega \mathrm{d}\boldsymbol{\omega}$$

Proof. PLACEHOLDER also statement □

ix Curl of curl

For a vector field $\mathbf{F} : U \subseteq \mathbb{R}^3 \to \mathbb{R}^3$, at a point where it's C^2,

$$\nabla \times (\nabla \times \mathbf{F}) = \nabla(\nabla \cdot \mathbf{F}) - \nabla^2 \mathbf{F}.$$

Proof.

$$\begin{aligned}
\nabla \times (\nabla \times \mathbf{F}) =& \nabla \times \left(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z}, \frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x}, \frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right) \\
=& \left(\frac{\partial \frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y}}{\partial y} - \frac{\partial \frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x}}{\partial z}, \frac{\partial \frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z}}{\partial z} - \frac{\partial \frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y}}{\partial x}, \frac{\partial \frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x}}{\partial x} - \frac{\partial \frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z}}{\partial y} \right) \\
=& \left(\frac{\partial^2 F_2}{\partial x \partial y} - \frac{\partial^2 F_1}{\partial y^2} + \frac{\partial^2 F_3}{\partial x \partial z} - \frac{\partial^2 F_1}{\partial z^2}, \frac{\partial^2 F_3}{\partial y \partial z} - \frac{\partial^2 F_2}{\partial z^2} + \frac{\partial^2 F_1}{\partial x \partial y} - \frac{\partial^2 F_2}{\partial x^2}, \frac{\partial^2 F_1}{\partial x \partial z} - \frac{\partial^2 F_3}{\partial x^2} + \frac{\partial^2 F_2}{\partial y \partial z} - \frac{\partial^2 F_3}{\partial y^2} \right) \\
\nabla(\nabla \cdot \mathbf{F}) =& \nabla \left(\frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z} \right) \\
=& \left(\frac{\partial^2 F_1}{\partial x^2} + \frac{\partial^2 F_2}{\partial x \partial y} + \frac{\partial^2 F_3}{\partial x \partial z}, \frac{\partial^2 F_1}{\partial x \partial y} + \frac{\partial^2 F_2}{\partial y^2} + \frac{\partial^2 F_3}{\partial y \partial z}, \frac{\partial^2 F_1}{\partial x \partial z} + \frac{\partial^2 F_2}{\partial y \partial z} + \frac{\partial^2 F_3}{\partial z^2} \right) \\
\nabla \times (\nabla \times \mathbf{F}) - \nabla(\nabla \cdot \mathbf{F}) =& \left(-\frac{\partial^2 F_1}{\partial x^2} - \frac{\partial^2 F_1}{\partial y^2} - \frac{\partial^2 F_1}{\partial z^2}, -\frac{\partial^2 F_2}{\partial x^2} - \frac{\partial^2 F_2}{\partial y^2} - \frac{\partial^2 F_2}{\partial z^2}, -\frac{\partial^2 F_3}{\partial x^2} - \frac{\partial^2 F_3}{\partial y^2} - \frac{\partial^2 F_3}{\partial z^2} \right) \\
=& -\left(\nabla^2 F_1, \nabla^2 F_2, \nabla^2 F_3 \right) \\
=& -\nabla^2 \mathbf{F}
\end{aligned}$$

□

XVII Conservative field

i Path independence equivalent to zero closed path integral

For an n-dimensional vector field $\mathbf{v}$ in $L^1(U)$ for an open subset U of $\mathbb{R}^n$, for any two piecewise C^1 paths $C, D \subseteq U$ with same initial and terminal points,

$$\int_C \mathbf{v} \cdot \mathrm{d}\mathbf{x} = \int_D \mathbf{v} \cdot \mathrm{d}\mathbf{x}$$

iff for any piecewise C^1 closed path $E \subseteq U$,

$$\oint_E \mathbf{v} \cdot \mathrm{d}\mathbf{x} = 0.$$

Proof. By $\int_C \mathbf{v} \cdot \mathrm{d}\mathbf{x} = -\int_{-C} \mathbf{v} \cdot \mathrm{d}\mathbf{x}$ and $\int_{C+D} \mathbf{v} \cdot \mathrm{d}\mathbf{x} = \int_C \mathbf{v} \cdot \mathrm{d}\mathbf{x} + \int_D \mathbf{v} \cdot \mathrm{d}\mathbf{x}$. □

ii Conservative equivalent to path independence

For a conservative field f on an open subset U of $\mathbb{R}^n$ and any two piecewise C^1 paths $C, D \subseteq U$ with same initial point a and terminal point b,

$$\int_C f \cdot \mathrm{d}\mathbf{x} = \int_D f \cdot \mathrm{d}\mathbf{x}.$$

Proof. Let $f = \nabla\varphi$. By Gradient theorem, both equal to $\varphi(b) - \varphi(a)$. □

iii One dimension

A one-dimensional field is conservative on an open interval iff it is continuous on it.

iv Two dimensions

If a two-dimensional field $f = (f_1, f_2)$ is conservative on a connected open set U, then

$$\frac{\partial f_2}{\partial x} - \frac{\partial f_1}{\partial y} = 0$$

on the subset of U where LHS exists.

Proof. Let $f = \nabla\varphi$. $f_1 = \varphi_x$, $f_2 = \varphi_y$,

$$\frac{\partial f_2}{\partial x} = \frac{\partial f_1}{\partial y} = \varphi_{xy}.$$

□

A differentiable, two-dimensional field $f = (f_1, f_2)$ is conservative on a simply-connected open set U iff

$$\frac{\partial f_2}{\partial x} - \frac{\partial f_1}{\partial y} = 0$$

on U.

Proof. If: Any simply-connected surface $S \subseteq U$ is bounded by a piecewise C^1, simple, closed curve C, by Green's theorem,

$$\iint_S \frac{\partial f_2}{\partial x} - \frac{\partial f_1}{\partial y}\,\mathrm{d}S = \oint_C f \cdot \mathrm{d}(x, y) = 0$$

□

v Three dimensions

If a three-dimensional field f is conservative on a connected open set U, then

$$\nabla \times f = \mathbf{0}$$

on the subset of U where LHS exists.

Proof. Let $f = \nabla\varphi = (f_1, f_2, f_3)$. $f_1 = \varphi_x$, $f_2 = \varphi_y$, $f_3 = \varphi_z$,

$$\begin{aligned}\nabla \times f &= \left(\frac{\partial f_3}{\partial y} - \frac{\partial f_2}{\partial z}, \frac{\partial f_1}{\partial z} - \frac{\partial f_3}{\partial x}, \frac{\partial f_2}{\partial x} - \frac{\partial f_1}{\partial y}\right)\\ &= \left(\varphi_{yz} - \varphi_{yz}, \varphi_{xz} - \varphi_{xz}, \varphi_{xy} - \varphi_{xy}\right)\\ &=(0, 0, 0)\end{aligned}$$

□

A differentiable, three-dimensional field f is conservative on a simply-connected open set U iff

$$\nabla \times f = \mathbf{0}$$

on U.

Proof. If: Any simply-connected surface $S \subseteq U$ is bounded by a piecewise C^1, simple, closed curve C, by Stokes' theorem,

$$\iint_S \nabla \times f \,\mathrm{d}S = \oint_C f \cdot \mathrm{d}(x, y, z) = 0$$

□

vi Divergence of curl

For a scalar field $\mathbf{F}$ C^2 on $U \subseteq \mathbb{R}^3$,

$$\nabla \cdot (\nabla \times \mathbf{F}) = 0.$$

Proof.

$$\begin{aligned}\nabla \cdot (\nabla \times \mathbf{F}) =& \nabla \cdot \left(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z}, \frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x}, \frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right) \\ =& \frac{\partial^2 F_3}{\partial x y^2} - \frac{\partial^2 F_2}{\partial x z^2} + \frac{\partial F_1}{\partial yz} - \frac{\partial F_3}{\partial xy} + \frac{\partial F_2}{\partial xz} - \frac{\partial F_1}{\partial yz} \\ =& 0\end{aligned}$$

□

XVIII Two-dimensional curve rotates about principal axis on the same plane

The area of the surface formed by rotating the simple curve $\mathbf{r}(t) = (x(t), y(t)), \quad t \in [a, b]$ about $y = d$ is

$$A = 2\pi \int_{t=a}^{b} |y(t) - d| \,\mathrm{d}s(t)\,,$$

where $s(t)$ is the arc length function of $\mathbf{r}$ from a to t.

XIX Solid of revolution

i Pappus's (centroid) theorem I or (Pappus–)Guldinus theorem I

The area of the surface formed by rotating a two-dimensional simple curve with arc length s and centroid $\mathbf{P}$ about a rotational axis on the same plane of the curve that doesn't cut the curve in a distance d from $\mathbf{P}$ is

$$A = 2\pi s d.$$

Proof. Let the curve be $\mathbf{r}(t) = (x(t), y(t)), \quad t \in [a, b]$, $s(t)$ be the arc length function of it, the centroid of it be the origin, that is,

$$\int \mathbf{r}(t) \,\mathrm{d}s(t) = (0, 0),$$

and the rotational axis is $y = d$.

$$\begin{aligned} A &= 2\pi \int_{t=a}^{b} |y(t) - d| \, \mathrm{d}s(t) \\ &= 2\pi \left| \int_{t=a}^{b} y(t) \, \mathrm{d}s(t) - \int_{t=a}^{b} d \, \mathrm{d}s(t) \right| \\ &= 2\pi s d \end{aligned}$$

□

ii Disc method or disc integration

The volume of the solid formed by rotating the area bounded by $y = f(x)$,$y = g(x)$, $x = a$, and $x = b$ with $a < b$ about the x-axis is given by

$$V = \pi \int_a^b \left| f(x)^2 - g(x)^2 \right| \mathrm{d}x \,,$$

aka Washer method.

If $g(x) = 0$, this reduces to:

$$V = \pi \int_a^b f(x)^2 \, \mathrm{d}x \,.$$

iii Shell method or shell integration

The volume of the solid formed by rotating the area bounded by $y = f(x)$,$y = g(x)$, $x = a$, and $x = b$ with $a < b$ about the y-axis is given by

$$V = 2\pi \int_a^b x \, |f(x) - g(x)| \, \mathrm{d}x \,.$$

iv Pappus's (centroid) theorem II or (Pappus–)Guldinus theorem II

The volume of the solid formed by rotating a plane region with area A and centroid **P** about a rotational axis on the same plane of the region that doesn't cut the region in a distance d from **P** is

$$V = 2\pi A d.$$

Proof. Let the rotational axis be the y-axis. Split the region to subregions such that each subregion is bounded by $y = f(x)$,$y = g(x)$, $x = a$, and $x = b$, for some functions $f(x)$ and $g(x)$ and some constant $a < b$.

For each subregion,

$$V = 2\pi \int_a^b x \, |f(x) - g(x)| \, \mathrm{d}x \,.$$

$$Ad = \int_a^b x \, |f(x) - g(x)| \, \mathrm{d}x \,.$$

$$V = 2\pi A d.$$

The sum of the volume of the solid formed by rotating the subregions is the volume of the solid formed by rotating the original region. The average of the centroids $\mathbf{P}_i = (d_i, y_i)$ of the subregions weighted by their areas A_i is the centroid $\mathbf{P} = (d, p_y)$ of the original region.

$$\mathbf{P} = \frac{\sum_i A_i \mathbf{P}_i}{A}$$

$$V = \sum_i V_i = \sum_i A_i d_i = A\frac{\sum_i A_i d_i}{A} = Ad.$$

□

XX Numerical integration (數值積分)

i Left and right endpoint rule

Let f be a continuous real-valued function on $[a, b]$. The left endpoint rule gives the approximation of $\int_a^b f(x)\,\mathrm{d}x$

$$L_n = \frac{b-a}{n}\sum_{i=0}^{n-1} f\left(a + \frac{i}{n}(b-a)\right).$$

The right endpoint rule gives the approximation of $\int_a^b f(x)\,\mathrm{d}x$

$$R_n = \frac{b-a}{n}\sum_{i=1}^{n} f\left(a + \frac{i}{n}(b-a)\right).$$

ii Midpoint rule

Let f be a continuous real-valued function on $[a, b]$. The midpoint rule gives the approximation of $\int_a^b f(x)\,\mathrm{d}x$

$$M_n = \frac{b-a}{n}\sum_{i=0}^{n-1} f\left(a + \frac{2i+1}{2n}(b-a)\right).$$

The error is

$$E_M = \int_a^b f(x)\,\mathrm{d}x - M_n.$$

If f'' exists a.e. on $[a, b]$ and $\left|f''(x)\right| \le K$ for all $x \in [a, b]$ such that $f''(x)$ exist, then

$$\left|E_M\right| \le \frac{K(b-a)^3}{24n^2}.$$

If $f \in C^2([a, b])$, then

$$E_M = \frac{(b-a)^3}{24n^2} f''(\xi)$$

for some $\xi \in [a, b]$.

iii Trapezoidal rule

Let f be a continuous real-valued function on $[a,b]$. The trapezoidal rule gives the approximation of $\int_a^b f(x)\,\mathrm{d}x$

$$T_n = \frac{b-a}{2n}\left(f(a) + f(b) + 2\sum_{i=1}^{n-1} f\left(a + \frac{i}{n}(b-a)\right)\right).$$

The error is

$$E_T = \int_a^b f(x)\,\mathrm{d}x - T_n.$$

If f'' exists a.e. on $[a,b]$ and $|f''(x)| \le K$ for all $x \in [a,b]$ such that $f''(x)$ exist, then

$$|E_T| \le \frac{K(b-a)^3}{12n^2}.$$

If $f \in C^2([a,b])$, then

$$E_T = -\frac{(b-a)^3}{12n^2} f''(\xi)$$

for some $\xi \in [a,b]$.

iv Simpson's rule or Simpson's 1/3 rule

Let f be a continuous real-valued function on $[a,b]$, the Simpson's rule gives the approximation of $\int_a^b f(x)\,\mathrm{d}x$

$$S_n = \frac{b-a}{3n}\left(f(a) + f(b) + 2\sum_{i=1}^{n-1} f\left(a + \frac{i(i \mod 2 + 1)}{n}(b-a)\right)\right), \quad \frac{n}{2} \in \mathbb{N}.$$

The error is

$$E_S = \int_a^b f(x)\,\mathrm{d}x - S_n.$$

If $f^{(4)}$ exists a.e. on $[a,b]$ and $|f^{(4)}(x)| \le K$ for all $x \in [a,b]$ such that $f^{(4)}(x)$ exist, then

$$|E_S| \le \frac{K(b-a)^5}{180n^4}.$$

If $f \in C^4([a,b])$, then

$$E_S = -\frac{(b-a)^5}{180n^4} f^{(4)}(\xi)$$

for some $\xi \in [a,b]$.

XXI Symmetry function integral

Suppose $f : \mathbb{R} \to \mathbb{R}$ is such that $f(x) = f(-x)$ and $f \in L^1(\mathbb{R})$. Let

$$I = \int_{-\infty}^{\infty} f(x)\,\mathrm{d}x$$

and

$$F(x) = \int_{-\infty}^{x} f(t)\,\mathrm{d}t$$

Then,

$$F(x) + F(-x) = I$$

Proof. For $x \geq 0$,

$$F(x) = \frac{I}{2} + \int_{0}^{x} f(t)\,\mathrm{d}t$$

$$F(-x) = \int_{-\infty}^{-x} f(t)\,\mathrm{d}t = \int_{x}^{\infty} f(t)\,\mathrm{d}t$$

□

XXII Multiple integral theorems

i Multiple integral to multiplication of integrals

For a rectangle

$$R = \{(x_1, x_2, \dots, x_n) \mid a_1 \leq x_1 \leq b_1 \wedge a_2 \leq x_2 \leq b_2 \wedge \dots a_n \leq x_n \leq b_n\},$$

$$\int_R \prod_{i=1}^{n} f_i(x_i)\,\mathrm{d}V = \prod_{i=1}^{n} \int_{a_i}^{b_i} f_i(x)\,\mathrm{d}x\,.$$

ii Multiple integral to iterated integral theorem

For a region D in $\mathbb{R}^n$ of the form

$$D = \left\{(x_1, x_2, \dots, x_n) \middle| \bigwedge_{i=0}^{n-1} f_i\left(x_{\sigma(1)}, x_{\sigma(2)}, \dots, x_{\sigma(i)}\right) \leq x_{\sigma(i+1)} \leq g_i\left(x_{\sigma(1)}, x_{\sigma(2)}, \dots, x_{\sigma(i)}\right)\right\},$$

where σ is any permutation on $[1, n] \cap \mathbb{N}$, f_0, g_0 are real numbers, and f_i, g_i are i-ary functions continuous on $\left[f_{i-1}\left(x_{\sigma(1)}, x_{\sigma(2)}, \dots, x_{\sigma(i-1)}\right), g_{i-1}\left(x_{\sigma(1)}, x_{\sigma(2)}, \dots, x_{\sigma(i-1)}\right)\right]$, and function $F \in L^1(D)$,

$$\int_D F(x_1, x_2, \dots, x_n)\,\mathrm{d}\mathbf{x} = \left(\int_{f_i\left(x_{\sigma(1)}, x_{\sigma(2)}, \dots, x_{\sigma(i)}\right)}^{g_i\left(x_{\sigma(1)}, x_{\sigma(2)}, \dots, x_{\sigma(i)}\right)}\right)_{i=0}^{n-1} F(x_1, x_2, \dots, x_n)\left(\mathrm{d}x_{\sigma(n-i)}\right)_{i=0}^{n-1}.$$

14 Total Variation Theorems

I Bounded

If real function is BV over an interval, it's bounded on it.

Proof. Obviously. □

II Implying or implied by bounded variation

i Monotonicity

If a real function is monotone on a closed interval, it is BV over it.

If a real function is monotone and bounded on an interval, it is BV over it.

Proof. Obviously. □

ii Absolute continuity

If a real function is absolutely continuous on a closed interval, it is BV over it.

If a real function is absolutely continuous and bounded on an interval, it is BV over it.

Proof. Suppose a real function f is absolutely continuous on an interval $[a, b]$.

For every positive number ε, there exists a positive number δ such that for any finite sequence of disjoint subintervals $[x_k, y_k]$ of $[a, b]$,

$$\sum_k |y_k - x_k| < \delta$$

implies

$$\sum_k |f(y_k) - f(x_k)| < \varepsilon.$$

Fix $\varepsilon = 1$ and get obtain a δ. Let N be the smallest integer such that $N > \dfrac{b-a}{\delta}$. Let $x_k = a + \dfrac{k(b-a)}{N}$ and $I_k = [x_{k-1}, x_k]$ for $k \in \mathbb{Z} \cap [0, N]$. Then

$$\sum_{j=1}^{N} |f(x_k) - f(x_{k-1})| < 1.$$

Take any fixed I_k. For any partition $P(t, m)$ of I_k, since $\sum_{j=1}^{m} (t_j - t_{j-1}) < \delta$,

$$\sum_{j=1}^{m} |f(t_j) - f(t_{j-1})| < 1.$$

Thus

$$V_{I_k}(f) < 1$$

Thus

$$V_a^b(f) < N < \infty.$$

□

iii Differentiability

A function of bouned variation on an interval is differentiable a.e. on it.

Proof. Both functions in the Jordan decomposition is monotone and thus differentiable a.e. □

III Total variation of differentiable function theorem

i Total variation of differentiable real function

If a real function f is differentiable on $[a, b]$ and f' is Riemann integrable over $[a, b]$, then the total variation of f over $[a, b]$ is

$$V_a^b(f) = \int_a^b \left|f'(x)\right| \mathrm{d}x\,.$$

Proof. Obviously. □

15 Series and Improper Integral Theorems and Methods

I Limit of real series theorems

i Integral test (for convergence) or Maclaurin–Cauchy test

Suppose $f(x)\colon [a, \infty) \to \mathbb{R}$ for some $a \in \mathbb{N}$ is monotone and $\lim_{x\to\infty} f(x) = 0$. Then $\int_a^\infty f(x)\,\mathrm{d}x$ is convergent iff $\sum_{k=a}^{\infty} f(k)$ is convergent. And if $\int_a^\infty f(x)\,\mathrm{d}x$ is convergent and $f(x)$ is monotone decreasing, for any $n \in \mathbb{N}_{\geq a}$,

$$\int_n^\infty f(x)\,\mathrm{d}x \leq \sum_{k=n}^{\infty} f(k) \leq f(n) + \int_n^\infty f(x)\,\mathrm{d}x\,,$$

called remainder estimate for integral test; (Note that for $n \in \mathbb{N}_{>a}$, $f(n) + \int_n^\infty f(x)\,\mathrm{d}x \leq \int_{n-1}^\infty f(x)\,\mathrm{d}x$.)

if $\int_a^\infty f(x)\,\mathrm{d}x$ is convergent and $f(x)$ is monotone increasing, for any $n \in \mathbb{N}_{\geq a}$,

$$\int_n^\infty f(x)\,\mathrm{d}x \geq \sum_{k=n}^{\infty} f(k) \geq f(n) + \int_n^\infty f(x)\,\mathrm{d}x\,,$$

called remainder estimate for integral test. (Note that for $n \in \mathbb{N}_{>a}$, $f(n) + \int_n^\infty f(x)\,\mathrm{d}x \geq \int_{n-1}^\infty f(x)\,\mathrm{d}x$.)

Proof. Consider the case where $f(x)$ is monotone increasing. The case where $f(x)$ is monotone decreasing can be proven similarly.

First consider the case where $\int_a^\infty f(x)\,\mathrm{d}x$ is convergent. This implies that $f(x)$ is non-positive. Thus, for any fixed $n \in \mathbb{N}_{\geq a}$,

$$\int_n^{n+1} f(x)\,\mathrm{d}x \geq f(n),$$

and for any $k \geq n+1$,

$$\int_k^{k+1} f(x)\,\mathrm{d}x \geq f(k) \geq \int_{k-1}^{k} f(x)\,\mathrm{d}x\,.$$

For any $N \in \mathbb{N}_{\geq n}$, summing over all $k \in \mathbb{N}_{\geq n}$, we obtain,

$$\int_n^{N+1} f(x)\,\mathrm{d}x \geq \sum_{k=n}^{N} f(k) \geq f(n) + \int_n^{N} f(x)\,\mathrm{d}x\,.$$

Take limit $\lim\limits_{N\to\infty}$, we obtain

$$\int_n^{\infty} f(x)\,\mathrm{d}x \geq \sum_{k=n}^{\infty} f(n) \geq f(n) + \int_n^{\infty} f(x)\,\mathrm{d}x\,.$$

Second, consider the case where $\int_a^{\infty} f(x)\,\mathrm{d}x$ is divergent. Since $f(x)$ is monotone. Either $\int_a^{\infty} f(x)\,\mathrm{d}x = -\infty$ or $\int_a^{\infty} f(x)\,\mathrm{d}x = \infty$. For the former, the supremum of $f(x)$ is negative, and $\sum\limits_{k=a}^{\infty} f(k) = -\infty$. For the latter, there exists $x_0 > a$ such that $f(x) > 0$ for all $x \geq x_0$, and thus $\sum\limits_{k=a}^{\infty} f(k) = \infty$. □

ii Alternating series test or Leibniz's test

If a series $\sum\limits_{n=1}^{\infty}(-1)^{n-1}b_n$ satisfies that

$$\lim_{n\to\infty} b_n = 0$$

and for some $m \in \mathbb{N}$, for all $n \in \mathbb{N}_{\geq m}$,

$$b_n \geq b_{n+1} \geq 0,$$

then it is convergent, and for all $n \in \mathbb{N}_{\geq m}$,

$$0 \leq \left|\sum_{k=n}^{\infty}(-1)^{k-1}b_k\right| \leq b_n,$$

where the latter is called alternating series estimation theorem.

Proof. Let

$$c_1 = b_1,$$

$$c_n = b_{2n-1} - b_{2n-2}, \quad n \in \mathbb{N} \wedge n > 1,$$

$$S_n = \sum_{k=1}^{n}(-1)^{k-1}b_k,$$

$$S = \sum_{n=1}^{\infty}(-1)^{n-1}b_n,$$

$$S_{2n-1} = T_n = \sum_{k=1}^{n} c_n.$$

T_n is non-increasing since

$$c_n \le 0, \quad \forall n \in \mathbb{N} \wedge n > 1.$$

T_n is bounced-below by 0 since

$$c_1 \ge b_2 \ge -\sum_{k=2}^{\infty} c_k,$$

$$T_n \ge 0.$$

By monotone convergence theorem, T_n is convergent, and thus S_n is convergent.

$$b_{2n-2} \ge -\sum_{k=n}^{\infty} c_k = S - T_{n-1} = S - S_{2n-3}.$$

Thus for even n,

$$b_n \ge \sum_{k=n}^{\infty}(-1)^{k-1} b_k = \left|\sum_{k=n}^{\infty}(-1)^{k-1} b_k\right|.$$

For odd n,

$$\left|\sum_{k=n}^{\infty}(-1)^{k-1} b_k\right| = -\sum_{k=n}^{\infty}(-1)^{k-1} b_k = -b_n + \sum_{k=n+1}^{\infty}(-1)^{k-1} b_k \ge -b_n.$$

□

iii Series conditional convergence implies squared-index-multiplied-termed series divergence

For a real sequence $(a_n)_{n=1}^{\infty}$, if $\sum_{n=1}^{\infty}$ is conditionally convergent, then $\sum_{n=1}^{\infty} n^2 a_n$ is divergent.

Proof. Prove by contradiction. Assume $\sum_{n=1}^{\infty} n^2 a_n$ is convergent. Then $\lim_{n\to\infty} n^2 a_n = 0$. Hence there exists $N \in \mathbb{N}$ such that $n \ge N \implies |n^2 a_n| < 1$, that is, $n \ge N \implies |a_n| < \frac{1}{n^2}$. By comparison test, $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is convergent, then $\sum_{n=1}^{\infty} |a_n|$ is convergent, that is, $\sum_{n=1}^{\infty} a_n$ is absolutely convergent, which contradicts that $\sum_{n=1}^{\infty} a_n$ is conditionally convergent. □

iv Series conditional convergence can not determine index-multiplied-termed series divergence

For a real sequence $(a_n)_{n=1}^{\infty}$, the fact that $\sum_{n=1}^{\infty}$ is conditionally convergent can not determine whether $\sum_{n=1}^{\infty} n a_n$ is conditionally convergent or divergent.

Proof. An example of conditionally convergent $\sum_{n=1}^{\infty} a_n$ with convergent $\sum_{n=1}^{\infty} na_n$ is

$$a_n = \frac{(-1)^{n-1}}{n \ln n},$$

where both $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} na_n = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{\ln n}$ are convergent by alternating series test.

An example of conditionally convergent $\sum_{n=1}^{\infty} a_n$ with convergent $\sum_{n=1}^{\infty} na_n$ is

$$a_n = \frac{(-1)^{n-1}}{n},$$

where $\sum_{n=1}^{\infty} a_n$ is convergent by alternating series test and $\sum_{n=1}^{\infty} na_n = \sum_{n=1}^{\infty} (-1)^{n-1}$ is divergent since $\lim_{n\to\infty} a_n$ is divergent. □

v (d'Alembert's ratio or Cauchy) ratio test

1. If $\limsup_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| < 1$, then the series $\sum_{n=1}^{\infty} a_n$ is absolutely convergent.
2. If $\liminf_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| > 1$, then the series $\sum_{n=1}^{\infty} a_n$ is divergent.
3. Otherwise, the test is inconclusive.

Proof. For the first statement, choose an arbitrary r such that $\limsup_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| < r < 1$. There exists an integer N such that

$$\left|\frac{a_{n+1}}{a_n}\right| < r, \quad \forall n \geq N.$$

Thus,

$$\left|a_{N+k}\right| < \left|a_N\right| r^k, \quad \forall k \geq 1.$$

Since

$$\sum_{k=1}^{\infty} \left|a_N\right| r^k$$

is convergent, by comparison test, $\sum_{n=1}^{\infty} \left|a_n\right|$ is convergent.

For the second statement,

$$\lim_{n\to\infty} a_n \neq 0,$$

thus $\sum_{n=1}^{\infty} a_n$ is divergent. □

vi Ratio test for power series

Suppose that the power series $\sum_{n=0}^{\infty} c_n(x-a)^n$ satisfies $c_n \neq 0$ for all n. If $\lim_{n\to\infty} \left|c_n/c_{n+1}\right|$ exists in $\bar{\mathbb{R}}_{\geq 0}$, then it is equal to the radius of convergence of $\sum_{n=0}^{\infty} c_n(x-a)^n$.

Proof. If

$$L := \lim_{n\to\infty} \left|c_n/c_{n+1}\right| \in \bar{\mathbb{R}}_{>0},$$

$$\lim_{n\to\infty} \left|\frac{c_{n+1}(x-a)^{n+1}}{c_n(x-a)^n}\right| = \frac{|x-a|}{L}.$$

Use ratio test. For $|x-a| < L$, $\frac{|x-a|}{L} < 1$, thus the series is convergent. For $|x-a| > L$, $\frac{|x-a|}{L} > 1$, thus the series is divergent.

If

$$\lim_{n\to\infty} \left|c_n/c_{n+1}\right| = 0,$$

for all $x \neq a$,

$$\lim_{n\to\infty} \left|\frac{c_{n+1}(x-a)^{n+1}}{c_n(x-a)^n}\right| = \infty.$$

Use ratio test. For all $x \neq a$, the series is divergent.

Thus, L is the radius of convergence of $\sum_{n=0}^{\infty} c_n(x-a)^n$. □

vii (Cauchy) root test

1. If $\limsup_{n\to\infty} \sqrt[n]{|a_n|} < 1$, then the series $\sum_{n=1}^{\infty} a_n$ is absolutely convergent.
2. If $\liminf_{n\to\infty} \sqrt[n]{|a_n|} > 1$, then the series $\sum_{n=1}^{\infty} a_n$ is divergent.
3. Otherwise, the test is inconclusive.

Proof. For the first statement, choose an arbitrary r such that $\limsup_{n\to\infty} \sqrt[n]{|a_n|} < r < 1$. There exists an integer N such that

$$\sqrt[n]{|a_n|} < r, \quad \forall n \geq N.$$

Thus,

$$|a_n| < r^n, \quad \forall n \geq N.$$

Since

$$\sum_{n=1}^{\infty} r^n$$

is convergent, by comparison test, $\sum_{n=1}^{\infty} |a_n|$ is convergent.

For the second statement,

$$\neg \lim_{n\to\infty} a_n = 0.$$

Thus $\sum_{n=1}^{\infty} a_n$ is divergent. □

viii Raabe's test or Rabbe–Duhamel test

For a series $\sum_{n=1}^{\infty} a_n$ of positive terms, define

$$R_n = n\left(\frac{a_n}{a_{n+1}} - 1\right)$$

- If $\limsup_{n\to\infty} R_n < 1$, then the series converges.
- If $\liminf_{n\to\infty} R_n > 1$, then the series converges.
- Otherwise, the test is inconclusive.

Proof. Case $\limsup_{n\to\infty} R_n < 0$: $a_{n+1} \geq a_n$ for sufficiently large n, so the series diverges.

Case $0 \leq \limsup_{n\to\infty} R_n < 1$: There exists $R < 1$ and $N \in \mathbb{N}$ such that $R_n \leq R$ for all $n \geq N$, that is,

$$\frac{a_n}{a_{n+1}} \leq \left(1 + \frac{R}{n}\right) \leq e^{\frac{R}{n}}.$$

Thus

$$a_{n+1} \geq a_n e^{-\frac{R}{n}}$$

This implies that

$$a_{n+1} \geq a_N e^{-R\sum_{i=N}^{n} i^{-1}} \geq c a_N e^{-R\ln n} = c a_N n^{-R}$$

Since $R < 1$, by p-series test, the series diverges.

Case $\liminf_{n\to\infty} R_n > 1$: There exists $R > 1$ and $N \in \mathbb{N}$ such that $R_n \geq R$ for all $n \geq N$, that is,

$$\frac{a_n}{a_{n+1}} \geq 1 + \frac{R}{n}$$

Thus

$$\ln a_n - \ln a_{n+1} \geq \ln\left(1 + \frac{R}{n}\right)$$

$$\ln a_{n+1} \leq \ln a_N - \sum_{i=N}^{n} \ln\left(1 + \frac{R}{i}\right) \leq -R\ln N + O(1)$$

This implies that

$$a_{n+1} \leq cN^{-R}$$

Since $R > 1$, by p-series test, the series diverges. □

ix Riemann series theorem or Riemann rearrangement theorem

Let $\langle a_n \rangle_{n=1}^{\infty}$ be a real sequence, and that $\sum_{n=1}^{\infty} a_n$ is conditionally convergent. Then, for any extended-real number M, there exists a permutation σ on $\mathbb{N}$ such that

$$\sum_{n=1}^{\infty} a_{\sigma(n)} = M.$$

Proof. Since $\sum_{n=1}^{\infty} a_n$ is conditionally convergent, $\langle a_n \rangle$ has infinitely many positive terms and infinitely many negative terms. Let

$$a_n^+ = \begin{cases} a_n, & a_n > 0 \\ 0, & a_n \leq 0 \end{cases},$$

$$a_n^- = \begin{cases} a_n, & a_n < 0 \\ 0, & a_n \geq 0 \end{cases}.$$

Since $\sum_{n=1}^{\infty} a_n$ is conditionally convergent,

$$\sum_{n=1}^{\infty} a_n^+ = \infty,$$

$$\sum_{n=1}^{\infty} a_n^- = -\infty.$$

For any real number M, let p_1 be the smallest positive integer such that

$$M < \sum_{n=1}^{p_1} a_n^+,$$

and let q_1 be the smallest positive integer such that

$$M > \sum_{n=1}^{p_1} a_n^+ + \sum_{n=1}^{q_1} a_n^-.$$

Now recursively, for all $k \in \mathbb{N}_{>1}$, let p_k be the smallest integer such that $p_k > p_{k-1}$ and

$$M < \sum_{n=1}^{p_k} a_n^+ + \sum_{n=1}^{q_{k-1}} a_n^-,$$

and let q_k be the smallest integer such that $q_k > q_{k-1}$ and

$$M > \sum_{n=1}^{p_k} a_n^+ + \sum_{n=1}^{q_k} a_n^-.$$

The new sequence is defined as

$$\langle a_n^+ \rangle_{n=1 \wedge a_n \geq 0}^{p_1} \langle a_n^- \rangle_{n=1 \wedge a_n < 0}^{q_1} \langle \langle a_n^+ \rangle_{n=p_k+1 \wedge a_n \geq 0}^{p_{k+1}} \langle a_n^- \rangle_{n=q_k+1 \wedge a_n < 0}^{q_{k+1}} \rangle_{k=1}^{\infty}.$$

We have

$$\sum_{n=1}^{p_{k+1}-1} a_n^+ + \sum_{n=1}^{q_k} a_n^- \le M < \sum_{n=1}^{p_{k+1}} a_n^+ + \sum_{n=1}^{q_k} a_n^-,$$

$$0 < \left(\sum_{n=1}^{p_{k+1}} a_n^+ + \sum_{n=1}^{q_k} a_n^-\right) - M \le a_{p_{k+1}}^+.$$

Since $\sum_{n=1}^{\infty} a_n$ is conditionally convergent,

$$\lim_{k\to\infty} a_{p_{k+1}}^+ = 0.$$

By squeeze theorem,

$$\lim_{k\to\infty} \sum_{n=1}^{p_{k+1}} a_n^+ + \sum_{n=1}^{q_k} a_n^- = M.$$

Similarly,

$$\sum_{n=1}^{p_k} a_n^+ + \sum_{n=1}^{q_k-1} a_n^- \ge M > \sum_{n=1}^{p_k} a_n^+ + \sum_{n=1}^{q_k} a_n^-,$$

$$0 \ge M - \left(\sum_{n=1}^{p_k} a_n^+ + \sum_{n=1}^{q_k-1} a_n^-\right) > a_{q_k}^-.$$

Since $\sum_{n=1}^{\infty} a_n$ is convergent,

$$\lim_{k\to\infty} a_{q_k}^- = 0.$$

By squeeze theorem,

$$\lim_{k\to\infty} \sum_{n=1}^{p_k} a_n^+ + \sum_{n=1}^{q_k} a_n^- = M.$$

Since for all $p_k < p < p_{k+1}$,

$$\sum_{n=1}^{p_k} a_n^+ + \sum_{n=1}^{q_k} a_n^- \le \sum_{n=1}^{p} a_n^+ + \sum_{n=1}^{q_k} a_n^- \le \sum_{n=1}^{p_{k+1}} a_n^+ + \sum_{n=1}^{q_k} a_n^-,$$

and for all $q_k < q < q_{k+1}$,

$$\sum_{n=1}^{p_{k+1}} a_n^+ + \sum_{n=1}^{q_k} a_n^- \ge \sum_{n=1}^{p_{k+1}} a_n^+ + \sum_{n=1}^{q} a_n^- \ge \sum_{n=1}^{p_{k+1}} a_n^+ + \sum_{n=1}^{q_{k+1}} a_n^-,$$

by squeeze theorem, the new sequence converges to M.

For ∞ case ($-\infty$ case can be proven similarly), let p_1 be the smallest positive integer such that

$$1 < \sum_{n=1}^{p_1} a_n^+,$$

and let q_1 be the smallest positive integer such that

$$1 > \sum_{n=1}^{p_1} a_n^+ + \sum_{n=1}^{q_1} a_n^-.$$

Now recursively, for all $k \in \mathbb{N}_{>1}$, let p_k be the smallest integer such that $p_k > p_{k-1}$ and

$$k < \sum_{n=1}^{p_k} a_n^+ + \sum_{n=1}^{q_{k-1}} a_n^-,$$

and let q_k be the smallest integer such that $q_k > q_{k-1}$ and

$$k > \sum_{n=1}^{p_k} a_n^+ + \sum_{n=1}^{q_k} a_n^-.$$

The new sequence is defined as

$$\langle a_n^+ \rangle_{n=1 \wedge a_n \geq 0}^{p_1} \langle a_n^- \rangle_{n=1 \wedge a_n < 0}^{q_1} \langle \langle a_n^+ \rangle_{n=p_k+1 \wedge a_n \geq 0}^{p_{k+1}} \langle a_n^- \rangle_{n=q_k+1 \wedge a_n < 0}^{q_{k+1}} \rangle_{k=1}^{\infty}.$$

Thus for arbitrary large k, we have

$$k < \sum_{n=1}^{p_k} a_n^+ + \sum_{n=1}^{q_{k-1}} a_n^-,$$

and

$$k < \sum_{n=1}^{p_k} a_n^- + \sum_{n=1}^{q_k - 1} a_n^+.$$

Since for all $p_k < p < p_{k+1}$,

$$\sum_{n=1}^{p_k} a_n^+ + \sum_{n=1}^{q_k} a_n^- \leq \sum_{n=1}^{p} a_n^+ + \sum_{n=1}^{q_k} a_n^- \leq \sum_{n=1}^{p_{k+1}} a_n^+ + \sum_{n=1}^{q_k} a_n^-,$$

and for all $q_k - 1 < q < q_{k+1} - 1$,

$$\sum_{n=1}^{p_{k+1}} a_n^+ + \sum_{n=1}^{q_k - 1} a_n^- \geq \sum_{n=1}^{p_{k+1}} a_n^+ + \sum_{n=1}^{q} a_n^- \geq \sum_{n=1}^{p_{k+1}} a_n^+ + \sum_{n=1}^{q_{k+1}-1} a_n^-,$$

by squeeze theorem, the new sequence has limit ∞. $\square$

x Cauchy–Hadamard theorem

For any given power series $\sum_{n=0}^{\infty} c_n (x-a)^n$, let

$$L = \limsup_{n \to \infty} \sqrt[n]{|c_n|} \in \bar{\mathbb{R}}_{\geq 0},$$

then,

$$R = \begin{cases} \infty, & L = 0 \\ \dfrac{1}{L}, & L \in \mathbb{R}_{>0} \\ 0, & L = \infty \end{cases},$$

called radius of convergence, is such that the series is absolute convergent if $|x-a| < R$ and divergent if $|x-a| > R$, and if $R = 0$, the series is absolutely convergent at $x = a$ and divergent elsewhere.

Proof. Let $r = |x - a|$.

$$\limsup_{n\to\infty} \sqrt[n]{\left|c_n(x-a)^n\right|} = \begin{cases} Lr, & r \neq 0 \\ 0, & r = 0 \end{cases}.$$

If $L = 0$, then the series is absolutely convergent for all x, thus $R = \infty$.

Else if $L \in \mathbb{R}_{>0}$, then the series is absolutely convergent for $r < \frac{1}{L}$ and divergent for $r > \frac{1}{L}$, thus $R = \frac{1}{L}$.

Else if $L = \infty$, then the series is absolutely convergent at $x = a$ and divergent elsewhere, thus $R = 0$. □

xi Power series term-by-term differentiation and integration theorem

If the power series $\sum_{n=0}^{\infty} c_n(x-a)^n$ has radius of convergence $R > 0$, then the function f defined as

$$f(x) = \sum_{n=0}^{\infty} c_n(x-a)^n$$

is differentiable on $(a - R, a + R)$, and

$$f'(x) = \sum_{n=0}^{\infty} nc_n(x-a)^{n-1},$$

$$\int f(x)\,\mathrm{d}x = C + \sum_{n=0}^{\infty} c_n \frac{(x-a)^{n+1}}{n+1},$$

and the radii of convergence of both of them are R.

Proof. PLACEHOLDER □

xii Power series multiplying rational function of n does not affect radius of convergence

If the power series $\sum_{n=0}^{\infty} c_n(x-a)^n$ has radius of convergence R and $Q(n)$ is a rational function of n, then the power series $\sum_{n=0}^{\infty} Q(n)c_n(x-a)^n$ also has radius of convergence R.

Proof. Let

$$Q(n) = \frac{P(n)}{S(n)}$$

where $P(n)$ and $S(n)$ are polynomial functions and $S(n) \neq 0$.

Let

$$m = \deg P - \deg S.$$

There exists $A, B \in \mathbb{R}$ such that

$$An^m \leq Q(n) \leq Bn^m$$

for sufficiently large n.

$$\lim_{n\to\infty} \sqrt[n]{|n^m|} = \lim_{n\to\infty} n^{m/n} = \lim_{n\to\infty} e^{m\ln n/n} = 1$$

$$\lim_{n\to\infty} \sqrt[n]{|A|} = \lim_{n\to\infty} \sqrt[n]{|B|} = 1$$

Thus

$$\lim_{n\to\infty} \sqrt[n]{|Q(n)|} = 1$$

$$\limsup_{n\to\infty} \sqrt[n]{|Q(n)c_n|} = \limsup_{n\to\infty} \sqrt[n]{|c_n|} = \frac{1}{R}$$

By Cauchy–Hadamard theorem, the statement holds. □

xiii Cauchy's convergence test

A series $\sum_{n=0}^{\infty} a_n$ converges iff for every $\varepsilon > 0$, there is a natural number N such that

$$\left|\sum_{i=n+1}^{n+p} a_i\right| < \varepsilon$$

for all $n \geq N$ and $p \geq 1$.

xiv Mertens' theorem of Cauchy product

If $\sum_{n=0}^{\infty} a_n$ converges to A, $\sum_{n=0}^{\infty} b_n$ converges to B, and at least one of them converges absolutely, then their Cauchy product converges to AB.

If $\sum_{n=0}^{\infty} a_n$ converges to A, $\sum_{n=0}^{\infty} b_n$ converges to B, and both of them converge absolutely, then their Cauchy product converges absolutely to AB.

Proof. PLACEHOLDER □

xv Multiplication of power series

If $f(x) = \sum_{n=0}^{\infty} a_n x^n$ and $g(x) = \sum_{n=0}^{\infty} b_n x^n$ for $x \in R$ then for $x \in R$,

$$f(x)g(x) = \sum_{n=0}^{\infty} a_n b_n x^n.$$

Proof. PLACEHOLDER □

xvi Division of power series

If $f(x) = \sum_{n=0}^{\infty} a_n x^n$ and $g(x) = \sum_{n=0}^{\infty} b_n x^n$ for $x \in R$ and $b_0 \neq 0$, then, with

$$c_0 = \frac{a_0}{b_0},$$

$$c_n = \frac{1}{b_0}\left(a_n - \sum_{k=1}^{n} b_k c_{n-k}\right),$$

for $x \in R$,

$$\frac{f(x)}{g(x)} = \sum_{n=0}^{\infty} c_n x^n.$$

Proof. PLACEHOLDER □

II p-test and $\frac{1}{x^p(\ln x)^q}$-test

i p-test (for improper integral)

For any fixed $a > 0$, $\int_a^\infty \frac{1}{x^p}\,\mathrm{d}x$ converges iff $p > 1$, and if so,

$$\int_a^\infty \frac{1}{x^p}\,\mathrm{d}x = \frac{1}{(p-1)a^{p-1}};$$

$\int_0^a \frac{1}{x^p}\,\mathrm{d}x$ converges iff $p < 1$, and if so,

$$\int_0^a \frac{1}{x^p}\,\mathrm{d}x = -\frac{1}{(p-1)a^{p-1}}.$$

Proof.

$$\int \frac{1}{x^p}\,\mathrm{d}x = \begin{cases} -\frac{1}{(p-1)x^{p-1}}, & p \neq 1 \\ \ln(x), & p = 1 \end{cases}$$

$$\forall p < 1 : \nexists \lim_{x\to\infty} -\frac{1}{(p-1)x^{p-1}}$$

$$\nexists \lim_{x\to\infty} \ln(x)$$

$$\forall p > 1 : \lim_{x\to\infty} -\frac{1}{(p-1)x^{p-1}} = 0 \quad \int_a^\infty \frac{1}{x^p}\,\mathrm{d}x = \frac{1}{(p-1)a^{p-1}}$$

$$\forall p > 1 : \nexists \lim_{x\to 0^+} -\frac{1}{(p-1)x^{p-1}}$$

$$\nexists \lim_{x\to 0^+} \ln(x)$$

$$\forall p < 1 : \lim_{x\to 0^+} -\frac{1}{(p-1)x^{p-1}} = 0 \quad \int_0^a \frac{1}{x^p}\,\mathrm{d}x = -\frac{1}{(p-1)a^{p-1}}$$

□

ii p-test for series or p-series test

For any fixed $a \in \mathbb{N}$, $\sum_{n=a}^{\infty} \frac{1}{n^p}$, called p-series, converges iff $p > 1$.

Proof. For $p \leq 0$, $\sum_{n=a}^{\infty} \frac{1}{n^p} = \infty$. For $p > 0$, $f(x) = \frac{1}{x^p}$ is monotone decreasing on $[1, \infty)$. By integral test and p-test, the statement holds. □

iii $\frac{1}{x^p (\ln x)^q}$-test (for improper integral)

For any fixed $a > 1$, $\int_a^{\infty} \frac{1}{x^p (\ln x)^q} \, dx$ converges iff $p > 1 \vee (p = 1 \wedge q > 1)$.

For any fixed $0 < a < 1$, $\int_0^{a} \frac{1}{x^p (\ln x)^q} \, dx$ converges iff $p < 1 \vee (p = 1 \wedge q < 1)$.

Proof. For $p > 1$, take a ε such that $p - 1 > \varepsilon > 0$. $\frac{1}{x^{p-\varepsilon}} \geq \frac{1}{x^p (\ln x)^q}$ for all $x \geq b$ and $\frac{1}{x^{p-\varepsilon}} \leq \frac{1}{x^p (\ln x)^q}$ for all $0 < x \leq b$ for some $b > 0$. $\int_a^{\infty} \frac{1}{x^p (\ln x)^q} \, dx$ converges by p-test. Thus, by comparison theorem, $\int_a^{\infty} \frac{1}{x^p (\ln x)^q} \, dx$ converges. $\int_0^{a} \frac{1}{x^p (\ln x)^q} \, dx$ diverges by p-test. Thus, by comparison theorem, $\int_0^{a} \frac{1}{x^p (\ln x)^q} \, dx$ diverges.

For $p < 1$, take a ε such that $1 - p > \varepsilon > 0$. $\frac{1}{x^{p-\varepsilon}} \leq \frac{1}{x^p (\ln x)^q}$ for all $x \geq b$ and $\frac{1}{x^{p-\varepsilon}} \geq \frac{1}{x^p (\ln x)^q}$ for all $0 < x \leq b$ for some $b > 0$. $\int_a^{\infty} \frac{1}{x^p (\ln x)^q} \, dx$ diverges by p-test. Thus, by comparison theorem, $\int_a^{\infty} \frac{1}{x^p (\ln x)^q} \, dx$ diverges. $\int_0^{a} \frac{1}{x^p (\ln x)^q} \, dx$ converges by p-test. Thus, by comparison theorem, $\int_0^{a} \frac{1}{x^p (\ln x)^q} \, dx$ converges.

For $p = 1$,

$$u = \ln x, \quad du = \frac{1}{x} \, dx .$$

$$\int_a^{\infty} \frac{1}{x (\ln x)^q} \, dx = \int_{\ln a}^{\infty} \frac{1}{u^q} \, du ,$$

which converges iff $q > 1$ by p-test;

$$\int_0^{a} \frac{1}{x (\ln x)^q} \, dx = \int_0^{\ln a} \frac{1}{u^q} \, du ,$$

which converges iff $q < 1$ by p-test. □

iv $\frac{1}{x^p (\ln x)^q}$-test for series or $\frac{1}{x^p (\ln x)^q}$-series test

For any fixed $a \in \mathbb{N}$, $\sum_{n=a}^{\infty} \frac{1}{n^p (\ln n)^q}$ converges iff $p > 1 \vee (p = 1 \wedge q > 1)$.

Proof. For $p \le 0$, it's obviously ∞. For $p > 0$, $f(x) = \dfrac{1}{x^p(\ln x)^q}$ is monotone decreasing on $[b, \infty)$ for some $b > 1$. By integral test and $\dfrac{1}{x^p(\ln x)^q}$-test for improper integral, the statement holds. □

III Improper integral theorems

i Convergence of improper integral of monotone increasing/decreasing integrand implies non-positive/non-negative integrand

If function $f : [a, \infty) \to \mathbb{R}$ for some $a \in \mathbb{R}$ is monotone increasing and such that $\displaystyle\int_a^\infty f(x)\,\mathrm{d}x$ converges, then $f(x)$ is non-positive.

Proof. Use proof by contradiction. Suppose there exists $x_0 > a$ such that $f(x_0) > 0$. Then $f(x) > 0$ for all $x \ge x_0$ and

$$\int_{x_0}^\infty f(x)\,\mathrm{d}x = \infty.$$

Since $f(x)$ is monotone on $[a, x_0]$, it's BV and bounded on it, thus $\displaystyle\int_a^{x_0} f(x)\,\mathrm{d}x \in \mathbb{R}$. □

If function $f : [a, \infty) \to \mathbb{R}$ is monotone decreasing and such that $\displaystyle\int_a^\infty f(x)\,\mathrm{d}x$ converges, then $f(x)$ is non-negative.

ii Convergence of improper integral of monotone integrand implies limit of product of integrand and independent variable converges to 0

If function $f : [a, \infty) \to \mathbb{R}$ for some $a \in \mathbb{R}$ is motone and $\displaystyle\int_a^\infty f(x)\,\mathrm{d}x$ converges, then $\displaystyle\lim_{x\to\infty} x f(x) = 0$. (And $\displaystyle\lim_{x\to\infty} f(x) = 0$.)

Proof. $f(x)$ is either non-negative and non-increasing or non-positive and non-decreasing. Consider the former case.

Since $\displaystyle\int_a^\infty f(x)\,\mathrm{d}x$ and $f(x)$ is non-negative, $\forall \varepsilon > 0$, there exists $M > a$ such that $\forall A > M$:

$$0 \le \int_A^\infty f(x)\,\mathrm{d}x < \varepsilon.$$

Thus for $t > 2M$,

$$0 \le \int_{t/2}^t f(x)\,\mathrm{d}x = \int_{t/2}^\infty f(x)\,\mathrm{d}x - \int_t^\infty f(x)\,\mathrm{d}x < 2\varepsilon.$$

Since $f(x)$ is non-increasing,

$$f(x) \ge f(t), \quad \forall x \in [t/2, t]$$

$$\int_{t/2}^t f(x)\,\mathrm{d}x \ge \int_{t/2}^t f(t)\,\mathrm{d}x = \frac{t}{2} f(t)$$

$$t f(t) < 4\varepsilon.$$

Thus, $\forall \varepsilon > 0$, there exists $2M > 0$ such that $\forall t > 2M$,

$$t f(t) < \varepsilon,$$

that is,

$$\lim_{x\to\infty} x f(x) = 0.$$

□

IV Absolute convergence implies convergence

i For real series

A real series is convergent if it is absolutely convergent.

Proof. Suppose $\sum_{n=1}^{\infty} |a_n|$ is convergent.

$$0 \le \sum_{n=1}^{\infty} a_n + |a_n| \le \sum_{n=1}^{\infty} 2\,|a_n|$$

By comparison test theorem, $\sum_{n=1}^{\infty} a_n + |a_n|$ is convergent.

$$\sum_{n=1}^{\infty} a_n + |a_n| - \sum_{n=1}^{\infty} |a_n| = \sum_{n=1}^{\infty} a_n$$

□

ii For improper integral

An improper integral is convergent if it is absolutely convergent.

Proof. Suppose

$$\int_a^b |f(x)|\,\mathrm{d}x$$

is a convergent improper integral and

$$\int_a^c |f(x)|\,\mathrm{d}x$$

is proper integral for any $a < c < b$.

$$0 \le \int_a^b f(x) + |f(x)|\,\mathrm{d}x \le \int_a^b 2\,|f(x)|\,\mathrm{d}x$$

By comparison test theorem, $\int_a^b f(x) + |f(x)|\,\mathrm{d}x$ is convergent.

$$\int_a^b f(x) + |f(x)|\,\mathrm{d}x - \int_a^b |f(x)|\,\mathrm{d}x = \int_a^b f(x)\,\mathrm{d}x$$

□

V Deleting head

i For real series

For real sequence $\langle a_i \rangle_{i=m}^{\infty}$, $\sum_{i=m}^{\infty} a_i$ converges iff $\sum_{i=m+n}^{\infty} a_i$ converges, for all $n \in \mathbb{N}$.

ii For improper integral

If function $f : [a, b) \to \mathbb{R}$ is measurable with $b \in (a, \infty]$, $\int_a^b f(x)\,\mathrm{d}x$ is improper integral, and $\int_a^c f(x)\,\mathrm{d}x$ is proper integral for any $a < c < b$, then $\int_a^b f(x)\,\mathrm{d}x$ converges iff $\int_c^b f(x)\,\mathrm{d}x$ converges, for all $a < c < b$.

VI Term test

i For real series

If $\sum_{n=0}^{\infty} a_n$ converges, then $\lim_{n\to\infty} a_n = 0$.

Proof. Let

$$S_n = \sum_{k=0}^{n} a_k$$

and $S_{-1} = 0$.

$$a_n = S_n - S_{n-1}.$$

Suppose S_n converges to S. Then S_{n-1} converges to S. Then $S_n - S_{n-1}$ converges to $S - S = 0$. □

ii For improper integral

If function $f : [a, \infty) \to \mathbb{R}$ is measurable and of bounded variation, $\int_a^b f(x)\,\mathrm{d}x$ is proper integral for any $b > a$, and $\int_a^{\infty} f(x)\,\mathrm{d}x$ converges to some real number, then

$$\lim_{x\to\infty} f(x) = 0.$$

Proof. Let f^+ and f^- be the Jordan decomposition of f such that $f = f^+ - f^-$.

$$\int_a^{\infty} f(x)\,\mathrm{d}x = \int_a^{\infty} f^+(x)\,\mathrm{d}x - \int_a^{\infty} f^-(x)\,\mathrm{d}x$$

and that LHS converges iff both terms of RHS converges.

$\lim_{x\to\infty} f(x) = 0$ iff $\lim_{x\to\infty} f^+(x) = \lim_{x\to\infty} f^-(x)$.

Assume for contradiction that there exists $\varepsilon > 0$ and $M \geq a$ such that

$$\forall x \geq M : f^+(x) > \varepsilon.$$

Then, $\int_M^\infty f^+(x)$ diverges.

Assume for contradiction that there exists $\varepsilon > 0$ and $M \geq a$ such that

$$\forall x \geq M \;:\; f^-(x) > \varepsilon.$$

Then, $\int_M^\infty f^-(x)$ diverges. □

VII (Direct) comparison (test) (theorem)

i For real series

If real sequences $\langle a_n \rangle_{n=1}^\infty$ and $\langle b_n \rangle_{n=1}^\infty$ are such that $a_n \geq b_n \geq 0$ for all $n \in \mathbb{N}$, then

1. $\sum_{n=1}^\infty a_n$ is convergent implies $\sum_{n=1}^\infty b_n$ is convergent.
2. $\sum_{n=1}^\infty b_n$ is divergent implies $\sum_{n=1}^\infty a_n$ is divergent.

Proof. Let $\left\langle s_n = \sum_{i=1}^n a_i \right\rangle_{i=1}^\infty$ and $\left\langle t_n = \sum_{i=1}^n b_i \right\rangle_{i=1}^\infty$ be the sequences of partial sums of $\langle a_n \rangle_{n=1}^\infty$ and $\langle b_n \rangle_{n=1}^\infty$.

s_n and t_n are monotone increasing for all $n \in \mathbb{N}$ since $a_n, b_n \geq 0$. $s_n \geq t_n$ for all $n \in \mathbb{N}$ since $a_n \geq b_n$.

Statement 1:

$$\sum_{n=1}^\infty a_n = \lim_{n\to\infty} s_n = L.$$

Since

$$0 \leq t_n \leq s_n,$$

then,

$$\sum_{n=1}^\infty b_n = \lim_{n\to\infty} t_n$$

exists and $\in [0, L]$.

Statement 2 is the contrapositive of statement 1. □

ii For improper integral

If functions $f : [a, b) \to \mathbb{R}$ and $g : [a, b) \to \mathbb{R}$ are measurable with $b \in (a, \infty]$, $\int_a^b f(x)\,\mathrm{d}x$ and $\int_a^b g(x)\,\mathrm{d}x$ are improper integrals, $\int_a^c f(x)\,\mathrm{d}x$ and $\int_a^c g(x)\,\mathrm{d}x$ are proper integrals for any $a < c < b$, and $f(x) \geq g(x) \geq 0$ a.e. for $x \in [a, b)$, then

1. $\int_a^b f(x)\,\mathrm{d}x$ is convergent implies $\int_a^b g(x)\,\mathrm{d}x$ is convergent.

2. $\int_a^b g(x)\,\mathrm{d}x$ is divergent implies $\int_a^b f(x)\,\mathrm{d}x$ is divergent.

Proof. Let

$$F(t) = \int_a^t f(x)\,\mathrm{d}x\,, \quad t \in [a,b),$$
$$G(t) = \int_a^t g(x)\,\mathrm{d}x\,, \quad t \in [a,b).$$

F and G are monotone increasing since $f(x), g(x) \geq 0$ a.e. for $x \in [a,b)$.

$F(x) \geq G(x)$ since $f(x) \geq g(x)$ a.e. for $x \in [a,b)$.

Statement 1:

$$\int_a^b f(x)\,\mathrm{d}x = \lim_{t\to b} F(t) = L.$$

Since

$$0 \leq G(x) \leq F(x),$$

then,

$$\int_a^b g(x)\,\mathrm{d}x = \lim_{t\to b} G(t)$$

exists and $\in [0, L]$.

Statement 2 is the contrapositive of statement 1. □

VIII Limit comparison (test) (theorem) (LCT)

i For real series

If real sequences $\langle a_n\rangle_{n=1}^{\infty}$ and $\langle b_n\rangle_{n=1}^{\infty}$ are such that $a_n \geq 0$ and $b_n > 0$ for all $n \geq m$ for some $m \in \mathbb{N}$, if

$$L = \lim_{n\to\infty} \frac{a_n}{b_n} \in [0, \infty],$$

then:

1. If $0 < L < \infty$, then $\sum_{n=1}^{\infty} a_n$ is convergent iff $\sum_{n=1}^{\infty} b_n$ is convergent. (classical LCT)
2. if $L = 0$, then:
   (a) $\sum_{n=1}^{\infty} b_n$ is convergent implies $\sum_{n=1}^{\infty} a_n$ is convergent.
   (b) $\sum_{n=1}^{\infty} a_n$ is divergent implies $\sum_{n=1}^{\infty} b_n$ is divergent.

   (one-sided LCT)
3. If $L = \infty$, then:
   (a) $\sum_{n=1}^{\infty} a_n$ is convergent implies $\sum_{n=1}^{\infty} b_n$ is convergent.

(b) $\sum_{n=1}^{\infty} b_n$ is divergent implies $\sum_{n=1}^{\infty} a_n$ is divergent.

(one-sided LCT)

Proof. Statement 1:

For every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that for all $n \geq N$, $\left|\frac{a_n}{b_n} - L\right| < \varepsilon$, equivalently,

$$(L-\varepsilon)b_n < a_n < (L+\varepsilon)b_n.$$

As $L > 0$, choose ε such that $L - \varepsilon > 0$, so that

$$\frac{a_n}{L+\varepsilon} < b_n < \frac{a_n}{L-\varepsilon}.$$

$\sum_{n=1}^{\infty} \frac{a_n}{L+\varepsilon}$ is convergent iff $\sum_{n=1}^{\infty} a_n$ is convergent iff $\sum_{n=1}^{\infty} \frac{a_n}{L-\varepsilon}$ is convergent.

And by direct comparison test,

- $\sum_{n=1}^{\infty} \frac{a_n}{L-\varepsilon}$ is convergent implies $\sum_{n=1}^{\infty} b_n$ is convergent,
- $\sum_{n=1}^{\infty} b_n$ is convergent implies $\sum_{n=1}^{\infty} \frac{a_n}{L+\varepsilon}$ is convergent.

Statement 2:

There exists $N \in \mathbb{N}$ such that for all $n \geq N$, $0 \leq \frac{a_n}{b_n} < 1$, equivalently,

$$0 \leq a_n < b_n.$$

By direct comparison test of $\sum_{n=N}^{\infty} a_n$ and $\sum_{n=N}^{\infty} b_n$, the statements hold.

Statement 3:

There exists $N \in \mathbb{N}$ such that for all $n \geq N$, $\frac{a_n}{b_n} > 1$, equivalently,

$$a_n > b_n > 0.$$

By direct comparison test of $\sum_{n=N}^{\infty} a_n$ and $\sum_{n=N}^{\infty} b_n$, the statements hold. □

ii For improper integral

If functions $f : [a,b) \to \mathbb{R}$ and $g : [a,b) \to \mathbb{R}$ are measurable with $b \in (a,\infty]$, $\int_a^b f(x)\,\mathrm{d}x$ and $\int_a^b g(x)\,\mathrm{d}x$ are improper integrals, $\int_a^c f(x)\,\mathrm{d}x$ and $\int_a^c g(x)\,\mathrm{d}x$ are proper integrals for any $a < c < b$, $f(x) \geq 0$ and $g(x) > 0$ for $x \in [a,b)$, if

$$L = \lim_{x\to b^-} \frac{f(x)}{g(x)} \in [0,\infty],$$

then:

1. If $0 < L < \infty$, then $\int_a^b f(x)\,\mathrm{d}x$ is convergent iff $\int_a^b g(x)\,\mathrm{d}x$ is convergent. (classical LCT)
2. if $L = 0$, then:
   (a) $\int_a^b g(x)\,\mathrm{d}x$ is convergent implies $\int_a^b f(x)\,\mathrm{d}x$ is convergent.
   (b) $\int_a^b f(x)\,\mathrm{d}x$ is divergent implies $\int_a^b g(x)\,\mathrm{d}x$ is divergent.

   (one-sided LCT)
3. If $L = \infty$, then:
   (a) $\int_a^b g(x)\,\mathrm{d}x$ is divergent implies $\int_a^b f(x)\,\mathrm{d}x$ is divergent.
   (b) $\int_a^b f(x)\,\mathrm{d}x$ is convergent implies $\int_a^b g(x)\,\mathrm{d}x$ is convergent.

   (one-sided LCT)

Proof. Statement 1:

For every $\varepsilon > 0$, there exists $a \le M < b$ such that for all $b > x \ge M$, $\left|\frac{f(x)}{g(x)} - L\right| < \varepsilon$, equivalently,

$$(L-\varepsilon)g(x) < f(x) < (L+\varepsilon)g(x).$$

As $L > 0$, we can choose ε such that $L - \varepsilon > 0$, so that

$$\frac{f(x)}{L+\varepsilon} < g(x) < \frac{f(x)}{L-\varepsilon}.$$

$\int_a^b \frac{f(x)}{L+\varepsilon}\,\mathrm{d}x$ is convergent iff $\int_a^b f(x)\,\mathrm{d}x$ is convergent iff $\int_a^b \frac{f(x)}{L-\varepsilon}\,\mathrm{d}x$ is convergent.

And by direct comparison test,

- $\int_a^b \frac{f(x)}{L-\varepsilon}\,\mathrm{d}x$ is convergent implies $\int_a^b g(x)\,\mathrm{d}x$ is convergent,
- $\int_a^b g(x)\,\mathrm{d}x$ is convergent implies $\int_a^b \frac{f(x)}{L+\varepsilon}\,\mathrm{d}x$ is convergent.

Statement 2:

There exists $a \le M < b$ such that for all $b > x \ge M$, $0 \le \frac{f(x)}{g} < 1$, equivalently,

$$0 \le f(x) < g(x).$$

By direct comparison test of $\int_M^b f(x)\,\mathrm{d}x$ and $\int_M^b g(x)\,\mathrm{d}x$, the statements hold.

Statement 3:

There exists $a \le M < b$ such that for all $b > x \ge M$, $\frac{f(x)}{g(x)} > 1$, equivalently,

$$f(x) > g(x) > 0.$$

By direct comparison test of $\int_M^b f(x)\,\mathrm{d}x$ and $\int_M^b g(x)\,\mathrm{d}x$, the statements hold. □

IX Dirichlet test or Abel's test

i For real series

If $\langle a_n \rangle_{n=1}^{\infty}$ is a sequence such that the sequence of partial sums $\langle s_n = \begin{cases} 0, & n = 0 \\ \sum_{i=0}^{n} a_i, & n \in \mathbb{N} \end{cases} \rangle_{n=0}^{\infty}$ is bounded and $\langle b_n \rangle_{n=1}^{\infty}$ is a monotone sequence such that $\lim_{n\to\infty} b_n = 0$, then

$$\sum_{n=1}^{\infty} a_n b_n$$

converges.

Proof. Choose M such that $\langle |s_n| \rangle_{n=0}^{\infty}$ is bounded above by M.

For any $\varepsilon > 0$, there is $N \in \mathbb{N}$ such that $|b_m| < \frac{\varepsilon}{2M}$ for all $m \geq N$.

For any $N \leq p \leq q$, using Abel transformation,

$$\begin{aligned}
\left|\sum_{n=p}^{q} a_n b_n\right| &= \left|\sum_{n=p}^{q} \left(s_{n+1} - s_n\right) b_n\right| \\
&= \left|s_{q+1} b_{q+1} - s_p b_p - \sum_{n=p}^{q} s_{n+1} \left(b_{n+1} - b_n\right)\right| \\
&\leq M \left(|b_{q+1}| + |b_p| + \sum_{n=p}^{q} |b_{n+1} - b_n| \right) \\
&= M \left(|b_{q+1}| + |b_p| + |b_{q+1} - b_p| \right) \\
&= M \left(|b_{q+1}| + |b_p| + |b_p| - |b_{q+1}| \right) \\
&= 2M |b_p| \\
&\leq \varepsilon
\end{aligned}$$

□

ii For improper integral

If $f : [a, b) \to \mathbb{R}$ with $b \in (a, \infty]$ is a function such that the function

$$F(x) = \int_a^x f(t) \,\mathrm{d}t, \quad b > x \geq a$$

is bounded, $g : [a, b) \to \mathbb{R}$ is a monotone and measurable function such that $\lim_{x\to b^-} g(x) = 0$, then

$$\int_a^b f(x) g(x) \,\mathrm{d}x$$

converges.

Proof. Rewrite as Riemann–Stieltjes integral

$$\int_a^b f(x) g(x) \,\mathrm{d}x = \int_{x=a}^{b} g(x) \,\mathrm{d}F(x).$$

Choose M such that $|F(x)|$ is bounded above by M.

For any $\varepsilon > 0$, there is $a \leq N < b$ such that $|g(N)| < \dfrac{\varepsilon}{2M}$ for all $m \geq N$.

For any $N \leq p \leq q < b$, using integration by parts of Riemann–Stieltjes integral,

$$\begin{aligned}\left|\int_{x=p}^{q} g(x)\,\mathrm{d}F(x)\right| &= \left|F(q)g(q) - F(p)g(p) - \int_{x=p}^{q} f(x)\,\mathrm{d}g(x)\right| \\ &\leq M\left(|g(q)| + |g(p)| + \int_{x=p}^{q} \mathrm{d}g(x)\right) \\ &= M\left(|g(q)| + |g(p)| + |g(q) - g(p)|\right) \\ &= M\left(|g(q)| + |g(p)| + |g(p)| - |g(q)|\right) \\ &= 2M\,|g(p)| \\ &\leq \varepsilon\end{aligned}$$

□

X Cesàro-summability theorem

i For real sequence

If a real sequence converges to $L \in \mathbb{R}$, then the sequence is Cesàro-summable and its Cesàro sum is L.

Proof. PLACEHOLDER □

ii For real function

If a real function f and $a \in \mathbb{R}$ is such that $\lim_{x\to\infty} f(x) = L \in \mathbb{R}$ and $\int_a^b f(x)\,\mathrm{d}x$ is a proper integral for all $b \in [a, \infty)$, then the function is Cesàro-summable over $[a, \infty)$ and its Cesàro sum over $[a, \infty)$ is L.

Proof. PLACEHOLDER □

If a real function f and $b \in \mathbb{R}$ is such that $\lim_{x\to-\infty} f(x) = L \in \mathbb{R}$ and $\int_a^b f(x)\,\mathrm{d}x$ is a proper integral for all $a \in (-\infty, b)$, then the function is Cesàro-summable over $(-\infty, b]$ and its Cesàro sum over $(-\infty, b]$ is L.

XI Harmonic series (調和級數) and alternating harmonic series

i Harmonic series

$$H_M = \sum_{n=1}^{M} \frac{1}{n}.$$

ii Alternating harmonic series

$$S_M = \sum_{n=1}^{M} \frac{(-1)^{n+1}}{n}.$$

iii Relation

$$S_{2M} = H_{2M} - H_M.$$

Proof.

$$S_{2M} = H_{2M} - 2\sum_{n=1}^{M} \frac{1}{2n} = H_{2M} - H_M$$

□

iv Euler's constant or Euler–Mascheroni constant

Let

$$H_n = \sum_{n=1}^{\infty} \frac{1}{n}$$

$$t_n = H_n - \ln n$$

$$\lim_{n\to\infty} t_n = \int_1^{\infty} \frac{1}{\lfloor x \rfloor} - \frac{1}{x} \, \mathrm{d}x = \gamma \in \mathbb{R},$$

which is called Euler's constant or Euler–Mascheroni constant and approximately 0.57721.

Proof.

$$t_n = \sum_{k=1}^{n} \frac{1}{k} - \sum_{k=1}^{n-1} \ln(k+1) - \ln k = \sum_{k=1}^{n-1} \int_k^{k+1} \frac{1}{k} - \frac{1}{x} \, \mathrm{d}x + \frac{1}{n} = \int_1^{n} \frac{1}{\lfloor x \rfloor} - \frac{1}{x} \, \mathrm{d}x + \frac{1}{n}$$

$$\lim_{n\to\infty} \frac{1}{n} = 0$$

$$\frac{1}{\lfloor x \rfloor} - \frac{1}{x} \geq 0$$

$$\frac{1}{n} > 0$$

$$t_n > 0$$

$$t_n - t_{n+1} = \ln(n+1) - \ln n - \frac{1}{n+1} = \int_n^{n+1} \frac{1}{x} - \frac{1}{n+1} \, \mathrm{d}x$$

Since $\frac{1}{x} \geq \frac{1}{n+1}$ for $x \in [n, n+1]$,

$$\int_n^{n+1} \frac{1}{x} - \frac{1}{n+1} \, \mathrm{d}x > 0$$

t_n is bounded below by 0, and t_n is decreasing, hence t_n is convergent. □

It's unknown whether γ is irrational.

v Alternating harmonic series convergence

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} = \ln 2.$$

Proof.

$$\begin{aligned}
\ln 2 &= \ln(1+1) \\
&= \int_0^1 \frac{1}{1+x} \, \mathrm{d}x \\
&= \int_0^1 \sum_{n=0}^{\infty} (-x)^n \, \mathrm{d}x \\
&= \sum_{n=0}^{\infty} \int_0^1 (-x)^n \, \mathrm{d}x \\
&= \sum_{n=0}^{\infty} (-1)^n \frac{1}{n+1} \\
&= \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}
\end{aligned}$$

□

XII Riemann zeta function or Euler–Riemann zeta function

i Definition

The Riemann zeta function or Euler–Riemann zeta function $\zeta(s)$ is a complex function defined as the analytic extension of

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}, \quad \forall \Re(s) > 1$$

such that it is meromorphic and with an only pole at $s = 1$.

PLACEHOLDER

ii With Gamma function

$$\frac{1}{\Gamma(s)} \int_0^{\infty} \frac{x^{s-1}}{e^x - 1} \, \mathrm{d}x \, .$$

Proof. PLACEHOLDER □

Functional equation

Euler product formula

PLACEHOLDER

Basel Problem (巴塞爾問題):

$$\zeta(2) = \frac{\pi^2}{6}$$

PLACEHOLDER

Other Even Positive Integers:

$$\zeta(4) = \frac{\pi^4}{90}$$

$$\zeta(6) = \frac{\pi^6}{945}$$

$$\zeta(8) = \frac{\pi^8}{9450}$$

$$\zeta(10) = \frac{\pi^{10}}{93555}$$

$$\zeta(12) = \frac{691\pi^{12}}{638512875}$$

$$\zeta(14) = \frac{2\pi^{14}}{18243225}$$

PLACEHOLDER

Infinity:

$$\lim_{n\to\infty} \zeta(n) = 1$$

PLACEHOLDER

Derivative

digamma function ψ

$$\psi(z) = \frac{\mathrm{d}\ln(\Gamma(z))}{\mathrm{d}z}$$

PLACEHOLDER

16 Mean

I Abstract mean

A map M from a bounded function whose domain is a subset of $\mathbb{R}$ to a real number is an abstract mean iff for all f, g that are bounded functions whose domains are subsets of $\mathbb{R}$ and $a, b \in \mathbb{R}$ such that both sides are defined,

- Linearity:

$$M(af + bg) = aM(f) + bM(g).$$

- Positivity:

$$f(x) \geq 0 \forall x \in D_f \implies M(f) \geq 0.$$

- Normalization:

$$f(x) = a \forall x \in D_f \implies M(f) = a.$$

II Arithmetic mean (AM)

i Sequence

For a finite sequence $\langle a_i \rangle_{i=l}^{n}$, the arithmetic mean $\bar{a}$ of it is defined as

$$\bar{a} = \frac{1}{n-l+1} \sum_{i=l}^{n} a_i .$$

ii Real-domain function

For a function $f(x)$ defined on a nonempty interval $[a,b] \subseteq D_f$, the arithmetic mean $\bar{f}$ of it is defined as

$$\bar{f} = \frac{1}{b-a} \int_a^b f(x) \, \mathrm{d}x \, .$$

iii Measurable function

For a measurable function f defined on a measurable set $\Omega \subseteq D_f$ with $\mu(\Omega) \in \mathbb{R}$, the arithmetic mean $\bar{f}$ of it is defined as

$$\bar{f} = \frac{1}{\mu(\Omega)} \int_\Omega f \, \mathrm{d}\mu \, .$$

III Geometric mean (GM)

i Sequence

For a finite sequence $\langle a_i \rangle_{i=l}^{n}$ of positive real number, the geometric mean of it is defined as

$$\exp\left(\frac{1}{n-l+1} \sum_{i=l}^{n} \ln a_i \right).$$

ii Real-domain function

For a positive real-valued function $f(x)$ defined on a nonempty interval $[a,b] \subseteq D_f$, the geometric mean of it is defined as

$$\exp\left(\frac{1}{b-a} \int_a^b \ln(f(x)) \, \mathrm{d}x \right).$$

IV Harmonic mean (HM)

i Sequence

For a finite sequence $\langle a_i \rangle_{i=l}^{n}$ of nonzero number, the harmonic mean of it is defined as

$$\frac{n-l+1}{\sum_{i=l}^{n} \frac{1}{a_i}}.$$

ii Real-domain function

For a nonzero real-valued function $f(x)$ defined on a nonempty interval $[a,b] \subseteq D_f$, the harmonic mean of it is defined as

$$\frac{b-a}{\int_a^b \frac{1}{f(x)}\,\mathrm{d}x}.$$

V Cesàro sum or Cesàro mean

i Sequence

For an infinite sequence $\langle a_i \rangle_{i=l}^{\infty}$, the Cesàro sum or Cesàro mean of it is defined as

$$\lim_{n\to\infty} \frac{1}{n} \sum_{i=l}^{n} a_i.$$

If the Cesàro sum is finite, the sequence is called Cesàro-summable.

ii Real-domain function

For a function f defined on $[a,\infty)$ for $a \in \mathbb{R}$ with $\displaystyle\int_a^b f(x)\,\mathrm{d}x$ being a proper integral for all $b \in (a,\infty)$, the Cesàro sum or Cesàro mean of it is defined as

$$\lim_{n\to\infty} \frac{1}{n} \int_a^n f(x)\,\mathrm{d}x\,.$$

If the Cesàro sum is finite, the function is called Cesàro-summable over $[a,\infty)$.

For a function f defined on $(-\infty,b]$ for $b \in \mathbb{R}$ with $\displaystyle\int_a^b f(x)\,\mathrm{d}x$ being a proper integral for all $a \in (-\infty,b)$, the Cesàro sum or Cesàro mean of it is defined as

$$\lim_{n\to\infty} \frac{1}{n} \int_{-n}^b f(x)\,\mathrm{d}x\,.$$

If the Cesàro sum is finite, the function is called Cesàro-summable over $(-\infty,b]$.

For a function f defined on $(-\infty,\infty)$ with $\displaystyle\int_a^b f(x)\,\mathrm{d}x$ being a proper integral for all $a < b \wedge a, b \in \mathbb{R}$, the Cesàro sum or Cesàro mean of it is defined as

$$\lim_{n\to\infty} \frac{1}{2n} \int_{-n}^n f(x)\,\mathrm{d}x\,.$$

If the Cesàro sum is finite, the function is called Cesàro-summable over $(-\infty,\infty)$.

VI Root mean square (RMS or rms) or quadratic mean

i Sequence

For a finite sequence $\langle a_i \rangle_{i=l}^{n}$, the RMS a_{rms} of it is defined as

$$a_{rms} = \sqrt{\frac{1}{n-l+1} \sum_{i=l}^{n} a_i^{\,2}}.$$

For an infinite sequence $\langle a_i \rangle_{i=l}^{\infty}$, the RMS a_{rms} of it is defined as

$$a_{rms} = \sqrt{\lim_{n\to\infty} \frac{1}{n-l+1} \sum_{i=l}^{n} {a_i}^2}.$$

ii Real-domain function

For a function $f(x)$ defined on a nonempty interval $[a, b]$, the RMS f_{rms} of it is defined as

$$f_{rms} = \sqrt{\frac{1}{b-a} \int_a^b (f(x))^2 \, \mathrm{d}x}.$$

For a function $f(x)$ defined on $[a, \infty)$ with $\displaystyle\int_a^b f(x) \, \mathrm{d}x$ being a proper integral for all $b \in (a, \infty)$, the RMS f_{rms} of it is defined as

$$f_{rms} = \sqrt{\lim_{b\to\infty} \frac{1}{b-a} \int_a^b (f(x))^2 \, \mathrm{d}x}.$$

For a function $f(x)$ defined on $(-\infty, b]$ with $\displaystyle\int_a^b f(x) \, \mathrm{d}x$ being a proper integral for all $a \in (-\infty, b)$, the RMS f_{rms} of it is defined as

$$f_{rms} = \sqrt{\lim_{a\to-\infty} \frac{1}{b-a} \int_a^b (f(x))^2 \, \mathrm{d}x}.$$

For a function $f(x)$ defined on $(-\infty, \infty)$ with $\displaystyle\int_a^b f(x) \, \mathrm{d}x$ being a proper integral for all $a < b \wedge a, b \in \mathbb{R}$, the RMS f_{rms} of it is defined as

$$f_{rms} = \sqrt{\lim_{n\to\infty} \frac{1}{2n} \int_{-n}^{n} (f(x))^2 \, \mathrm{d}x}.$$

17 Lists

I List of Sequences

i Arithmetic progression or sequence (等差數列)

An arithmetic sequence is a sequence $\langle a_n \rangle = \langle a_1 + (n-1)d \rangle$.

Given a and b, $\dfrac{a+b}{2}$ is called the median of an arithmetic sequence (等差中項).

$$\nexists \lim_{n\to\infty} a_n, \quad d \neq 0$$

$$\lim_{n\to\infty} a_n = a_1, \quad d = 0$$

ii Geometric progression or sequence (等比 or 幾何數列)

A geometric sequence is a sequence $\langle a_n \rangle = \langle a_1 \cdot r^{n-1} \rangle$, where $a_1 r \neq 0$.

Given a and b, $\pm\sqrt{ab}$ is called the median of an geometric sequence (等比中項).

$$\nexists \lim_{n\to\infty} a_n, \quad |d| \geq 1 \wedge d \neq 1$$

$$\lim_{n\to\infty} a_n = a_1, \quad d = 1$$

$$\lim_{n\to\infty} a_n = 0, \quad |d| < 1$$

iii $\left(1 - \frac{1}{n}\right)^n$

$$\lim_{n\to\infty} \left(1 - \frac{1}{n}\right)^n = e^{-1}$$

Proof. Let

$$\lim_{n\to\infty} \left(1 - \frac{1}{n}\right)^n = L$$

$$\lim_{n\to\infty} n \ln\left(1 - \frac{1}{n}\right) = \ln L$$

Near 0,

$$\ln(1 - x) = -x + O(x^2)$$

$$\lim_{n\to\infty} n \left(-\frac{1}{n} + O\left(\frac{1}{n^2}\right)\right) = -1 = \ln L$$

$$L = e^{-1}$$

□

II List of Series

i Arithmetic series (等差級數)

An arithmetic series is a series $S_n = \sum_{i=1}^{n} a_i$, where $\langle a_n \rangle$ is an arithmetic sequence.

$$S_n = \frac{n}{2}\left(a_1 + a_n\right) = \frac{n}{2}\left(2a_1 + (n-1)d\right) = na_1 + \frac{n(n-1)d}{2}$$

$$\nexists \lim_{n\to\infty} S_n, \quad a_1 \neq 0 \vee d \neq 0$$

$$\lim_{n\to\infty} S_n = 0, \quad a_1 = 0 \wedge d = 0$$

ii Geometric series (等比 or 幾何級數)

A geometric series is a series $S_n = \sum_{i=1}^{n} a_i$, where $\langle a_n \rangle$ is a geometric sequence.

$$S_n = \frac{a_1\left(1 - r^n\right)}{1 - r}, \quad r \neq 1$$

$$
\begin{aligned}
S_n &= na_1, \quad r = 1 \\
\lim_{n\to\infty} S_n &= \frac{a_1}{1-r}, \quad |r| < 1 \\
\nexists \lim_{n\to\infty} S_n&, \quad |r| \geq 1
\end{aligned}
$$

iii Power series (冪級數)

$$
\begin{aligned}
\sum_{i=1}^{n} i &= \frac{n(n+1)}{2} \\
\sum_{i=1}^{n} i^2 &= \frac{n(n+1)(2n+1)}{6} \\
\sum_{i=1}^{n} i^3 &= \left(\frac{n(n+1)}{2}\right)^2 \\
\sum_{i=1}^{n} i^r &= n + \sum_{k=1}^{n-1}(n-k)((k+1)^r - k^r) \\
&= n + \sum_{k=1}^{n-1}(n-k)\sum_{j=0}^{r-1}\binom{r}{j}k^j
\end{aligned}
$$

III List of piecewise Functions

i Absolute value (絕對值) or modulus (模)

The absolute value or modulus function $|\cdot| : \mathbb{R} \to \mathbb{R}$ is defined as:

$$
|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}.
$$

ii Sign function or signum function (符號函數)

The sign function $\operatorname{sgn} : \mathbb{R} \to \mathbb{R}$ is defined as:

$$
\operatorname{sgn}(x) = \begin{cases} \dfrac{x}{|x|}, & x \neq 0 \\ 0, & x = 0 \end{cases}.
$$

iii Floor function (下取整函數), integral part, integer part (整數部份), greatest integer, entier, rounding down (無條件捨去), or rounding toward negative infinity

The floor function $\lfloor\cdot\rfloor : \mathbb{R} \to \mathbb{R}$ or $[\cdot] : \mathbb{R} \to \mathbb{R}$ is defined as:

$$
\lfloor x \rfloor = [x] = \max\{m \in \mathbb{Z} \mid m \leq x\},
$$

in which $[\cdot]$ is called the Gauss sign (高斯符號).

iv Ceiling function (上取整函數), rounding up (無條件進位), or rounding toward positive infinity

The ceiling function $\lceil \cdot \rceil : \mathbb{R} \to \mathbb{R}$ is defined as:

$$\lceil x \rceil = \min\{m \in \mathbb{Z} \mid m \geq x\}.$$

v Truncation, rounding toward zero, or rounding away from infinity

The truncation $\text{truncate} : \mathbb{R} \to \mathbb{R}$ is defined as:

$$\text{truncate}(x) = \begin{cases} \lfloor x \rfloor, & x \geq 0 \\ \lceil x \rceil, & x < 0 \end{cases}$$

vi Rounding half up (四捨五入) or rounding half toward positive infinity

The rounding half up $\text{roundhalfup} : \mathbb{R} \to \mathbb{R}$ is defined as:

$$\text{roundhalfup}(x) = \left\lfloor x + \frac{1}{2} \right\rfloor.$$

vii Rounding half down (五捨六入) or rounding half toward negative infinity

The rounding half down $\text{roundhalfdown} : \mathbb{R} \to \mathbb{R}$ is defined as:

$$\text{roundhalfdown}(x) = \left\lceil x - \frac{1}{2} \right\rceil.$$

viii Rounding half toward zero or rounding half away from infinity

The rounding half toward zero $\text{roundhalftowardzero} : \mathbb{R} \to \mathbb{R}$ is defined as:

$$\text{roundhalftowardzero}(x) = \text{sgn}(x) \left\lceil |x| - \frac{1}{2} \right\rceil.$$

ix Rounding half away from zero or rounding half toward infinity

The rounding half away from zero $\text{roundhalfawayfromzero} : \mathbb{R} \to \mathbb{R}$ is defined as:

$$\text{roundhalfawayfromzero}(x) = \text{sgn}(x) \left\lfloor |x| + \frac{1}{2} \right\rfloor.$$

x Rounding half to even (四捨六入五成雙) or banker's rounding

The rounding half to even $\text{roundhalftoeven} : \mathbb{R} \to \mathbb{R}$ is defined as:

$$\text{roundhalftoeven}(x) = \begin{cases} \left\lfloor x + \frac{1}{2} \right\rfloor, & \frac{1}{2}\left\lfloor x + \frac{1}{2} \right\rfloor \in \mathbb{Z} \\ \left\lceil x - \frac{1}{2} \right\rceil, & \frac{1}{2}\left\lfloor x + \frac{1}{2} \right\rfloor \notin \mathbb{Z} \end{cases}.$$

xi Rounding half to odd (四捨六入五成單)

The rounding half to odd $\text{roundhalftoodd} : \mathbb{R} \to \mathbb{R}$ is defined as:

$$\text{roundhalftoodd}(x) = \begin{cases} \left\lfloor x + \frac{1}{2} \right\rfloor, & \frac{1}{2}\left\lfloor x + \frac{1}{2} \right\rfloor \notin \mathbb{Z} \\ \left\lceil x - \frac{1}{2} \right\rceil, & \frac{1}{2}\left\lfloor x + \frac{1}{2} \right\rfloor \in \mathbb{Z} \end{cases}.$$

xii Pulse wave, pulse train, or rectangular wave

A pulse wave with period T, duty cycle D, crest A_1, and trough A_0 is

$$f(t) = A_0 + \left(A_1 - A_0\right)\left(\left\lfloor \frac{t}{T} + D \right\rfloor - \left\lfloor \frac{t}{T} \right\rfloor\right).$$

The ratio of the high period to the total period of a pulse wave is called the duty cycle.

When the duty cycle is 50%, the pulse wave is called square wave.

xiii Triangular wave or triangle wave

A triangular wave with period T, crest A_1, and trough A_0 is

$$f(t) = A_0 + 2\left(A_1 - A_0\right)\left|\frac{t}{T} - \left\lfloor \frac{t}{T} + \frac{1}{2} \right\rfloor\right|.$$

xiv Sawtooth wave or saw wave

A sawtooth wave with period T, crest A_1, and trough A_0 is

$$f(t) = \frac{A_0 + A_1}{2} + \left(A_1 - A_0\right)\left(\frac{t}{T} - \left\lfloor \frac{t}{T} + \frac{1}{2} \right\rfloor\right).$$

xv Heaviside step function (黑維塞階躍函數) or Unit step function (單位階躍函數)

The Heaviside step function or unit step function $u(x): \mathbb{R} \to \mathbb{R}$, also denoted as $H(x)$, is defined as

$$u(x) = \begin{cases} 0, & x < 0 \\ 1, & x \geq 0 \end{cases}.$$

Some sources define $u(0) = 0$, $u(0) = \frac{1}{2}$, or u not defined at 0.

xvi Ramp function

The ramp function $R(x): \mathbb{R} \to \mathbb{R}$ is defined as

$$R(x) = \begin{cases} 0, & x < 0 \\ x, & x \geq 0 \end{cases}.$$

IV List of Limits of Real Functions

i Limit of sinc function at zero

$$\lim_{h \to 0} \frac{\sin h}{h} = 1.$$

Proof. On a unit circle with center O, consider the arc $\overset{\frown}{AB} = h$ subtended by acute angle h (i.e. $\frac{\pi}{2} > h > 0$), the chord length $\overline{BC} = \sin(h)$ with C be the intersection of the chord and $\overrightarrow{OA}$, and the tangent length $\overline{AD} = \tan(h)$ with D be the intersection of the tangent and $\overrightarrow{OB}$.

We can observe that

$$\overline{BC} < \overset{\frown}{AB}.$$

Take tangent of the circle at B. Let it intersects $\overline{AD}$ at E. We can observe that the adjacent $\overline{BE}$ is less than the hypothenuse $\overline{DE}$. Thus

$$\overset{\frown}{AB} < \overline{BE} + \overline{AE} < \overline{DE} + \overline{AE} = \overline{AD}.$$

Thus,

$$\sin h < h < \tan h.$$
$$1 < \frac{h}{\sin h} < \frac{1}{\cos h}.$$
$$\cos h < \frac{\sin h}{h} < 1.$$
$$\lim_{h\to 0} \cos h = 1.$$

By the squeeze theorem:

$$\lim_{h\to 0} \frac{\sin h}{h} = 1.$$

□

ii Limit of power over factorial

$$\lim_{n\to\infty} \frac{x^n}{n!} = 0, \quad x \in \mathbb{R}$$

Proof. PLACEHOLDER □

V List of Derivatives of Real Functions and Real Functions as Definite Integrals

i Power function

$$\frac{\mathrm{d}x^n}{\mathrm{d}x} = nx^{n-1}.$$
$$x^n = n\int_0^x t^{n-1}\,\mathrm{d}t\,.$$

Proof.

$$\frac{\mathrm{d}x^n}{\mathrm{d}x} = \lim_{h\to 0} \frac{(x+h)^n - x^n}{h} = \lim_{h\to 0} \frac{\sum_{k=1}^{\infty} \binom{n}{k} x^{n-k} h^k}{h} = nx^{n-1}.$$

□

ii Absolute value function

$$\frac{\mathrm{d}|x|}{\mathrm{d}x} = \mathrm{sgn}(x), \quad x \neq 0.$$

iii Logarithmic function

$$\frac{\mathrm{d}\ln(x)}{\mathrm{d}x} = \frac{1}{x}.$$

$$\ln(x) = \int_1^x \frac{1}{t}\,\mathrm{d}t\,, \quad x > 0.$$

iv Exponential function

$$\frac{\mathrm{d}a^x}{\mathrm{d}x} = \ln(a)a^x.$$

$$a^x = 1 + \int_0^x \ln(a)a^t\,\mathrm{d}t\,.$$

Proof.

$$\begin{aligned}
\frac{\mathrm{d}a^x}{\mathrm{d}x} &= \lim_{h\to 0}\frac{a^{x+h}-a^x}{h} \\
&= a^x\lim_{h\to 0}\frac{a^h-1}{h} \\
&= a^x\lim_{h\to 0}\frac{e^{\ln(a)h}-1}{h} \\
&= a^x\lim_{h\to 0}\frac{\lim_{n\to\infty}\sum_{k=0}^{n}\frac{(\ln(a)h)^k}{k!}-1}{h} \\
&= a^x\lim_{h\to 0}\frac{\lim_{n\to\infty}\sum_{k=1}^{n}\frac{(\ln(a)h)^k}{k!}}{h} \\
&= \ln(a)a^x
\end{aligned}$$

□

v Sine function

$$\frac{\mathrm{d}\sin(x)}{\mathrm{d}x} = \cos(x).$$

$$\sin(x) = \int_0^x \cos(t)\,\mathrm{d}t\,.$$

Proof.

$$\begin{aligned}
\frac{\mathrm{d}\sin(x)}{\mathrm{d}x} &= \lim_{h\to 0}\frac{\sin(x+h)-\sin(x)}{h} \\
&= \lim_{h\to 0}\frac{\sin(x)(\cos(h)-1)+\cos(x)\sin(h)}{h}
\end{aligned}$$

$$\cos(h)-1 = -2\sin^2\left(\frac{h}{2}\right),$$

$$\frac{\sin^2\left(\frac{h}{2}\right)}{h} = \frac{\sin\left(\frac{h}{2}\right)}{\frac{h}{2}}\cdot\frac{\sin\left(\frac{h}{2}\right)}{2}.$$

By

$$\lim_{h\to 0}\frac{\sin(h)}{h} = 1.$$

we have:

$$\lim_{h\to 0}\frac{\cos(h)-1}{h}=0.$$

$$\begin{aligned}&=\lim_{h\to 0}\sin(x)\cdot 0+\cos(x)\cdot 1\\&=\cos(x).\end{aligned}$$

□

vi Cosine function

$$\frac{\mathrm{d}\cos(x)}{\mathrm{d}x}=-\sin(x).$$

$$\cos(x)=1-\int_0^x \sin(t)\,\mathrm{d}t\,.$$

vii Tangent function

$$\frac{\mathrm{d}\tan(x)}{\mathrm{d}x}=\sec^2(x).$$

$$\tan(x)=\int_{\mathrm{round}\left(\frac{x}{\pi}\right)\pi}^{x}\sec^2(t)\,\mathrm{d}t\,,\quad x\in\mathbb{R}\setminus\{\frac{\pi}{2}+k\pi\mid k\in\mathbb{Z}\},$$

in which $\mathrm{round}(x)=\left\lfloor x+\frac{1}{2}\right\rfloor$.

Proof.

$$\begin{aligned}\frac{\mathrm{d}\tan(x)}{\mathrm{d}x}&=\frac{\mathrm{d}}{\mathrm{d}x}\frac{\sin(x)}{\cos(x)}\\&=\frac{\cos^2(x)+\sin^2(x)}{\cos^2(x)}\\&=\sec^2(x).\end{aligned}$$

□

viii Cotangent function

$$\frac{\mathrm{d}\cot(x)}{\mathrm{d}x}=-\csc^2(x).$$

$$\cot(x)=-\int_{\frac{\pi}{2}+\mathrm{round}\left(\frac{x}{\pi}-\frac{1}{2}\right)\pi}^{x}\csc^2(t)\,\mathrm{d}t\,,\quad x\in\mathbb{R}\setminus\{k\pi\mid k\in\mathbb{Z}\},$$

in which $\mathrm{round}(x)=\left\lfloor x+\frac{1}{2}\right\rfloor$.

Proof.

$$\cot(x)=\frac{\cos(x)}{\sin(x)}.$$

$$\begin{aligned}\frac{\mathrm{d}\cot(x)}{\mathrm{d}x}&=\frac{\mathrm{d}}{\mathrm{d}x}\frac{\cos(x)}{\sin(x)}\\&=\frac{-\sin^2(x)-\cos^2(x)}{\sin^2(x)}\\&=-\csc^2(x).\end{aligned}$$

□

ix Secant function

$$\frac{\mathrm{d}\sec(x)}{\mathrm{d}x} = \tan(x)\sec(x).$$

$$\sec(x) = \int_{\mathrm{round}\left(\frac{x}{\pi}\right)\pi}^{x} \tan(t)\sec(t)\,\mathrm{d}t\,, \quad x \in \mathbb{R} \setminus \{\frac{\pi}{2} + k\pi \mid k \in \mathbb{Z}\},$$

in which $\mathrm{round}(x) = \left\lfloor x + \frac{1}{2} \right\rfloor$.

Proof.

$$\begin{aligned}\frac{\mathrm{d}\sec(x)}{\mathrm{d}x} &= \frac{\mathrm{d}}{\mathrm{d}x}\frac{1}{\cos(x)} \\ &= \frac{\sin(x)}{\cos^2(x)} \\ &= \tan(x)\sec(x).\end{aligned}$$

□

x Cosecant function

$$\frac{\mathrm{d}\csc(x)}{\mathrm{d}x} = -\cot(x)\csc(x).$$

$$\csc(x) = -\int_{\frac{\pi}{2}+\mathrm{round}\left(\frac{x}{\pi}-\frac{1}{2}\right)\pi}^{x} \cot(t)\csc(t)\,\mathrm{d}t\,,$$

in which $\mathrm{round}(x) = \left\lfloor x + \frac{1}{2} \right\rfloor$.

Proof.

$$\begin{aligned}\frac{\mathrm{d}\csc(x)}{\mathrm{d}x} &= \frac{\mathrm{d}}{\mathrm{d}x}\frac{1}{\sin(x)} \\ &= \frac{-\cos(x)}{\sin^2(x)} \\ &= -\cot(x)\csc(x).\end{aligned}$$

□

xi Inverse sine function

$$\frac{\mathrm{d}\arcsin(x)}{\mathrm{d}x} = \frac{1}{\sqrt{1-x^2}}.$$

$$\arcsin(x) = \int_0^x \frac{1}{\sqrt{1-t^2}}\,\mathrm{d}t\,, \quad |x| \le 1.$$

Proof. Let $y = \arcsin(x)$. $\sin(y) = x$.

$$\frac{\mathrm{d}}{\mathrm{d}x}\sin(y) = \frac{\mathrm{d}}{\mathrm{d}x}(x).$$

Using the chain rule on the left:

$$\cos(y)\frac{\mathrm{d}y}{\mathrm{d}x} = 1.$$

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{\cos(y)} = \frac{1}{\sqrt{1-x^2}}.$$

□

xii Inverse cosine function

$$\frac{\mathrm{d}\arccos(x)}{\mathrm{d}x} = -\frac{1}{\sqrt{1-x^2}}.$$

$$\arccos(x) = \int_x^1 \frac{1}{\sqrt{1-t^2}}\,\mathrm{d}t\,, \quad |x| \leq 1.$$

Proof. Let $y = \arccos(x)$. $\cos(y) = x$.

$$\frac{\mathrm{d}}{\mathrm{d}x}\cos(y) = \frac{\mathrm{d}}{\mathrm{d}x}(x).$$

Using the chain rule on the left:

$$-\sin(y)\frac{\mathrm{d}y}{\mathrm{d}x} = 1.$$

$$\frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{1}{\cos(y)} = -\frac{1}{\sqrt{1-x^2}}.$$

□

xiii Inverse tangent function

$$\frac{\mathrm{d}\arctan(x)}{\mathrm{d}x} = \frac{1}{1+x^2}.$$

$$\arctan(x) = \int_0^x \frac{1}{1+t^2}\,\mathrm{d}t\,.$$

Proof. Let $y = \arctan(x)$. $\tan(y) = x$.

$$\frac{\mathrm{d}}{\mathrm{d}x}\tan(y) = \frac{\mathrm{d}}{\mathrm{d}x}(x).$$

Using the chain rule on the left:

$$\sec^2(y)\frac{\mathrm{d}y}{\mathrm{d}x} = 1.$$

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \cos^2(y) = \frac{1}{1+x^2}.$$

□

xiv Inverse cotangent function

$$\frac{\mathrm{d}\,\mathrm{arccot}(x)}{\mathrm{d}x} = -\frac{1}{1+x^2}.$$

$$\mathrm{arccot}(x) = \int_x^\infty \frac{1}{1+t^2}\,\mathrm{d}t\,.$$

Proof. Let $y = \mathrm{arccot}(x)$. $\cot(y) = x$.

$$\frac{\mathrm{d}}{\mathrm{d}x}\cot(y) = \frac{\mathrm{d}}{\mathrm{d}x}(x).$$

Using the chain rule on the left:

$$-\csc^2(y)\frac{\mathrm{d}y}{\mathrm{d}x} = 1.$$

$$\frac{\mathrm{d}y}{\mathrm{d}x} = -\sin^2(y) = -\frac{1}{1+x^2}.$$

□

xv Inverse secant function

$$\frac{\mathrm{d}\,\mathrm{arcsec}(x)}{\mathrm{d}x} = \frac{1}{|x|\sqrt{x^2-1}}.$$

$$\mathrm{arcsec}(x) = \mathrm{sgn}(x)\int_1^{|x|} \frac{1}{t\sqrt{t^2-1}}\,\mathrm{d}t + (1-\mathrm{sgn}(x))\frac{\pi}{2}, \quad |x| \geq 1.$$

Proof.

$$\begin{aligned}\frac{\mathrm{d}\,\mathrm{arcsec}(x)}{\mathrm{d}x} &= \frac{\mathrm{d}\arccos\left(\frac{1}{x}\right)}{\mathrm{d}x}\\ &= -\frac{1}{\sqrt{1-x^{-2}}}\left(-x^{-2}\right)\\ &= \frac{1}{|x|\sqrt{x^2-1}}\end{aligned}$$

□

xvi Inverse cosecant function

$$\frac{\mathrm{d}\,\mathrm{arccsc}(x)}{\mathrm{d}x} = -\frac{1}{|x|\sqrt{x^2-1}}.$$

$$\mathrm{arccsc}(x) = \mathrm{sgn}(x)\int_{|x|}^{\infty} \frac{1}{t\sqrt{t^2-1}}\,\mathrm{d}t\,, \quad |x| \geq 1.$$

Proof.

$$\begin{aligned}\frac{\mathrm{d}\,\mathrm{arccsc}(x)}{\mathrm{d}x} &= \frac{\mathrm{d}\arcsin\left(\frac{1}{x}\right)}{\mathrm{d}x}\\ &= \frac{1}{\sqrt{1-x^{-2}}}\left(-x^{-2}\right)\\ &= -\frac{1}{|x|\sqrt{x^2-1}}\end{aligned}$$

□

xvii Hyperbolic sine function

$$\frac{\mathrm{d}\sinh(x)}{\mathrm{d}x} = \cosh(x).$$

Proof.

$$\begin{aligned}\frac{\mathrm{d}\sinh(x)}{\mathrm{d}x} &= \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{e^x-e^{-x}}{2}\right)\\ &= \frac{e^x+e^{-x}}{2} = \cosh(x)\end{aligned}$$

□

xviii Hyperbolic cosine function

$$\frac{\mathrm{d}\cosh(x)}{\mathrm{d}x} = \sinh(x).$$

Proof.

$$\begin{aligned}\frac{\mathrm{d}\cosh(x)}{\mathrm{d}x} &= \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{e^x + e^{-x}}{2}\right)\\ &= \frac{e^x - e^{-x}}{2} = \sinh(x)\end{aligned}$$

□

xix Hyperbolic tangent function

$$\frac{\mathrm{d}\tanh(x)}{\mathrm{d}x} = \operatorname{sech}^2(x).$$

Proof.

$$\begin{aligned}\frac{\mathrm{d}\tanh(x)}{\mathrm{d}x} &= \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{e^{2x}-1}{e^{2x}+1}\right)\\ &= \frac{2e^{2x}\left(e^{2x}+1\right) - \left(e^{2x}-1\right)2e^{2x}}{\left(e^{2x}+1\right)^2}\\ &= \frac{4e^{2x}}{\left(e^{2x}+1\right)^2}\\ &= \left(\frac{2}{e^x + e^{-x}}\right)^2\\ &= \operatorname{sech}^2(x)\end{aligned}$$

□

xx Hyperbolic cotangent function

$$\frac{\mathrm{d}\coth(x)}{\mathrm{d}x} = -\operatorname{csch}^2(x).$$

Proof.

$$\begin{aligned}\frac{\mathrm{d}\coth(x)}{\mathrm{d}x} &= \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{e^{2x}+1}{e^{2x}-1}\right)\\ &= \frac{2e^{2x}\left(e^{2x}-1\right) - \left(e^{2x}+1\right)2e^{2x}}{\left(e^{2x}-1\right)^2}\\ &= \frac{-4e^{2x}}{\left(e^{2x}-1\right)^2}\\ &= -\left(\frac{2}{e^x - e^{-x}}\right)^2\\ &= -\operatorname{csch}^2(x)\end{aligned}$$

□

xxi Hyperbolic secant function

$$\frac{\mathrm{d}\,\mathrm{sech}(x)}{\mathrm{d}x} = -\tanh(x)\,\mathrm{sech}(x).$$

Proof.

$$\begin{aligned}
\frac{\mathrm{d}\,\mathrm{sech}(x)}{\mathrm{d}x} &= \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{2}{e^x+e^{-x}}\right)\\
&= \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{-2\left(e^x-e^{-x}\right)}{\left(e^x+e^{-x}\right)^2}\right)\\
&= -\tanh(x)\,\mathrm{sech}(x)
\end{aligned}$$

□

xxii Hyperbolic cosecant function

$$\frac{\mathrm{d}\,\mathrm{csch}(x)}{\mathrm{d}x} = -\coth(x)\,\mathrm{csch}(x).$$

Proof.

$$\begin{aligned}
\frac{\mathrm{d}\,\mathrm{csch}(x)}{\mathrm{d}x} &= \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{2}{e^x-e^{-x}}\right)\\
&= \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{-2\left(e^x+e^{-x}\right)}{\left(e^x-e^{-x}\right)^2}\right)\\
&= -\coth(x)\,\mathrm{csch}(x)
\end{aligned}$$

□

xxiii Inverse hyperbolic sine function

$$\frac{\mathrm{d}\,\mathrm{arcsinh}(x)}{\mathrm{d}x} = \frac{1}{\sqrt{x^2+1}}.$$

Proof.

$$\begin{aligned}
\frac{\mathrm{d}\,\mathrm{arcsinh}(x)}{\mathrm{d}x} &= \frac{\mathrm{d}}{\mathrm{d}x}\ln\left(x+\sqrt{x^2+1}\right)\\
&= \frac{1+\frac{x}{\sqrt{x^2+1}}}{x+\sqrt{x^2+1}}\\
&= \frac{1}{\sqrt{x^2+1}}
\end{aligned}$$

□

xxiv Inverse hyperbolic cosine function

$$\frac{\mathrm{d}\,\mathrm{arccosh}(x)}{\mathrm{d}x} = \frac{1}{\sqrt{x^2-1}}.$$

Proof.

$$\begin{aligned}\frac{\mathrm{d}\operatorname{arccosh}(x)}{\mathrm{d}x} &= \frac{\mathrm{d}}{\mathrm{d}x}\ln\left(x+\sqrt{x^2-1}\right)\\ &= \frac{1+\frac{x}{\sqrt{x^2-1}}}{x+\sqrt{x^2-1}}\\ &= \frac{1}{\sqrt{x^2-1}}\end{aligned}$$

□

xxv Inverse hyperbolic tangent function

$$\frac{\mathrm{d}\operatorname{arctanh}(x)}{\mathrm{d}x} = \frac{1}{1-x^2}, \quad |x| < 1.$$

Proof.

$$\begin{aligned}\frac{\mathrm{d}\operatorname{arctanh}(x)}{\mathrm{d}x} &= \frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}x}\ln\left(\frac{1+x}{1-x}\right)\\ &= \frac{1}{2}\frac{1-x}{1+x}\frac{(1-x)-(-1)(1+x)}{(1-x)^2}\\ &= \frac{1}{2}\frac{1}{1+x}\frac{2}{1-x}\\ &= \frac{1}{1-x^2}\end{aligned}$$

□

xxvi Inverse hyperbolic cotangent function

$$\frac{\mathrm{d}\operatorname{arccoth}(x)}{\mathrm{d}x} = \frac{1}{1-x^2}, \quad |x| > 1.$$

Proof.

$$\begin{aligned}\frac{\mathrm{d}\operatorname{arccoth}(x)}{\mathrm{d}x} &= \frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}x}\ln\left(\frac{x+1}{x-1}\right)\\ &= \frac{1}{2}\frac{x-1}{x+1}\frac{(x-1)-(x+1)}{(x-1)^2}\\ &= \frac{1}{2}\frac{1}{x+1}\frac{-2}{x-1}\\ &= \frac{1}{1-x^2}\end{aligned}$$

□

xxvii Inverse hyperbolic secant function

$$\frac{\mathrm{d}\operatorname{arcsech}(x)}{\mathrm{d}x} = -\frac{1}{x\sqrt{1-x^2}}.$$

Proof.

$$\begin{aligned}
\frac{\mathrm{d}\,\operatorname{arcsech}(x)}{\mathrm{d}x} &= \frac{\mathrm{d}}{\mathrm{d}x}\ln\left(\frac{1}{x}+\frac{\sqrt{1-x^2}}{x}\right)\\
&= \frac{x}{1+\sqrt{1-x^2}}\,\frac{\frac{-x}{\sqrt{1-x^2}}x-\left(1+\sqrt{1-x^2}\right)}{x^2}\\
&= \frac{1}{1+\sqrt{1-x^2}}\,\frac{-x^2-\sqrt{1-x^2}-1+x^2}{x\sqrt{1-x^2}}\\
&= -\frac{1}{x\sqrt{1-x^2}}
\end{aligned}$$

□

xxviii Inverse hyperbolic cosecant function

$$\frac{\mathrm{d}\,\operatorname{arccsch}(x)}{\mathrm{d}x} = -\frac{1}{|x|\sqrt{1+x^2}}.$$

Proof.

$$\begin{aligned}
\frac{\mathrm{d}\,\operatorname{arccsch}(x)}{\mathrm{d}x} &= \frac{\mathrm{d}}{\mathrm{d}x}\ln\left(\frac{1}{x}+\frac{\sqrt{1+x^2}}{|x|}\right)\\
&= \frac{\mathrm{d}}{\mathrm{d}x}\ln\left(\frac{1+\operatorname{sgn}(x)\sqrt{1+x^2}}{x}\right)\\
&= \frac{x}{1+\operatorname{sgn}(x)\sqrt{1+x^2}}\,\frac{\frac{\operatorname{sgn}(x)x}{\sqrt{1+x^2}}x-\left(1+\operatorname{sgn}(x)\sqrt{1+x^2}\right)}{x^2}\\
&= \frac{1}{1+\operatorname{sgn}(x)\sqrt{1+x^2}}\,\frac{-\sqrt{1+x^2}-\operatorname{sgn}(x)}{x\sqrt{1+x^2}}\\
&= \frac{-\operatorname{sgn}(x)}{x\sqrt{1+x^2}}\\
&= -\frac{1}{|x|\sqrt{1+x^2}}
\end{aligned}$$

□

VI List of Integrals of Real Functions

i Power function

$$\int x^{-1}\,\mathrm{d}x = \ln|x| + C.$$

$$\int x^n\,\mathrm{d}x = \frac{1}{n+1}x^{n+1} + C, \quad n \neq -1$$

ii Logarithmic function

$$\int \ln(x)\,\mathrm{d}x = x\ln(x) - x + C.$$

Proof.

$$\int \ln(x)\,\mathrm{d}x = x\ln(x) - \int x\frac{1}{x}\,\mathrm{d}x = x\ln(x) - x + C.$$

□

$$\int \left(\ln(x)\right)^n \mathrm{d}x = x\left(\ln(x)\right)^n - n\int \left(\ln(x)\right)^{n-1}\mathrm{d}x\,,\quad n\in\mathbb{N}.$$

iii Exponential function

$$\int e^{nx}\,\mathrm{d}x = \frac{1}{n}e^{nx} + C,\quad n\neq 0.$$

iv Sine function

$$\int \sin(x)\,\mathrm{d}x = -\cos(x) + C.$$

$$\int \sin^2(x)\,\mathrm{d}x = \frac{x}{2} - \frac{\sin(2x)}{4} + C.$$

Proof.

$$\sin^2(x) = \frac{1}{2}\left(1-\cos(2x)\right).$$

$$\int \sin^2(x)\,\mathrm{d}x = \frac{x}{2} - \frac{1}{2}\int \cos(2x)\,\mathrm{d}x = \frac{x}{2} - \frac{\sin(2x)}{4} + C.$$

□

$$\int \sin^n(x)\,\mathrm{d}x = -\frac{1}{n}\sin^{n-1}(x)\cos(x) + \frac{n-1}{n}\int \sin^{n-2}(x)\,\mathrm{d}x\,,\quad n\in\mathbb{N}_{>2}.$$

Proof.

$$\begin{aligned}\int \sin^n(x)\,\mathrm{d}x &= -\sin^{n-1}(x)\cos(x) - (n-1)\int \sin^{n-2}(x)\cos^2(x)\,\mathrm{d}x\\ &= -\sin^{n-1}(x)\cos(x) - (n-1)\int \sin^{n-2}(x)\,\mathrm{d}x + (n-1)\int \sin^n(x)\,\mathrm{d}x\\ &= -\frac{1}{n}\sin^{n-1}(x)\cos(x) + \frac{n-1}{n}\int \sin^{n-2}(x)\,\mathrm{d}x\end{aligned}$$

□

v Cosine function

$$\int \cos(x)\,\mathrm{d}x = \sin(x) + C.$$

$$\int \cos^2(x)\,\mathrm{d}x = \frac{x}{2} + \frac{\sin(2x)}{4} + C.$$

Proof.

$$\sin^2(x) = \frac{1}{2}(1+\cos(2x)).$$

$$\int \sin^2(x)\,\mathrm{d}x = \frac{x}{2} + \frac{1}{2}\int \cos(2x)\,\mathrm{d}x = \frac{x}{2} + \frac{\sin(2x)}{4} + C.$$

□

$$\int \cos^n(x)\,\mathrm{d}x = \frac{1}{n}\cos^{n-1}(x)\sin(x) + \frac{n-1}{n}\int \cos^{n-2}(x)\,\mathrm{d}x\,, \quad n \in \mathbb{N}_{>2}.$$

Proof.

$$\begin{aligned}\int \cos^n(x)\,\mathrm{d}x &= \cos^{n-1}(x)\sin(x) + (n-1)\int \cos^{n-2}(x)\sin^2(x)\,\mathrm{d}x\\ &= \cos^{n-1}(x)\sin(x) + (n-1)\int \cos^{n-2}(x)\,\mathrm{d}x - (n-1)\int \cos^n(x)\,\mathrm{d}x\\ &= \frac{1}{n}\cos^{n-1}(x)\sin(x) + \frac{n-1}{n}\int \cos^{n-2}(x)\,\mathrm{d}x\end{aligned}$$

□

vi Tangent function

$$\int \tan(x)\,\mathrm{d}x = -\ln|\cos(x)| + C = \ln|\sec(x)| + C.$$

Proof.

$$\begin{aligned}\int \tan(x)\,\mathrm{d}x &= -\int \frac{-\sin(x)}{\cos(x)}\,\mathrm{d}x\\ &= -\ln|\cos(x)| + C = \ln|\sec(x)| + C\end{aligned}$$

□

$$\int \tan^2(x)\,\mathrm{d}x = \tan(x) - x + C.$$

Proof.

$$\begin{aligned}\int \tan^2(x)\,\mathrm{d}x &= \int \sec^2(x) - 1\,\mathrm{d}x\\ &= \tan(x) - x + C\end{aligned}$$

□

$$\int \tan^n(x)\,\mathrm{d}x = \frac{1}{n-1}\tan^{n-1}(x) - \int \tan^{n-2}(x)\,\mathrm{d}x\,, \quad n \in \mathbb{N}_{>2}.$$

Proof.

$$\int \tan^n(x)\,\mathrm{d}x = \int \tan^{n-2}(x)\sec^2(x)\,\mathrm{d}x - \int \tan^{n-2}(x)\,\mathrm{d}x$$

$$u = \tan^{n-2}(x), \quad \mathrm{d}v = \sec^2(x)\,\mathrm{d}x\,, \mathrm{d}u = (n-2)\tan^{n-3}(x)\sec^2(x), \quad v = \tan(x).$$

$$
\begin{aligned}
\int \tan^{n-2}(x)\sec^2(x)\,\mathrm{d}x &= \tan^{n-1}(x) - (n-2)\int \tan^{n-2}(x)\sec^2(x)\,\mathrm{d}x \\
&= \frac{1}{n-1}\tan^{n-1}(x) \\
\int \tan^{n}(x)\,\mathrm{d}x &= \frac{1}{n-1}\tan^{n-1}(x) - \int \tan^{n-2}(x)\,\mathrm{d}x\,.
\end{aligned}
$$

□

vii Cotangent function

$$
\int \cot(x)\,\mathrm{d}x = \ln|\sin(x)| + C.
$$

Proof.

$$
\begin{aligned}
\int \cot(x)\,\mathrm{d}x &= \int \frac{-\cos(x)}{\sin(x)}\,\mathrm{d}x \\
&= \ln|\sin(x)| + C
\end{aligned}
$$

□

$$
\int \cot^2(x)\,\mathrm{d}x = -\cot(x) - x + C.
$$

Proof.

$$
\begin{aligned}
\int \cot^2(x)\,\mathrm{d}x &= \int \csc^2(x) - 1\,\mathrm{d}x \\
&= -\cot(x) - x + C
\end{aligned}
$$

□

$$
\int \cot^n(x)\,\mathrm{d}x = -\frac{1}{n-1}\cot^{n-1}(x) - \int \cot^{n-2}(x)\,\mathrm{d}x\,, \quad n \in \mathbb{N}_{>2}.
$$

Proof.

$$
\int \cot^n(x)\,\mathrm{d}x = \int \cot^{n-2}(x)\csc^2(x)\,\mathrm{d}x - \int \cot^{n-2}(x)\,\mathrm{d}x
$$

$$
u = \cot^{n-2}(x), \quad \mathrm{d}v = \csc^2(x)\,\mathrm{d}x\,, \mathrm{d}u = -(n-2)\cot^{n-3}(x)\csc^2(x), \quad v = -\cot(x).
$$

$$
\begin{aligned}
\int \cot^{n-2}(x)\csc^2(x)\,\mathrm{d}x &= -\cot^{n-1}(x) + (n-2)\int \cot^{n-2}(x)\csc^2(x)\,\mathrm{d}x \\
&= \frac{1}{n-1}\cot^{n-1}(x) \\
\int \cot^n(x)\,\mathrm{d}x &= -\frac{1}{n-1}\cot^{n-1}(x) - \int \cot^{n-2}(x)\,\mathrm{d}x\,.
\end{aligned}
$$

□

viii Secant function

$$\int \sec(x)\,\mathrm{d}x = \ln|\tan(x)+\sec(x)| + C = \ln\left|\tan\left(\frac{x}{2}+\frac{\pi}{4}\right)\right| + C.$$

Proof.

$$\begin{aligned}
\int \sec(x)\,\mathrm{d}x &= \int \sec(x)\frac{\tan(x)+\sec(x)}{\tan(x)+\sec(x)}\,\mathrm{d}x \\
&= \int u^{-1}\,\mathrm{d}u\,, \quad u = \tan(x)+\sec(x), \quad \mathrm{d}u = \sec(x)\,(\tan(x)+\sec(x))\,\mathrm{d}x \\
&= \ln|u| + C = \ln|\tan(x)+\sec(x)| + C \\
&= \ln\left|\tan\left(\frac{x}{2}+\frac{\pi}{4}\right)\right| + C
\end{aligned}$$

□

$$\int \sec^2(x)\,\mathrm{d}x = \tan(x) + C.$$

$$\int \sec^n(x)\,\mathrm{d}x = \frac{1}{n-1}\tan(x)\sec^{n-2}(x) + \frac{n-2}{n-1}\int \sec^{n-2}(x)\,\mathrm{d}x\,, \quad n \in \mathbb{N}_{>2}.$$

Proof.

$$\begin{aligned}
\int \sec^n(x)\,\mathrm{d}x &= \tan(x)\sec^{n-2}(x) - \int \tan(x)(n-2)\sec^{n-3}(x)\tan(x)\sec(x)\,\mathrm{d}x \\
&= \tan(x)\sec^{n-2}(x) - (n-2)\int \sec^n(x) - \sec^{n-2}(x)\,\mathrm{d}x \\
&= \frac{1}{n-1}\tan(x)\sec^{n-2}(x) + \frac{n-2}{n-1}\int \sec^{n-2}(x)\,\mathrm{d}x
\end{aligned}$$

□

ix Cosecant function

$$\int \csc(x)\,\mathrm{d}x = -\ln|\cot(x)+\csc(x)| + C = -\ln\left|\cot\left(\frac{x}{2}\right)\right| + C.$$

Proof.

$$\begin{aligned}
\int \csc(x)\,\mathrm{d}x &= \int \csc(x)\frac{\cot(x)+\csc(x)}{\cot(x)+\csc(x)}\,\mathrm{d}x \\
&= -\int u^{-1}\,\mathrm{d}u\,, \quad u = \cot(x)+\csc(x), \quad \mathrm{d}u = -\cot(x)\,(\cot(x)+\csc(x))\,\mathrm{d}x \\
&= -\ln|u| + C \\
&= -\ln|\cot(x)+\csc(x)| + C \\
&= -\ln\left|\cot\left(\frac{x}{2}\right)\right| + C
\end{aligned}$$

□

$$\int \csc^2(x)\,\mathrm{d}x = -\cot(x) + C.$$

$$\int \csc^n(x)\,\mathrm{d}x = -\frac{1}{n-1}\cot(x)\csc^{n-2}(x) + \frac{n-2}{n-1}\int \csc^{n-2}(x)\,\mathrm{d}x\,, \quad n \in \mathbb{N}_{>2}.$$

Proof.

$$\begin{aligned}\int \csc^n(x)\,\mathrm{d}x &= -\cot(x)\csc^{n-2}(x) + \int \cot(x)(n-2)\csc^{n-3}(x)(-1)\cot(x)\csc(x)\,\mathrm{d}x\\ &= -\cot(x)\csc^{n-2}(x) - (n-2)\int \csc^n(x) - \csc^{n-2}(x)\,\mathrm{d}x\\ &= -\frac{1}{n-1}\cot(x)\csc^{n-2}(x) + \frac{n-2}{n-1}\int \csc^{n-2}(x)\,\mathrm{d}x\end{aligned}$$

□

x Inverse sine function

$$\int \arcsin(x)\,\mathrm{d}x = x\arcsin(x) + \sqrt{1-x^2} + C.$$

xi Inverse cosine function

$$\int \arccos(x)\,\mathrm{d}x = x\arccos(x) - \sqrt{1-x^2} + C.$$

xii Inverse tangent function

$$\int \arctan(x)\,\mathrm{d}x = x\arctan(x) - \frac{1}{2}\ln\left(1+x^2\right) + C.$$

xiii Inverse cotangent function

$$\int \operatorname{arccot}(x)\,\mathrm{d}x = x\operatorname{arccot}(x) + \frac{1}{2}\ln\left(1+x^2\right) + C.$$

xiv Inverse secant function

$$\int \operatorname{arcsec}(x)\,\mathrm{d}x = x\operatorname{arcsec}(x) - \operatorname{sgn}(x)\ln\left|x + \sqrt{x^2-1}\right| + C, \quad |x| \geq 1.$$

xv Inverse cosecant function

$$\int \operatorname{arccsc}(x)\,\mathrm{d}x = x\operatorname{arccsc}(x) + \operatorname{sgn}(x)\ln\left|x + \sqrt{x^2-1}\right| + C, \quad |x| \geq 1.$$

xvi Hyperbolic sine function

$$\int \sinh(x)\,\mathrm{d}x = \cosh(x) + C.$$

$$\int \sinh^2(x)\,\mathrm{d}x = -\frac{x}{2} + \frac{\sinh(2x)}{4} + C.$$

Proof. PLACEHOLDER □

$$\int \sinh^n(x)\,\mathrm{d}x = \frac{1}{n}\sinh^{n-1}(x)\cosh(x) - \frac{n-1}{n}\int \sinh^{n-2}(x)\,\mathrm{d}x\,, \quad n \in \mathbb{N}_{>2}.$$

Proof. PLACEHOLDER □

xvii Hyperbolic cosine function

$$\int \cosh(x)\,\mathrm{d}x = \sinh(x) + C.$$

$$\int \cosh^2(x)\,\mathrm{d}x = \frac{x}{2} + \frac{\sinh(2x)}{4} + C.$$

Proof. PLACEHOLDER □

$$\int \cosh^n(x)\,\mathrm{d}x = \frac{1}{n}\cosh^{n-1}(x)\sinh(x) + \frac{n-1}{n}\int \cosh^{n-2}(x)\,\mathrm{d}x\,, \quad n \in \mathbb{N}_{>2}.$$

Proof. PLACEHOLDER □

xviii Hyperbolic tangent function

$$\int \tanh(x)\,\mathrm{d}x = \ln\big(\cosh(x)\big) + C.$$

Proof.

$$\begin{aligned}\int \tanh(x)\,\mathrm{d}x &= \int \frac{\sinh(x)}{\cosh(x)}\,\mathrm{d}x \\ &= \ln\big(\cosh(x)\big) + C\end{aligned}$$

□

$$\int \tanh^2(x)\,\mathrm{d}x = x - \tanh(x) + C.$$

Proof. PLACEHOLDER □

$$\int \tanh^n(x)\,\mathrm{d}x = \frac{1}{n-1}\tanh^{n-1}(x) - \int \tanh^{n-2}(x)\,\mathrm{d}x\,, \quad n \in \mathbb{N}_{>2}.$$

Proof. PLACEHOLDER □

xix Hyperbolic cotangent function

$$\int \coth(x)\,\mathrm{d}x = \ln\left|\sinh(x)\right| + C.$$

Proof.

$$\begin{aligned}\int \coth(x)\,\mathrm{d}x &= \int \frac{\cosh(x)}{\sinh(x)}\,\mathrm{d}x \\ &= \ln\left|\sinh(x)\right| + C\end{aligned}$$

□

$$\int \coth^2(x)\,\mathrm{d}x = x - \coth(x) + C.$$

Proof. PLACEHOLDER □

$$\int \coth^n(x)\,\mathrm{d}x = -\frac{1}{n-1}\coth^{n-1}(x) - \int \coth^{n-2}(x)\,\mathrm{d}x\,, \quad n \in \mathbb{N}_{>2}.$$

Proof. PLACEHOLDER □

xx Hyperbolic secant function

$$\int \operatorname{sech}(x)\,\mathrm{d}x = 2\arctan\left(\tanh\left(\frac{x}{2}\right)\right) + C = \arctan\left(\sinh(x)\right) + C = \arcsin\left(\tanh(x)\right) + C.$$

Proof.

$$\begin{aligned}\int \operatorname{sech}(x)\,\mathrm{d}x &= \int \frac{1-t^2}{1+t^2}\,\mathrm{d}x\,, \quad t = \tanh\left(\frac{x}{2}\right) \\ &= 2\int \frac{1}{1+t^2}\,\mathrm{d}t\,, \quad \mathrm{d}t = \frac{1}{2}\operatorname{sech}^2\left(\frac{x}{2}\right)\mathrm{d}x = \frac{1}{2}\left(1 - \tanh^2\left(\frac{x}{2}\right)\right)\mathrm{d}x \\ &= 2\arctan\left(\tanh\left(\frac{x}{2}\right)\right) + C \\ &= \arctan\left(\sinh(x)\right) + C \\ &= \arcsin\left(\tanh(x)\right) + C\end{aligned}$$

□

$$\int \operatorname{sech}^2(x)\,\mathrm{d}x = \tanh(x) + C.$$

$$\int \operatorname{sech}^n(x)\,\mathrm{d}x = \frac{1}{n-1}\tanh(x)\operatorname{sech}^{n-1}(x) + \frac{n-2}{n-1}\int \operatorname{sech}^{n-2}(x)\,\mathrm{d}x\,, \quad n \in \mathbb{N}_{>2}.$$

Proof. PLACEHOLDER □

xxi Hyperbolic cosecant function

$$\int \operatorname{csch}(x)\,\mathrm{d}x = \ln\left|\tanh\left(\frac{x}{2}\right)\right| + C = \ln\left|\coth(x) - \operatorname{csch}(x)\right| + C.$$

Proof.

$$\begin{aligned}\int \operatorname{csch}(x)\,\mathrm{d}x &= \int \frac{1-t^2}{2t}\,\mathrm{d}x\,, \quad t = \tanh\left(\frac{x}{2}\right)\\ &= \int \frac{1}{t}\,\mathrm{d}t\,, \quad \mathrm{d}t = \frac{1}{2}\operatorname{sech}^2\left(\frac{x}{2}\right)\mathrm{d}x = \frac{1}{2}\left(1 - \tanh^2\left(\frac{x}{2}\right)\right)\mathrm{d}x\\ &= \ln\left|\tanh\left(\frac{x}{2}\right)\right| + C\\ &= \ln\left|\coth(x) - \operatorname{csch}(x)\right| + C\end{aligned}$$

□

$$\int \operatorname{csch}^2(x)\,\mathrm{d}x = -\coth(x) + C.$$

$$\int \operatorname{csch}^n(x)\,\mathrm{d}x = -\frac{1}{n-1}\coth(x)\operatorname{csch}^{n-1}(x) + \frac{n-2}{n-1}\int \operatorname{csch}^{n-2}(x)\,\mathrm{d}x\,, \quad n \in \mathbb{N}_{>2}.$$

Proof. PLACEHOLDER □

xxii Inverse hyperbolic sine function

$$\int \operatorname{arcsinh}(x)\,\mathrm{d}x = x\operatorname{arcsinh}(x) - \sqrt{x^2+1} + C.$$

xxiii Inverse hyperbolic cosine function

$$\int \operatorname{arccosh}(x)\,\mathrm{d}x = x\operatorname{arccosh}(x) - \sqrt{x^2-1} + C.$$

xxiv Inverse hyperbolic tangent function

$$\int \operatorname{arctanh}(x)\,\mathrm{d}x = x\operatorname{arctanh}(x) + \frac{1}{2}\ln\left(1-x^2\right) + C.$$

xxv Inverse hyperbolic cotangent function

$$\int \operatorname{arccoth}(x)\,\mathrm{d}x = x\operatorname{arccoth}(x) + \frac{1}{2}\ln\left(x^2-1\right) + C.$$

xxvi Inverse hyperbolic secant function

$$\int \operatorname{arcsech}(x)\,\mathrm{d}x = x\operatorname{arcsech}(x) + \arctan\left(\frac{\sqrt{1-x^2}}{x}\right) + C.$$

xxvii Inverse hyperbolic cosecant function

$$\int \operatorname{arccsch}(x)\,\mathrm{d}x = x\operatorname{arccsch}(x) + \ln\left|x + \sqrt{x^2+1}\right| + C.$$

xxviii Derivative over function

$$\int \frac{f'(x)}{f(x)}\,\mathrm{d}x = \ln|f(x)| + C, \quad f(x) \neq 0.$$

xxix $\frac{1}{x^2+a^2}$

$$\int \frac{1}{x^2+a^2}\,\mathrm{d}x = \frac{1}{a}\arctan\left(\frac{x}{a}\right) + C, \quad a > 0.$$

Proof.

$$\begin{aligned}\int \frac{1}{x^2+a^2}\,\mathrm{d}x &= \frac{1}{a}\int \frac{1}{u^2+1^2}\,\mathrm{d}u, \quad u = \frac{x}{a}, \quad \mathrm{d}u = \frac{1}{a}\,\mathrm{d}x \\ &= \frac{1}{a}\arctan\left(\frac{x}{a}\right) + C\end{aligned}$$

□

xxx $\frac{1}{x^2-a^2}$

$$\int \frac{1}{x^2-a^2}\,\mathrm{d}x = \frac{1}{2a}\ln\left(\frac{x-a}{x+a}\right) + C = \frac{1}{a}\operatorname{arccoth}\left(\frac{x}{a}\right) + C, \quad a > 0 \wedge x^2 - a^2 > 0.$$

Proof.

$$\begin{aligned}\int \frac{1}{x^2-a^2}\,\mathrm{d}x &= \int \frac{1}{2a}\left(\frac{1}{x-a} - \frac{1}{x+a}\right)\mathrm{d}x \\ &= \frac{1}{2a}\ln\left(\frac{x-a}{x+a}\right) + C\end{aligned}$$

□

xxxi $\frac{1}{a^2-x^2}$

$$\int \frac{1}{a^2-x^2}\,\mathrm{d}x = \frac{1}{2a}\ln\left(\frac{a-x}{a+x}\right) + C = \frac{1}{a}\operatorname{arctanh}\left(\frac{x}{a}\right) + C, \quad a > 0 \wedge a^2 - x^2 > 0.$$

Proof.

$$\begin{aligned}\int \frac{1}{a^2-x^2}\,\mathrm{d}x &= \int \frac{1}{2a}\left(\frac{1}{a-x} - \frac{1}{a+x}\right)\mathrm{d}x \\ &= \frac{1}{2a}\ln\left(\frac{a-x}{a+x}\right) + C\end{aligned}$$

□

xxxii $ax+b$

$$\int \frac{1}{ax+b}\,\mathrm{d}x = \frac{1}{a}\ln|ax+b| + C, \quad a > 0.$$

Proof.

$$\begin{aligned}\int \frac{1}{ax+b}\,\mathrm{d}x &= \frac{1}{a}\int \frac{1}{u}\,\mathrm{d}u, \quad u = ax+b, \quad \mathrm{d}u = a\,\mathrm{d}x \\ &= \frac{1}{a}\ln|ax+b| + C\end{aligned}$$

□

$$\int (ax+b)^n \,\mathsf{d}x = \frac{1}{a(n+1)}(ax+b)^{n+1} + C, \quad a \neq 0 \wedge n \neq -1.$$

Proof.

$$\begin{aligned}\int (ax+b)^n \,\mathsf{d}x &= \frac{1}{a}\int u^n \,\mathsf{d}u, \quad u = ax+b, \quad \mathsf{d}u = a\,\mathsf{d}x\\ &= \frac{1}{a}u^{n+1} + C\\ &= \frac{1}{a(n+1)}(ax+b)^{n+1} + C\end{aligned}$$

□

$$\int \frac{x}{ax+b}\,\mathsf{d}x = \frac{x}{a} - \frac{b}{a^2}\ln|ax+b| + C, \quad a > 0.$$

Proof.

$$\begin{aligned}\int \frac{x}{ax+b}\,\mathsf{d}x &= \int \frac{u-b}{a^2u}\,\mathsf{d}u, \quad u = ax+b, \quad \mathsf{d}u = a\,\mathsf{d}x\\ &= \frac{ax+b}{a^2} - \frac{b}{a^2}\ln|ax+b| + C\\ &= \frac{x}{a} - \frac{b}{a^2}\ln|ax+b| + C\end{aligned}$$

□

$$\int \frac{x}{(ax+b)^2}\,\mathsf{d}x = \frac{1}{a^2}\ln|ax+b| + \frac{b}{a^2(ax+b)} + C, \quad a > 0.$$

Proof.

$$\begin{aligned}\int \frac{x}{(ax+b)^2}\,\mathsf{d}x &= \int \frac{u-b}{a^2u^2}\,\mathsf{d}u, \quad u = ax+b, \quad \mathsf{d}u = a\,\mathsf{d}x\\ &= \frac{1}{a^2}\ln|ax+b| + \frac{b}{a^2(ax+b)}\end{aligned}$$

□

$$\int x(ax+b)^n \,\mathsf{d}x = \frac{a(n+1)x-b}{a^2(n+1)(n+2)}(ax+b)^{n+1} + C, \quad a \neq 0 \wedge n \notin \{-1,-2\}.$$

Proof.

$$\begin{aligned}\int x(ax+b)^n \,\mathsf{d}x &= \int \frac{u-b}{a^2}u^n \,\mathsf{d}u, \quad u = ax+b, \quad \mathsf{d}u = a\,\mathsf{d}x\\ &= \frac{(ax+b)^{n+2}}{a^2(n+2)} - \frac{b(ax+b)^{n+1}}{a^2(n+1)}\\ &= \frac{(ax+b)(n+1)-b(n+2)}{a^2(n+1)(n+2)}(ax+b)^{n+1} + C\\ &= \frac{a(n+1)x-b}{a^2(n+1)(n+2)}(ax+b)^{n+1} + C\end{aligned}$$

□

xxxiii $\sqrt{a^2+x^2}$

Let $r = \sqrt{a^2+x^2}$.

$$\int r\,\mathrm{d}x = \frac{1}{2}\left(xr + a^2\ln(x+r)\right) + C.$$

Proof.

$$\begin{aligned}
\int r\,\mathrm{d}x &= a^2\int\sqrt{1+u^2}\,\mathrm{d}u\,, \quad u = \frac{x}{|a|}, \quad \mathrm{d}u = \frac{1}{|a|}\,\mathrm{d}x \\
&= a^2\int\sqrt{1+\sinh^2 t}\cosh t\,\mathrm{d}t\,, \quad u = \sinh t, \quad \mathrm{d}u = \cosh t\,\mathrm{d}t \\
&= a^2\int\cosh^2 t\,\mathrm{d}t \\
&= a^2\int\frac{\cosh(2t)+1}{2}\,\mathrm{d}t \\
&= \frac{a^2\sinh(2t)}{4} + \frac{a^2 t}{2} + C \\
&= \frac{a^2 u\sqrt{1+u^2}}{2} + \frac{a^2}{2}\ln\left(u+\sqrt{1+u^2}\right) + C \\
&= \frac{xr}{2} + \frac{a^2}{2}\ln\left(\frac{x}{|a|} + \frac{1}{|a|}r\right) + C \\
&= \frac{xr}{2} + \frac{a^2}{2}\ln(x+r) + C \\
&= \frac{1}{2}\left(xr + a^2\ln(x+r)\right) + C
\end{aligned}$$

Alternatively,

$$\begin{aligned}
\int r\,\mathrm{d}x &= xr - \int\frac{x^2}{r}\,\mathrm{d}x \\
&= xr - a^2\int\frac{u^2}{\sqrt{1+u^2}}\,\mathrm{d}u\,, \quad u = \frac{x}{|a|}, \quad \mathrm{d}u = \frac{1}{|a|}\,\mathrm{d}x \\
&= xr - a^2\int\tan^2 t\sec t\,\mathrm{d}t\,, \quad u = \tan t, \quad \mathrm{d}u = \sec^2 t\,\mathrm{d}t \\
&= xr - a^2\int\sec^3 t - \sec t\,\mathrm{d}t \\
&= xr - \frac{a^2}{2}\tan(t)\sec(t) - \frac{a^2}{2}\ln|\tan t + \sec t| + a^2\ln|\tan t + \sec t| + C \\
&= xr - \frac{a^2}{2}u\sqrt{1+u^2} + \frac{a^2}{2}\ln\left|u+\sqrt{1+u^2}\right| + C \\
&= \frac{xr}{2} + \frac{a^2}{2}\ln\left(\frac{x}{|a|} + \frac{1}{|a|}r\right) + C \\
&= \frac{xr}{2} + \frac{a^2}{2}\ln(x+r) + C \\
&= \frac{1}{2}\left(xr + a^2\ln(x+r)\right) + C
\end{aligned}$$

□

$$\int xr\,\mathrm{d}x = \frac{1}{3}r^3 + C.$$

Proof.

$$\begin{aligned}\int xr\,\mathrm{d}x &= \int r^2\,\mathrm{d}r\,, \quad \mathrm{d}r = \frac{x}{r}\,\mathrm{d}x \\ &= \frac{1}{3}r^3 + C\end{aligned}$$

□

$$\int \frac{1}{r}\,\mathrm{d}x = \operatorname{arcsinh}\left(\frac{x}{|a|}\right) + C = \ln(x+r) + C.$$

Proof.

$$\begin{aligned}\int \frac{1}{r}\,\mathrm{d}x &= \int \frac{1}{\sqrt{1+u^2}}\,\mathrm{d}x\,, \quad u = \frac{x}{|a|}, \quad \mathrm{d}u = \frac{1}{|a|}\,\mathrm{d}x \\ &= \int 1\,\mathrm{d}t\,, \quad u = \sinh t, \quad \mathrm{d}u = \cosh t\,\mathrm{d}t \\ &= t + C = \operatorname{arcsinh} u + C = \operatorname{arcsinh}\left(\frac{x}{|a|}\right) + C \\ &= \ln\left(\frac{x}{|a|} + \sqrt{1 + \left(\frac{x}{|a|}\right)^2}\right) \\ &= \ln(x+r) + C\end{aligned}$$

□

$$\int \frac{x}{r}\,\mathrm{d}x = r + C.$$

xxxiv $\sqrt{x^2 - a^2}$

Let $s = \sqrt{x^2 - a^2}, \quad x^2 > a^2$.

$$\int s\,\mathrm{d}x = \frac{1}{2}\left(xs - a^2\ln(x+s)\right) + C.$$

Proof.

$$
\begin{aligned}
\int s \,\mathrm{d}x &= a^2 \int \sqrt{u^2-1}\,\mathrm{d}u\,, \quad u = \frac{x}{|a|}, \quad \mathrm{d}u = \frac{1}{|a|}\,\mathrm{d}x \\
&= a^2 \int \sinh^2 t\,\mathrm{d}t\,, \quad u = \cosh t, \quad \mathrm{d}u = \sinh t\,\mathrm{d}t \\
&= a^2 \int \frac{\cosh(2t)-1}{2}\,\mathrm{d}t \\
&= \frac{a^2\sinh(2t)}{4} - \frac{a^2 t}{2} + C \\
&= \frac{a^2 u\sqrt{u^2-1}}{2} - \frac{a^2}{2}\ln\left(u + \sqrt{u^2-1}\right) + C \\
&= \frac{xs}{2} + \frac{a^2}{2}\ln\left(\frac{x}{|a|} + \frac{1}{|a|}s\right) + C \\
&= \frac{xs}{2} - \frac{a^2}{2}\ln(x+s) + C \\
&= \frac{1}{2}\left(xs - a^2\ln(x+s)\right) + C
\end{aligned}
$$

□

$$\int xs\,\mathrm{d}x = \frac{1}{3}s^3 + C.$$

Proof.

$$
\begin{aligned}
\int xs\,\mathrm{d}x &= \int s^2\,\mathrm{d}s\,, \quad \mathrm{d}s = \frac{x}{s}\,\mathrm{d}x \\
&= \frac{1}{3}s^3 + C
\end{aligned}
$$

□

$$\int \frac{1}{s}\,\mathrm{d}x = \arcsin\left(\frac{x}{|a|}\right) + C.$$

Proof.

$$
\begin{aligned}
\int \frac{1}{s}\,\mathrm{d}x &= \int \frac{1}{\sqrt{u^2-1}}\,\mathrm{d}x\,, \quad u = \frac{x}{|a|}, \quad \mathrm{d}u = \frac{1}{|a|}\,\mathrm{d}x \\
&= \int 1\,\mathrm{d}t\,, \quad u = \cosh t, \quad \mathrm{d}u = \sinh t\,\mathrm{d}t \\
&= t + C = \operatorname{sgn}(u)\operatorname{arccosh} u + C \\
&= \operatorname{sgn}(x)\operatorname{arccosh}\left|\frac{x}{a}\right| + C \\
&= \ln\left|\frac{x}{a} + \sqrt{\left(\frac{x}{a}\right)^2 - 1}\right| + C \qquad = \ln|x+s| + C
\end{aligned}
$$

□

$$\int \frac{x}{s}\,\mathrm{d}x = -s + C.$$

xxxv $\sqrt{a^2 - x^2}$

Let $q = \sqrt{a^2 - x^2}, \quad a^2 \geq x^2.$

$$\int q \, \mathrm{d}x = \frac{1}{2}\left(q + a^2 \arcsin\left(\frac{x}{|a|}\right)\right) + C.$$

Proof.

$$\begin{aligned}
\int q \, \mathrm{d}x &= a^2 \int \sqrt{1-u^2} \, \mathrm{d}u, \quad u = \frac{x}{|a|}, \quad \mathrm{d}u = \frac{1}{|a|} \, \mathrm{d}x \\
&= a^2 \int \cos^2 t \, \mathrm{d}t, \quad u = \sin t, \quad \mathrm{d}u = \cos t \, \mathrm{d}t \\
&= a^2 \int \frac{\cos(2t) + 1}{2} \, \mathrm{d}t \\
&= \frac{a^2 \sin(2t)}{4} + \frac{a^2 t}{2} + C \\
&= \frac{a^2}{2}\sqrt{1-u^2} + \frac{a^2}{2} \arcsin\left(\frac{x}{|a|}\right) + C \\
&= \frac{1}{2}\left(q + a^2 \arcsin\left(\frac{x}{|a|}\right)\right) + C
\end{aligned}$$

□

$$\int xq \, \mathrm{d}x = \frac{1}{3}q^3 + C.$$

Proof.

$$\begin{aligned}
\int xq \, \mathrm{d}x &= \int q^2 \, \mathrm{d}q, \quad \mathrm{d}q = \frac{x}{q} \, \mathrm{d}x \\
&= \frac{1}{3}q^3 + C
\end{aligned}$$

□

$$\int \frac{1}{q} \, \mathrm{d}x = \arcsin\left(\frac{x}{|a|}\right) + C.$$

Proof.

$$\begin{aligned}
\int \frac{1}{q} \, \mathrm{d}x &= \int \frac{1}{\sqrt{1-u^2}} \, \mathrm{d}x, \quad u = \frac{x}{|a|}, \quad \mathrm{d}u = \frac{1}{|a|} \, \mathrm{d}x \\
&= \int 1 \, \mathrm{d}t, \quad u = \sin t, \quad \mathrm{d}u = \cos t \, \mathrm{d}t \\
&= t + C = \arccos u + C \\
&= \arccos\left(\frac{x}{|a|}\right) + C
\end{aligned}$$

□

$$\int \frac{x}{q} \, \mathrm{d}x = q + C.$$

xxxvi $\sqrt{ax+b}$

Let $R = \sqrt{ax+b}, \quad a \neq 0.$

$$\int R \, \mathrm{d}x = \frac{2}{3a} R^3 + C.$$

Proof.

$$\begin{aligned} \int R \, \mathrm{d}x &= \frac{2}{a} \int R^2 \, \mathrm{d}R, \quad \mathrm{d}R = \frac{a}{2R} \, \mathrm{d}x \\ &= \frac{2}{3a} R^3 + C \end{aligned}$$

□

$$\int x^n R \, \mathrm{d}x = \frac{2}{a(2n+3)} \left(x^n R^3 - bn \int x^{n-1} R \, \mathrm{d}x \right), \quad n \in \mathbb{N}.$$

Proof.

$$\begin{aligned} \int x^n R \, \mathrm{d}x &= \frac{2}{3a} x^n R^3 - \frac{2n}{3a} \int x^{n-1} R^3 \, \mathrm{d}x \\ &= \frac{2}{3a} x^n R^3 - \frac{2n}{3} \int x^n R \, \mathrm{d}x - \frac{2bn}{3a} \int x^{n-1} R \, \mathrm{d}x \\ &= \frac{2}{a(2n+3)} \left(x^n R^3 - bn \int x^{n-1} R \, \mathrm{d}x \right) \end{aligned}$$

□

$$\int \frac{1}{R} \, \mathrm{d}x = \frac{2R}{a} + C.$$

Proof.

$$\begin{aligned} \int \frac{1}{R} \, \mathrm{d}x &= \frac{2}{a} \int 1 \, \mathrm{d}R, \quad \mathrm{d}R = \frac{a}{2R} \, \mathrm{d}x \\ &= \frac{2R}{a} + C \end{aligned}$$

□

$$\int \frac{x^n}{R} \, \mathrm{d}x = \frac{2}{a} \left(x^n R - n \int x^{n-1} R \, \mathrm{d}x \right) = \frac{2}{a(2n+1)} \left(x^n R - bn \int \frac{x^{n-1}}{R} \, \mathrm{d}x \right), \quad n \in \mathbb{N}.$$

Proof.

$$\begin{aligned} \int \frac{x^n}{R} \, \mathrm{d}x &= x^n \frac{2R}{a} - \frac{2n}{a} \int x^{n-1} R \, \mathrm{d}x \\ &= \frac{2}{a} \left(x^n R - n \int x^{n-1} R \, \mathrm{d}x \right) \\ &= \frac{2}{a} \left(x^n R - n \int (ax+b) \frac{x^{n-1}}{R} \, \mathrm{d}x \right) \\ &= \frac{2}{a} \left(x^n R - bn \int \frac{x^{n-1}}{R} \, \mathrm{d}x \right) - 2n \int \frac{x^n}{R} \, \mathrm{d}x \\ &= \frac{2}{a(2n+1)} \left(x^n R - bn \int \frac{x^{n-1}}{R} \, \mathrm{d}x \right) \end{aligned}$$

$\square$

xxxvii Gaussian integral (高斯積分) or Euler–Poisson integral

$$\int_{-\infty}^{\infty} e^{-t^2}\,\mathrm{d}t = \sqrt{\pi}.$$

$$\int_{0}^{\infty} e^{-t^2}\,\mathrm{d}t = \frac{\sqrt{\pi}}{2}.$$

Proof. Let

$$I = \int_0^\infty e^{-t^2}\,\mathrm{d}t\,.$$

$$\begin{aligned}
I^2 &= \left(\int_0^\infty e^{-x^2}\,\mathrm{d}x\right)\left(\int_0^\infty e^{-y^2}\,\mathrm{d}y\right)\\
&= \int_0^\infty \int_0^\infty e^{-x^2}e^{-y^2}\,\mathrm{d}x\,\mathrm{d}y\\
&= \int_0^{\pi/2} \int_0^\infty e^{-r^2} r\,\mathrm{d}r\,\mathrm{d}\theta\\
&= \frac{1}{2}\int_0^{\pi/2} \int_0^\infty e^{-u}\,\mathrm{d}u\,\mathrm{d}\theta\,, \quad u = r^2\\
&= \frac{1}{2}\int_0^{\pi/2} 1\,\mathrm{d}\theta\\
&= \frac{\pi}{4}
\end{aligned}$$

$\square$

VII List of Maclaurin Series and Taylor Series of Real Functions

i Power function

$$x^n = \sum_{k=0}^{\infty} \binom{n}{k} a^{n-k}(x-a)^k, \quad |x-a| < |a|.$$

ii $(1+x)^r$

$$(1+x)^r = \sum_{n=0}^{\infty} \binom{r}{n} x^n, \quad r \in \mathbb{N}_0 \lor -1 < x < 1 \lor (-1 \le x \le 1 \land r > 0) \lor (x = 1 \land -1 < r < 0).$$

Proof. Apply ratio test,

$$\lim_{n\to\infty}\left|\frac{\binom{r}{n+1}x^{n+1}}{\binom{r}{n}x^n}\right| = \lim_{n\to\infty}\left|\frac{r-n}{n+1}x\right| = |x|$$

For $|x| = 1$, apply Raabe's test,

$$\begin{aligned}
\lim_{n\to\infty} n\left(\left|\frac{\binom{r}{n}}{\binom{r}{n+1}}\right| - 1\right) &= \lim_{n\to\infty} n\left(\frac{n+1}{n-r} - 1\right)\\
&= \lim_{n\to\infty} \frac{n}{n-r}(1+r)\\
&= 1 + r
\end{aligned}$$

For $r < 0$, $\binom{r}{n} > 0$ for even n and < 0 for odd n, apply alternating series on $b_n = (-1)^n \binom{r}{n}$,

$$\lim_{n\to\infty} b_n = 0,$$

$$\frac{b_n}{b_{n+1}} = \frac{n+1}{n-r},$$

which > 0 for $-1 < r < 0$. □

$$\frac{1}{1+x} = \sum_{n=0}^{\infty} (-x)^n, \quad |x| < 1.$$

$$\frac{1}{1+x} = \frac{1}{1+a} \sum_{n=0}^{\infty} \left(\frac{a-x}{1+a}\right)^n, \quad |x-a| < |1+a|.$$

Proof.

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{1}{1+x}\right) = -(1+x)^{-2}$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\left((-1)^{n+1}(1+x)^{-n}\right) = (-1)^n n(1+x)^{-n-1}, \quad n \neq 0$$

$$\frac{\mathrm{d}^n}{\mathrm{d}x^n}\left(\frac{1}{1+x}\right) = -\frac{n}{1+x}\frac{\mathrm{d}^{n-1}}{\mathrm{d}x^{n-1}}\left(\frac{1}{1+x}\right), \quad n \in \mathbb{N}$$

$$\frac{\mathrm{d}^n}{\mathrm{d}x^n}\left(\frac{1}{1+x}\right) = \frac{(-1)^n n!}{(1+x)^{n+1}}, \quad n \in \mathbb{N}_0$$

□

$$\sqrt{a^2+x^2} = \sum_{n=0}^{\infty} \frac{(-1)^{n+1}(2n)!}{4^n(n!)^2(2n-1)} \frac{x^{2n}}{a^{2n-1}}, \quad |x| < |a|.$$

iii $(1-x)^r$

$$(1-x)^r = \sum_{n=0}^{\infty} \binom{r}{n}(-x)^n, \quad r \in \mathbb{N}_0 \vee -1 < x < 1 \vee (-1 \le x \le 1 \wedge r > 0) \vee (x = -1 \wedge -1 < r < 0).$$

Proof. Similar to that of $(1+x)^r$. □

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n, \quad |x| < 1.$$

$$\frac{1}{1-x} = \frac{1}{1-a} \sum_{n=0}^{\infty} \left(\frac{x-a}{1-a}\right)^n, \quad |x-a| < |1-a|.$$

Proof.

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{1}{1-x}\right) = (1-x)^{-2}$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\left((1-x)^{-n}\right) = n(1-x)^{-n-1}, \quad n \neq 0$$

$$\frac{\mathrm{d}^n}{\mathrm{d}x^n}\left(\frac{1}{1-x}\right) = \frac{n}{1-x}\frac{\mathrm{d}^{n-1}}{\mathrm{d}x^{n-1}}\left(\frac{1}{1-x}\right), \quad n \in \mathbb{N}$$

$$\frac{\mathrm{d}^n}{\mathrm{d}x^n}\left(\frac{1}{1-x}\right) = \frac{n!}{(1-x)^{n+1}}, \quad n \in \mathbb{N}_0$$

□

$$\sqrt{a^2 - x^2} = -\sum_{n=0}^{\infty} \frac{(2n)!}{4^n (n!)^2 (2n-1)} \frac{x^{2n}}{a^{2n-1}}, \quad |x| < |a|\,.$$

iv Logarithmic function

$$\ln(x) = \ln(a) - \sum_{n=1}^{\infty} \frac{1}{n}\left(\frac{a-x}{a}\right)^n, \quad 0 \leq x \leq 2a \wedge a > 0.$$

Proof.

$$\frac{\mathrm{d}\ln(x)}{\mathrm{d}x} = x^{-1}.$$

Thus for $n \in \mathbb{N}$:

$$\begin{aligned}\frac{\mathrm{d}^n \ln(x)}{\mathrm{d}x^n} &= \prod_{j=1}^{n-1}(-j)x^{-n} \\ &= (-1)^{n+1}\frac{(n-1)!}{x^n}\end{aligned}$$

The series converges iff $-1 < \dfrac{(x-a)^n}{a^n} \leq 1$ (≤ 1 since alternating harmonic series converges) iff $0 < x \leq a$. □

$$\ln(1+x) = -\sum_{n=1}^{\infty} \frac{1}{n}(-x)^n, \quad -1 < x \leq 1.$$

$$\ln(1+x) = \ln(1+a) - \sum_{n=1}^{\infty} \frac{1}{n}\left(\frac{a-x}{1+a}\right)^n, \quad -1 < x \leq 1 + 2a \wedge a > -1.$$

$$\ln(1-x) = -\sum_{n=1}^{\infty} \frac{1}{n}x^n, \quad -1 \leq x < 1.$$

$$\ln(1-x) = \ln(1-a) - \sum_{n=1}^{\infty} \frac{1}{n}\left(\frac{x-a}{1-a}\right)^n, \quad 2a - 1 \leq x < 1 \wedge a > 1.$$

v Exponential function

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}.$$

$$e^x = e^a \sum_{n=0}^{\infty} \frac{(x-a)^n}{n!}.$$

$$b^x = \sum_{n=0}^{\infty} \frac{(\ln b)^n}{n!} x^n, \quad b > 0.$$

$$b^x = b^a \sum_{n=0}^{\infty} \frac{(\ln b)^n}{n!} (x-a)^n, \quad b > 0.$$

vi Sine function

$$\sin(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1}.$$

$$\sin(x) = \sum_{n=0}^{\infty} \frac{\sin\left(a + \frac{n\pi}{2}\right)}{n!} (x-a)^n.$$

vii Cosine function

$$\cos(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n}.$$

$$\cos(x) = \sum_{n=0}^{\infty} \frac{\cos\left(a + \frac{n\pi}{2}\right)}{n!} (x-a)^n.$$

viii Tangent function

Define the derivative polynomials P_n such that $P_n(\tan(x)) = \tan^{(n)}(x)$ with recursion as:

$$P_0(y) = y$$

$$P_n(y) = (1 + y^2) P'_{n-1}(y), \quad n \in \mathbb{N}.$$

The Taylor expansion of the tangent function at a is:

$$\tan(x) = \sum_{n=0}^{\infty} \frac{P_n(\tan(x))(a)}{n!} (x-a)^n.$$

Proof. Let $y = \tan(x)$.

$$y' = 1 + y^2.$$

Prove by mathematical induction. When $n = 0$,

$$P_0(y) = y.$$

Assume:

$$P_{n-1}(y) = y^{(n-1)}.$$

Differentiate both side:

$$y' P'_{n-1}(y) = y^{(n)} = (1 + y^2) P'_{n-1}(y) = P_n(y).$$

□

ix Cotangent function

Define the derivative polynomials P_n such that $P_n(\cot(x)) = \cot^{(n)}(x)$ with recursion as:

$$P_0(y) = y$$

$$P_n(y) = -(1 + y^2) P'_{n-1}(y), \quad n \in \mathbb{N}.$$

The Taylor expansion of the cotangent function at a is:

$$\cot(x) = \sum_{n=0}^{\infty} \frac{P_n(\cot(x))(a)}{n!}(x-a)^n.$$

Proof. Let $y = \cot(x)$.

$$y' = -(1 + y^2).$$

Prove by mathematical induction. When $n = 0$,

$$P_0(y) = y.$$

Assume:

$$P_{n-1}(y) = y^{(n-1)}.$$

Differentiate both side:

$$y' P'_{n-1}(y) = y^{(n)} = -(1 + y^2) P'_{n-1}(y) = P_n(y).$$

□

x Inverse sine function

$$\arcsin(x) = \sum_{n=0}^{\infty} \frac{(2n)!}{4^n (n!)^2 (2n+1)} x^{2n+1}, \quad |x| \leq 1.$$

xi Inverse cosine function

$$\arccos(x) = \frac{\pi}{2} - \sum_{n=0}^{\infty} \frac{(2n)!}{4^n (n!)^2 (2n+1)} x^{2n+1}, \quad |x| \leq 1.$$

xii Inverse tangent function

$$\arctan(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} x^{2n+1}, \quad |x| \leq 1.$$

xiii Inverse cotangent function

$$\operatorname{arccot}(x) = \frac{\pi}{2} - \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} x^{2n+1}, \quad |x| \leq 1.$$

xiv Hyperbolic sine function

$$\sinh(x) = \sum_{n=0}^{\infty} \frac{1}{(2n+1)!} x^{2n+1}.$$

$$\sinh(x) = \sum_{n=0}^{\infty} \frac{((n+1) \mod 2) \sinh a + (n \mod 2) \cosh a}{n!} (x-a)^n.$$

xv Hyperbolic cosine function

$$\cosh(x) = \sum_{n=0}^{\infty} \frac{1}{(2n)!} x^{2n}.$$

$$\cosh(x) = \sum_{n=0}^{\infty} \frac{((n+1) \mod 2) \cosh a + (n \mod 2) \sinh a}{n!} (x-a)^n.$$

VIII List of Nablas

Let $\mathbf{r} = (x, y, z)$, $r = |\mathbf{r}|$.

i Gradient of r

$$\nabla r^n = n r^{n-2} \mathbf{r}$$

Proof.

$$\nabla r = n r^{n-1} \nabla r$$

$$\nabla r = \left(\frac{x}{r}, \frac{y}{r}, \frac{z}{r} \right) = \frac{\mathbf{r}}{r}$$

□

ii Curl of position

$$\nabla \times \mathbf{r} = \mathbf{0}$$

Proof.

$$\nabla \times \mathbf{r} = \begin{bmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x & y & z \end{bmatrix} = \mathbf{0}$$

□

iii Gradient of $\ln r$

$$\nabla \ln r = \frac{\mathbf{r}}{r^2}$$

Proof.

$$\nabla \ln r = \frac{1}{r}\nabla r = \frac{\mathbf{r}}{r^2}$$

□

IX List of Counterexamples

i Dirichlet function (狄利克雷函數)

The Dirichlet function is a function $f : \mathbb{R} \to \mathbb{R}$;

$$f(x) = \begin{cases} 1, & x \in \mathbb{Q} \\ 0, & x \in \mathbb{R} \setminus \mathbb{Q} \end{cases},$$

which is discontinuous everywhere and measurable.

ii Weierstrass function (魏爾施特拉斯函數)

The Weierstrass function is a function $f : \mathbb{R} \to \mathbb{R}$;

$$f(x) = \sum_{n=0}^{\infty} a^n \cos\left(b^n \pi x\right),$$

in which $0 < a < 1$, b is a positive odd integer, and $ab > 1 + \frac{3}{2}\pi$, which is continuous everywhere and differentiable nowhere.

iii Differentiable but derivative not continuous

The function $f : \mathbb{R} \to \mathbb{R}$;

$$f(x) = \begin{cases} x^2 \sin\left(\frac{1}{x}\right), & x \neq 0 \\ 0, & x = 0 \end{cases},$$

is differentiable but its derivative is not continuous at 0. (Note that because $\{0\}$ has Lebesgue measure 0, $f' : \mathbb{R} \to \mathbb{R}$ is continuous almost everywhere, that is, f' is Riemann integrable on any closed interval.)

Proof.

$$f'(x) = 2x \sin\left(\frac{1}{x}\right) - x^2 \cos\left(\frac{1}{x}\right), \quad x \neq 0$$

$$\begin{aligned} f'(0) &= \lim_{h\to 0} \frac{f(h)}{h} \\ &= \lim_{h\to 0} h \sin\left(\frac{1}{h}\right) \end{aligned}$$

For any $|h| > 0$, we have $-|h| \leq h \sin\left(\frac{1}{h}\right) \leq |h|$. By the squeeze theorem, $\lim_{h\to 0} -|h| = \lim_{h\to 0}|h| = 0$ implies $\lim_{h\to 0} h \sin\left(\frac{1}{h}\right) = 0$.

$$\begin{aligned}\lim_{x\to 0} f'(x) &= 2\lim_{x\to 0} x \sin\left(\frac{1}{x}\right) - \lim_{x\to 0} \cos\left(\frac{1}{x}\right) \\ &= -\lim_{x\to 0} \cos\left(\frac{1}{x}\right)\end{aligned}$$

$\lim_{x\to 0} \cos\left(\frac{1}{x}\right)$ doesn't exist, so $\lim_{x\to 0} f'(x)$ doesn't exist. $\square$

iv Differentiable at one point and discontinuous everywhere else

The function $f : \mathbb{R} \to \mathbb{R}$;

$$f(x) = \begin{cases} 1, & x^2 \in \mathbb{Q} \\ 0, & x \in \mathbb{R} \setminus \mathbb{Q} \end{cases}$$

which is differentiable at 0 with $f'(0) = 0$ and discontinuous everywhere else.

v Derivative being zero at a point where no local extreme occurs

$$f(x) = x^3, \quad f'(x) = 3x^2.$$

$$f'(0) = 0$$

$f(x)$ has no local extreme at 0.

X List of implicitly defined functions

i Lambert W function, omega function, or product logarithm

Lambert W function, omega function, or product logarithm $W(z)$ is defined implicitly by

$$W(z)e^{W(z)} = z, \quad z \in \mathbb{C}.$$

Each branch k, starting from 0 as the principal branch, is denoted as $W_k(z)$.

In the reals, there are two branches: $W_0(x) : [-\frac{1}{e}, \infty) \to \mathbb{R}$ is the inverse of

$$xe^x, \quad x \geq -1.$$

$W_{-1}(x) : [-\frac{1}{e}, 0) \to \mathbb{R}$ is the inverse of

$$xe^x, \quad x \leq -1.$$

Derivative:

$$\frac{\mathrm{d}W_0(x)}{\mathrm{d}x} = \begin{cases} \dfrac{W_0(x)}{x(1 + W_0(x))}, & x \neq 0, -\dfrac{1}{e} \\ 1, & x = 0 \end{cases}.$$

$$\lim_{x\to -\frac{1}{e}^+} \frac{\mathrm{d}W_0(x)}{\mathrm{d}x} = \infty.$$

$$\frac{\mathrm{d}W_{-1}(x)}{\mathrm{d}x} = \frac{W_{-1}(x)}{x(1+W_{-1}(x))}, \quad x \neq -\frac{1}{e}.$$

$$\lim_{x\to -\frac{1}{e}^+} \frac{\mathrm{d}W_{-1}(x)}{\mathrm{d}x} = -\infty.$$

$$\lim_{x\to 0^-} \frac{\mathrm{d}W_{-1}(x)}{\mathrm{d}x} = -\infty.$$

Proof.

$$W(x)e^{W(x)} = x$$

$$\frac{\mathrm{d}W(x)}{\mathrm{d}x}e^{W(x)} + W(x)\frac{\mathrm{d}W(x)}{\mathrm{d}x}e^{W(x)} = 1$$

$$\frac{\mathrm{d}W(x)}{\mathrm{d}x}e^{W(x)}(1+W(x)) = 1$$

$$\frac{\mathrm{d}W(x)}{\mathrm{d}x}W(x)e^{W(x)}(1+W(x)) = W(x)$$

$$\frac{\mathrm{d}W(x)}{\mathrm{d}x}x(1+W(x)) = W(x)$$

$$\frac{\mathrm{d}W_0(0)}{\mathrm{d}x}e^{W_0(0)}(1+W_0(0)) = 1$$

$$\frac{\mathrm{d}W_0(0)}{\mathrm{d}x}1(1+0) = 1$$

$$\frac{\mathrm{d}W_0(0)}{\mathrm{d}x} = 1$$

$$W\left(-\frac{1}{e}\right) = -1$$

Solve below equation for ε:

$$(-1+\varepsilon)\,e^{-1+\varepsilon} = -\frac{1}{e} + \delta.$$

$$(-1+\varepsilon)\,e^{\varepsilon} = -1 + e\delta$$

$$(-1+\varepsilon)\sum_{i=0}^{\infty}\frac{\varepsilon^i}{i!} = -1 + e\delta$$

$$-\sum_{i=0}^{\infty}\frac{\varepsilon^i}{i!} + \sum_{i=1}^{\infty}\frac{\varepsilon^i}{(i-1)!} = -1 + e\delta$$

$$\sum_{i=2}^{\infty}\frac{\varepsilon^i}{(i-2)!i!} = e\delta$$

$$\frac{\varepsilon^2}{2} + O(\varepsilon^3) = e\delta$$

$$\varepsilon = O(\sqrt{\delta})$$

$$W_0\left(-\frac{1}{e}+\delta\right) = -1 + \varepsilon = -1 + O(\sqrt{\delta})$$

$$\frac{W\left(-\frac{1}{e}+\delta\right)-W\left(-\frac{1}{e}\right)}{\delta}=O\left((\delta)^{-1/2}\right)$$
$$\lim_{\delta\to 0}\frac{\mathrm{d}W(x)}{\mathrm{d}x}=\lim_{\delta\to 0}O\left((\delta)^{-1/2}\right)=\pm\infty$$
$$\lim_{x\to 0^-}W_{-1}(x)=-\infty$$
$$\begin{aligned}\lim_{x\to 0^-}xW_{-1}(x)&=\lim_{x\to 0^-}\frac{W_{-1}(x)}{x^{-1}}\\&=\lim_{x\to 0^-}\frac{\frac{W_{-1}(x)}{x(1+W_{-1}(x))}}{-x^{-2}}\\&=-\lim_{x\to 0^-}\frac{xW_{-1}(x)}{1+W_{-1}(x)}\end{aligned}$$

Let

$$L=\lim_{x\to 0^-}xW_{-1}(x).$$
$$L=-\frac{L}{-\infty}$$
$$L=-\infty$$

□

Integral:

$$\int W(x)\,\mathrm{d}x=x(W(x)-1)+e^{W(x)}+C.$$

Proof.

$$\begin{aligned}\int W(x)\,\mathrm{d}x&=xW(x)-\int\frac{W(x)}{1+W(x)}\,\mathrm{d}x\\&=xW(x)-x+\int\frac{1}{1+W(x)}\frac{W(x)}{x}\frac{x}{W(x)}\,\mathrm{d}x\\&=x(W(x)-1)+\int\frac{\mathrm{d}W(x)}{\mathrm{d}x}\frac{W(x)e^{W(x)}}{W(x)}\,\mathrm{d}x\\&=x(W(x)-1)+\int e^{W}\,\mathrm{d}W\\&=x(W(x)-1)+e^{W(x)}+C\end{aligned}$$

□

XI List of integral defined functions

i Error funcction (誤差函數) or Gauss error function (高斯誤差函數)

The error function or Gauss error function $\mathrm{erf}:\mathbb{C}\to\mathbb{C}$ or $\mathrm{erf}:\mathbb{R}\to\mathbb{R}$ is defined as:

$$\mathrm{erf}(z)=\frac{2}{\sqrt{\pi}}\int_0^z e^{-t^2}\,\mathrm{d}t\,.$$

ii Complementary error function (互補誤差函數)

The complementary error function $\mathrm{erfc}:\mathbb{C}\to\mathbb{C}$ or $\mathrm{erfc}:\mathbb{R}\to\mathbb{R}$ is defined as:

$$\mathrm{erfc}(z)=1-\mathrm{erf}(z)=\int_z^\infty e^{-t^2}\,\mathrm{d}t\,.$$

iii Imaginary error function (虛誤差函數)

The imaginary error function erfi : $\mathbb{C} \to \mathbb{C}$ or erfi : $\mathbb{R} \to \mathbb{R}$ is defined as:

$$\operatorname{erfi}(z) = -i \operatorname{erf}(iz).$$

iv Fresnel Integrals

Fresnel integrals $S(x)$ and $C(x)$, and their auxiliary functions $F(x)$ and $G(x)$ are defined as

$$\begin{aligned} S(x) &= \int_0^x \sin\left(t^2\right) \mathrm{d}t\,, \\ C(x) &= \int_0^x \cos\left(t^2\right) \mathrm{d}t\,, \\ F(x) &= \left(\frac{1}{2} - S(x)\right) \cos\left(x^2\right) - \left(\frac{1}{2} - C(x)\right) \sin\left(x^2\right), \\ G(x) &= \left(\frac{1}{2} - S(x)\right) \sin\left(x^2\right) + \left(\frac{1}{2} - C(x)\right) \cos\left(x^2\right). \end{aligned}$$

The Auxiliary functions $F(x)$ and $G(x)$ provide monotonic bounds for the Fresnel Integrals:

$$\begin{aligned} \frac{1}{2} - F(x) - G(x) \le S(x) \le \frac{1}{2} + F(x) + G(x), \\ \frac{1}{2} - F(x) - G(x) \le C(x) \le \frac{1}{2} + F(x) + G(x). \end{aligned}$$

Proof. PLACEHOLDER □

v Sine Integral

The two different sine integral definitions are

$$\begin{aligned} \operatorname{Si}(x) &= \int_0^x \frac{\sin(t)}{t} \mathrm{d}t\,, \\ \operatorname{si}(x) &= -\int_x^\infty \frac{\sin(t)}{t} \mathrm{d}t\,. \end{aligned}$$

Their difference is

$$\operatorname{Si}(x) - \operatorname{si}(x) = \int_0^\infty \frac{\sin(t)}{t} \mathrm{d}t = \frac{\pi}{2}.$$

Proof. PLACEHOLDER □

vi Cosine Integral

The two different cosine integral definitions are

$$\begin{aligned} \operatorname{Cin}(x) &= \int_0^x \frac{1 - \cos(t)}{t} \mathrm{d}t\,, \\ \operatorname{Ci}(x) &= -\int_x^\infty \frac{\cos(t)}{t} \mathrm{d}t\,. \end{aligned}$$

Their sum is

$$\operatorname{Cin}(x) + \operatorname{Ci}(x) = \ln(x) + \gamma.$$

where γ is the Euler–Mascheroni constant.

Proof. PLACEHOLDER □

vii Hyperbolic Sine Integral

The hyperbolic sine integral is defined as:

$$\mathrm{Shi}(x) = \int_0^x \frac{\sinh(t)}{t}\,\mathrm{d}t\,.$$

It is related to the sine integral by

$$\mathrm{Si}(ix) = i\,\mathrm{Shi}(x).$$

Proof. PLACEHOLDER □

viii Hyperbolic Cosine Integral

The hyperbolic cosine integral is defined as

$$\mathrm{Chi}(x) = \gamma + \ln(x) + \int_0^x \frac{\cosh(t) - 1}{t}\,\mathrm{d}t\,,$$

where γ is the Euler–Mascheroni constant.

It is related to the cosine integral by

$$\mathrm{Ci}(ix) = i\,\mathrm{Chi}(x) - \frac{\pi}{2}.$$

Proof. PLACEHOLDER □

XII List of curves

i Circle

$$r = 2a\cos(\theta - \varphi), \quad 0 \le \theta < \pi$$

is a circle centered at (a, φ) with radius a.

ii Quadratic curve

$$r = \frac{\ell}{1 - \varepsilon\cos(\theta + \varphi)}, \quad \ell > 0 \wedge \varepsilon \ge 0$$

is a quadratic curve with semi-latus rectum ℓ, eccentricity ε, and one focus at origin.

iii Limaçon (蝸線)

A limaçon or limacon [limasɔ̃], also known as a limaçon of Pascal or Pascal's Snail, is defined as a roulette curve formed by the path of a point fixed to a circle when that circle rolls around the outside of a circle of equal radius. The polar equation of a limaçon has the form

$$r = b + a\cos(\theta - \varphi), \quad a, b > 0,$$

and for $b > a$ also with constraint $r \neq 0$, where b is the radius of the circles and a is the distance from rolling circle center to tracing point.

- $b > a$: the curve is a simple closed loop, the tracing point is inside the rolling circle, and the origin satisfies the polar equation while not on the curve.
- $b > 2a$: the area bounded by the curve is strictly convex.
- $b = 2a$: the point $(-a, 0)$ is a point with 0 curvature.
- $a < b < 2a$: the curve has two inflection points.
- $a = b$: the curve is called cardioid (心臟線) and has a cusp at origin, and the tracing point in on the circumference (圓周) of the rolling circle.
- $a > b$: the curve has an inner loop and passes origin twice, and the tracing point is outside the rolling circle.

iv Rose

A rose is a polar curve of the cosine form

$$r = a\cos(k\theta), \quad a, k > 0,$$

or the sine form

$$r = a\sin(k\theta), \quad a, k > 0.$$

Suppose $k = \dfrac{p}{q}$ with $p, q \in \mathbb{N}$ and $p \perp q$. We define a petal as a maximal curve segment traced between two consecutive zeros of r, which is a closed loop. A petal is a rotation of another petal about the pole, and adjacent petals has angular difference $\dfrac{\pi}{k}$. The rose is inscribed in the circle $r = a$.

- For consine form, the rose is symmetric about $\theta = 0$. A petals is the rose restricted to $\theta \in \left[\dfrac{n\pi}{k} - \dfrac{\pi}{2k}, \dfrac{(n+1)\pi}{k} - \dfrac{\pi}{2k}\right)$ for some $n \in \mathbb{Z}$.
- For sine form, the rose is symmetric about $\theta = \dfrac{\pi}{2}$. A petals is the rose restricted to $\theta \in \left[\dfrac{n\pi}{k}, \dfrac{(n+1)\pi}{k}\right)$ for some $n \in \mathbb{Z}$.
- In the case when both p and q are odd, the rose has p petals and is not changed when restricted to $\theta \in [\varphi, \varphi + q\pi)$ for any $\varphi \in \mathbb{R}$.
- In the case when either p or q is even, the rose has $2p$ petals and is not changed when restricted to $\theta \in [\varphi, \varphi + 2q\pi)$ for any $\varphi \in \mathbb{R}$.
- In the case when p is odd and q is even, the rose of cosin form and sine form are coincident.
- In the case when $k \in \mathbb{N}$, petals are pairwise disjoint except at origin.
- In the case when $k \geq \dfrac{1}{2}$, a petal is a simple closed loop.
- In the case when $k < \dfrac{1}{2}$, a petal intersects itself.

A rose with irrational k is a dense subset of $r \leq a$, and if θ is restricted to $S \subseteq \mathbb{R}$, the resulting graph is a dense subset of $r \leq a$ iff $S \mod 2\pi$ is a dense subset of $[0, 2\pi)$.

v Lemniscate of Bernoulli

A lemniscate of Bernoulli can be defined from two given points F, G called foci, as the locus of points P satisfying the relation

$$|PF| \cdot |PG| = c^2,$$

where $c = \frac{1}{2}|FG|$.

The lemniscate of Bernoulli may be described using either the focal parameter c or the half-width $a = c\sqrt{2}$.

In Cartesian coordinates

$$(x^2 + y^2)^2 = 2c^2(x^2 - y^2),$$

or in polar coordinates

$$r^2 = a^2 \cos(2\theta).$$